Epic-FHIR — API category: Clinical Diagnostics (sections 9 – 9.3.6)

Source: Epic on FHIR (fhir.epic.com) documentation, mirrored 2026-11. Text-only edition for AI ingestion (NotebookLM); pages follow the source order (its table of contents when one exists).

9 API category: Clinical Diagnostics

File: site/apis/09-Clinical-Diagnostics/index.html

R4 (22)

API | Method | URL template | Type
DiagnosticReport.Read (Results) (R4) | GET | api/FHIR/R4/DiagnosticReport/{ID} | USCDI
DiagnosticReport.Search (Results) (R4) | GET | api/FHIR/R4/DiagnosticReport?patient={patient}&identifier={identifier}&code={code}&date={date}&category={category} | USCDI
Observation.Create (Flowsheets) (R4) | POST | /api/FHIR/R4/Observation`Observation*** | Industry-Standard
Observation.Create (Lines, Drains, Airways) (R4) | POST | api/FHIR/R4/Observation | Industry-Standard
Observation.Create (Vital Signs) (R4) | POST | api/FHIR/R4/Observation | Industry-Standard
Observation.Read (Activities of Daily Living) (R4) | GET | /api/FHIR/R4/Observation/{ID} | Industry-Standard
Observation.Read (Core Characteristics) (R4) | GET | api/FHIR/R4/Observation/{ID} | Industry-Standard
Observation.Read (Flowsheets) (R4) | GET | /api/FHIR/R4/Observation/{ID} | Industry-Standard
Observation.Read (Labs) (R4) | GET | /api/FHIR/R4/Observation/{ID} | USCDI
Observation.Read (Lines, Drains, Airways) (R4) | GET | api/FHIR/R4/Observation/{ID} | Industry-Standard
Observation.Read (SmartData Elements) (R4) | GET | /api/FHIR/R4/Observation/{ID} | USCDI
Observation.Read (Study Finding) (R4) | GET | /api/FHIR/R4/Observation/{ID} | USCDI
Observation.Read (Vital Signs) (R4) | GET | api/FHIR/R4/Observation/{ID} | USCDI
Observation.Search (Activities of Daily Living) (R4) | GET | /api/FHIR/R4/Observation?category={category}&patient={patient}&subject={subject} | Industry-Standard
Observation.Search (Core Characteristics) (R4) | GET | api/FHIR/R4/Observation?category={category}&code={code}&patient={patient} | Industry-Standard
Observation.Search (Flowsheets) (R4) | GET | /api/FHIR/R4/Observation?based-on={based-on}&category={category}&code={code}&date={date}&focus={focus}&issued={issued}&patient={patient}&subject={subject}&_count={_count}&_id={_id}&_include={_include}&_lastupdated={_lastupdated}&_revinclude={_revinclude} | Industry-Standard
Observation.Search (Labs) (R4) | GET | api/FHIR/R4/Observation?patient={patient}&subject={subject}&category={category}&code={code}&date={date} | USCDI
Observation.Search (Lines, Drains, Airways) (R4) | GET | api/FHIR/R4/Observation?category={category}&code={code}&date={date}&patient={patient}&subject={subject} | Industry-Standard
Observation.Search (SmartData Elements) (R4) | GET | /api/FHIR/R4/Observation?category={category}&code={code}&focus={focus}&patient={patient} | USCDI
Observation.Search (Study Finding) (R4) | GET | /api/FHIR/R4/Observation?patient={patient}&category={category} | USCDI
Observation.Search (Vital Signs) (R4) | GET | api/FHIR/R4/Observation?patient={patient}&subject={subject}&category={category}&code={code}&date={date} | USCDI
Observation.Update (Lines, Drains, Airways) (R4) | PUT | api/FHIR/R4/Observation/{ID} | Industry-Standard

STU3 (19)

API | Method | URL template | Type
DiagnosticReport.Read (Results) (STU3) | GET | api/FHIR/STU3/DiagnosticReport/{ID} | USCDI
DiagnosticReport.Search (Results) (STU3) | GET | api/FHIR/STU3/DiagnosticReport?patient={patient}&code={code}&date={date} | USCDI
Observation.$lastn (Activities of Daily Living) (STU3) | GET | /api/FHIR/STU3/Observation/$lastn?patient={patient}&category={category} | Industry-Standard
Observation.$lastn (Labs) (STU3) | GET | /api/FHIR/STU3/Observation/$lastn?patient={patient}&category={category}&max={max}&subject={subject} | Industry-Standard
Observation.$lastn (Social History) (STU3) (described under Clinical General) | GET | /api/FHIR/STU3/Observation/$lastn?category={category}&code={code}&max={max}&patient={patient}&subject={subject} | Industry-Standard
Observation.$lastn (Vitals) (STU3) | GET | /api/FHIR/STU3/Observation/$lastn?category={category}&code={code}&max={max}&patient={patient}&subject={subject} | Industry-Standard
Observation.Create (LDA-W) (STU3) | POST | api/FHIR/STU3/Observation | Industry-Standard
Observation.Create (Vitals) (STU3) | POST | api/FHIR/STU3/Observation | Industry-Standard
Observation.Read (Activities of Daily Living) (STU3) | GET | /api/FHIR/STU3/Observation/{ID} | Industry-Standard
Observation.Read (Core Characteristics) (STU3) | GET | /api/FHIR/STU3/Observation/{ID} | Industry-Standard
Observation.Read (Labs) (STU3) | GET | api/FHIR/STU3/Observation/{ID} | USCDI
Observation.Read (LDA-W) (STU3) | GET | api/FHIR/STU3/Observation/{ID} | Industry-Standard
Observation.Read (Vitals) (STU3) | GET | api/FHIR/STU3/Observation/{ID} | USCDI
Observation.Search (Activities of Daily Living) (STU3) | GET | /api/FHIR/STU3/Observation?category={category}&patient={patient}&subject={subject} | Industry-Standard
Observation.Search (Core Characteristics) (STU3) | GET | /api/FHIR/STU3/Observation?patient={patient}&category={category}&code={code} | Industry-Standard
Observation.Search (Labs) (STU3) | GET | api/FHIR/STU3/Observation?patient={patient}&subject={subject}&category={category}&code={code}&date={date} | USCDI
Observation.Search (LDA-W) (STU3) | GET | api/FHIR/STU3/Observation?patient={patient}&subject={subject}&code={code}&category={category} | Industry-Standard

Observation.Search (Vitals) (STU3) | GET | api/FHIR/STU3/Observation?patient={patient}&subject={subject}&code={code}&date={date}&category={category} | USCDI
Observation.Update (LDA-W) (STU3) | PUT | api/FHIR/STU3/Observation | Industry-Standard

DSTU2 (6)

API | Method | URL template | Type
DiagnosticReport.Read (Results) (DSTU2) | GET | api/FHIR/DSTU2/DiagnosticReport/{ID} | USCDI
DiagnosticReport.Search (Results) (DSTU2) | GET | api/FHIR/DSTU2/DiagnosticReport?patient={patient}&subject={subject}&code={code}&date={date} | USCDI
Observation.Read (Labs) (DSTU2) | GET | api/FHIR/DSTU2/Observation/{ID} | USCDI
Observation.Read (Vitals) (DSTU2) | GET | api/FHIR/DSTU2/Observation/{ID} | USCDI
Observation.Search (Labs) (DSTU2) | GET | api/FHIR/DSTU2/Observation?code={code}&date={date}&patient={patient}&subject={subject}&category={category} | USCDI
Observation.Search (Vitals) (DSTU2) | GET | /api/FHIR/DSTU2/Observation?code={code}&patient={patient}&date={date}&subject={subject}&category={category} | USCDI

9.1 Clinical Diagnostics: R4

File: site/apis/09-Clinical-Diagnostics/1-R4/index.html

* DiagnosticReport.Read (Results) (R4)
* DiagnosticReport.Search (Results) (R4)
* Observation.Create (Flowsheets) (R4)
* Observation.Create (Lines, Drains, Airways) (R4)
* Observation.Create (Vital Signs) (R4)
* Observation.Read (Activities of Daily Living) (R4)
* Observation.Read (Core Characteristics) (R4)
* Observation.Read (Flowsheets) (R4)
* Observation.Read (Labs) (R4)
* Observation.Read (Lines, Drains, Airways) (R4)
* Observation.Read (SmartData Elements) (R4)
* Observation.Read (Study Finding) (R4)
* Observation.Read (Vital Signs) (R4)
* Observation.Search (Activities of Daily Living) (R4)
* Observation.Search (Core Characteristics) (R4)
* Observation.Search (Flowsheets) (R4)
* Observation.Search (Labs) (R4)
* Observation.Search (Lines, Drains, Airways) (R4)
* Observation.Search (SmartData Elements) (R4)
* Observation.Search (Study Finding) (R4)
* Observation.Search (Vital Signs) (R4)
* Observation.Update (Lines, Drains, Airways) (R4)

9.1.1 DiagnosticReport.Read (Results) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/988-DiagnosticReport-Read-Results-R4.html
Epic API ID | 988
FHIR version | R4
Categories | Clinical Diagnostics
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Laboratory;Tests;Values;Results
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | February 2020
HTTP method | GET
URL template | api/FHIR/R4/DiagnosticReport/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=988

Description

The DiagnosticReport resource represents a resulted, patient-related, non-medication order and data associated with the order and the patient.

This resource returns textual results and discrete result components. It can return lab, imaging, cardiology and endoscopy results. It can also return pathology results that are not a scan. Scans filed into Epic are not returned by this resource.

Starting in February 2024, DiagnosticReport can return EKG data using the Midmark third-party integration. If the EKG order collected heart rate and blood pressure data as well, this data is returned as Observations contained within the response.

Starting in May 2024, DiagnosticReport can return audiology exam finding data. Data within the audiology exam is returned as observations contained within the response.

The read interaction enables the look up of a DiagnosticReport resource by a constant server ID. The read interaction allows clients to store only the server ID, and with a single request, retrieve the most up-to-date result data on a patient. Read interactions typically begin with a client having previously established a relationship, often through querying for DiagnosticReports through the search interaction.

Remarks

Heart rate and blood pressure data associated with a Midmark result are represented as full Vital Signs Observations within the DiagnosticReport. Unlike other Observations associated with the order, these are contained resources, and are only accessible embedded within the DiagnosticReport. These resources are associated with the following LOINC codes:

* Heart rate: 8867-4

* Diastolic blood pressure: 8462-4

* Systolic blood pressure: 8480-6

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | | The DiagnosticReport FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
DiagnosticReport (R4) | DiagnosticReport
DiagnosticReport (R4).basedOn | Reference | | Yes | 0..* | A reference to a ServiceRequest.
DiagnosticReport (R4).category | CodeableConcept | | Yes | 1..* | The Epic order type is always returned, and US Core Category Code is also returned if the order matches one of the categories. Also includes the LOINC or DiagnosticServiceSectionId codes when available. https://build.fhir.org/ig/HL7/US-Core/ValueSet-us-core-diagnosticreport-category.html
DiagnosticReport (R4).code | CodeableConcept | | | 0..1 | The name and code for this DiagnosticReport.
DiagnosticReport (R4).conclusionCode | CodeableConcept | | Yes | 0..* | The order diagnosis. Starting in the November 2024 version of Epic, each code also sends its display name.
DiagnosticReport (R4).effectiveDateTime | dateTime | | | 0..1 | Clinically relevant date and time for the order.
DiagnosticReport (R4).encounter | Reference | | | 1..1 | Reference to an Encounter resource associated with the procedure.
DiagnosticReport (R4).extension (SmartData) (also: extension (SmartData)) | Extension | | Yes | 0..* | Custom discrete SmartData Elements stored on the order. URL: http://open.epic.com/FHIR/StructureDefinition/extension/smartdata
DiagnosticReport (R4).identifier | Identifier | | Yes | 1..* | The business identifiers for the report, including Epic order ID, accession numbers, and external order identifiers.
DiagnosticReport (R4).issued | instant | | | 1..1 | Date and time of the result.
DiagnosticReport (R4).media | BackboneElement | | Yes | 0..* | Key images associated with the report.
DiagnosticReport (R4).performer | Reference | | Yes | 1..* | References representing the participants involved in producing the report. This always includes one or more references to Practitioner resources. This element can also include a reference to an Organization resource.
DiagnosticReport (R4).presentedForm | Attachment | | Yes | 1..* | Binary attachments with HTML textual results and/or PDFs.
DiagnosticReport (R4).presentedForm.contentType | String | | | 1..1 | Mime content type code. This is text/html for textual results and application/pdf for PDFs.
DiagnosticReport (R4).presentedForm.title | String | | | 1..1 | Label to display in place of the data.
DiagnosticReport (R4).presentedForm.url | String | | | 1..1 | Binary link for the textual content or report PDF.
DiagnosticReport (R4).result | Reference | | Yes | 0..* | References to the Observation resources representing the result components for the DiagnosticReport.
DiagnosticReport (R4).resultsInterpreter | Reference | | Yes | 0..* | The reading providers.
DiagnosticReport (R4).status | code | | | 1..1 | The status of the report. These values can be returned: registered preliminary final amended If a test was performed but ended in an unsuccessful attempt, the status will be cancelled. For example, if a mammography is performed, and the patient was exposed to radiation, and images were captured, but the images were too blurry to make a clinical determination, that would be considered an unsuccessful attempt.
DiagnosticReport (R4).subject | Reference | | | 1..1 | A reference to the Patient resource representing the subject of the report.

Sample request

```
https://hostname/instance/api/FHIR/R4/DiagnosticReport/e43lhnx.XUNbUL3v-.PDcNrEifQ9LIJG0xmvd-KPr8CM3
```

Sample response


```
{
    "resourceType": "DiagnosticReport",
    "id": "e43lhnx.XUNbUL3v-.PDcNrEifQ9LIJG0xmvd-KPr8CM3",
    "identifier": [
        {
```

```
        "use": "official",
        "type": {
            "coding": [
                {
                    "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
                    "code": "PLAC",
                    "display": "Placer Identifier"
                }
            ],
            "text": "Placer Identifier"
        },
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.798268",
        "value": "1012233558"
    },
    {
        "use": "official",
        "type": {
            "coding": [
                {
                    "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
                    "code": "FILL",
                    "display": "Filler Identifier"
                }
            ],
            "text": "Filler Identifier"
        },
        "system": "https://open.epic.com/FHIR/861/order-accession-number/ALB%20LAB",
        "value": "28550"
    }
],
"basedOn": [
    {
        "reference": "ServiceRequest/e-zJTgPbE4Mrf-nekgQJ.rRI5zqqeoxzDs4llNQSo69Y3",
        "type": "ServiceRequest"
    }
],
"status": "final",
"category": [
    {
        "coding": [
            {
                "system": "urn:oid:1.2.840.114350.1.13.861.1.7.10.798268.30",
                "code": "Imaging"
            }
        ],
        "text": "Imaging"
    },
    {
        "coding": [
            {
                "system": "http://loinc.org",
                "code": "LP29684-5",
                "display": "Radiology"
            }
        ],
        "text": "Radiology"
    }
],
"code": {
    "coding": [
        {
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.696580",
            "code": "52665",
            "display": "GENERIC RADIANT IMAGING EAP"
        }
    ],
    "text": "X-RAY HIP UNILAT 1 VW"
},
"subject": {
    "reference": "Patient/epnvY-g-k4v8aI71KfQzYYg3",
    "display": "Test, Patient"
},
"encounter": {
    "identifier": {
        "use": "usual",
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
        "value": "10001456304"
    }
},
"effectiveDateTime": "2019-10-11T05:00:00Z",
"issued": "2021-06-11T19:50:05Z",
"performer": [
    {
        "reference": "Practitioner/eNRT8GK-ZDKFtQwHDbar9EA3",
        "display": "Becky Black, MD"
    }
],
"resultsInterpreter": [
    {
```

```
            "reference": "Practitioner/eNRT8GK-ZDKFtQwHDbar9EA3",
            "type": "Practitioner",
            "display": "Becky Black, MD"
        },
        {
            "reference": "Practitioner/eNRT8GK-ZDKFtQwHDbar9EA3",
            "type": "Practitioner",
            "display": "Becky Black, MD"
        }
    ],
    "result": [
        {
            "reference": "Observation/eavL3N-ZwVkoAEn48KHfezsXPoMlYiWPL7dT3bapLBes3",
            "display": "Narrative"
        }
    ],
    "imagingStudy": [
        {"reference": "ImagingStudy/ed0KtzxnrBpuTfBLPnOOr2XpQ2Xu8S4jXS2A5GAFGOX83"}
    ],
    "presentedForm": [
        {
            "contentType": "text/html",
            "url": "Binary/eTmCM2E.Uhk9L7Sp2NXPO6w8cPHVTfoFcslhYxPN7Mrw3",
            "title": "Narrative"
        }
```


Change log

Epic Version | Description | Type
May 2024 | Code systems for the following code set now returns a URL in place of an OID: System URL OID Conclusion.Code http://HL7.ORG/FHIR/SID/ICD-10-CM 2.16.840.1.113883.6.90 | Breaking change
System | URL | OID
Conclusion.Code | http://HL7.ORG/FHIR/SID/ICD-10-CM | 2.16.840.1.113883.6.90
November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4
ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90
ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94
ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

9.1.2 DiagnosticReport.Search (Results) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/989-DiagnosticReport-Search-Results-R4.html
Epic API ID | 989
FHIR version | R4
Categories | Clinical Diagnostics
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Laboratory;Tests;Values;Results
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | February 2020
HTTP method | GET
URL template | api/FHIR/R4/DiagnosticReport?patient={patient}&identifier={identifier}&code={code}&date={date}&category={category}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=989

Description

The DiagnosticReport FHIR resource represents resulted diagnostic orders and relevant data associated with the order and the patient. Supported diagnostic services include laboratory, radiology, and cardiology, audiology, and endoscopy. Within DiagnosticReport, the Observation FHIR resource returns individual components from laboratory tests and text-based content, such as narratives or impressions, from radiology and cardiology results. This resource returns textual results and discrete result components. It can also return pathology results that are not a scan. Scans filed into Epic are not returned by this resource.

Starting in the February 2024 release of Epic, DiagnosticReport can return EKG data using the Midmark third-party integration. If the EKG order collected heart rate and blood pressure data as well, this data is returned as Observations contained within the report.

Starting in the May 2024 release of Epic, DiagnosticReport can return audiology exam finding data. Data within the audiology exam is returned as Observations contained within the response.

Remarks

Text based result content, such as narratives, impressions, or addendums, will be represented as related Observations rather than in the text element inherited by DomainResource. If the results from a Search return medication orders, an error will be returned for those orders.

Heart rate and blood pressure data associated with a Midmark result are represented as full Vital Signs Observations within the DiagnosticReport. Unlike other Observations associated with the order, these are contained resources, and are only accessible embedded within the DiagnosticReport. These resources are associated with the following LOINC codes:

* Heart rate: 8867-4

* Diastolic blood pressure: 8462-4

* Systolic blood pressure: 8480-6

Request parameters

Name | Type | Optional | Array | Cardinality | Description

_count | number | Optional | Yes | | Number of results per page.

_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revInclude. All others will be ignored.

_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.

_lastUpdated (also: _lastUpdated) | date | Optional | Yes | | Filter to a date range during which a result was most recently updated.

_revinclude | string | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.

category | token | Optional | Yes | | The service category. This can be either an Epic order type or a value from the US Core valueset for DiagnosticReport. For the category system of http://loinc.org, supported codes are: Cardiology: LP29708-2 Pathology: LP7839-6 Radiology: LP29684-5 For the category system of http://terminology.hl7.org/CodeSystem/v2-0074, a code of LAB is supported. This parameter is formatted with or without a system specified (code=[code] or code=[system]|[code]). If specifying a system, the corresponding code must be provided.

code | token | Optional | Yes | | The code for the DiagnosticReport. This element cannot search on codes for referenced Observation resources. This can be any code type associated with the procedures, including LOINC, CPT, SNOMED, or other code types mapped to a procedure in Epic. This parameter is formatted with or without a system specified (code=[code] or code=[system]|[code]). If specifying a system, the corresponding code must be provided.

date | date | Optional | Yes | | Clinically relevant time/time period for report. Matches to the effectiveDateTime response element.

identifier | token | Optional | Yes | | The business identifiers for the report. These can return the Epic internal order ID, an external order ID, or a specimen ID for the order. The system value for each concept will vary between Epic environments.

patient | reference | Conditional | Yes | | The FHIR ID of the Patient for the DiagnosticReport.

subject | reference | Conditional | Yes | | The FHIR ID of the Patient for the DiagnosticReport

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description

based-on | reference | Optional | Yes | | A reference to the ServiceRequest resource associated with the order represented by the DiagnosticReport resource.

conclusion | token | Optional | Yes | | The order diagnosis.

encounter | reference | Optional | Yes | | Reference to the FHIR ID of the encounter where the test was ordered.

issued | date | Optional | Yes | | DateTime the version of this result was released.

media | reference | Optional | Yes | | Key images associated with the report.

performer | reference | Optional | Yes | | The FHIR ID of the participants involved in producing the report. This can be a Practitioner or Organization resource FHIR ID.

result | reference | Optional | Yes | | An Observation resource FHIR ID within a DiagnosticReport.

results-interpreter | reference | Optional | Yes | | The FHIR ID for the Practitioner resource representing a reading physician for the order.

Response elements

Name | Type | Optional | Array | Cardinality | Description

DiagnosticReport (R4) | DiagnosticReport

DiagnosticReport (R4).basedOn | Reference | | Yes | 0..* | A reference to a ServiceRequest.

DiagnosticReport (R4).category | CodeableConcept | | Yes | 1..* | The Epic order type is always returned, and US Core Category Code is also returned if the order matches one of the categories. Also includes the LOINC or DiagnosticServiceSectionId codes when available. https://build.fhir.org/ig/HL7/US-Core/ValueSet-us-core-diagnosticreport-category.html

DiagnosticReport (R4).code | CodeableConcept | | | 0..1 | The name and code for this DiagnosticReport.

DiagnosticReport (R4).conclusionCode | CodeableConcept | | Yes | 0..* | The order diagnosis. Starting in the November 2024 version of Epic, each code also sends its display name.

DiagnosticReport (R4).effectiveDateTime | dateTime | | | 0..1 | Clinically relevant date and time for the order.

DiagnosticReport (R4).encounter | Reference | | | 0..1 | Reference to an Encounter resource associated with the procedure.

DiagnosticReport (R4).extension (SmartData) (also: extension (SmartData)) | Extension | | Yes | 0..* | Custom discrete SmartData Elements stored on the order. URL: http://open.epic.com/FHIR/StructureDefinition/extension/smartdata

DiagnosticReport (R4).identifier | Identifier | | Yes | 1..* | The business identifiers for the report, including Epic order ID, accession numbers, and external order identifiers.

DiagnosticReport (R4).issued | instant | | | 1..1 | Date and time of the result.

DiagnosticReport (R4).media | BackboneElement | | Yes | 0..* | Key images associated with the report.

DiagnosticReport (R4).meta | Meta | Required | | 1..1 | Metadata elements common to most resources.

DiagnosticReport (R4).meta.lastUpdated | String | Required | | 1..1 | The instant that the result was last updated.

DiagnosticReport (R4).performer | Reference | | Yes | 1..* | References representing the participants involved in producing the report. This always includes one or more references to Practitioner resources. This element can also include a reference to an Organization resource.

DiagnosticReport (R4).presentedForm | Attachment | | Yes | 0..* | Binary attachments with HTML textual results and/or PDFs.

DiagnosticReport (R4).presentedForm.contentType | String | | | 1..1 | Mime content type code. This is text/html for textual results and application/pdf for PDFs.

DiagnosticReport (R4).presentedForm.title | String | | | 1..1 | Label to display in place of the data.

DiagnosticReport (R4).presentedForm.url | String | | | 1..1 | Binary link for the textual content or report PDF.

DiagnosticReport (R4).result | Reference | | Yes | 0..* | References to the Observation resources representing the result components for the DiagnosticReport.

DiagnosticReport (R4).resultsInterpreter | Reference | | Yes | 0..* | The reading providers.

DiagnosticReport (R4).status | code | | | 1..1 | The status of the report. These values can be returned: registered preliminary final amended If a test was performed but ended in an unsuccessful attempt, the status will be cancelled. For example, if a mammography is performed, and the patient was exposed to radiation, and images were captured, but the images were too blurry to make a clinical determination, that would be considered an unsuccessful attempt.

DiagnosticReport (R4).subject | Reference | | | 1..1 | A reference to the Patient resource representing the subject of the report.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/DiagnosticReport?category=1.2.840.114350.1.13.0.1.7.10.798268.30%7CImaging&pat
    ient=epnvY-g-k4v8aI71KfQzYYg3
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 2,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/DiagnosticReport?category=1.2.840.114350.1.13.861.1.7.10.798268.30%7
    CImaging&patient=epnvY-g-k4v8aI71KfQzYYg3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/DiagnosticReport/e5usz9Yv5FedaHMXrx-GLJj9MkK.pa4IM954m0zHWcwo3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/DiagnosticReport/e5usz9Yv5FedaHMXrx-GLJj9MkK.pa4IM954m0zHWcwo3",
      "resource": {
        "resourceType": "DiagnosticReport",
        "id": "e5usz9Yv5FedaHMXrx-GLJj9MkK.pa4IM954m0zHWcwo3",
        "meta": {
          "lastUpdated": "2021-06-11T19:49:40Z"
        },
        "identifier": [
          {
            "use": "official",
            "type": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
                  "code": "PLAC",
                  "display": "Placer Identifier"
                }
              ],
              "text": "Placer Identifier"
            },
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.798268",
            "value": "1012228165"
          },
          {
            "use": "official",
            "type": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
                  "code": "FILL",
                  "display": "Filler Identifier"
                }
              ],
              "text": "Filler Identifier"
            },
            "system": ""https: //open.epic.com/FHIR/861/order-accession-number/ALB%20LAB",
            "value": "CTC000123"
          }
        ],
        "basedOn": [
          {
            "reference": "ServiceRequest/eVkErihwIBI9jdRYuWq8DJotu.RME5xX6extRRqj7myI3",
            "type": "ServiceRequest"
          }
```

```
    ],
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.10.798268.30",
            "code": "Imaging"
          }
        ],
        "text": "Imaging"
      },
      {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "LP29684-5",
            "display": "Radiology"
          }
        ],
        "text": "Radiology"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "urn:oid:2.16.840.1.113883.6.12",
          "code": "73500",
          "display": "X-RAY HIP UNILAT 1 VW"
        }
      ],
      "text": "X-RAY HIP UNILAT 1 VW"
    },
    "subject": {
      "reference": "Patient/epnvY-g-k4v8aI71KfQzYYg3",
      "display": "Test, Patient"
    },
    "encounter": {
      "reference": "Encounter/eDWAJJqZKd3cp7uTJJQY8RLq9GPLTvkG9kg.fdAetIgs3",
      "identifier": {
        "use": "usual",
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
        "value": "10001370420"
      }
    },
    "effectiveDateTime": "2019-10-11T05:00:00Z",
    "issued": "2021-06-11T19:49:40Z",
    "performer": [
      {
        "reference": "Practitioner/euJOUNeZyX6mofWELWG0Ibg3",
        "display": "Epic User, MD"
      },
      {
        "reference": "Practitioner/ezQhNivITGtqYbx5JeL79Kw3",
        "display": "Otto Rioja Persona Ris Rad, MD"
      }
    ],
    "resultsInterpreter": [
      {
        "reference": "Practitioner/ezQhNivITGtqYbx5JeL79Kw3",
        "type": "Practitioner",
        "display": "Otto Rioja Persona Ris Rad, MD"
      },
      {
        "reference": "Practitioner/euJOUNeZyX6mofWELWG0Ibg3",
        "type": "Practitioner",
        "display": "Epic User, MD"
      }
    ],
    "result": [
      {
        "reference": "Observation/eNnxYsrKX152K8OelRteq9xqbrdPpv8frLm0Wg0eq.g43",
        "display": "Narrative"
      }
    ],
    "imagingStudy": [
      {
        "reference": "ImagingStudy/eUJc6.P8x0S8rHLurJXQuh.QRfPzdiRH4yLo8B-F-z.o3"
      }
    ],
    "presentedForm": [
      {
        "contentType": "text/html",
        "url": "Binary/eyxaMBLEsnTpnmI4c2YhNyZ6iY7Ei1s-SWHDtoh53NnU3",
        "title": "Narrative"
      }
    ]
  },
  "search": {
```

```
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/DiagnosticReport/e43lhnx.XUNbUL3v-.PDcNrEifQ9LIJG0xmvd-KPr8CM3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/DiagnosticReport/e43lhnx.XUNbUL3v-.PDcNrEifQ9LIJG0xmvd-KPr8CM3",
  "resource": {
    "resourceType": "DiagnosticReport",
    "id": "e43lhnx.XUNbUL3v-.PDcNrEifQ9LIJG0xmvd-KPr8CM3",
    "meta": {
      "lastUpdated": "2021-06-11T19:50:05Z"
    },
    "identifier": [
      {
        "use": "official",
        "type": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
              "code": "PLAC",
              "display": "Placer Identifier"
            }
          ],
          "text": "Placer Identifier"
        },
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.798268",
        "value": "1012233558"
      },
      {
        "use": "official",
        "type": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
              "code": "FILL",
              "display": "Filler Identifier"
            }
          ],
          "text": "Filler Identifier"
        },
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.798268.320",
        "value": "28550"
      }
    ],
    "basedOn": [
      {
        "reference": "ServiceRequest/e-zJTgPbE4Mrf-nekgQJ.rRI5zqqeoxzDs4llNQSo69Y3",
        "type": "ServiceRequest"
      }
    ],
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.10.798268.30",
            "code": "Imaging"
          }
        ],
        "text": "Imaging"
      },
      {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "LP29684-5",
            "display": "Radiology"
          }
        ],
        "text": "Radiology"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.696580",
          "code": "52665",
          "display": "GENERIC RADIANT IMAGING EAP"
        }
      ],
      "text": "X-RAY HIP UNILAT 1 VW"
    },
    "subject": {
      "reference": "Patient/epnvY-g-k4v8aI71KfQzYYg3",
```

```
          "display": "Test, Patient"
        },
        "encounter": {
          "identifier": {
            "use": "usual",
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
            "value": "10001456304"
          }
        },
        "effectiveDateTime": "2019-10-11T05:00:00Z",
        "issued": "2021-06-11T19:50:05Z",
        "performer": [
          {
            "reference": "Practitioner/eNRT8GK-ZDKFtQwHDbar9EA3",
            "display": "Becky Black, MD"
          }
        ],
        "resultsInterpreter": [
          {
            "reference": "Practitioner/eNRT8GK-ZDKFtQwHDbar9EA3",
            "type": "Practitioner",
            "display": "Becky Black, MD"
          },
          {
            "reference": "Practitioner/eNRT8GK-ZDKFtQwHDbar9EA3",
            "type": "Practitioner",
            "display": "Becky Black, MD"
          }
        ],
        "result": [
          {
            "reference": "Observation/eavL3N-ZwVkoAEn48KHfezsXPoMlYiWPL7dT3bapLBes3",
            "display": "Narrative"
          }
        ],
        "imagingStudy": [
          {
            "reference": "ImagingStudy/ed0KtzxnrBpuTfBLPnOOr2XpQ2Xu8S4jXS2A5GAFGOX83"
          }
        ],
        "presentedForm": [
          {
            "contentType": "text/html",
            "url": "Binary/eTmCM2E.Uhk9L7Sp2NXPO6w8cPHVTfoFcslhYxPN7Mrw3",
            "title": "Narrative"
          },
          {
            "contentType": "application/pdf",
            "url": "Binary/fsFqDRZrfdiJlA5Rwwc8wfw4",
            "title": "Study Result Document"
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


Change log

Epic Version | Description | Type
May 2024 | Code systems for the following code set now returns a URL in place of an OID: System URL OID Conclusion.Code http://HL7.ORG/FHIR/SID/ICD-10-CM 2.16.840.1.113883.6.90 | Breaking change
System | URL | OID
Conclusion.Code | http://HL7.ORG/FHIR/SID/ICD-10-CM | 2.16.840.1.113883.6.90
November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4
ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90
ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94
ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

9.1.3 Observation.Create (Flowsheets) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/49210-Observation-Create-Flowsheets-R4.html
Epic API ID | 49210
FHIR version | R4

Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | POST
URL template | /api/FHIR/R4/Observation`Observation***
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=49210

Description

This API files a single value to a flowsheet row in the patient's chart.

To write to a vital sign flowsheet row without needing the flowsheet row id, apps can instead use Observation.Create (Vital Signs) (R4).

When a patient-facing app uses this API, it writes to patient-entered flowsheets. Before that can be done, a user in Epic must place an order to create the episode and patient-entered flowsheets for the resource to file to. Clinician or backend apps using this API file to the same flowsheets that clinicians can document in.

An error is returned if a reading already exists. The server assigns an ID to the newly created resource.

Alternatives:

For patient-entered information, there is an equivalent HL7v2 Incoming Clinical Documentation Flowsheet Data - Patient Entered Interface. For non-patient-entered information, there is an equivalent HL7v2 Incoming Clinical Documentation Flowsheet Data Interface.

Encounter Considerations:

For provider and backend applications, there are several limitations with this web service compared to the HL7v2 Interface alternative.

* Must match to an existing encounter using an Encounter FHIR ID (no date or visit type based matching).

* Cannot write to closed encounters.

* Cannot write to future appointments.

Patient-context Considerations:

This service can only file one reading at a time. For patient-entered information, this can trigger multiple notifications to the provider. If this is a concern, consider using an HL7v2 Interface alternative, such as the Incoming Clinical Documentation Flowsheet interface or Incoming Device Data interface, that can file multiple readings at a time.

Filing Readings to Same Timestamp:

This service can only file a reading if one doesn't already exist at the same timestamp. The HL7v2 Interface alternative is able to overwrite and update the existing value at the same timestamp.

The response of a Create depends on the Prefer header:

Prefer: return=minimal

returns the full URL to the resource in the Location header.

Prefer: return=representation

returns both the full URL to the resource in the Location header, as well as the full FHIR resource in the response body.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
Observation (R4) | Observation | | | | An instance of the R4 Observation resource.
Observation (R4).category | CodeableConcept | Conditional | Yes | 0..1 | The category code of "vital-signs" is accepted for vital sign flowsheet rows. No category is required. If multiple categories are provided, only the first valid category is processed.
Observation (R4).category.coding | Coding | Required | Yes | 1..1 | A reference to a code defined by a terminology system.
Observation (R4).category.coding.code | String | Required | | 1..1 | Must be "vital-signs".
Observation (R4).code | CodeableConcept | Required | | 1..1 | What is being measured. The Epic Encoded Flowsheet ID or LOINC code.
Observation (R4).code.coding | Coding | Required | Yes | 1..1 | A reference to a code defined by a terminology system.
Observation (R4).code.coding.code | String | Required | | 1..1 | The coded value identifying the flowsheet being filed to.
Observation (R4).code.coding.system | String | Required | | 1..1 | Valid code systems include the following: LOINC Code System Value: http://loinc.org Epic Encoded Flowsheet Identifier System Value: http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id Note: The encoded flowsheet identifier must be generated by IT admins at an Epic organization.
Observation (R4).component | BackboneElement | Optional | Yes | 0..* | Component level information about the Observation. Can be used for: Blood Pressure to file Systolic and Diastolic pressures Patient-Entered Flowsheet Metadata to file strict or loose metadata for patient-entered flowsheets.
Observation (R4).component.code | CodeableConcept | Required | | 1..1 | The code and system for the component.
Observation (R4).component.code.coding | Coding | Required | Yes | 1..1 | A reference to a code defined by a terminology system.
Observation (R4).component.code.coding.code | String | Required | | 1..1 | Code values vary based on the component being filed. Valid codes include: Blood Pressure 8480-6: Systolic pressure 8462-4: Diastolic pressure Patient-Entered Flowsheet Metadata 1-UUID 2-Blood Glucose Meal Times 5-VO2 Max Test Type 6-Manually Entered

Observation (R4).component.code.coding.system | String | Required | | 1..1 | System varies based on the component being filed. Valid systems include: Blood Pressure LOINC: http://loinc.org Patient-Entered Flowsheet Metadata For Code = 6 - Manually Entered: http://open.epic.com/FHIR/StructureDefinition/patient-vitals-metadata An Epic-generated OID for the patient-entered metadata key.

Observation (R4).component.valueBoolean | boolean | | | 0..1 | Only used for Patient-Entered Flowsheet Metadata values. Valid values include: For Code = 6-Manually Entered false true

Observation (R4).component.valueQuantity | Quantity | Optional | | 0..1 | Only used for Blood Pressure measurements for systolic and diastolic pressures. If a unit is provided, it will be ignored and assumed to be mm[Hg].

Observation (R4).component.valueQuantity.value | Decimal | Required | | 1..1 | Numeric value of the reading. Only this element is required for valueQuantity to be processed for systolic and diastolic pressures.

Observation (R4).component.valueString | string | Optional | | 0..1 | Only used for Patient-Entered Flowsheet Metadata values. Valid values include: For Code = 1-UUID The unique Reading ID For Code = 2-Blood Glucose Meal Times 1-Before Meal 2-After Meal For Code = 5-VO2 Max Test Type 1-Max exercise test 2-Sub-max exercise prediction 3-Non exercise prediction

Observation (R4).effectiveDateTime | dateTime | Required | | 1..1 | The date and time the Observation was taken. A time zone can be specified using ISO 8601 formatting. Refer to this document on how to format a date or dateTime range. If no time zone is provided, the time zone of the encounter is assumed.

Observation (R4).encounter | Reference | Conditional | | 0..1 | The encounter that this Observation belongs to. This is not required if filing to patient-entered flowsheets.

Observation (R4).encounter.reference | String | Required | | 1..1 | Reference to the FHIR ID of the Encounter. Formatted as Encounter/<Encounter FHIR ID>.

Observation (R4).extension (template-id) (also: extension (template-id)) | Extension | Optional | Yes | 0..1 | Specify an array containing the numeric ID of a specific flowsheet template to which data should file.

Observation (R4).extension (template-id).url | uri | Required | | 1..1 | Must be "http://open.epic.com/FHIR/StructureDefinition/extension/template-id".

Observation (R4).extension (template-id).valueIdentifier | Identifier | Required | | 1..1 | The valueIdentifier container for the template-id extension.

Observation (R4).extension (template-id).valueIdentifier.system | String | Required | | 1..1 | The Epic generated OID of the Epic flowsheet template database. Varies based on the Epic environment.

Observation (R4).extension (template-id).valueIdentifier.value | String | Required | | 1..1 | The Epic flowsheet template identifier.

Observation (R4).note | Annotation | Optional | Yes | 0..1 | The comment associated with the value - limited to 254 characters.

Observation (R4).note.text | String | Required | | 1..1 | The textual content of the comment.

Observation (R4).performer | Reference | Optional | Yes | 0..1 | Who filed the Observation. For apps making the request in the provider-facing context, the authenticated user is used. For apps making the request in the patient-facing context, the patient reference is used if it exists. If none is supplied, the authenticated patient is used.

Observation (R4).performer.reference | String | Required | | 1..1 | For apps calling the web service in a patient context, this is a reference to the FHIR ID of the Patient, formatted as Patient/<Patient FHIR ID>. If it is not supplied, the authenticated patient is used. This is ignored for apps calling the web service in a user context, and the authenticated user is used instead.

Observation (R4).status | code | Optional | | 0..1 | The status of the Observation. Only a value of "final" is supported. Data in an unfinalized status is not supported.

Observation (R4).subject | Reference | Required | | 1..1 | The FHIR ID of the patient the Observation is about.

Observation (R4).subject.reference | String | Required | | 1..1 | Reference to the FHIR ID of the Patient. Formatted as Patient/<Patient FHIR ID>.

Observation (R4).valueCodeableConcept | CodeableConcept | Conditional | | 0..1 | Use this element if the data type is a custom list or category type. Only one of value* elements is required.

Observation (R4).valueDateTime | dateTime | Conditional | | 0..1 | Use this element if data type is date. Only one of value* elements is required.

Observation (R4).valueQuantity | Quantity | Conditional | | 0..1 | Use this element if a value has units or the data type is numeric. Only one of value* elements is required.

Observation (R4).valueQuantity.unit | String | Optional | | 0..1 | The unit of measure for the valueQuantity. For height, weight, and temperature readings, the unit is validated and attempted to be converted based on most common forms of expressing these units (e.g. both "g" and "grams" are valid for the unit of grams). For other readings or if a unit is not provided, a default unit is used if applicable. The default unit varies between the type of reading and between organizations.

Observation (R4).valueQuantity.value | Decimal | Required | | 1..1 | Numeric value of the reading. Only this element is required for valueQuantity to be processed.

Observation (R4).valueString | string | Conditional | | 0..1 | Use this element if a value doesn't have units or the data type is a string. Only one of value* elements is required.

Observation (R4).valueTime | time | Conditional | | 0..1 | Use this element if data type is time. Only one of value* elements is required.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Observation (R4) | Observation

Observation (R4).bodySite | CodeableConcept | Conditional | | 0..1 | Applicable only to organizations in the Netherlands. The site on the body where the Observation was made. Used for the location where a blood pressure reading was obtained.

Observation (R4).category | CodeableConcept | | Yes | 0..* | When a flowsheet row is mapped to an HL7 observation category like "vital-signs", "survey", or "care-experience-preference", then that category is used. Flowsheet rows which are not mapped to a specific category do not have a category element in the response.

Observation (R4).code | CodeableConcept | Required | | 1..1 | Coded value for what is being measured. The following code systems can be returned: OID for internal Epic flowsheet ID Encoded value of the internal Epic flowsheet ID External concept mappings such as LOINC and SNOMED

Observation (R4).component | BackboneElement | | Yes | 0..* | Includes extra information about the Observation. Populated only for: Blood Pressure: Systolic (LOINC 8480-6) and diastolic (LOINC 8462-4) pressures are always returned as components if available. Can include components with a SNOMED code of 70665002 and LOINC code of 8358-4 for the cuff type or size, and a SNOMED code of 424724000 and a LOINC code of 8361-8 for cuff position. Average Blood Pressure (starting in the November 2026 version of Epic): Average systolic (LOINC 96608-5) and average diastolic (LOINC 96609-3) are always returned as components if available. Can include a component with a SNOMED code of 246432004 for number of occurrences. SpO2 can include components with LOINC code of 3151-8 for inhaled oxygen flow rate, 3150-0 for inhaled oxygen concentration, and 74206-4 for whether oxygen therapy was given. Pulse Rate can include a component with LOINC code of 44969-4 for heart rate rhythm palpitation. Weight can include a component with a code of 8352-7 for how the patient was clothed for weight reading. Patient-Entered Flowsheet Metadata Represents strict metadata if the coding sub-element is populated. The following strict metadata can be returned: 1-UUID The unique reading ID 2-Blood Glucose Meal Times 1-Before meal 2-After meal 3-Insulin Delivery Reason 1-Basal 2-Bolus 5-VO2 Max Test Type 1-Max exercise test 2-Sub-max exercise prediction 3-Non exercise prediction 6-Manually Entered - Whether or not the patient manually entered this reading into the system (as opposed to automatically entered by a device). false true Represents loose metadata if only the text sub-element is populated. UUID,

Blood Glucose Meal times, Insulin Delivery Reason, and VO2 Max Test Type must have their value specified in component.valueString. Manually Entered is specified as a valueBoolean.

Observation (R4).component.code | CodeableConcept | | | 0..1 | Code for the extra information for the measurement. The following extra information can be returned for each measurement, with the associated code: Blood Pressure 8480-6: Systolic pressure 8462-4: Diastolic pressure SNOMED 70665002 / LOINC 8358-4: Cuff type or size SNOMED 424724000 / LOINC 8361-8: Cuff position Average Blood Pressure (starting in the November 2026 version of Epic) 96608-5: Systolic blood pressure mean 96609-3: Diastolic blood pressure mean SNOMED 246432004: Number of occurrences SpO2 3151-8: Inhaled oxygen flow rate 3150-0: Inhaled oxygen concentration 74206-4: Whether oxygen therapy was given Pulse Rate 44969-4: Heart rate rhythm palpitation. Weight 8352-7: Whether patient was clothed

Observation (R4).component.valueBoolean | boolean | | | 0..1 | Only used for Patient-Entered Flowsheet Metadata values. Valid values include: For Code = 6-Manually Entered false true

Observation (R4).component.valueCodeableConcept | CodeableConcept | | | 0..1 | The value as a CodeableConcept for the extra information for the measurement.

Observation (R4).component.valueQuantity | Quantity | | | 0..1 | If the extra information for the measurement is numeric, this is the numeric value. For example, it could be the flow rate (L/min) or concentration (%) for an SpO2 measurement. Starting in the November 2026 version of Epic, this is also the number of occurrences for average blood pressure.

Observation (R4).component.valueString | string | | | 0..1 | Values for the patient-entered flowsheet metadata.

Observation (R4).dataAbsentReason | CodeableConcept | | | 0..1 | Indicates that an Observation value is missing. This is always "unknown" for systolic blood pressure values in Observations for diastolic-only blood pressure flowsheets, or for diastolic blood pressure value in Observations for systolic-only blood pressure flowsheets.

Observation (R4).effectiveDateTime | dateTime | | | 0..1 | The date and time the Observation was taken. This element is omitted when effectivePeriod is used.

Observation (R4).effectivePeriod | Period | | | 0..1 | Starting in the November 2026 version of Epic, average blood pressure observations can use this element to return the begin and end instants for the average. Only one of the effective* elements is returned.

Observation (R4).encounter | Reference | Conditional | | 1..1 | Reference to an Encounter resource that this Observation belongs to.

Observation (R4).extension (temperature-in-fahrenheit) (also: extension (temperature-in-fahrenheit)) | Extension | | Yes | 0..1 | If the measurement is a temperature value, this element contains the temperature value quantity in Fahrenheit. This element is excluded for other data types. URL: http://open.epic.com/FHIR/STU3/StructureDefinition/temperature-in-fahrenheit

Observation (R4).extension (temperature-in-fahrenheit).valueQuantity | Quantity | | | 1..1 | The temperature value in degrees Fahrenheit [degF]. This represents the same temperature measurement as Observation.valueQuantity, which can use different units.

Observation (R4).extension (template-id) (also: extension (template-id)) | Extension | | Yes | 0..1 | This is the specific flowsheet template (a collection of flowsheet rows and groups) to which data was filed. URL: http://open.epic.com/FHIR/StructureDefinition/extension/template-id

Observation (R4).extension (template-id).valueIdentifier | Identifier | | | 1..1 | The system and value for the flowsheet template.

Observation (R4).extension (template-id).valueIdentifier.system | String | | | 1..1 | The OID for the Epic internal flowsheet template identifier. This varies based on the specific Epic environment.

Observation (R4).extension (template-id).valueIdentifier.value | String | | | 1..1 | The Epic internal identifier of the flowsheet template.

Observation (R4).interpretation | CodeableConcept | | Yes | 0..* | Returns a system of http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation and value of "A" for flowsheet values with an abnormal status. Returns an environment-specific OID system with a value of "SIG" for flowsheet values with a significant status. In Netherlands locales, these interpretation values can also be returned, alongside a system value of http://snomed.info/sct: For heart rate profile Observations, possible values include: SNOMED code 271636001 (Heart regular) for normal heart rate readings SNOMED code 248650006 (Heart irregular) for abnormal heart rate readings For blood glucose profile Observations, possible values include: SNOMED code 281302008 (Above reference range) for blood glucose readings higher than the reference range SNOMED code 11896004 (Intermediate) for blood glucose readings within the reference range (inclusive of the reference range values) SNOMED code 281300000 (Below reference range) for blood glucose readings below the reference range

Observation (R4).issued | instant | | | 1..1 | The date and time the Observation was documented.

Observation (R4).method | CodeableConcept | | | 0..1 | For patient-entered flowsheets, this element is returned with a text value of "Patient-entered". In Netherlands locales with blood pressure vitals Observations, SNOMED (system http://snomed.info/sct) codes can be returned: Blood pressure taken with an invasive method: 10179008 Blood pressure taken with a non-invasive method: 22762002

Observation (R4).note | Annotation | | Yes | 0..1 | The comment associated with the value.

Observation (R4).performer | Reference | | Yes | 1..* | Reference to a FHIR resource for who filed the Observation. If the flowsheet value was filed by a patient, this references a Patient resource. Otherwise, it references a Practitioner resource.

Observation (R4).performer.extension (performerFunction) (also: extension (performerFunction)) | Extension | | Yes | 0..1 | Additional Practitioners associated with the Observation value, if applicable. Possible values include: "data entry person" for Practitioners who transcribed the Observation into the system "verifier" for Practitioners who cosigned the Observation URL: http://hl7.org/fhir/StructureDefinition/event-performerFunction

Observation (R4).referenceRange | BackboneElement | Conditional | Yes | 0..1 | Applicable to organizations in the Netherlands starting in the November 2022 version of Epic. This is populated only for blood glucose readings. This element indicates the abnormal minimum and maximum values defined for the blood glucose reading.

Observation (R4).referenceRange.high | Quantity | | | 0..1 | Applicable to organizations in the Netherlands starting in the November 2022 version of Epic. The higher reference limit for blood glucose readings.

Observation (R4).referenceRange.low | Quantity | | | 0..1 | Applicable to organizations in the Netherlands starting in the November 2022 version of Epic. The lower reference limit for blood glucose readings.

Observation (R4).status | code | | | 1..1 | This element has the following possible values: "preliminary" - value is pended "final" - value has been filed "amended" - value has been edited "entered-in-error" - value has been deleted

Observation (R4).subject | Reference | Required | | 1..1 | The FHIR ID for the patient the Observation is about.

Observation (R4).valueCodeableConcept | CodeableConcept | | | 0..1 | This element is populated if the data type is a category or custom list. Only one of the value* elements is returned.

Observation (R4).valueDateTime | dateTime | | | 0..1 | This element is populated if data type is date. Only one of the value* elements is returned.

Observation (R4).valueQuantity | Quantity | | | 0..1 | This element is populated if a value has units or the data type is numeric. Only one of the value* elements is returned.

Observation (R4).valueString | string | | | 0..1 | This element is populated if a value doesn't have units or the data type is string. Only one of the value* elements is returned.

Observation (R4).valueTime | time | | | 0..1 | This element is populated if data type is time. Only one of the value* elements is returned.

Sample request

```
POST https://hostname/instance/api/FHIR/R4/Observation
{
  "resourceType": "Observation",
```

```
  "code": {
    "coding": [
      {
        "system": "http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id",
        "code": "tFPAQItwnOAotqq9GHG-ZAg0",
        "display": "Precautions"
      }
    ],
    "text": "Precautions"
  },
  "subject": {
    "reference": "Patient/eZO66HeGXB6VCJemL-ciSKA3",
    "display": "Meiko TEST"
  },
  "encounter": {
    "reference": "Encounter/e.R4dWbUDqF01.Z6BEIfEkZecZklxAhL82tS5wzzlD0o3"
  },
  "effectiveDateTime": "2019-09-05T20:00:00Z",
  "valueCodeableConcept": {
    "coding": [
      {
        "code": "Other (Comment)",
        "display": "Other (Comment)"
      }
    ],
    "text": "Other (Comment)"
  },
  "note": [
    {
      "text": "Example comment"
    }
  ]
}
```


Sample response


```
Status code: Created (201)
Content-Type: application/json; charset=utf-8
Location: Observation/eF7uc-VX129EjpVE35In7ZroUUTPOudM9oOQTmIWiPZS9O3pXRr6snBCvQVadYl6VdkVo.PCeyixt2muU4qpWPQ3
```


9.1.4 Observation.Create (Lines, Drains, Airways) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/962-Observation-Create-Lines-Drains-Airways-R4.html
Epic API ID | 962
FHIR version | R4
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Keywords | Flowsheets;LDA
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | POST
URL template | api/FHIR/R4/Observation
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=962

Description

The FHIR Observation resource defines measurements and assertions about a patient, including LDA-W (lines, drains, airways, and wound) data.

This web service cannot write to closed encounters. Ensure that your app doesn't need to write to closed encounters before you begin development.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False
Observation.category | CodeableConcept | False | Yes | | "LDA" to create LDA or LDA assessment.
Observation.code | CodeableConcept | False | | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. This element can't be modified by Observation.Update. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.
Observation.component | BackboneElement | True | Yes
Observation.component.code | CodeableConcept | False | | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. This element can't be modified by Observation.Update. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.

Observation.component.extension (also: LDA-Property-Comment) | Extension | True | Yes | | Comment for the property row.
Observation.component.valueQuantity | Quantity | False | | | Value of the property row.
Observation.effectivePeriod | Period | True | | | Start instant and/or end instant of the LDA.
Observation.encounter | Reference | False | | | Reference to an encounter resource. This element can't be modified by Observation.Update.
Observation.note | Annotation | True | Yes | | The comment. Limited to 254 characters in the August 2025 version of Epic and later. Limited to 60 characters in earlier versions of Epic.
Observation.extension (also: sequelTo) | Extension | False | Yes | | sequelTo extension that specifies the parent LDA FHIR ID.
Observation.status | code | True | | | The status of the observation. Only a value of "final" is supported. We do not support filing data that isn't finalized.
Observation.subject | Reference | False | | | Reference to a patient resource.
Observation.valueQuantity | Quantity | False | | | Value of the assessment. For image rows, a comma-delimited list of external DCS IDs is required. valueQuantity, valueString, valueCodeableConcept, valueTime, and valueDateTime are also accepted.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | An instance of the R4 Observation resource.
Observation.id | id | False | | | The FHIR ID of the newly created observation resource. Optionally, the client can set the return preference header to "representation", and the server will return the entire resource in a format similar to Observation.Read.

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation
<Observation>
<extension url="http://hl7.org/fhir/StructureDefinition/observation-sequelTo">
<valueReference>
<reference value="Observation/ePgszz-JpoXhhBHDW-tMoBw3" />
<display value="FLOQA Line LDA Arm Anterior Left" />
</valueReference>
</extension>
<category>
<coding>
<system value="http://open.epic.com/FHIR/StructureDefinition/observation-category" />
<code value="LDA" />
<display value="Line, Drains, and Airways" />
</coding>
<text value="Line, Drains, and Airways" />
</category>
<code>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679" />
<code value="911010659" />
<display value="Output" />
</coding>
<text value="Output" />
</code>
<subject>
<reference value="Patient/eZO66HeGXB6VCJemL-ciSKA3" />
<display value="Meiko Lufhir" />
</subject>
<encounter>
<reference value="Encounter/e.R4dWbUDqF01.Z6BEIfEkZecZklxAhL82tS5wzzlD0o3" />
</encounter>
<effectiveDateTime value="2019-09-05T20:00:00Z" />
<valueQuantity>
<value value="29" />
<unit value="mL" />
<system value="http://unitsofmeasure.org" />
<code value="mL" />
</valueQuantity>
</Observation>
```

Sample response

```
HTTP/1.1 201 Created
Cache-Control: no-cache,no-store
Pragma: no-cache
Content-Type: application/xml; charset=utf-8
Expires: -1
Location: Observation/enxsPphZGTNXTDZIieZH1p4mvEXVoWJ9YKNzSbW3DttjBvJ.G3ri-HWxh6nyNqegU3
Server: Microsoft-IIS/8.5
ServerMetrics: {"BlockReads":8,"BlockWrites":25,"BlocksAllocated":4,"DBTime":149,"ECFNetworkTime":1,"ECFRequestCount":0,
    "ECFRequestTime":150,"GREF":12298,"JournalEntries":38,"LocksFailed":0,"LocksGranted":29,"MCommands":159688,"MemoryDi
    fference":0,"WorkflowEventBlockReads":8,"WorkflowEventDBTime":149,"WorkflowEventECFNetworkTime":0,"WorkflowEventECFR
    equestCount":0,"WorkflowEventGREF":12298}
X-AspNet-Version: 4.0.30319
X-Powered-By: ASP.NET
Date: Thu, 05 Sep 2019 21:53:20 GMT
Content-Length: 0
```

Change log

Epic Version | Description | Type

November 2022 | Epic has updated the behavior of certain modifier elements in FHIR write APIs to ensure the data in the incoming request body matches the data filed to the database and to comply with the HL7 FHIR specification. Previously, the modifier element was ignored in certain situations, and an undocumented default value was saved in its place. Now, if the modifier element is included and is set to an unaccepted value, a fatal error logs, preventing the request from filing to the database. The FHIR request returns an OperationOutcome with the invalid element and what value is accepted for that element. Element Accepted Incoming Value Default Value status final final | Substantive change
Element | Accepted Incoming Value | Default Value
status | final | final

9.1.5 Observation.Create (Vital Signs) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/963-Observation-Create-Vital-Signs-R4.html
Epic API ID | 963
FHIR version | R4
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Keywords | Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | POST
URL template | api/FHIR/R4/Observation
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=963

Description

This API files a single value to a flowsheet row in the patient's chart. It is primarily intended for flowsheet rows mapped as vital signs.

Starting in the November 2026 version of Epic, this API no longer files to flowsheet rows that aren't mapped as vital signs; use Observation.Create (Flowsheets) (R4) instead.

When a patient-facing app uses this API, it writes to patient-entered flowsheets. Before that can be done, a user in Epic must place an order to create the episode and patient-entered flowsheets for the resource to file to. Clinician or backend apps using this API file to the same flowsheets that clinicians can document in.

An error is returned if a reading already exists. If an ID is included in the request, an error is returned. The server assigns an ID to the newly created resource.

Alternatives:

For patient-entered information, there is an equivalent HL7v2 Incoming Clinical Documentation Flowsheet Data - Patient Entered Interface. For non-patient-entered information, there is an equivalent HL7v2 Incoming Clinical Documentation Flowsheet Data Interface.

Encounter Considerations:

For provider and backend applications, there are several limitations with this web service compared to the HL7v2 Interface alternative.

* Must match to an existing encounter using an Encounter FHIR ID (no date or visit type based matching).

* Cannot write to closed encounters.

* Cannot write to future appointments.

Patient-context Considerations:

This service can only file one reading at a time. For patient-entered information, this can trigger multiple notifications to the provider. If this is a concern, consider using an HL7v2 Interface alternative, such as the Incoming Clinical Documentation Flowsheet interface or Incoming Device Data interface, that can file multiple readings at a time.

Filing Readings to Same Timestamp:

This service can only file a reading if one doesn't already exist at the same timestamp. The HL7v2 Interface alternative is able to overwrite and update the existing value at the same timestamp.

The response of a Create depends on the Prefer header:

Prefer: return=minimal

returns the full URL to the resource in the Location header.

Prefer: return=representation

returns both the full URL to the resource in the Location header, as well as the full FHIR resource in the response body.

Remarks

To see an example of how the component.code element is structured for patient-entered flowsheet data, refer to the sample request section of the Observation.Create (Vitals) (STU3) API entry.

Request parameters

Name | Type | Optional | Array | Cardinality | Description

Observation (R4) | Observation | | | | An instance of the R4 Observation resource.

Observation (R4).category | CodeableConcept | Required | Yes | 1..1 | The code for the category must be "vital-signs". If multiple categories are provided, only the first valid category is processed.

Observation (R4).category.coding | Coding | Required | Yes | 1..1 | A reference to a code defined by a terminology system.

Observation (R4).category.coding.code | String | Required | | 1..1 | Must be "vital-signs".

Observation (R4).category.coding.system | String | Required | | 1..1 | Must be "http://hl7.org/fhir/observation-category".

Observation (R4).code | CodeableConcept | Required | | 1..1 | What is being measured. The Epic Flowsheet ID, Epic Encoded Flowsheet ID, or LOINC code.

Observation (R4).code.coding | Coding | Required | Yes | 1..1 | A reference to a code defined by a terminology system.

Observation (R4).code.coding.code | String | Required | | 1..1 | The coded value identifying the flowsheet being filed to.

Observation (R4).code.coding.system | String | Required | | 1..1 | Valid code systems include the following: LOINC Code System Value: http://loinc.org Epic Encoded Flowsheet Identifier - Beginning in Epic version November 2026, this only works if the flowsheet is considered a vital sign. Use Observation.Create (Flowsheets) (R4) to file to other flowsheet rows. System Value: http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id Note: The encoded flowsheet identifier must be generated by IT admins at an Epic organization. Epic Internal Flowsheet Identifier - This only works if the flowsheet is considered a vital sign. System Value: Epic generated OID, varies in each Epic environment.

Observation (R4).component | BackboneElement | Optional | Yes | 0..* | Component level information about the Observation. Can be used for: Blood Pressure to file Systolic and Diastolic pressures Patient-Entered Flowsheet Metadata to file strict or loose metadata for patient-entered flowsheets.

Observation (R4).component.code | CodeableConcept | Required | | 1..1 | The code and system for the component.

Observation (R4).component.code.coding | Coding | Required | Yes | 1..1 | A reference to a code defined by a terminology system.

Observation (R4).component.code.coding.code | String | Required | | 1..1 | Code values vary based on the component being filed. Valid codes include: Blood Pressure 8480-6: Systolic pressure 8462-4: Diastolic pressure Patient-Entered Flowsheet Metadata 1-UUID 2-Blood Glucose Meal Times 5-VO2 Max Test Type 6-Manually Entered

Observation (R4).component.code.coding.system | String | Required | | 1..1 | System varies based on the component being filed. Valid systems include: Blood Pressure LOINC: http://loinc.org Patient-Entered Flowsheet Metadata For Code = 6 - Manually Entered: http://open.epic.com/FHIR/StructureDefinition/patient-vitals-metadata An Epic-generated OID for the patient-entered metadata key.

Observation (R4).component.valueBoolean | boolean | | | 0..1 | Only used for Patient-Entered Flowsheet Metadata values. Valid values include: For Code = 6-Manually Entered false true

Observation (R4).component.valueQuantity | Quantity | Optional | | 0..1 | Only used for Blood Pressure measurements for systolic and diastolic pressures. If a unit is provided, it will be ignored and assumed to be mm[Hg].

Observation (R4).component.valueQuantity.value | Decimal | Required | | 1..1 | Numeric value of the reading. Only this element is required for valueQuantity to be processed for systolic and diastolic pressures.

Observation (R4).component.valueString | string | Optional | | 0..1 | Only used for Patient-Entered Flowsheet Metadata values. Valid values include: For Code = 1-UUID The unique Reading ID For Code = 2-Blood Glucose Meal Times 1-Before Meal 2-After Meal For Code = 5-VO2 Max Test Type 1-Max exercise test 2-Sub-max exercise prediction 3-Non exercise prediction

Observation (R4).effectiveDateTime | dateTime | Required | | 1..1 | The date and time the Observation was taken. A time zone can be specified using ISO 8601 formatting. Refer to this document on how to format a date or dateTime range. If no time zone is provided, the time zone of the encounter is assumed.

Observation (R4).encounter | Reference | Conditional | | 0..1 | The encounter that this Observation belongs to. This is not required if filing to patient-entered flowsheets.

Observation (R4).encounter.reference | String | Required | | 1..1 | Reference to the FHIR ID of the Encounter. Formatted as Encounter/<Encounter FHIR ID>.

Observation (R4).extension (template-id) (also: extension (template-id)) | Extension | Optional | Yes | 0..1 | Starting in the May 2024 version of Epic, or February 2024 with a special update, specify an array containing the numeric ID of a specific flowsheet template to which data should file.

Observation (R4).extension (template-id).url | uri | Required | | 1..1 | Must be "http://open.epic.com/FHIR/StructureDefinition/extension/template-id".

Observation (R4).extension (template-id).valueIdentifier | Identifier | Required | | 1..1 | The valueIdentifier container for the template-id extension.

Observation (R4).extension (template-id).valueIdentifier.system | String | Required | | 1..1 | The Epic generated OID of the Epic flowsheet template database. Varies based on the Epic environment.

Observation (R4).extension (template-id).valueIdentifier.value | String | Required | | 1..1 | The Epic flowsheet template identifier.

Observation (R4).note | Annotation | Optional | Yes | 0..1 | The comment associated with the value - limited to 254 characters. Prior to the August 2025 version of Epic, this was limited to 60 characters.

Observation (R4).note.text | String | Required | | 1..1 | The textual content of the comment.

Observation (R4).performer | Reference | Optional | Yes | 0..1 | Who filed the Observation. For apps making the request in the provider-facing context, the authenticated user is used. For apps making the request in the patient-facing context, the patient reference is used if it exists. If none is supplied, the authenticated patient is used.

Observation (R4).performer.reference | String | Required | | 1..1 | For apps calling the web service in a patient context, this is a reference to the FHIR ID of the Patient, formatted as Patient/<Patient FHIR ID>. If it is not supplied, the authenticated patient is used. This is ignored for apps calling the web service in a user context, and the authenticated user is used instead.

Observation (R4).status | code | Optional | | 0..1 | The status of the Observation. Only a value of "final" is supported. Data in an unfinalized status is not supported.

Observation (R4).subject | Reference | Required | | 1..1 | The FHIR ID of the patient the Observation is about.

Observation (R4).subject.reference | String | Required | | 1..1 | Reference to the FHIR ID of the Patient. Formatted as Patient/<Patient FHIR ID>.

Observation (R4).valueCodeableConcept | CodeableConcept | Conditional | | 0..1 | Use this element if the data type is a custom list or category type. Only one of value* elements is required.

Observation (R4).valueDateTime | dateTime | Conditional | | 0..1 | Use this element if data type is date. Only one of value* elements is required.

Observation (R4).valueQuantity | Quantity | Conditional | | 0..1 | Use this element if a value has units or the data type is numeric. Only one of value* elements is required.

Observation (R4).valueQuantity.unit | String | Optional | | 0..1 | The unit of measure for the valueQuantity. For height, weight, and temperature readings, the unit is validated and attempted to be converted based on most common forms of expressing these units (e.g. both "g" and "grams" are valid for the unit of grams). For other readings or if a unit is not provided, a default unit is used if applicable. The default unit varies between the type

of reading and between organizations.

Observation (R4).valueQuantity.value | Decimal | Required | | 1..1 | Numeric value of the reading. Only this element is required for valueQuantity to be processed.

Observation (R4).valueString | string | Conditional | | 0..1 | Use this element if a value doesn't have units or the data type is a string. Only one of value* elements is required.

Observation (R4).valueTime | time | Conditional | | 0..1 | Use this element if data type is time. Only one of value* elements is required.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Observation (R4) | Observation

Observation (R4).bodySite | CodeableConcept | Conditional | | 0..1 | Applicable only to organizations in the Netherlands. The site on the body where the Observation was made. Used for the location where a blood pressure reading was obtained.

Observation (R4).category | CodeableConcept | | Yes | 1..1 | This element always returns "vital-signs".

Observation (R4).code | CodeableConcept | Required | | 0..* | Coded value for what is being measured. The following code systems can be returned: OID for internal Epic flowsheet ID Encoded value of the internal Epic flowsheet ID External concept mappings such as LOINC and SNOMED

Observation (R4).component | BackboneElement | | Yes | 0..* | Includes extra information about the Observation. Populated only for: Blood Pressure: Systolic (LOINC 8480-6) and diastolic (LOINC 8462-4) pressures are always returned as components if available. Can include components with a SNOMED code of 70665002 and LOINC code of 8358-4 for the cuff type or size, and a SNOMED code of 424724000 and a LOINC code of 8361-8 for cuff position. Average Blood Pressure (starting in the November 2026 version of Epic): Average systolic (LOINC 96608-5) and average diastolic (LOINC 96609-3) are always returned as components if available. Can include a component with a SNOMED code of 246432004 for number of occurrences. SpO2 can include components with LOINC code of 3151-8 for inhaled oxygen flow rate, 3150-0 for inhaled oxygen concentration, and 74206-4 for whether oxygen therapy was given. Pulse Rate can include a component with LOINC code of 44969-4 for heart rate rhythm palpitation. Weight can include a component with a code of 8352-7 for how the patient was clothed for weight reading. Patient-Entered Flowsheet Metadata Represents strict metadata if the coding sub-element is populated. The following strict metadata can be returned: 1-UUID The unique reading ID 2-Blood Glucose Meal Times 1-Before meal 2-After meal 3-Insulin Delivery Reason 1-Basal 2-Bolus 5-VO2 Max Test Type 1-Max exercise test 2-Sub-max exercise prediction 3-Non exercise prediction 6-Manually Entered - Whether or not the patient manually entered this reading into the system (as opposed to automatically entered by a device). false true Represents loose metadata if only the text sub-element is populated. UUID, Blood Glucose Meal times, Insulin Delivery Reason, and VO2 Max Test Type must have their value specified in component.valueString. Manually Entered is specified as a valueBoolean.

Observation (R4).component.code | CodeableConcept | | | 0..1 | Code for the extra information for the measurement. The following extra information can be returned for each measurement, with the associated code: Blood Pressure 8480-6: Systolic pressure 8462-4: Diastolic pressure SNOMED 70665002 / LOINC 8358-4: Cuff type or size SNOMED 424724000 / LOINC 8361-8: Cuff position Average Blood Pressure (starting in the November 2026 version of Epic) 96608-5: Systolic blood pressure mean 96609-3: Diastolic blood pressure mean SNOMED 246432004: Number of occurrences SpO2 3151-8: Inhaled oxygen flow rate 3150-0: Inhaled oxygen concentration 74206-4: Whether oxygen therapy was given Pulse Rate 44969-4: Heart rate rhythm palpitation. Weight 8352-7: Whether patient was clothed

Observation (R4).component.valueBoolean | boolean | | | 0..1 | Only used for Patient-Entered Flowsheet Metadata values. Valid values include: For Code = 6-Manually Entered false true

Observation (R4).component.valueCodeableConcept | CodeableConcept | | | 0..1 | The value as a CodeableConcept for the extra information for the measurement.

Observation (R4).component.valueQuantity | Quantity | | | 0..1 | If the extra information for the measurement is numeric, this is the numeric value. For example, it could be the flow rate (L/min) or concentration (%) for an SpO2 measurement. Starting in the November 2026 version of Epic, this is also the number of occurrences for average blood pressure.

Observation (R4).component.valueString | string | | | 0..1 | Values for the patient-entered flowsheet metadata.

Observation (R4).dataAbsentReason | CodeableConcept | | | 0..1 | Indicates that an Observation value is missing. This is always "unknown" for systolic blood pressure values in Observations for diastolic-only blood pressure flowsheets, or for diastolic blood pressure value in Observations for systolic-only blood pressure flowsheets.

Observation (R4).effectiveDateTime | dateTime | | | 0..1 | The date and time the Observation was taken. Starting in the November 2026 version of Epic, this element is omitted when effectivePeriod is used.

Observation (R4).effectivePeriod | Period | | | 0..1 | Starting in the November 2026 version of Epic, average blood pressure observations can use this element to return the begin and end instants for the average as an effective time range. If effectivePeriod is used, then effectiveDateTime is omitted.

Observation (R4).encounter | Reference | Conditional | | 1..1 | Reference to an Encounter resource that this Observation belongs to.

Observation (R4).extension (temperature-in-fahrenheit) (also: extension (temperature-in-fahrenheit)) | Extension | | Yes | 0..1 | If the measurement is a temperature value, this element contains the temperature value quantity in Fahrenheit. This element is excluded for other data types. URL: http://open.epic.com/FHIR/STU3/StructureDefinition/temperature-in-fahrenheit

Observation (R4).extension (temperature-in-fahrenheit).valueQuantity | Quantity | | | 1..1 | The temperature value in degrees Fahrenheit [degF]. This represents the same temperature measurement as Observation.valueQuantity, which can use different units.

Observation (R4).extension (template-id) (also: extension (template-id)) | Extension | | Yes | 0..1 | Starting in the May 2024 version of Epic, or February 2024 with a special update, this is the specific flowsheet template (a collection of flowsheet rows and groups) to which data was filed. URL: http://open.epic.com/FHIR/StructureDefinition/extension/template-id

Observation (R4).extension (template-id).valueIdentifier | Identifier | | | 1..1 | The system and value for the flowsheet template.

Observation (R4).extension (template-id).valueIdentifier.system | String | | | 1..1 | The OID for the Epic internal flowsheet template identifier. This varies based on the specific Epic environment.

Observation (R4).extension (template-id).valueIdentifier.value | String | | | 1..1 | The Epic internal identifier of the flowsheet template.

Observation (R4).interpretation | CodeableConcept | | Yes | 0..* | Returns a system of http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation and value of "A" for vitals with an abnormal status. Returns an environment-specific OID system with a value of "SIG" for vitals with a significant status. In Netherlands locales starting in the November 2022 release of Epic, these interpretation values can also be returned, alongside a system value of http://snomed.info/sct: For heart rate profile Observations, possible values include: SNOMED code 271636001 (Heart regular) for normal heart rate readings SNOMED code 248650006 (Heart irregular) for abnormal heart rate readings For blood glucose profile Observations, possible values include: SNOMED code 281302008 (Above reference range) for blood glucose readings higher than the reference range SNOMED code 11896004 (Intermediate) for blood glucose readings within the reference range (inclusive of the reference range values) SNOMED code 281300000 (Below reference range) for blood glucose readings below the reference range

Observation (R4).issued | instant | | | 1..1 | The date and time the Observation was documented.

Observation (R4).method | CodeableConcept | | | 0..1 | For patient-entered flowsheets, this element is returned with a text value of "Patient-entered". In Netherlands locales with blood pressure vitals Observations, SNOMED (system http://snomed.info/sct) codes can be returned: Blood pressure taken with an invasive method: 10179008 Blood pressure taken with a non-invasive method: 22762002

Observation (R4).note | Annotation | | Yes | 0..1 | The comment associated with the value.

Observation (R4).performer | Reference | | Yes | 1..* | Reference to a FHIR resource for who filed the Observation. If the vital was filed by a patient, this references a Patient resource. Otherwise, it references a Practitioner resource.

Observation (R4).performer.extension (performerFunction) (also: extension (performerFunction)) | Extension | | Yes | 0..1 | Additional Practitioners associated with the Observation value, if applicable. Possible values include: "data entry person" for Practitioners who transcribed the Observation into the system "verifier" for Practitioners who cosigned the Observation URL: http://hl7.org/fhir/StructureDefinition/event-performerFunction

Observation (R4).referenceRange | BackboneElement | Conditional | Yes | 0..1 | Applicable to organizations in the Netherlands starting in the November 2022 version of Epic. This is populated only for blood glucose readings. This element indicates the abnormal minimum and maximum values defined for the blood glucose reading.

Observation (R4).referenceRange.high | Quantity | | | 0..1 | Applicable to organizations in the Netherlands starting in the November 2022 version of Epic. The higher reference limit for blood glucose readings.

Observation (R4).referenceRange.low | Quantity | | | 0..1 | Applicable to organizations in the Netherlands starting in the November 2022 version of Epic. The lower reference limit for blood glucose readings.

Observation (R4).status | code | | | 1..1 | This element has the following possible values: "preliminary" - value is pended "final" - value has been filed "amended" - value has been edited "entered-in-error" - value has been deleted

Observation (R4).subject | Reference | Required | | 1..1 | The FHIR ID for the patient the Observation is about.

Observation (R4).valueCodeableConcept | CodeableConcept | | | 0..1 | This element is populated if the data type is a category or custom list. Only one of the value* elements is returned.

Observation (R4).valueDateTime | dateTime | | | 0..1 | This element is populated if data type is date. Only one of the value* elements is returned.

Observation (R4).valueQuantity | Quantity | | | 0..1 | This element is populated if a value has units or the data type is numeric. Only one of the value* elements is returned.

Observation (R4).valueString | string | | | 0..1 | This element is populated if a value doesn't have units or the data type is string. Only one of the value* elements is returned.

Observation (R4).valueTime | time | | | 0..1 | This element is populated if data type is time. Only one of the value* elements is returned.

Sample request

```
POST https://hostname/instance/api/FHIR/R4/Observation
{
  "resourceType": "Observation",
  "category": [
    {
      "coding": [
        {
          "system": "http://hl7.org/fhir/observation-category",
          "code": "vital-signs",
          "display": "Vital Signs"
        }
      ],
      "text": "Vital Signs"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679",
        "code": "6",
        "display": "Temp"
      },
      {
        "system": "http://loinc.org",
        "code": "8310-5",
        "display": "Body temperature"
      },
      {
        "system": "http://loinc.org",
        "code": "8716-3",
        "display": "Vital signs"
      }
    ],
    "text": "Temp"
  },
  "subject": {
    "reference": "Patient/eZO66HeGXB6VCJemL-ciSKA3",
    "display": "Meiko TEST"
  },
  "encounter": {
    "reference": "Encounter/e.R4dWbUDqF01.Z6BEIfEkZecZklxAhL82tS5wzzlD0o3"
  },
  "effectiveDateTime": "2019-09-05T20:00:00Z",
  "valueQuantity": {
    "value": 37.23,
    "unit": "Cel",
    "system": "http://unitsofmeasure.org",
    "code": "Cel"
  }
}
```

Sample response

```
Status code: Created (201)
Content-Type: application/json; charset=utf-8
Location: Observation/eizmjygdN9YY0-abviOrMfcbd8esh7G-wPYmSzcav8lwa7LUxXgzh4ZySR8Y1jt2TMBBlVBz2DX-KZlAEGhOP0g3
```

Change log

Epic Version | Description | Type

November 2022 | Epic has updated the behavior of certain modifier elements in FHIR write APIs to ensure the data in the incoming request body matches the data filed to the database and to comply with the HL7 FHIR specification. Previously, the modifier element was ignored in certain situations, and an undocumented default value was saved in its place. Now, if the modifier element is included and is set to an unaccepted value, a fatal error logs, preventing the request from filing to the database. The FHIR request returns an OperationOutcome with the invalid element and what value is accepted for that element. Element Accepted Incoming Value Default Value status final final | Substantive change

Element | Accepted Incoming Value | Default Value

status | final | final

May 2024 (February 2024 by special update) | Epic has created a new template ID extension element that allows specifying the flowsheet template to which vitals data is filed. When used in the Observation.Create request, the extension allows incoming data to be filed to a specific template, rather than to a default template specified at the department level (I DEP 24015) or the system level (I LSD 34050), or to Epic-released flowsheet template 1-Vital Signs. | Substantive Changes

9.1.6 Observation.Read (Activities of Daily Living) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/10122-Observation-Read-Activities-of-Daily-Living-R4.html

Epic API ID | 10122

FHIR version | R4

Categories | Clinical Diagnostics

API type | Industry-Standard

USCDI | No

Available in inpatient and other contexts | Yes

ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base

ONC certification versions | August 2021

HTTP method | GET

URL template | /api/FHIR/R4/Observation/{ID}

Supports OAuth 2.0 | True

Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients

Performs scope validation | False

Source | https://fhir.epic.com/Specifications?api=10122

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description

ID | token | False | | | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Observation | Observation | False | | | An instance of the R4 Observation resource.

Observation.category | CodeableConcept | False | Yes | | Type of observation. Always includes SNOMED codes 118228005 and 384821006, and custom code “functional-mental status”.

Observation.code | CodeableConcept | False | | | The type of observation. Mapped values. If no mapping exists, only the text value is returned.

Observation.effectivePeriod | Period | True | | | Clinically relevant time period for the observation.

Observation.note | Annotation | True | Yes | | Comments about the observation. A comment if flowsheet data or the associated note if inpatient goals. If using cascading flowsheets, the note element contains information about the additional flowsheet, if applicable, instead of the note. For example, a nurse documents flowsheet “daily activities” as abnormal, which opens flowsheet “mobility.” The nurse documents the value “highly mobile” in the flowsheet. The note element for “daily activities” is returned as “daily activities;highly mobile” if flowsheet “mobility” isn’t included in the Epic organization's configuration.

Observation.status | code | False | | | Always "final"

Observation.subject | Reference | False | | | Who the observation is about. Reference to the patient resource.

Observation.valueCodeableConcept | CodeableConcept | True | | | Result of the observation

Sample request

```
https://hostname/instance/api/FHIR/r4/Observation/eK.YYu4CGX.bgDi.i48Ry0rwjGC3r1K2R6gMLL-cvPyTD.BkmsblugeJXH0AR7lq43
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "eK.YYu4CGX.bgDi.i48Ry0rwjGC3r1K2R6gMLL-cvPyTD.BkmsblugeJXH0AR7lq43",
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
          "code": "functional-mental-status",
          "display": "Functional and Mental Status"
        }
      ]
    },
```

```
    {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "118228005",
          "display": "Functional finding"
        }
      ]
    },
    {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "384821006",
          "display": "Mental state, behavior and/or psychosocial function finding"
        }
      ]
    }
  ],
  "code": {
    "text": "Body Movement in Desired Range"
  },
  "subject": {
    "reference": "Patient/enoCp373txQSghUte78RDIQ3",
    "display": "FHIR, Automation"
  },
  "effectivePeriod": {
    "start": "2021-05-04",
    "end": "2021-05-07"
  },
  "valueCodeableConcept": {
    "text": "Progressing"
  }
}
```


9.1.7 Observation.Read (Core Characteristics) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/964-Observation-Read-Core-Characteristics-R4.html
Epic API ID | 964
FHIR version | R4
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Keywords | Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | GET
URL template | api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=964

Description

The FHIR Observation resource defines measurements and assertions about a patient, including vital signs, laboratory data, imaging results, devices measurements, clinical assessment tools, personal characteristics, social history, and core characteristics.

This FHIR Observation resource returns a core characteristic such as gestational age or birth weight given the associated FHIR ID.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | The read interaction uses the given Observation FHIR ID to retrieve the most up-to-date information about core characteristics.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | An instance of the R4 Observation resource.
Observation.category | CodeableConcept | False | Yes | | Always set to "core characteristics".
Observation.code | CodeableConcept | False | | | This is the LOINC code corresponding to value[x] returned.
Observation.issued | instant | False | | | The instant when this observation was last edited.
Observation.performer | Reference | False | Yes | | Who documented the observation.
Observation.status | code | False | | | Always set to "final".
Observation.subject | Reference | False | | | The patient.
Observation.valueCodeableConcept | CodeableConcept | One value[x] element is always returned. Which type depends on the data in the specific instance of the resource. | | | Available starting in February 2022. The following values, depending on the type of data returned: Delivery method:

LOINC 73762-7 Labor duration: LOINC 32396-4 Hospital name: LOINC 62330-6 Hospital Location: LOINC 62331-4

Observation.valueInteger | integer | One value[x] element is always returned. Which type depends on the data in the specific instance of the resource. | | | The following values, depending on the type of data returned: Birth APGAR scores APGAR 1: LOINC 9272-6 APGAR 5: LOINC 9274-2 APGAR 10: LOINC 9271-8

Observation.valueQuantity | Quantity | One value[x] element is always returned. Which type depends on the data in the specific instance of the resource. | | | The following values, depending on the type of data returned: Birth APGAR scores APGAR 1: LOINC 9272-6 APGAR 5: LOINC 9274-2 APGAR 10: LOINC 9271-8

Observation.valueString | string | One value[x] element is always returned. Which type depends on the data in the specific instance of the resource. | | | The following values, depending on the type of data returned: Delivery method: LOINC 73762-7 Labor duration: LOINC 32396-4 Hospital name: LOINC 62330-6 Hospital location: LOINC 62331-4

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation/eJW3Cy0gsPSma08Z6jFDVDPI7tUavtRv8uzTVG5MX4PQ3
```

Sample response

```
<Observation xmlns="http://hl7.org/fhir">
  <id value="eJW3Cy0gsPSma08Z6jFDVDPI7tUavtRv8uzTVG5MX4PQ3"/>
  <status value="final"/>
  <category>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/observation-category"/>
      <code value="core-characteristics"/>
      <display value="Core Characteristics"/>
    </coding>
    <text value="Core Characteristics"/>
  </category>
  <code>
    <coding>
      <system value="http://loinc.org"/>
      <code value="8339-4"/>
      <display value="Birth weight measured"/>
    </coding>
    <text value="Birth weight measured"/>
  </code>
  <subject>
    <reference value="Patient/eZO66HeGXB6VCJemL-ciSKA3"/>
    <display value="Meiko Lufhir"/>
  </subject>
  <issued value="2019-09-05T21:32:37Z"/>
  <performer>
    <reference value="Practitioner/euJOUNeZyX6mofWELWG0Ibg3"/>
    <display value="Starter Provider, MD"/>
  </performer>
  <valueQuantity>
    <value value="2.999"/>
    <unit value="kg"/>
    <system value="http://unitsofmeasure.org"/>
    <code value="kg"/>
  </valueQuantity>
</Observation>
```

9.1.8 Observation.Read (Flowsheets) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/49209-Observation-Read-Flowsheets-R4.html
Epic API ID | 49209
FHIR version | R4
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=49209

Description

This web service retrieves general non-duplicable data found in the patient's flowsheets across all encounters. This web service can only be queried using an Observation FHIR ID. This sub-resource of Observation is suitable for flowsheet rows which are not configured as lines, drains, or airways.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Observation (R4) | Observation

Observation (R4).bodySite | CodeableConcept | Conditional | | 0..1 | Applicable only to organizations in the Netherlands. The site on the body where the Observation was made. Used for the location where a blood pressure reading was obtained.

Observation (R4).category | CodeableConcept | | Yes | 0..* | When a flowsheet row is mapped to an HL7 observation category like "vital-signs", "survey", or "care-experience-preference", then that category is used. Flowsheet rows which are not mapped to a specific category do not have a category element in the response.

Observation (R4).code | CodeableConcept | Required | | 1..1 | Coded value for what is being measured. The following code systems can be returned: OID for internal Epic flowsheet ID Encoded value of the internal Epic flowsheet ID External concept mappings such as LOINC and SNOMED

Observation (R4).component | BackboneElement | | Yes | 0..* | Includes extra information about the Observation. Populated only for: Blood Pressure: Systolic (LOINC 8480-6) and diastolic (LOINC 8462-4) pressures are always returned as components if available. Can include components with a SNOMED code of 70665002 and LOINC code of 8358-4 for the cuff type or size, and a SNOMED code of 424724000 and a LOINC code of 8361-8 for cuff position. Average Blood Pressure (starting in the November 2026 version of Epic): Average systolic (LOINC 96608-5) and average diastolic (LOINC 96609-3) are always returned as components if available. Can include a component with a SNOMED code of 246432004 for number of occurrences. SpO2 can include components with LOINC code of 3151-8 for inhaled oxygen flow rate, 3150-0 for inhaled oxygen concentration, and 74206-4 for whether oxygen therapy was given. Pulse Rate can include a component with LOINC code of 44969-4 for heart rate rhythm palpitation. Weight can include a component with a code of 8352-7 for how the patient was clothed for weight reading. Patient-Entered Flowsheet Metadata Represents strict metadata if the coding sub-element is populated. The following strict metadata can be returned: 1-UUID The unique reading ID 2-Blood Glucose Meal Times 1-Before meal 2-After meal 3-Insulin Delivery Reason 1-Basal 2-Bolus 5-VO2 Max Test Type 1-Max exercise test 2-Sub-max exercise prediction 3-Non exercise prediction 6-Manually Entered - Whether or not the patient manually entered this reading into the system (as opposed to automatically entered by a device). false true Represents loose metadata if only the text sub-element is populated. UUID, Blood Glucose Meal times, Insulin Delivery Reason, and VO2 Max Test Type must have their value specified in component.valueString. Manually Entered is specified as a valueBoolean.

Observation (R4).component.code | CodeableConcept | | | 0..1 | Code for the extra information for the measurement. The following extra information can be returned for each measurement, with the associated code: Blood Pressure 8480-6: Systolic pressure 8462-4: Diastolic pressure SNOMED 70665002 / LOINC 8358-4: Cuff type or size SNOMED 424724000 / LOINC 8361-8: Cuff position Average Blood Pressure (starting in the November 2026 version of Epic) 96608-5: Systolic blood pressure mean 96609-3: Diastolic blood pressure mean SNOMED 246432004: Number of occurrences SpO2 3151-8: Inhaled oxygen flow rate 3150-0: Inhaled oxygen concentration 74206-4: Whether oxygen therapy was given Pulse Rate 44969-4: Heart rate rhythm palpitation. Weight 8352-7: Whether patient was clothed

Observation (R4).component.valueBoolean | boolean | | | 0..1 | Only used for Patient-Entered Flowsheet Metadata values. Valid values include: For Code = 6-Manually Entered false true

Observation (R4).component.valueCodeableConcept | CodeableConcept | | | 0..1 | The value as a CodeableConcept for the extra information for the measurement.

Observation (R4).component.valueQuantity | Quantity | | | 0..1 | If the extra information for the measurement is numeric, this is the numeric value. For example, it could be the flow rate (L/min) or concentration (%) for an SpO2 measurement. Starting in the November 2026 version of Epic, this is also the number of occurrences for average blood pressure.

Observation (R4).component.valueString | string | | | 0..1 | Values for the patient-entered flowsheet metadata.

Observation (R4).dataAbsentReason | CodeableConcept | | | 0..1 | Indicates that an Observation value is missing. This is always "unknown" for systolic blood pressure values in Observations for diastolic-only blood pressure flowsheets, or for diastolic blood pressure value in Observations for systolic-only blood pressure flowsheets.

Observation (R4).effectiveDateTime | dateTime | | | 0..1 | The date and time the Observation was taken. This element is omitted when effectivePeriod is used.

Observation (R4).effectivePeriod | Period | | | 0..1 | Starting in the November 2026 version of Epic, average blood pressure observations can use this element to return the begin and end instants for the average. Only one of the effective* elements is returned.

Observation (R4).encounter | Reference | Conditional | | 1..1 | Reference to an Encounter resource that this Observation belongs to.

Observation (R4).extension (supportingInfo) (also: extension (supportingInfo)) | Extension | Conditional | Yes | 0..* | If the measurement is linked to a note written by a clinician, this returns a reference to the DocumentReference resource. URL: http://hl7.org/fhir/StructureDefinition/workflow-supportingInfo

Observation (R4).extension (temperature-in-fahrenheit) (also: extension (temperature-in-fahrenheit)) | Extension | | Yes | 0..1 | If the measurement is a temperature value, this element contains the temperature value quantity in Fahrenheit. This element is excluded for other data types. URL: http://open.epic.com/FHIR/STU3/StructureDefinition/temperature-in-fahrenheit

Observation (R4).extension (temperature-in-fahrenheit).valueQuantity | Quantity | | | 1..1 | The temperature value in degrees Fahrenheit [degF]. This represents the same temperature measurement as Observation.valueQuantity, which can use different units.

Observation (R4).extension (template-id) (also: extension (template-id)) | Extension | | Yes | 0..1 | This is the specific flowsheet template (a collection of flowsheet rows and groups) to which data was filed. URL: http://open.epic.com/FHIR/StructureDefinition/extension/template-id

Observation (R4).extension (template-id).valueIdentifier | Identifier | | | 1..1 | The system and value for the flowsheet template.

Observation (R4).extension (template-id).valueIdentifier.system | String | | | 1..1 | The OID for the Epic internal flowsheet template identifier. This varies based on the specific Epic environment.

Observation (R4).extension (template-id).valueIdentifier.value | String | | | 1..1 | The Epic internal identifier of the flowsheet template.

Observation (R4).interpretation | CodeableConcept | | Yes | 0..* | Returns a system of http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation and value of "A" for flowsheet values with an abnormal status. Returns an environment-specific OID system with a value of "SIG" for flowsheet values with a significant status. In Netherlands locales, these interpretation values can also be returned, alongside a system value of http://snomed.info/sct: For heart rate profile Observations, possible values include: SNOMED code 271636001 (Heart regular) for normal heart rate readings SNOMED code 248650006 (Heart irregular) for abnormal heart rate readings For blood glucose profile Observations, possible values include: SNOMED code 281302008 (Above reference range) for blood glucose readings higher than the reference range SNOMED code 11896004 (Intermediate) for blood glucose readings within the reference range (inclusive of the reference range values) SNOMED code 281300000 (Below reference range) for blood glucose readings below the reference range

Observation (R4).issued | instant | | | 1..1 | The date and time the Observation was documented.

Observation (R4).method | CodeableConcept | | | 0..1 | For patient-entered flowsheets, this element is returned with a text value of "Patient-entered". In Netherlands locales with blood pressure vitals Observations, SNOMED (system http://snomed.info/sct) codes can be returned: Blood pressure taken with an invasive method: 10179008 Blood pressure taken with a non-invasive method: 22762002

Observation (R4).note | Annotation | | Yes | 0..1 | The comment associated with the value.

Observation (R4).performer | Reference | | Yes | 1..* | Reference to a FHIR resource for who filed the Observation. If the flowsheet value was filed by a patient, this references a Patient resource. Otherwise, it references a Practitioner resource.

Observation (R4).performer.extension (performerFunction) (also: extension (performerFunction)) | Extension | | Yes | 0..1 | Additional Practitioners associated with the Observation value, if applicable. Possible values include: "data entry person" for Practitioners who transcribed the Observation into the system "verifier" for Practitioners who cosigned the Observation URL: http://hl7.org/fhir/StructureDefinition/event-performerFunction

Observation (R4).referenceRange | BackboneElement | Conditional | Yes | 0..1 | Applicable to organizations in the Netherlands starting in the November 2022 version of Epic. This is populated only for blood glucose readings. This element indicates the abnormal minimum and maximum values defined for the blood glucose reading.

Observation (R4).referenceRange.high | Quantity | | | 0..1 | Applicable to organizations in the Netherlands starting in the November 2022 version of Epic. The higher reference limit for blood glucose readings.

Observation (R4).referenceRange.low | Quantity | | | 0..1 | Applicable to organizations in the Netherlands starting in the November 2022 version of Epic. The lower reference limit for blood glucose readings.

Observation (R4).status | code | | | 1..1 | This element has the following possible values: "preliminary" - value is pended "final" - value has been filed "amended" - value has been edited "entered-in-error" - value has been deleted

Observation (R4).subject | Reference | Required | | 1..1 | The FHIR ID for the patient the Observation is about.

Observation (R4).valueCodeableConcept | CodeableConcept | | | 0..1 | This element is populated if the data type is a category or custom list. Only one of the value* elements is returned.

Observation (R4).valueDateTime | dateTime | | | 0..1 | This element is populated if data type is date. Only one of the value* elements is returned.

Observation (R4).valueQuantity | Quantity | | | 0..1 | This element is populated if a value has units or the data type is numeric. Only one of the value* elements is returned.

Observation (R4).valueString | string | | | 0..1 | This element is populated if a value doesn't have units or the data type is string. Only one of the value* elements is returned.

Observation (R4).valueTime | time | | | 0..1 | This element is populated if data type is time. Only one of the value* elements is returned.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Observation/eF7uc-
    VX129EjpVE35In7ZroUUTPOudM9oOQTmIWiPZS9O3pXRr6snBCvQVadYl6VdkVo.PCeyixt2muU4qpWPQ3
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "eF7uc-VX129EjpVE35In7ZroUUTPOudM9oOQTmIWiPZS9O3pXRr6snBCvQVadYl6VdkVo.PCeyixt2muU4qpWPQ3",
  "extension": [
    {
      "valueIdentifier": {
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.707684",
        "value": "46130"
      },
      "url": "http://open.epic.com/FHIR/StructureDefinition/extension/template-id"
    }
  ],
  "status": "final",
  "code": {
    "coding": [
      {
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.707679",
        "code": "7096230",
        "display": "Precautions"
      },
      {
        "system": "http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id",
        "code": "tFPAQItwnOAotqq9GHG-ZAg0",
        "display": "Precautions"
      }
    ],
    "text": "Precautions"
  },
  "subject": {
    "reference": "Patient/eZO66HeGXB6VCJemL-ciSKA3",
    "display": "Meiko TEST"
  },
  "encounter": {
    "reference": "Encounter/e.R4dWbUDqF01.Z6BEIfEkZecZklxAhL82tS5wzzlD0o3"
  },
  "effectiveDateTime": "2019-09-05T20:00:00Z",
  "issued": "2019-09-05T20:49:27Z",
  "performer": [
    {
      "reference": "Practitioner/euJOUNeZyX6mofWELWG0Ibg3",
      "display": "Starter Provider, MD"
    }
  ],
  "valueCodeableConcept": {
    "coding": [
      {
        "code": "Other (Comment)",
        "display": "Other (Comment)"
      }
    ],
    "text": "Other (Comment)"
  },
  "note": [
    {
      "text": "Example comment"
    }
```

```
  ]
}
```

9.1.9 Observation.Read (Labs) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/998-Observation-Read-Labs-R4.html
Epic API ID | 998
FHIR version | R4
Categories | Clinical Diagnostics, Labs
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | BMI percentile per age and sex for youth 2-20;Weights for age per length and sex;Occipital-frontal circumference for children < 3 years old;Vital Signs;Diastolic blood pressure;Systolic blood pressure;Body height;Body weight;Heart rate;Respiratory rate;Body temperature;Pulse oximetry;Inhaled oxygen concentration;Diagnostic Reports;Imaging;Laboratory;Pathology;Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | GET
URL template | /api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=998

Description

An Observation (Labs) resource can represent component-level, isolate-level, method-level, and sensitivity-level result details, as well as narrative or impression Observations.

Component-level, narrative, or impression Observation references can be found in

DiagnosticReport.Search (Results) (R4)

DiagnosticReport.Read (Results) (R4)

responses.

Isolate-level Observation references can be found in

Observation.Search (Labs) (R4)

responses or Observation.Read (Labs) responses for component-level reads, within the hasMember element. They can also be found in Observation.Read (Labs) responses for sensitivity-level or method-level reads, within the derivedFrom element.

Method-level Observation references can be found in Observation.Read (Labs) responses for isolate-level reads, within the hasMember element, if applicable (see structure outline below). They can also be found in Observation.Read (Labs) responses for sensitivity-level reads, within the derivedFrom element.

Sensitivity-level Observation references can be found in Observation.Read (Labs) responses for isolate-level or method-level reads within the hasMember element, depending on the applicable structure from below.

When processing values, be sure to consider all elements. Classification of value is subject to change from system to system and version to version.

For microbiology results, there are two potential structures:

Microbiology results without methods:

Culture component (Observation)

Isolate A (Observation)

* Sensitivity results for antibiotic X (Observation)

* Sensitivity results for antibiotic Y (Observation)

Isolate B (Observation)

* Sensitivity results for antibiotic X (Observation)

* Sensitivity results for antibiotic Y (Observation)

Microbiology results with methods:

Culture component (Observation)

Isolate A (Observation)

Method 1 (Observation)

* Sensitivity results for antibiotic X (Observation)

* Sensitivity results for antibiotic Y (Observation)

Method 2 (Observation)

* Sensitivity results for antibiotic X (Observation)

* Sensitivity results for antibiotic Y (Observation)

Isolate B (Observation)

Method 1 (Observation)

* Sensitivity results for antibiotic X (Observation)

* Sensitivity results for antibiotic Y (Observation)

Method 2 (Observation)

* Sensitivity results for antibiotic X (Observation)

* Sensitivity results for antibiotic Y (Observation)

Starting in Epic version February 2024, this sub-resource can also read result components linked to Midmark EKG orders.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation (R4) | Observation
Observation (R4).basedOn | Reference | | Yes | 1..* | A ServiceRequest reference for the Epic order associated with the results.
Observation (R4).category | CodeableConcept | | Yes | | The Observation type. All lab results have a category of "laboratory" System: http://terminology.hl7.org/CodeSystem/observation-category Category.code: laboratory Results for imaging orders have a category of "imaging" System: http://terminology.hl7.org/CodeSystem/observation-category Category.code: imaging All results have a category for the procedure order type System: <Order Type OID> (ends in .10.798268.30) Category.code: <Order Type Abbreviation> Additional information: <Order Type OID> and <Order Type Abbreviation> vary by Epic organization. Work with the organization to obtain these values. If doing a Read on a component Observation and the Epic organization has mapped result components to SNOMED codes, they are returned in this element as well System: http://snomed.info/sct Category.code: <SNOMED code> Additional information: For a search to be able to return SNOMED codes for result components, the Epic organization must map the result components to a SNOMED code.
Observation (R4).code | CodeableConcept | | | | The code element value depends on the Obersvation being read: Component: Component name, MPI IDs, and associated LOINC code Isolate: Organism name, SNOMED code, and MPI IDs Method: Method procedure name and MPI IDs Sensitivity: Antibiotic LOINC Code Narrative: "Narrative" Impression: "Impression"
Observation (R4).dataAbsentReason | CodeableConcept | | | 0..1 | The reason why content of the data element is missing. System: http://terminology.hl7.org/CodeSystem/data-absent-reason If a previous result contact on the order had result data but the most current result contact does not: code: "error" text: "Data for specified Observation no longer exists" In patient-facing contexts if the results aren't ready yet, or if the patient is missing security to view results: code: "temp-unknown" text: "Data for specified Observation is temporarily unavailabile" If the result data is missing from an Observation for any other reason: code: "unknown" text: "No value exists"
Observation (R4).derivedFrom | Reference | | Yes | | The derivedFrom element value depends on the Obersvation being read: Component: value is not returned Isolate: value is not returned Method: an Observation reference to the associated Isolate Sensitivity: Observation references to the associated Isolate and Method
Observation (R4).effectiveDateTime | dateTime | | | 1..1 | This represents the most relevant date/time for an order. This is typically the specimen collection date/time for resulted orders.
Observation (R4).encounter | Reference | | | 0..1 | The Encounter FHIR ID associated with the order.
Observation (R4).extension (CopyIndicator) (also: extension (CopyIndicator)) | Extension | | Yes | 0..1 | Indicates whether this response originates from the local system (false) or if it is from an external source (true). URL: http://nictiz.nl/fhir/StructureDefinition/ext-CopyIndicator
Observation (R4).extension (historical-fhir-id) (also: extension (historical-fhir-id)) | Extension | | Yes | 0..1 | This extension contains a historical FHIR ID in cases where the historical ID exceeded the 64-character ID length specified in the HL7 FHIR standard. This extension is only returned if the requesting client is configured to use 64-character-limited FHIR IDs as described in the FHIR ID Generation Scheme topic. This element is returned starting in the November 2025 version of Epic and in August 2025 by special update. URL: http://open.epic.com/FHIR/StructureDefinition/extension/historical-fhir-id
Observation (R4).extension (Lab-e-signature) (also: extension (Lab-e-signature)) | Extension | | Yes | 0..* | A string containing the person(s) who verified the result component and when they did so. This element will not be present in patient-facing contexts. URL: http://open.epic.com/FHIR/StructureDefinition/extension/lab-e-signature
Observation (R4).hasMember | Reference | | Yes | | The hasMember element value depends on the Obersvation being read: Component: Observation references to the associated Isolates Isolate: Observation references to the associated Methods Method: Observation references to the associated Sensitivities Sensitivity: value is not returned
Observation (R4).id | id | | | | The FHIR ID for the Observation being read.
Observation (R4).interpretation | CodeableConcept | | Yes | | The interpretation element value depends on the Obersvation being read: Component: The level of abnormality for the result Isolate: value is not returned Method: value is not returned Sensitivity: the sensitivity of the organism to the antibiotic
Observation (R4).issued | instant | | | | The issued element value depends on the Obersvation being read: Component: Component analysis instant Isolate: value is not returned Method: value is not returned Sensitivity: Sensitivity analysis instant
Observation (R4).meta | Meta | Required | | 1..1 | Metadata common to most resources.
Observation (R4).meta.lastUpdated | String | Required | | 1..1 | The instant that the result was last updated.
Observation (R4).method | CodeableConcept | | | | The method element value depends on the Obersvation being read: Component: value is not returned Isolate: value is not returned Method: the SNOMED code mapped to the method procedure and the name of the method procedure Sensitivity: the SNOMED code mapped to the method procedure and the name of the method procedure
Observation (R4).note | Annotation | | Yes | 0..* | Comments about the result.
Observation (R4).performer | Reference | | Yes | 0..* | References to the Practitioner FHIR ID for the resulting agency director and the Organization FHIR ID for the resulting lab.
Observation (R4).referenceRange | BackboneElement | | Yes | 0..* | The reference range for the result, if applicable.
Observation (R4).referenceRange.high | Quantity | | | 0..1 | The high value of the reference range.
Observation (R4).referenceRange.low | Quantity | | | 0..1 | The low value of the reference range.
Observation (R4).referenceRange.text | String | | | 0..1 | Display version of reference range.

Observation (R4).specimen | Reference | | | 0..1 | A reference to the Specimen resource.

Observation (R4).status | code | | | 1..1 | The lab status of the associated order or component. Possible values are: registered preliminary final amended corrected unknown entered-in-error cancelled pending (for organizatons in the Netherlands only) If a test was performed but ended in an unsuccessful attempt, the status will be cancelled. For example, if a mammography is performed, and the patient was exposed to radiation, and images were captured, but the images were too blurry to make a clinical determination, that would be considered an unsuccessful attempt.

Observation (R4).subject | Reference | | | 1..1 | The Patient FHIR ID linked to the order.

Observation (R4).valueCodeableConcept | CodeableConcept | | | 0..1 | The SNOMED code for the organism associated with the result, when applicable.

Observation (R4).valueQuantity | Quantity | | | 0..1 | The result value if the result is numeric and represents a quantity.

Observation (R4).valueRange | Range | | | 0..1 | The result value if the result is structured numeric and the operator is "-".

Observation (R4).valueRatio | Ratio | | | 0..1 | The result value if the result is structured numeric and the operator is ":".

Observation (R4).valueString | string | | | 0..1 | The result value if the result is a string, or the text from the narrative or impression.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Observation/eWilfwxg9QID48NN3HNHxVhG8H0eLgpAhiK7YzVHn.oVJvO5ZhL4KgFHVctbztVEor
    bsjpSDX-4sgAeTsTUsIjw3
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "eWilfwxg9QID48NN3HNHxVhG8H0eLgpAhiK7YzVHn.oVJvO5ZhL4KgFHVctbztVEorbsjpSDX-4sgAeTsTUsIjw3",
  "meta": {
    "lastUpdated": "2022-04-29T21:54:13Z"
  },
  "basedOn": [
    {
      "reference": "ServiceRequest/eBLjU8nQwwFY4BK83wFL5scqCtdl6Qao5ETVzpO.VW2k3",
      "display": "Blood culture"
    }
  ],
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "19851009",
          "display": "Microbiology procedure (procedure)"
        }
      ],
      "text": "Microbiology procedure (procedure)"
    },
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "laboratory",
          "display": "Laboratory"
        }
      ],
      "text": "Laboratory"
    },
    {
      "coding": [
        {
          "system": "urn:oid:1.2.840.114350.1.13.11511.1.7.10.798268.30",
          "code": "Microbiology",
          "display": "Microbiology"
        }
      ],
      "text": "Microbiology"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "urn:oid:1.2.840.114350.1.13.11511.1.7.2.768282",
        "code": "1802008",
        "display": "Blood Culture"
      }
    ],
    "text": "Blood Culture"
  },
  "subject": {
    "reference": "Patient/eUEKdnYPKuCXlSq8WIYaPTA3",
    "display": "Bugsy, Beau Dakota"
  },
  "encounter": {
    "reference": "Encounter/eKvkb8V4sgTKL9nm3MtDjXw3",
    "identifier": {
      "use": "usual",
      "system": "urn:oid:1.2.840.114350.1.13.11511.1.7.3.698084.8",
      "value": "40282"
    },
```

```
    "display": "Hospital Encounter"
  },
  "effectiveDateTime": "2022-04-29T21:53:00Z",
  "issued": "2022-04-29T21:54:13Z",
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "115329001"
      }
    ],
    "text": "Methicillin-Resistant Staphylococcus aureus"
  },
  "interpretation": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation",
          "code": "AA",
          "display": "Panic"
        }
      ],
      "text": "Panic"
    }
  ],
  "note": [
    {
      "text": "\r\n\r\nMethicillin-Resistant Staph aureus, patient may be an isolation risk."
    }
  ],
  "specimen": {
    "reference": "Specimen/eNKpY.ifDjG-raaJh60tz5w3",
    "display": "Specimen 22H-119M0006"
  },
  "hasMember": [
    {
      "reference": "Observation/eWilfwxg9QID48NN3HNHxVrWBVj0Kcd1GxT6JJOF9D8.NgYuvcuOUZTFDiDdanohh3",
      "display": "Isolate (1): STAPHYLOCOCCUS AUREUS METHICILLIN (OXACILLIN) RESISTANT"
    }
  ]
}
```


Change log

Epic Version | Description | Type
November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4
ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90
ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94
ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

9.1.10 Observation.Read (Lines, Drains, Airways) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/965-Observation-Read-Lines-Drains-Airways-R4.html
Epic API ID | 965
FHIR version | R4
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Keywords | Flowsheets;LDA
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | GET
URL template | api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=965

Description

The FHIR Observation resource defines measurements and assertions about a patient, including LDA-W (lines, drains, airways, and wound) data.

Request parameters

Name | Type | Optional | Array | Cardinality | Description

ID | token | False | | | The read interaction uses the given Observation FHIR ID to retrieve the most up-to-date information about a LDA (details and properties) or a LDA assessment.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Observation | Observation | False | | | An instance of the R4 Observation resource.

Observation.category | CodeableConcept | False | Yes | | The category is always "LDA". Starting in February 2022, this element also always returns a second category which clarifies the type of LDA data returned in the response: The category “LDA-property” is used for LDA properties, which are documented only once. For example, a nurse might insert a line and document properties such as the insertion time, the body site, and the method of insertion. The category “LDA-assessment” is used for LDA assessments, which are performed over time. For example, a nurse might inspect an inserted line regularly to make sure the site is clean or assess a wound to see if a new dressing should be applied. This element can't be modified by Observation.Update.

Observation.code | CodeableConcept | False | | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. This element can't be modified by Observation.Update. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.

Observation.component | BackboneElement | True | Yes | | Populated only for patient-entered flowsheets.

Observation.component.code | CodeableConcept | False | | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. This element can't be modified by Observation.Update. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.

Observation.component.interpretation | CodeableConcept | True | Yes | | Available starting in the February 2022 version of Epic. Possible values include: “Abnormal” for observation values outside the reference range or expected norm “Significant” for observation values that a clinician has manually flagged for extra attention

Observation.component.extension (also: LDA-Property-Comment) | Extension | True | Yes | | The comment associated with the LDA property observation value, if applicable.

Observation.component.valueCodeableConcept | CodeableConcept | False | | | This element is populated if the data type is category or custom list. Only one of the value* elements is required.

Observation.component.valueDateTime | String | False | | | This element is populated if the data type is category or custom list. Only one of the value* elements is required.

Observation.component.valueQuantity | Quantity | False | | | This element is populated if a value has units or the data type is numeric or a percentile. Only one of the value* elements is required.

Observation.component.valueString | string | False | | | This element is populated if a value doesn't have units or the data type is string. Only one of the value* elements is required.

Observation.component.valueTime | String | False | | | This element is populated if the data type is time. Only one of the value* elements is required.

Observation.effectiveDateTime | dateTime | True | | | This is a required element for LDA assessment observations. The date and time the LDA assessment was taken. This element can't be modified by Observation.Update.

Observation.effectivePeriod | Period | True | | | Start instant and/or end instant of the LDA. This element is only returned with LDA properties.

Observation.encounter | Reference | False | | | Reference to an encounter resource. This element can't be modified by Observation.Update.

Observation.interpretation | CodeableConcept | True | Yes | | Available starting in the February 2022 version of Epic. Possible values include: “Abnormal” for observation values outside the reference range or expected norm “Significant” for observation values that clinician has manually flagged for extra attention

Observation.issued | instant | True | | | This is a required element for LDA assessment observations. The date and time the observation was documented. This is always the current time.

Observation.note | Annotation | True | Yes | | The comment associated with the LDA assessment observation value, if applicable.

Observation.performer | Reference | True | Yes | | This is a required element for LDA assessment observations. The practitioner who filed the LDA assessment observation.

Observation.performer.extension (also: event-performerFunction) | Extension | True | Yes | | Available starting in the February 2022 version of Epic for LDA assessments. Additional practitioners associated with the LDA assessment observation value, if applicable. Possible values include: “data entry person” for practitioners who transcribed the observation into the system “verifier” for practitioners who cosigned the observation

Observation.extension (also: sequelTo) | Extension | True | Yes | | This is a required element for LDA assessment observations. sequelTo extension that is a reference to the parent LDA FHIR ID.

Observation.status | code | False | | | The status of the observation. The status is always "final" for LDA properties. Starting in February 2022, possible values for LDA assessments include: “final” if the data is filed at time of retrieval “preliminary” if the data has not yet been filed at the time of retrieval “amended” if the data has been modified at time of retrieval “entered-in-error” if the data had once existed, but has since been withdrawn at time of retrieval

Observation.subject | Reference | False | | | Reference to a patient resource. This element can't be modified by Observation.Update.

Observation.extension (also: supportingInfo) | Extension | True | Yes | | Available starting in the February 2022 version of Epic for LDA assessments. If an LDA assessment is linked to a note written by a clinician, this is a reference to the DocumentReference resource.

Observation.valueCodeableConcept | CodeableConcept | False | | | In LDA assessments, this element is populated if the data type is category or custom list. Only one of the value* elements is required.

Observation.valueDateTime | dateTime | False | | | In LDA assessments, this element is populated if the data type is date. Only one of the value* elements is required.

Observation.valueQuantity | Quantity | False | | | In LDA assessments, this element is populated if a value has units or the data type is numeric or a percentile. Only one of the value* elements is required.

Observation.valueString | string | False | | | In LDA properties, this is a required element which returns the LDA display name. In LDA assessments, this element is populated if a value doesn't have units or if the data type is string. For image rows, a comma-delimited list of external document identifiers is returned. For assessments that include an image that does not have external document identifiers, "Images linked" is returned. Only one of the value* elements is required.

Observation.valueTime | time | False | | | In LDA assessments, this element is populated if the data type is time. Only one of the value* elements is required.

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation/ehkjaNxykIByHIGRfK.dE67Wy8U5t9uYfVbCGnje260i7MQONTfGBM9nShk5Rf0z.g.Bau
    7vlKK4h704ls5sNHP2HmyLqalARyCORLLS1Zsc3
```

Sample response

```
<Observation xmlns="http://hl7.org/fhir">
  <id
    value="ehkjaNxykIByHIGRfK.dE67Wy8U5t9uYfVbCGnje260i7MQONTfGBM9nShk5Rf0z.g.Bau7vlKK4h704ls5sNHP2HmyLqalARyCORLLS1Zsc3
    "/>
  <extension url="http://hl7.org/fhir/StructureDefinition/observation-sequelTo">
    <valueReference>
      <reference value="Observation/eFPRBpmtzuDY7Twf3aBdgtg3"/>
      <display value="Peripheral IV 01/17/22 Right Hand"/>
    </valueReference>
  </extension>
  <status value="final"/>
  <category>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/observation-category"/>
      <code value="LDA"/>
      <display value="Line, Drains, and Airways"/>
    </coding>
    <text value="Line, Drains, and Airways"/>
  </category>
  <category>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/observation-category"/>
      <code value="LDA-assessment"/>
      <display value="Lines, Drains, and Airways Assessment"/>
    </coding>
    <text value="Lines, Drains, and Airways Assessment"/>
  </category>
  <code>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679"/>
      <code value="29550304840"/>
      <display value="Site Assessment"/>
    </coding>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id"/>
      <code value="tCdGNZ80BNoU0xFUxE7pFrQ0"/>
      <display value="Site Assessment"/>
    </coding>
    <text value="Site Assessment"/>
  </code>
  <subject>
    <reference value="Patient/eBFGq5CRApF1O-fv-r-9.G1kkDtJCFhZQm1YRU1ZAuuE3"/>
    <display value="Smith, John"/>
  </subject>
  <encounter>
    <reference value="Encounter/eymC2HHWDr3EXnMapj-LyHFgF.zx2pxKAengd7SoBaRE3"/>
    <display value="Office Visit"/>
  </encounter>
  <effectiveDateTime value="2022-01-17T21:24:00Z"/>
  <issued value="2022-01-17T21:26:36Z"/>
  <performer>
    <reference value="Practitioner/eRhB5No2Z8P6jEmZfgaqHvg3"/>
    <display value="Becky Lipsitz, MD"/>
  </performer>
  <performer>
    <extension url="http://hl7.org/fhir/StructureDefinition/event-performerFunction">
      <valueCodeableConcept>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
          <code value="VRF"/>
          <display value="verifier"/>
        </coding>
      </valueCodeableConcept>
    </extension>
    <reference value="Practitioner/e4PZNte6MtBGVltgJ30CD7A3"/>
    <display value="Mary Hale, RN"/>
  </performer>
  <performer>
    <extension url="http://hl7.org/fhir/StructureDefinition/event-performerFunction">
      <valueCodeableConcept>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
          <code value="ENT"/>
          <display value="data entry person"/>
        </coding>
      </valueCodeableConcept>
    </extension>
    <reference value="Practitioner/eB8EFBuK79tGFEZphFXeprg3"/>
    <display value="Henry Bratcher, MD"/>
  </performer>
  <valueCodeableConcept>
    <coding>
```

```
      <code value="Red;Tender"/>
      <display value="Red;Tender"/>
    </coding>
    <text value="Red;Tender"/>
  </valueCodeableConcept>
  <interpretation>
    <coding>
      <system value="http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"/>
      <code value="A"/>
      <display value="Abnormal"/>
    </coding>
  </interpretation>
</Observation>
```

Change log

Epic Version | Description | Type
August 2025 (May 2025, February 2025, November 2024 by special update) | Starting in August 2024, May 2025 with special update E11400862, February 2025 with special update E11306699, and November 2024 with special update E11211328, multi-select fields are handled separately, and if a SNOMED concept mapping exists for any individual selection, the SNOMED concept mapping is included in the FHIR response. Previously, when a user selected multiple options, the system treated the entire list as a single combined value. | Substantive Changes

9.1.11 Observation.Read (SmartData Elements) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/10120-Observation-Read-SmartData-Elements-R4.html
Epic API ID | 10120
FHIR version | R4
Categories | Clinical Diagnostics
API type | USCDI
A la carte | Industry-Standard Level 2 API
USCDI | Yes
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
HTTP method | GET
URL template | /api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10120

Description

This FHIR Observation resource returns data stored in SmartData elements with the following datatypes:

* string

* number

* date

* time

* category

* database

* element ID

* time with unit

* instant

Image and hierarchy datatypes are not returned.

SmartData elements are used to store discrete data about a patient and can be used in a wide variety of situations to store data related to the patient, a specific encounter, a note, an order, or another context.

Only SmartData elements stored in the following contexts are supported by this resource:

* Encounter

* Episode

* Note

* Order

* Organ

* Patient

* Stage

* Result

* Condition

* Problem

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | | A single instance of the Observation resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation (R4) | Observation
Observation (R4).category | CodeableConcept | | Yes | 1..1 | Always includes "smartdata" coding. Additional codings of "treatment-intervention-preference" and "care-experience-preference" can be included if the element qualifies as treatment intervention preference data or care experience preference data accordingly.
Observation (R4).code | CodeableConcept | | | 1..* | The codes that represent the type of data returned. This can include a SmartData element ID or SNOMED code. Starting in the August 2024 version of Epic, LOINC codes are also supported.
Observation (R4).component | BackboneElement | | Yes | 1..* | Additional information. The number of components returned equals the number of lines in the SmartData element.
Observation (R4).component.code | CodeableConcept | | | 1..1 | Information about the SmartData element including the element reference and display string. If a SNOMED mapping for the SmartData element exists, that mapping appears here as well. The code element value varies depending on what SmartData element is passed. The system value is different between Epic organizations as well as between production and non-production environments for the same organization.
Observation (R4).component.code.text | String | | | 1..1 | Returns the line number for the component within a SmartData element.
Observation (R4).effectiveDateTime | dateTime | | | 0..1 | Clinically relevant time/time-period for observation.
Observation (R4).encounter (also: encounter) | Reference | | | | A reference to the Encounter during which this SmartData Observation was documented. Only returned for Observations recorded in an encounter context.
Observation (R4).focus | Reference | | Yes | 1..* | The FHIR ID for the context record that the SmartData element is attached to.
Observation (R4).hasMember | Reference | | Yes | 0..* | Returns nested references to other Observation (SmartData Element) resources.
Observation (R4).issued | instant | | | 1..1 | The date and time the Observation was documented.
Observation (R4).note | Annotation | | Yes | 1..* | The comments for the SmartData element. If the component and comment are saved to separate SmartData elements, the comment is returned as a separate Observation and not as a note on the component's Observation.
Observation (R4).performer | Reference | | Yes | 1..* | A reference to the Practitioner resource for the user who filed the Observation.
Observation (R4).status | code | | | 1..1 | The value is "unknown" unless the data corresponds to a treatment intervention preference or a care experience preference, in which case the value is "final".
Observation (R4).subject | Reference | | | 1..1 | The FHIR ID for the patient the Observation is about.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Observation/eBBGgszv54EfAOrdWJqYDgbqMbaIyAua1WztUUZpc720A31rLzxozrJYKb9cFnbur-
    NbpRG6U8IlGmAEUpRZi5k7UpVT9imMQsLB.NDBYxpY3
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "eBBGgszv54EfAOrdWJqYDgbqMbaIyAua1WztUUZpc720A31rLzxozrJYKb9cFnbur-NbpRG6U8IlGmAEUpRZi5k7UpVT9imMQsLB.NDBYxpY3",
  "status": "unknown",
  "category": [
    {
      "coding": [
        {
          "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
          "code": "smartdata",
          "display": "SmartData"
        }
      ],
      "text": "SmartData"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.727688",
        "code": "A32881",
        "display": "STRING"
      }
    ],
    "text": "STRING"
  },
  "subject": {
    "reference": "Patient/eVe.LNtp0U.2HjbCdUxvA5g3",
    "display": "zzFhir, FhirGirl"
  },
  "focus": [
    {
      "reference": "Encounter/eSFXQ9FKvuZndGegQibaWniVhd0H4y.4AMYETyo5A6gc3",
      "display": "zzFhir, FhirGirl"
    }
  ],
  "issued": "2021-04-16T20:56:23Z",
  "performer": [
    {
      "reference": "Practitioner/eIOnxJsdFcfgJ4iy716I3MA3",
      "display": "USER, EPIC"
    }
```

```
  ],
  "component": [
    {
      "code": {
        "text": "Line 1"
      },
      "valueString": "FhirGirl"
    }
  ]
}
```


Change log

Epic Version	Description	Type
August 2024	FHIR IDs for Observation (SmartData Elements) (R4) change to be 64 characters or fewer.	Breaking change
February 2025	Observation (SmartData Elements) (R4) returns SNOMED codes without the SNOMED# prefix.	Breaking change

9.1.12 Observation.Read (Study Finding) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/10289-Observation-Read-Study-Finding-R4.html

Epic API ID	10289
FHIR version	R4
Categories	Clinical Diagnostics
API type	USCDI
A la carte	Industry-Standard Level 1 API
USCDI	Yes
Available in inpatient and other contexts	Yes
HTTP method	GET
URL template	/api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0	True
Supported user types	Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation	False
Source	https://fhir.epic.com/Specifications?api=10289

Description

This web service retrieves study result data from procedures, such as fetal and maternal measurement data from an obstetric ultrasound, or myocardial finding data from a cardiology study.

Starting in May 2024, this web service can also retrieve study result data from audiology exams and follicle measurement data from a follicular ultrasound for fertility treatments.

Response elements

Name	Type	Optional	Array	Cardinality	Description
Observation	Observation	False			An instance of the R4 Observation resource.
Observation.basedOn	Reference	False	Yes		Order service request.
Observation.bodySite	CodeableConcept	False			Contains the SNOMED code that identifies where on the body the study was performed.
Observation.category	CodeableConcept	False	Yes		This is always "study-finding." Imaging observations also have the “imaging” category. Audiology studies also have the “exam” and “audiology” category values.
Observation.code	CodeableConcept	False			What type of study finding this is. If this is an Estimated Fetal Weight study finding documented on an obstetric ultrasound, then this element also includes LOINC code 11727-5.
Observation.component	BackboneElement	False	Yes		Populated only for patient-entered flowsheets.
Observation.effectiveDateTime	dateTime	False			The date and time the exam ended.
Observation.encounter	Reference	False			Encounter associated with this observation value.
Observation.focus	Reference	True	Yes		The specific fetus the observation this is about. Epic currently only returns this element if the patient has data on a fetus documented in an ultrasound study in their medical chart.
Observation.hasMember	Reference	False	Yes		SmartData element documented on the procedure. Links to its own observation that contains more specific data about the SmartData element.
Observation.issued	instant	False			The date and time the observation was documented.
Observation.method	CodeableConcept	True			Method by which the observation was obtained. For an Estimated Fetal Weight study finding, this element returns the method used to determine the Estimated Fetal Weight.
Observation.referenceRange	BackboneElement	True	Yes		Text-based reference range for the observation.
Observation.status	code	False			The status of the study. This should be set to final.
Observation.subject	Reference	False			The FHIR ID for the patient the observation is about.
Observation.valueQuantity	Quantity	True			This element is populated if a value has units or the data type is numeric or a percentile.

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation/eJFgryT-06nUPbHLVFnpCw2igXjN97CZNFknSMjC5dKXF0uuTajDHLRZO.qy4c2Z83
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "eJFgryT-06nUPbHLVFnpCw2igXjN97CZNFknSMjC5dKXF0uuTajDHLRZO.qy4c2Z83",
  "extension": [
    {
```

```
    "valueReference": {
      "reference": "Observation/enQcDqvmReQd8FpmQFhQnqqA6pUFMqIc7o4tpiuAkTlPRxfWuubz7daE48l-1DPcsHmpN-
  71wLtBoAUcVOrOfOWFRfDXjX83iOeyiuM3O7MCCxBuAT1AEI3zQh4OI9QV1I2bm1q39h1RLUjTe31jG4w3",
      "display": "FINDINGS - TESTS - OB - ULTRASOUND - MOM-SPECIFIC - GESTATIONAL SAC"
    },
    "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
  },
  {
    "valueReference": {
      "reference": "Observation/esB6B.SD-hvSTfBjzYYOTp.v4LcgfBPaYOfLLVs3L3IAKohu2YKidZl4fdONqUVXj1MjUu0DFgUnlAGgx7420Q
  uPkzj1OSwIDjKbosJgEsWNPwowUoFN9WlYhWO6kIylBi41tPZqypMU-920SHuzhTA3",
      "display": "FINDINGS - TESTS - OB - ULTRASOUND - MOM-SPECIFIC - PLACENTA"
    },
    "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
  },
  {
    "valueReference": {
      "reference": "Observation/eik4yRQDgDm5wt8NsTyFB3dHj8n1qOk-Ena7e4wy-ofTSlCdl-BblYOGQCi0a3c1IgOeaUP-
  ukUFtAVORfH.nsbNhIkiNevLD4ee0QdmM8E8WgKuK5pEv22TL2Pr5gFZYU1HuMR4mcVmyq6JQ2vFgwg3",
      "display": "FINDINGS - TESTS - OB - ULTRASOUND - BABY-SPECIFIC - GESTATIONAL SAC"
    },
    "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
  },
  {
    "valueReference": {
      "reference": "Observation/eu-xLmCAr2qveOdu7-1IMDN.8RjAHquZg74nl.vwjgvERBu5BCwNnJFH.Lcm3cWH6Du2KZ-
  4DyTvKxxm4hX4hiAX8.-e9ufatl4LzUL5WefdkGultMFx1llj8.iVoC.C33zImchQyigOemvhC99uZ3Q3",
      "display": "FINDINGS - TESTS - OB - ULTRASOUND - BABY-SPECIFIC - YOLK SAC"
    },
    "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
  },
  {
    "valueReference": {
      "reference": "Observation/eoJiEtTMWX1L5nJLGzWvKMzU4tN99Zkpn7uzQ9yEu9p4Bj7gPRbxR685MpACniF9sHGDidYaBjOFlkoQRv3b.3
  jGT9EVqqYpaOawqSs2dj3w6e.wSFHjdsIVxtkLhQ8dn1KCjYIOpDkZWvZv64pkDbw3",
      "display": "FINDINGS - TESTS - OB - ULTRASOUND - BABY-SPECIFIC - PRESENTATION"
    },
    "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
  },
  {
    "valueReference": {
      "reference": "Observation/ewih1pOtcFiisTOOGsvavqITupGQ2VBKJlLqjX2qtjtT65nZv9JE6fnMpm1PNeU9E1KAEy7AlLB4uMQ-
  5fGudZWN80fFv0Auy7fynanPEy7En3PP2i6SrLNJ3ikxk19lLXw3SyGXgJqDHhkkoa6mgdw3",
      "display": "FINDINGS - TESTS - OB - ULTRASOUND - BABY-SPECIFIC - FETAL POLE"
    },
    "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
  },
  {
    "valueReference": {
      "reference": "Observation/e1lhXf37485X8FKVLjJ0yn1x356sxIvIQq1jjHtltCFkzggyzMcUbMdNJ-Q26HB.MRJZhdw-
  CYVahAPsyR569.1T8TUUVAbAltY2K6jQZgtCboLndTN3LoYoK-20BgzyG3VOp0SZ11eitBktFJd4mIA3",
      "display": "FINDINGS - TESTS - OB - ULTRASOUND - BABY-SPECIFIC - CARDIAC ACTIVITY"
    },
    "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
  },
  {
    "valueReference": {
      "reference": "Observation/eMYLKSfTIZEpouCOo2w1VmpOZFtRwh.O5AOc8FcYtr9WWJGN3z9g2BF.hwlzj8trk5rfrViXNArWzz88MAmy1q
  e67YmwKGfL8GZ08bJWHG0DcCdWrcuBEihzppYXN4imBFUspuNy4gBFLXL2J.uX0kg3",
      "display": "FINDINGS - TESTS - OB - ULTRASOUND - BABY-SPECIFIC - AMNIOTIC FLUID"
    },
    "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
  }
],
"basedOn": [
  {
    "reference": "ServiceRequest/eOS.q.8Pg0GxH6e3weglmO7Wxqv.bf9-vS64qqOw08e03",
    "display": "OB US + 14 WEEKS"
  }
],
"status": "final",
"category": [
  {
    "coding": [
      {
        "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
        "code": "study-finding",
        "display": "Study Finding"
      }
    ],
    "text": "Study Finding"
  }
],
"code": {
  "text": "Fetus A: Fetal Anatomy"
},
"subject": {
  "reference": "Patient/epzKvfQ.SNFEmP3QlFwh6DcSbNRycEOOdMejDcAvTiFI3",
  "display": "Simota, Noodle"
```

```
  },
  "focus": [
    {
      "identifier": {
        "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.826983",
        "value": "1001127844"
      },
      "display": "Fetus A"
    }
  ],
  "encounter": {
    "reference": "Encounter/eLcG44-s8Jg6QtUOoGqD62BWGbgGCtXxf50ZQrK-h8.E3",
    "display": "Office Visit"
  },
  "effectiveDateTime": "2021-09-17T14:30:11Z",
  "issued": "2021-09-17T14:32:58Z"
}
```


Change log

Epic Version	Description	Type
February 2024	SmartData Elements in the Observation (Study Finding) (R4) Resource moved from extension to hasMember	Breaking change

9.1.13 Observation.Read (Vital Signs) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/968-Observation-Read-Vital-Signs-R4.html	
Epic API ID	968
FHIR version	R4
Categories	Clinical Diagnostics
API type	USCDI
A la carte	Industry-Standard Level 1 API
USCDI	Yes
Keywords	Flowsheets
Available in inpatient and other contexts	Yes
ONC certification products	EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions	November 2019
HTTP method	GET
URL template	api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0	True
Supported user types	Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation	False
Source	https://fhir.epic.com/Specifications?api=968

Description

This web service retrieves vital sign data from a patient's chart, as well as any other non-duplicable data found in the patient's flowsheets across all encounters. This web service can only be queried using an Observation FHIR ID.

The following types of vital information and corresponding LOINC codes are supported by default. How specific types of data translate from within the Epic system to FHIR might vary depending on organizational configuration:

* 2708-6 Oxygen saturation in arterial blood

* 2710-2 Oxygen saturation in capillary blood by oximetry

* 29463-7 Body weight

* 3141-9 Body weight measured

* 3150-0 Inhaled oxygen concentration

* 3151-8 Inhaled oxygen flow rate

* 59408-5 Oxygen saturation in arterial blood by pulse oximetry (SpO2)

* 8287-5 Head occipital-frontal circumference by tape measure

* 8302-2 Body height

* 8306-3 Body height --lying

* 8310-5 Body temperature

* 8462-4 Diastolic blood pressure

* 8480-6 Systolic blood pressure

* 85354-9 Blood pressure systolic and diastolic

* 8716-3 Vital signs

* 8867-4 Heart rate

* 9279-1 Respiratory rate

Remarks

To see an example of how the component.code element is structured for patient-entered flowsheet data, refer to the sample request section of the Observation.Create (Vitals) (STU3) API entry.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation (R4) | Observation
Observation (R4).bodySite | CodeableConcept | Conditional | | 0..1 | Applicable only to organizations in the Netherlands. The site on the body where the Observation was made. Used for the location where a blood pressure reading was obtained.
Observation (R4).category | CodeableConcept | | Yes | 1..1 | This element always returns "vital-signs".
Observation (R4).code | CodeableConcept | Required | | 0..* | Coded value for what is being measured. The following code systems can be returned: OID for internal Epic flowsheet ID Encoded value of the internal Epic flowsheet ID External concept mappings such as LOINC and SNOMED
Observation (R4).component | BackboneElement | | Yes | 0..* | Includes extra information about the Observation. Populated only for: Blood Pressure: Systolic (LOINC 8480-6) and diastolic (LOINC 8462-4) pressures are always returned as components if available. Can include components with a SNOMED code of 70665002 and LOINC code of 8358-4 for the cuff type or size, and a SNOMED code of 424724000 and a LOINC code of 8361-8 for cuff position. Average Blood Pressure (starting in the November 2026 version of Epic): Average systolic (LOINC 96608-5) and average diastolic (LOINC 96609-3) are always returned as components if available. Can include a component with a SNOMED code of 246432004 for number of occurrences. SpO2 can include components with LOINC code of 3151-8 for inhaled oxygen flow rate, 3150-0 for inhaled oxygen concentration, and 74206-4 for whether oxygen therapy was given. Pulse Rate can include a component with LOINC code of 44969-4 for heart rate rhythm palpitation. Weight can include a component with a code of 8352-7 for how the patient was clothed for weight reading. Patient-Entered Flowsheet Metadata Represents strict metadata if the coding sub-element is populated. The following strict metadata can be returned: 1-UUID The unique reading ID 2-Blood Glucose Meal Times 1-Before meal 2-After meal 3-Insulin Delivery Reason 1-Basal 2-Bolus 5-VO2 Max Test Type 1-Max exercise test 2-Sub-max exercise prediction 3-Non exercise prediction 6-Manually Entered - Whether or not the patient manually entered this reading into the system (as opposed to automatically entered by a device). false true Represents loose metadata if only the text sub-element is populated. UUID, Blood Glucose Meal times, Insulin Delivery Reason, and VO2 Max Test Type must have their value specified in component.valueString. Manually Entered is specified as a valueBoolean.
Observation (R4).component.code | CodeableConcept | | | 0..1 | Code for the extra information for the measurement. The following extra information can be returned for each measurement, with the associated code: Blood Pressure 8480-6: Systolic pressure 8462-4: Diastolic pressure SNOMED 70665002 / LOINC 8358-4: Cuff type or size SNOMED 424724000 / LOINC 8361-8: Cuff position Average Blood Pressure (starting in the November 2026 version of Epic) 96608-5: Systolic blood pressure mean 96609-3: Diastolic blood pressure mean SNOMED 246432004: Number of occurrences SpO2 3151-8: Inhaled oxygen flow rate 3150-0: Inhaled oxygen concentration 74206-4: Whether oxygen therapy was given Pulse Rate 44969-4: Heart rate rhythm palpitation. Weight 8352-7: Whether patient was clothed
Observation (R4).component.valueBoolean | boolean | | | 0..1 | Only used for Patient-Entered Flowsheet Metadata values. Valid values include: For Code = 6-Manually Entered false true
Observation (R4).component.valueCodeableConcept | CodeableConcept | | | 0..1 | The value as a CodeableConcept for the extra information for the measurement.
Observation (R4).component.valueQuantity | Quantity | | | 0..1 | If the extra information for the measurement is numeric, this is the numeric value. For example, it could be the flow rate (L/min) or concentration (%) for an SpO2 measurement. Starting in the November 2026 version of Epic, this is also the number of occurrences for average blood pressure.
Observation (R4).component.valueString | string | | | 0..1 | Values for the patient-entered flowsheet metadata.
Observation (R4).dataAbsentReason | CodeableConcept | | | 0..1 | Indicates that an Observation value is missing. This is always "unknown" for systolic blood pressure values in Observations for diastolic-only blood pressure flowsheets, or for diastolic blood pressure value in Observations for systolic-only blood pressure flowsheets.
Observation (R4).effectiveDateTime | dateTime | | | 0..1 | The date and time the Observation was taken. Starting in the November 2026 version of Epic, this element is omitted when effectivePeriod is used.
Observation (R4).effectivePeriod | Period | | | 0..1 | Starting in the November 2026 version of Epic, average blood pressure observations can use this element to return the begin and end instants for the average as an effective time range. If effectivePeriod is used, then effectiveDateTime is omitted.
Observation (R4).encounter | Reference | Conditional | | 1..1 | Reference to an Encounter resource that this Observation belongs to.
Observation (R4).extension (supportingInfo) (also: extension (supportingInfo)) | Extension | Conditional | Yes | 0..* | If the measurement is linked to a note written by a clinician, this returns a reference to the DocumentReference resource. URL: http://hl7.org/fhir/StructureDefinition/workflow-supportingInfo
Observation (R4).extension (temperature-in-fahrenheit) (also: extension (temperature-in-fahrenheit)) | Extension | | Yes | 0..1 | If the measurement is a temperature value, this element contains the temperature value quantity in Fahrenheit. This element is excluded for other data types. URL: http://open.epic.com/FHIR/STU3/StructureDefinition/temperature-in-fahrenheit
Observation (R4).extension (temperature-in-fahrenheit).valueQuantity | Quantity | | | 1..1 | The temperature value in degrees Fahrenheit [degF]. This represents the same temperature measurement as Observation.valueQuantity, which can use different units.
Observation (R4).extension (template-id) (also: extension (template-id)) | Extension | | Yes | 0..1 | Starting in the May 2024 version of Epic, or February 2024 with a special update, this is the specific flowsheet template (a collection of flowsheet rows and groups) to which data was filed. URL: http://open.epic.com/FHIR/StructureDefinition/extension/template-id
Observation (R4).extension (template-id).valueIdentifier | Identifier | | | 1..1 | The system and value for the flowsheet template.
Observation (R4).extension (template-id).valueIdentifier.system | String | | | 1..1 | The OID for the Epic internal flowsheet template identifier. This varies based on the specific Epic environment.
Observation (R4).extension (template-id).valueIdentifier.value | String | | | 1..1 | The Epic internal identifier of the flowsheet template.
Observation (R4).interpretation | CodeableConcept | | Yes | 0..* | Returns a system of http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation and value of "A" for vitals with an abnormal status. Returns an environment-specific OID system with a value of "SIG" for vitals with a significant status. In Netherlands locales starting in the November 2022 release of Epic, these interpretation values can also be returned, alongside a system value of http://snomed.info/sct: For heart rate profile Observations, possible values include: SNOMED code 271636001 (Heart regular) for normal heart rate readings SNOMED code 248650006 (Heart irregular) for abnormal heart rate readings For blood glucose profile Observations, possible values include: SNOMED code 281302008 (Above reference range) for blood glucose readings higher than the reference range SNOMED code 11896004 (Intermediate) for blood glucose readings within the reference range (inclusive of the reference range values) SNOMED code 281300000 (Below reference range) for blood glucose readings below the reference range

Observation (R4).issued | instant | | | 1..1 | The date and time the Observation was documented.

Observation (R4).method | CodeableConcept | | | 0..1 | For patient-entered flowsheets, this element is returned with a text value of "Patient-entered". In Netherlands locales with blood pressure vitals Observations, SNOMED (system http://snomed.info/sct) codes can be returned: Blood pressure taken with an invasive method: 10179008 Blood pressure taken with a non-invasive method: 22762002

Observation (R4).note | Annotation | | Yes | 0..1 | The comment associated with the value.

Observation (R4).performer | Reference | | Yes | 1..* | Reference to a FHIR resource for who filed the Observation. If the vital was filed by a patient, this references a Patient resource. Otherwise, it references a Practitioner resource.

Observation (R4).performer.extension (performerFunction) (also: extension (performerFunction)) | Extension | | Yes | 0..1 | Additional Practitioners associated with the Observation value, if applicable. Possible values include: "data entry person" for Practitioners who transcribed the Observation into the system "verifier" for Practitioners who cosigned the Observation URL: http://hl7.org/fhir/StructureDefinition/event-performerFunction

Observation (R4).referenceRange | BackboneElement | Conditional | Yes | 0..1 | Applicable to organizations in the Netherlands starting in the November 2022 version of Epic. This is populated only for blood glucose readings. This element indicates the abnormal minimum and maximum values defined for the blood glucose reading.

Observation (R4).referenceRange.high | Quantity | | | 0..1 | Applicable to organizations in the Netherlands starting in the November 2022 version of Epic. The higher reference limit for blood glucose readings.

Observation (R4).referenceRange.low | Quantity | | | 0..1 | Applicable to organizations in the Netherlands starting in the November 2022 version of Epic. The lower reference limit for blood glucose readings.

Observation (R4).status | code | | | 1..1 | This element has the following possible values: "preliminary" - value is pended "final" - value has been filed "amended" - value has been edited "entered-in-error" - value has been deleted

Observation (R4).subject | Reference | Required | | 1..1 | The FHIR ID for the patient the Observation is about.

Observation (R4).valueCodeableConcept | CodeableConcept | | | 0..1 | This element is populated if the data type is a category or custom list. Only one of the value* elements is returned.

Observation (R4).valueDateTime | dateTime | | | 0..1 | This element is populated if data type is date. Only one of the value* elements is returned.

Observation (R4).valueQuantity | Quantity | | | 0..1 | This element is populated if a value has units or the data type is numeric. Only one of the value* elements is returned.

Observation (R4).valueString | string | | | 0..1 | This element is populated if a value doesn't have units or the data type is string. Only one of the value* elements is returned.

Observation (R4).valueTime | time | | | 0..1 | This element is populated if data type is time. Only one of the value* elements is returned.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Observation/eizmjygdN9YY0-abviOrMfcbd8esh7G-wPYmSzcav8lxekRZfdsGmzwU9x-
    gKdyEyW0mlnkIDi0BK83wSZnkSlA3
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "eizmjygdN9YY0-abviOrMfcbd8esh7G-wPYmSzcav8lxekRZfdsGmzwU9x-gKdyEyW0mlnkIDi0BK83wSZnkSlA3",
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://hl7.org/fhir/observation-category",
          "code": "vital-signs",
          "display": "Vital Signs"
        }
      ],
      "text": "Vital Signs"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679",
        "code": "14",
        "display": "Weight"
      },
      {
        "system": "http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id",
        "code": "ttjlASTxfaxgiecvotrb78w0",
        "display": "Weight"
      },
      {
        "system": "urn:oid:1.2.246.537.6.96",
        "code": "29463-7"
      },
      {
        "system": "http://loinc.org",
        "code": "29463-7",
        "display": "Body weight"
      },
      {
        "system": "http://loinc.org",
        "code": "3141-9",
        "display": "Body weight Measured"
      },
      {
        "system": "http://loinc.org",
        "code": "8716-3",
        "display": "Vital signs"
      }
    ],
```

```
    "text": "Weight"
  },
  "subject": {
    "reference": "Patient/eZO66HeGXB6VCJemL-ciSKA3",
    "display": "Meiko TEST"
  },
  "encounter": {
    "reference": "Encounter/e.R4dWbUDqF01.Z6BEIfEkZecZklxAhL82tS5wzzlD0o3"
  },
  "effectiveDateTime": "2019-09-05T20:00:00Z",
  "issued": "2019-09-05T20:49:27Z",
  "performer": [
    {
      "reference": "Practitioner/euJOUNeZyX6mofWELWG0Ibg3",
      "display": "Starter Provider, MD"
    }
  ],
  "valueQuantity": {
    "value": 50,
    "unit": "kg",
    "system": "http://unitsofmeasure.org",
    "code": "kg"
  }
}
```


Change log

Epic Version | Description | Type
May 2024 (February 2024 by special update) | A new “template-id” extension has been added. If vitals data was filed using the Observation.Create API, this extension is returned and includes the ID of the flowsheet template that vitals data was filed to. | Substantive Changes
August 2024 | Observation (Vitals) (R4) no longer returns Functional Status, Disability Status, Mental/Cognitive Status, or Social Determinants of Health (SDOH) assessments. | Breaking Change
August 2024 | Observation Sub-Resource Security Overrides Encoded Flowsheet ID | Breaking change
August 2025 (May 2025, February 2025, November 2024 by special update) | Starting in August 2024, May 2025 with special update E11400862, February 2025 with special update E11306699, and November 2024 with special update E11211328, multi-select fields are handled separately, and if a SNOMED concept mapping exists for any individual selection, the SNOMED concept mapping is included in the FHIR response. Previously, when a user selected multiple options, the system treated the entire list as a single combined value. | Substantive Changes

9.1.14 Observation.Search (Activities of Daily Living) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/10123-Observation-Search-Activities-of-Daily-Living-R4.html
Epic API ID | 10123
FHIR version | R4
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
HTTP method | GET
URL template | /api/FHIR/R4/Observation?category={category}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10123

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | False | | | Use "functional-mental-status", http://snomed.info/sct|384821006, and/or http://snomed.info/sct|118228005 to search for activities of daily living.
patient | reference | Either patient or subject must be provided. If both are provided, they must reference the same patient. | | | The patient to search data for.
subject | reference | Either patient or subject must be provided. If both are provided, they must reference the same patient. | | | The patient to search data for.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
based-on | String | True | | | Unsupported
code | token | True | | | The type of observation. Values mapped. If no mapping exists, only the text value is returned.
combo-code | token | True | | | Unsupported
combo-data-absent-reason | token | True | | | Unsupported
combo-value-concept | token | True | | | Unsupported
component-code | token | True | | | Unsupported
component-data-absent-reason | token | True | | | Unsupported

component-value-concept | token | True | | | Unsupported
data-absent-reason | token | True | | | Unsupported
date | date | True | | | Clinically relevant time period for the observation
derived-from | String | True | | | Unsupported
device | String | True | | | Unsupported
encounter | reference | True | | | Unsupported
focus | reference | True | | | Unsupported
has-member | reference | True | | | Unsupported
identifier | token | True | | | Unsupported
issued | date | True | | | Unsupported
method | token | True | | | Unsupported
part-of | String | True | | | Unsupported
performer | reference | True | | | Unsupported
specimen | String | True | | | Unsupported
status | token | True | | | Always "final"
value-concept | token | True | | | Result of the observation.
value-date | date | True | | | Unsupported
value-string | string | True | | | Result of the observation.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Observation | Observation | False | | | An instance of the R4 Observation resource.
Bundle.entry.resource.Observation.category | CodeableConcept | False | Yes | | Type of observation. Always includes SNOMED codes “118228005” and “384821006”, and custom code “functional-mental status.”
Bundle.entry.resource.Observation.code | CodeableConcept | False | | | The type of observation. Values mapped. If no mapping exists, only the text value is returned.
Bundle.entry.resource.Observation.effectivePeriod | Period | False | | | Clinically relevant time period for the observation
Bundle.entry.resource.Observation.note | Annotation | False | Yes | | Comments about the observation. A comment if flowsheet data or the associated note if inpatient goals. If using cascading flowsheets, the note element contains information about the additional flowsheet, if applicable, instead of the note. For example, a nurse documents flowsheet “daily activities” as abnormal, which opens flowsheet “mobility.” The nurse documents the value “highly mobile” in the flowsheet. The note element for “daily activities” is returned as “daily activities;highly mobile” if flowsheet “mobility” isn’t included in the Epic organization's configuration.
Bundle.entry.resource.Observation.status | code | False | | | Always "final"
Bundle.entry.resource.Observation.subject | Reference | False | | | Who the observation is about. Reference to the patient resource.
Bundle.entry.resource.Observation.valueCodeableConcept | CodeableConcept | False | | | Result of the observation. Values mapped. If not mapping exists, only the text value is returned.

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation?patient=enoCp373txQSghUte78RDIQ3&category=http://snomed.info/sct|38482
    1006
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 2,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation?patient=enoCp373txQSghUte78RDIQ3&category=http://snomed.
    info/sct%7C384821006,http://snomed.info/sct%7C118228005"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Observation/eK.YYu4CGX.bgDi.i48Ry0qqp31Vci5cpqDiXY17qo-
    lcGEO.J6Z8OJbctD0XBFWB3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/eK.YYu4CGX.bgDi.i48Ry0qqp31Vci5cpqDiXY17qo-
    lcGEO.J6Z8OJbctD0XBFWB3",
      "resource": {
        "resourceType": "Observation",
        "id": "eK.YYu4CGX.bgDi.i48Ry0qqp31Vci5cpqDiXY17qo-lcGEO.J6Z8OJbctD0XBFWB3",
        "status": "final",
        "category": [
          {
            "coding": [
              {
                "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
                "code": "functional-mental-status",
```

```
            "display": "Functional and Mental Status"
          }
        ]
      },
      {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "118228005",
            "display": "Functional finding"
          }
        ]
      },
      {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "384821006",
            "display": "Mental state, behavior and/or psychosocial function finding"
          }
        ]
      }
    ],
    "code": {
      "text": "Neuro (WDL)"
    },
    "subject": {
      "reference": "Patient/enoCp373txQSghUte78RDIQ3",
      "display": "FHIR, Automation"
    },
    "effectivePeriod": {
      "start": "2021-05-04T20:00:00Z"
    },
    "valueCodeableConcept": {
      "text": "Exceptions to WDL"
    },
    "note": [
      {
        "text": "Level of Consciousness: New agitation; Orientation Level: Disoriented to place; Cognition:
Impulsive"
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation/eK.YYu4CGX.bgDi.i48Ry0rwjGC3r1K2R6gMLL-
cvPyTD.BkmsblugeJXH0AR7lq43"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/eK.YYu4CGX.bgDi.i48Ry0rwjGC3r1K2R6gMLL-
cvPyTD.BkmsblugeJXH0AR7lq43",
  "resource": {
    "resourceType": "Observation",
    "id": "eK.YYu4CGX.bgDi.i48Ry0rwjGC3r1K2R6gMLL-cvPyTD.BkmsblugeJXH0AR7lq43",
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
            "code": "functional-mental-status",
            "display": "Functional and Mental Status"
          }
        ]
      },
      {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "118228005",
            "display": "Functional finding"
          }
        ]
      },
      {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "384821006",
            "display": "Mental state, behavior and/or psychosocial function finding"
          }
        ]
      }
```

```
        ],
        "code": {
          "text": "Body Movement in Desired Range"
        },
        "subject": {
          "reference": "Patient/enoCp373txQSghUte78RDIQ3",
          "display": "FHIR, Automation"
        },
        "effectivePeriod": {
          "start": "2021-05-04",
          "end": "2021-05-07"
        },
        "valueCodeableConcept": {
          "text": "Progressing"
        }
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


9.1.15 Observation.Search (Core Characteristics) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/969-Observation-Search-Core-Characteristics-R4.html
Epic API ID | 969
FHIR version | R4
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Keywords | Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | GET
URL template | api/FHIR/R4/Observation?category={category}&code={code}&patient={patient}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=969

Description

The FHIR Observation resource defines measurements and assertions about a patient, including vital signs, laboratory data, imaging results, devices measurements, clinical assessment tools, personal characteristics, social history, and core characteristics.

This FHIR Observation resource returns core characteristics for a patient including gestational age and birth weight.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | Either this or code must be provided, not both. | | | Use "core-characteristics" to search for characteristics observations.
code | token | Either this or category must be provided, not both. | | | LOINC code of observation. Can be 76516-4 for gestational age or 8339-4 for birth weight.
patient | reference | False | | | Reference to a patient resource. If both patient and subject are provided, they must match.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
based-on | String | True | | | Unsupported
combo-code | token | True | | | This is the LOINC code corresponding to value returned in this observation.
combo-data-absent-reason | token | True | | | Unsupported
combo-value-concept | token | True | | | Type of data returned. See the valueCodeableConcept response element for more information.
component-code | token | True | | | Unsupported
component-data-absent-reason | token | True | | | Unsupported
component-value-concept | token | True | | | Unsupported
data-absent-reason | token | True | | | Unsupported
date | date | True | | | Unsupported
derived-from | String | True | | | Unsupported
device | String | True | | | Unsupported
encounter | reference | True | | | Unsupported
focus | reference | True | | | Unsupported
has-member | reference | True | | | Unsupported
identifier | token | True | | | Unsupported
issued | date | True | | | The instant when this observation was last edited.
method | token | True | | | Unsupported

part-of | String | True | | | Unsupported
performer | reference | True | | | The party responsible for documenting the observation.
specimen | String | True | | | Unsupported
status | token | True | | | The status of the observation. Always use "final" when searching for this type of data.
value-concept | token | True | | | Type of data returned. See the valueCodeableConcept response element for more information.
value-date | date | True | | | Unsupported
value-string | string | True | | | Type of data returned. See the valueString response element for more information.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.response | Response | False
Bundle.entry.response.outcome | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.response.outcome.Observation | Observation | False | | | An instance of the R4 Observation resource.
Bundle.entry.response.outcome.Observation.category | CodeableConcept | False | Yes | | Always "core-characteristics".
Bundle.entry.response.outcome.Observation.code | CodeableConcept | False | | | This is the LOINC code corresponding to value[x] returned.
Bundle.entry.response.outcome.Observation.issued | instant | False | | | The instant when this observation was last edited.
Bundle.entry.response.outcome.Observation.performer | Reference | False | Yes | | Who documented the observation.
Bundle.entry.response.outcome.Observation.status | code | False | | | Always set to "final".
Bundle.entry.response.outcome.Observation.valueCodeableConcept | CodeableConcept | One value[x] element is always returned. Which type depends on the data in the specific instance of the resource. | | | Available starting in February 2022. The following values, depending on the type of data returned: Delivery method: LOINC 73762-7 Labor duration: LOINC 32396-4 Hospital name: LOINC 62330-6 Hospital location: LOINC 62331-4
Bundle.entry.response.outcome.Observation.valueInteger | integer | One value[x] element is always returned. Which type depends on the data in the specific instance of the resource. | | | The following values, depending on the type of data returned: Birth APGAR scores APGAR 1: LOINC 9272-6 APGAR 5: LOINC 9274 -2 APGAR 10: LOINC 9271-8
Bundle.entry.response.outcome.Observation.valueQuantity | Quantity | One value[x] element is always returned. Which type depends on the data in the specific instance of the resource. | | | Available starting in February 2022. The following values, depending on the type of data returned: Birth weight: LOINC code 8339-4 Gestational age: LOINC code 76516-4 Birth length: LOINC 8291-7 Birth head circumference: LOINC 8290-9 Hospital Days
Bundle.entry.response.outcome.Observation.valueString | string | One value[x] element is always returned. Which type depends on the data in the specific instance of the resource. | | | The following values, depending on the type of data returned: Delivery method: LOINC 73762-7 Labor duration: LOINC 32396-4 Hospital name: LOINC 62330-6 Hospital location: LOINC 62331-4

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation?patient=eZO66HeGXB6VCJemL-ciSKA3&category=core-characteristics
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="2"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/R4/Observation?patient=eZO66HeGXB6VCJemL-ciSKA3&amp;category=core-
    characteristics"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/R4/Observation/eJW3Cy0gsPSma08Z6jFDVDFZqbJqmUZ63rNfdLxd0GiU3"/>
    </link>
    <fullUrl value="https://hostname/instance/api/FHIR/R4/Observation/eJW3Cy0gsPSma08Z6jFDVDFZqbJqmUZ63rNfdLxd0GiU3"/>
    <resource>
      <Observation>
        <id value="eJW3Cy0gsPSma08Z6jFDVDFZqbJqmUZ63rNfdLxd0GiU3"/>
        <status value="final"/>
        <category>
          <coding>
            <system value="http://open.epic.com/FHIR/StructureDefinition/observation-category"/>
            <code value="core-characteristics"/>
            <display value="Core Characteristics"/>
          </coding>
          <text value="Core Characteristics"/>
        </category>
        <code>
          <coding>
            <system value="http://loinc.org"/>
            <code value="76516-4"/>
            <display value="Gestational age at birth"/>
          </coding>
          <text value="Gestational age at birth"/>
        </code>
        <subject>
          <reference value="Patient/eZO66HeGXB6VCJemL-ciSKA3"/>
          <display value="Meiko Lufhir"/>
        </subject>
        <issued value="2019-09-05T21:32:37Z"/>
        <performer>
          <reference value="Practitioner/euJOUNeZyX6mofWELWG0Ibg3"/>
```

```
        <display value="Starter Provider, MD"/>
      </performer>
      <valueQuantity>
        <value value="264"/>
        <unit value="d"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="d"/>
      </valueQuantity>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
  </search>
 </entry>
 <entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/R4/Observation/eJW3Cy0gsPSma08Z6jFDVDPI7tUavtRv8uzTVG5MX4PQ3"/>
  </link>
  <fullUrl value="https://hostname/instance/api/FHIR/R4/Observation/eJW3Cy0gsPSma08Z6jFDVDPI7tUavtRv8uzTVG5MX4PQ3"/>
  <resource>
    <Observation>
      <id value="eJW3Cy0gsPSma08Z6jFDVDPI7tUavtRv8uzTVG5MX4PQ3"/>
      <status value="final"/>
      <category>
        <coding>
          <system value="http://open.epic.com/FHIR/StructureDefinition/observation-category"/>
          <code value="core-characteristics"/>
          <display value="Core Characteristics"/>
        </coding>
        <text value="Core Characteristics"/>
      </category>
      <code>
        <coding>
          <system value="http://loinc.org"/>
          <code value="8339-4"/>
          <display value="Birth weight measured"/>
        </coding>
        <text value="Birth weight measured"/>
      </code>
      <subject>
        <reference value="Patient/eZO66HeGXB6VCJemL-ciSKA3"/>
        <display value="Meiko Lufhir"/>
      </subject>
      <issued value="2019-09-05T21:32:37Z"/>
      <performer>
        <reference value="Practitioner/euJOUNeZyX6mofWELWG0Ibg3"/>
        <display value="Starter Provider, MD"/>
      </performer>
      <valueQuantity>
        <value value="2.999"/>
        <unit value="kg"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="kg"/>
      </valueQuantity>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
  </search>
 </entry>
</Bundle>
```


9.1.16 Observation.Search (Flowsheets) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/49208-Observation-Search-Flowsheets-R4.html
Epic API ID | 49208
FHIR version | R4
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation?based-on={based-on}&category={category}&code={code}&date={date}&focus={focus}&issued={issued}&patient={patient}&subject={subject}&_count={_count}&_id={_id}&_include={_include}&_lastupdated={_lastupdated}&_revinclude={_revinclude}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=49208

Description

This web service retrieves general non-duplicable data found in the patient's flowsheets across all encounters.

Individual flowsheet rows can be categorized to more specific sub-resources such as Vital Signs and Assessments. Clients can use Epic's more specific APIs including Observation.Search (Vital Signs) (R4) and Observation.Search (Assessments) (R4) to find these flowsheets with a narrower scope. Observation.Search (Flowsheets) (R4) can search for flowsheet data among all categories except for lines, drains, and airways. Flowsheet rows not categorized under other sub-resources can only be retrieved through Observation.Search (Flowsheets) (R4) by querying with the code search parameter.

Request parameters

Name | Type | Optional | Array | Cardinality | Description

_count | number | Optional | Yes | | Number of results per page.

_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revInclude. All others will be ignored.

_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.

_revinclude | string | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.

category | token | Conditional | Yes | | Either category or code must be provided. Use "vital-signs" to search for vitals Observations. Use "care-experience-preference" to search for care experience preference Observations. Use "survey" to search for all assessments. Otherwise, use one of the following values to search for a specific type of assessment: * functional-status * disability-status * cognitive-status * sdoh * activity * social-history Note: Category assignments are subjective and can differ across organizations. When possible, searching by a specific code provides more consistent results. Alcohol use and substance use can be found by searching via social-history.

code | token | Conditional | Yes | | Formatted as <system>|<code>. If a system is not provided, the code is assumed to be a LOINC code. Valid codes and their systems include: LOINC Code System Value: http://loinc.org or no system value. Epic Encoded Flowsheet Identifier System Value: http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id Note: The encoded flowsheet identifier must be generated by IT admins at an Epic organization. Cancer Data Standards Registry and Repository (CADSR) Terminology System Value: http://wiki.nci.nih.gov/display/cadsr

date | date | Optional | Yes | | The date range for when the Observation was taken.

patient | reference | Conditional | Yes | | Reference to a Patient resource the Observation is about. Either this parameter or subject must be provided. If both are provided, they must match.

subject | reference | Conditional | Yes | | Reference to a Patient resource the Observation is about. Either this parameter or patient must be provided. If both are provided, they must match.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description

combo-code | token | Optional | Yes | | Matches to the code or component.code element in the response.

combo-data-absent-reason | token | Optional | Yes | | The reason why the expected value in the element Observation.value* or Observation.component.value* is missing.

combo-value-concept | token | Optional | Yes | | The value or component value of the observation, if that value is a CodeableConcept. Matches to valueCodeableConcept or component.valueCodeableConcept.

component-code | token | Optional | Yes | | Matches to the component.code element in the response.

component-data-absent-reason | token | Optional | Yes | | The reason why the expected value in the element Observation.component.value* is missing.

component-value-concept | token | Optional | Yes | | The component value of the Observation, if that value is a CodeableConcept. Matches to component.valueCodeableConcept.

data-absent-reason | token | Optional | Yes | | The reason why the expected value in the element Observation.value* is missing.

encounter | reference | Optional | Yes | | Encounter associated with this Observation value, if applicable.

issued | date | Optional | Yes | | The date and time the Observation was documented.

method | token | Optional | Yes | | Method by which the Observation was obtained. See the method response element for more information.

performer | reference | Optional | Yes | | Who filed the Observation. If the measurement is patient-reported, this is a reference to the Patient resource.

status | token | Optional | Yes | | The status of the Observation. Possible values include: "preliminary" - value is pended "final" - value has been filed "amended" - value has been edited "entered-in-error" - value has been deleted

value-concept | token | Optional | Yes | | The value of the Observation if it is a CodeableConcept. Matches to valueCodeableConcept.

value-date | date | Optional | Yes | | The value of the Observation, if the value is a date or period of time.

value-string | string | Optional | Yes | | The value of the Observation, if valueString is populated.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Observation (R4) | Observation

Observation (R4).bodySite | CodeableConcept | Conditional | | 0..1 | Applicable only to organizations in the Netherlands. The site on the body where the Observation was made. Used for the location where a blood pressure reading was obtained.

Observation (R4).category | CodeableConcept | | Yes | 0..* | When a flowsheet row is mapped to an HL7 observation category like "vital-signs", "survey", or "care-experience-preference", then that category is used. Flowsheet rows which are not mapped to a specific category do not have a category element in the response.

Observation (R4).code | CodeableConcept | Required | | 1..1 | Coded value for what is being measured. The following code systems can be returned: OID for internal Epic flowsheet ID Encoded value of the internal Epic flowsheet ID External concept mappings such as LOINC and SNOMED

Observation (R4).component | BackboneElement | | Yes | 0..* | Includes extra information about the Observation. Populated only for: Blood Pressure: Systolic (LOINC 8480-6) and diastolic (LOINC 8462-4) pressures are always returned as components if available. Can include components with a SNOMED code of 70665002 and LOINC code of 8358-4 for the cuff type or size, and a SNOMED code of 424724000 and a LOINC code of 8361-8 for cuff position. Average Blood Pressure (starting in the November 2026 version of Epic): Average systolic (LOINC 96608-5) and average diastolic (LOINC 96609-3) are always returned as components if available. Can include a component with a SNOMED code of 246432004 for number of occurrences. SpO2 can include components with LOINC code of 3151-8 for inhaled oxygen flow rate, 3150-0 for inhaled oxygen concentration, and 74206-4 for whether oxygen therapy was given. Pulse Rate can include a component with LOINC code of 44969-4 for heart rate rhythm palpitation. Weight can include a component with a code of 8352-7 for how the patient was clothed for weight reading. Patient-Entered Flowsheet Metadata Represents strict metadata if the coding sub-element is populated. The following strict metadata can be returned: 1-UUID The unique reading ID 2-Blood Glucose Meal Times 1-Before meal 2-After meal 3-Insulin Delivery Reason 1-Basal 2-Bolus 5-VO2 Max Test Type 1-Max exercise test 2-Sub-max exercise prediction 3-Non exercise prediction 6-Manually Entered - Whether or not the patient manually entered this reading into the system (as opposed to automatically entered by a device). false true Represents loose metadata if only the text sub-element is populated. UUID,

Blood Glucose Meal times, Insulin Delivery Reason, and VO2 Max Test Type must have their value specified in component.valueString. Manually Entered is specified as a valueBoolean.

Observation (R4).component.code | CodeableConcept | | | 0..1 | Code for the extra information for the measurement. The following extra information can be returned for each measurement, with the associated code: Blood Pressure 8480-6: Systolic pressure 8462-4: Diastolic pressure SNOMED 70665002 / LOINC 8358-4: Cuff type or size SNOMED 424724000 / LOINC 8361-8: Cuff position Average Blood Pressure (starting in the November 2026 version of Epic) 96608-5: Systolic blood pressure mean 96609-3: Diastolic blood pressure mean SNOMED 246432004: Number of occurrences SpO2 3151-8: Inhaled oxygen flow rate 3150-0: Inhaled oxygen concentration 74206-4: Whether oxygen therapy was given Pulse Rate 44969-4: Heart rate rhythm palpitation. Weight 8352-7: Whether patient was clothed

Observation (R4).component.valueBoolean | boolean | | | 0..1 | Only used for Patient-Entered Flowsheet Metadata values. Valid values include: For Code = 6-Manually Entered false true

Observation (R4).component.valueCodeableConcept | CodeableConcept | | | 0..1 | The value as a CodeableConcept for the extra information for the measurement.

Observation (R4).component.valueQuantity | Quantity | | | 0..1 | If the extra information for the measurement is numeric, this is the numeric value. For example, it could be the flow rate (L/min) or concentration (%) for an SpO2 measurement. Starting in the November 2026 version of Epic, this is also the number of occurrences for average blood pressure.

Observation (R4).component.valueString | string | | | 0..1 | Values for the patient-entered flowsheet metadata.

Observation (R4).dataAbsentReason | CodeableConcept | | | 0..1 | Indicates that an Observation value is missing. This is always "unknown" for systolic blood pressure values in Observations for diastolic-only blood pressure flowsheets, or for diastolic blood pressure value in Observations for systolic-only blood pressure flowsheets.

Observation (R4).effectiveDateTime | dateTime | | | 0..1 | The date and time the Observation was taken. This element is omitted when effectivePeriod is used.

Observation (R4).effectivePeriod | Period | | | 0..1 | Starting in the November 2026 version of Epic, average blood pressure observations can use this element to return the begin and end instants for the average. Only one of the effective* elements is returned.

Observation (R4).encounter | Reference | Conditional | | 1..1 | Reference to an Encounter resource that this Observation belongs to.

Observation (R4).extension (supportingInfo) (also: extension (supportingInfo)) | Extension | Conditional | Yes | 0..* | If the measurement is linked to a note written by a clinician, this returns a reference to the DocumentReference resource. URL: http://hl7.org/fhir/StructureDefinition/workflow-supportingInfo

Observation (R4).extension (temperature-in-fahrenheit) (also: extension (temperature-in-fahrenheit)) | Extension | | Yes | 0..1 | If the measurement is a temperature value, this element contains the temperature value quantity in Fahrenheit. This element is excluded for other data types. URL: http://open.epic.com/FHIR/STU3/StructureDefinition/temperature-in-fahrenheit

Observation (R4).extension (temperature-in-fahrenheit).valueQuantity | Quantity | | | 1..1 | The temperature value in degrees Fahrenheit [degF]. This represents the same temperature measurement as Observation.valueQuantity, which can use different units.

Observation (R4).extension (template-id) (also: extension (template-id)) | Extension | | Yes | 0..1 | This is the specific flowsheet template (a collection of flowsheet rows and groups) to which data was filed. URL: http://open.epic.com/FHIR/StructureDefinition/extension/template-id

Observation (R4).extension (template-id).valueIdentifier | Identifier | | | 1..1 | The system and value for the flowsheet template.

Observation (R4).extension (template-id).valueIdentifier.system | String | | | 1..1 | The OID for the Epic internal flowsheet template identifier. This varies based on the specific Epic environment.

Observation (R4).extension (template-id).valueIdentifier.value | String | | | 1..1 | The Epic internal identifier of the flowsheet template.

Observation (R4).interpretation | CodeableConcept | | Yes | 0..* | Returns a system of http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation and value of "A" for flowsheet values with an abnormal status. Returns an environment-specific OID system with a value of "SIG" for flowsheet values with a significant status. In Netherlands locales, these interpretation values can also be returned, alongside a system value of http://snomed.info/sct: For heart rate profile Observations, possible values include: SNOMED code 271636001 (Heart regular) for normal heart rate readings SNOMED code 248650006 (Heart irregular) for abnormal heart rate readings For blood glucose profile Observations, possible values include: SNOMED code 281302008 (Above reference range) for blood glucose readings higher than the reference range SNOMED code 11896004 (Intermediate) for blood glucose readings within the reference range (inclusive of the reference range values) SNOMED code 281300000 (Below reference range) for blood glucose readings below the reference range

Observation (R4).issued | instant | | | 1..1 | The date and time the Observation was documented.

Observation (R4).method | CodeableConcept | | | 0..1 | For patient-entered flowsheets, this element is returned with a text value of "Patient-entered". In Netherlands locales with blood pressure vitals Observations, SNOMED (system http://snomed.info/sct) codes can be returned: Blood pressure taken with an invasive method: 10179008 Blood pressure taken with a non-invasive method: 22762002

Observation (R4).note | Annotation | | Yes | 0..1 | The comment associated with the value.

Observation (R4).performer | Reference | | Yes | 1..* | Reference to a FHIR resource for who filed the Observation. If the flowsheet value was filed by a patient, this references a Patient resource. Otherwise, it references a Practitioner resource.

Observation (R4).performer.extension (performerFunction) (also: extension (performerFunction)) | Extension | | Yes | 0..1 | Additional Practitioners associated with the Observation value, if applicable. Possible values include: "data entry person" for Practitioners who transcribed the Observation into the system "verifier" for Practitioners who cosigned the Observation URL: http://hl7.org/fhir/StructureDefinition/event-performerFunction

Observation (R4).referenceRange | BackboneElement | Conditional | Yes | 0..1 | Applicable to organizations in the Netherlands starting in the November 2022 version of Epic. This is populated only for blood glucose readings. This element indicates the abnormal minimum and maximum values defined for the blood glucose reading.

Observation (R4).referenceRange.high | Quantity | | | 0..1 | Applicable to organizations in the Netherlands starting in the November 2022 version of Epic. The higher reference limit for blood glucose readings.

Observation (R4).referenceRange.low | Quantity | | | 0..1 | Applicable to organizations in the Netherlands starting in the November 2022 version of Epic. The lower reference limit for blood glucose readings.

Observation (R4).status | code | | | 1..1 | This element has the following possible values: "preliminary" - value is pended "final" - value has been filed "amended" - value has been edited "entered-in-error" - value has been deleted

Observation (R4).subject | Reference | Required | | 1..1 | The FHIR ID for the patient the Observation is about.

Observation (R4).valueCodeableConcept | CodeableConcept | | | 0..1 | This element is populated if the data type is a category or custom list. Only one of the value* elements is returned.

Observation (R4).valueDateTime | dateTime | | | 0..1 | This element is populated if data type is date. Only one of the value* elements is returned.

Observation (R4).valueQuantity | Quantity | | | 0..1 | This element is populated if a value has units or the data type is numeric. Only one of the value* elements is returned.

Observation (R4).valueString | string | | | 0..1 | This element is populated if a value doesn't have units or the data type is string. Only one of the value* elements is returned.

Observation (R4).valueTime | time | | | 0..1 | This element is populated if data type is time. Only one of the value* elements is returned.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Observation?patient=e63wRTbPfr1p8UW81d8Seiw3&code=http://open.epic.com/FHIR/St
    ructureDefinition/observation-flowsheet-id|tFPAQItwnOAotqq9GHG-ZAg0
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation?code=http%3A%2F%2Fopen.epic.com%2FFHIR%2FStructureDefini
    tion%2Fobservation-flowsheet-id|tFPAQItwnOAotqq9GHG-ZAg0&patient=e63wRTbPfr1p8UW81d8Seiw3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Observation/eF7uc-
    VX129EjpVE35In7ZroUUTPOudM9oOQTmIWiPZS9O3pXRr6snBCvQVadYl6VdkVo.PCeyixt2muU4qpWPQ3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/eF7uc-
    VX129EjpVE35In7ZroUUTPOudM9oOQTmIWiPZS9O3pXRr6snBCvQVadYl6VdkVo.PCeyixt2muU4qpWPQ3",
      "resource": {
        "resourceType": "Observation",
        "id": "eF7uc-VX129EjpVE35In7ZroUUTPOudM9oOQTmIWiPZS9O3pXRr6snBCvQVadYl6VdkVo.PCeyixt2muU4qpWPQ3",
        "extension": [
          {
            "valueIdentifier": {
              "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.707684",
              "value": "46130"
            },
            "url": "http://open.epic.com/FHIR/StructureDefinition/extension/template-id"
          }
        ],
        "status": "final",
        "code": {
          "coding": [
            {
              "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.707679",
              "code": "7096230",
              "display": "Precautions"
            },
            {
              "system": "http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id",
              "code": "tFPAQItwnOAotqq9GHG-ZAg0",
              "display": "Precautions"
            }
          ],
          "text": "Precautions"
        },
        "subject": {
          "reference": "Patient/e63wRTbPfr1p8UW81d8Seiw3",
          "display": "Mychart, Teddy"
        },
        "encounter": {
          "reference": "Encounter/e8hBVXYLK9-otx9sWEP7xJQ3",
          "identifier": {
            "use": "usual",
            "system": "urn:oid:1.2.840.114350.1.13.0.1.7.3.698084.8",
            "value": "1875"
          },
          "display": "Office Visit"
        },
        "effectiveDateTime": "2010-06-12T17:20:00Z",
        "issued": "2010-06-12T17:20:27Z",
        "performer": [
          {
            "reference": "Practitioner/eU2bhJHhLPs8.VF8LzIGbIw3",
            "display": "Nurse Family Medicine, RN"
          }
        ],
        "valueCodeableConcept": {
          "coding": [
            {
              "code": "Other (Comment)",
              "display": "Other (Comment)"
            }
          ],
          "text": "Other (Comment)"
        },
        "note": [
          {
            "text": "Example comment"
          }
```

```
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```

9.1.17 Observation.Search (Labs) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/999-Observation-Search-Labs-R4.html
Epic API ID | 999
FHIR version | R4
Categories | Clinical Diagnostics, Labs
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | BMI percentile per age and sex for youth 2-20;Weights for age per length and sex;Occipital-frontal circumference for children < 3 years old;Vital Signs;Diastolic blood pressure;Systolic blood pressure;Body height;Body weight;Heart rate;Respiratory rate;Body temperature;Pulse oximetry;Inhaled oxygen concentration;Diagnostic Reports;Imaging;Laboratory;Pathology;Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | GET
URL template | api/FHIR/R4/Observation?patient={patient}&subject={subject}&category={category}&code={code}&date={date}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=999

Description

The Observation.Search (Labs) web service returns component-level result details, as well as references to isolate Observations for microbiology results.

Observation.Read (Labs) (R4)

can be used to gather details about isolates, methods, and sensitivities for microbiology results, as well as narrative or impression result text.

This sub-resource can also return result components linked to Midmark EKG orders.

If the desire is to get all of the results for a particular procedure,

DiagnosticReport.Search (Results) (R4)

would be better suited for that. Examples where that might be more common are panel procedures like CBC or BMP, where there are multiple component-level results for each panel. Compared to HL7v2 result messages, DiagnosticReport is like an OBR segment and Observation is like an OBX segment.

When processing values, be sure to consider all elements. Classification of value is subject to change from system to system and version to version.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
_count | number | Optional | Yes | | Number of results per page.
_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revInclude. All others will be ignored.
_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.
_lastUpdated (also: _lastUpdated) | String | Optional | Yes | | Filter to a date range during which a result was most recently updated.
_revinclude | string | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.
category | token | Conditional | Yes | | The category of Observations to search for. Either category or code search parameters must be provided, but not both. The format for the category parameter is <system>|<category>. The accepted <category> values are listed below. Note: The <system> value is optional in each case, but providing it can improve performance Use case: Search for all component based-results System: http://terminology.hl7.org/CodeSystem/observation-category Category: laboratory Use case: Search for results by SNOMED code System: http://snomed.info/sct Category: 252275004 for Hematology 275711006 for Chemistry 68793005 for Serology 395124008 for Virology 314076009 for Toxicology 19851009 for Microbiology 405825005 for Molecular Genetics 49581000146104 for Laboratory Test Finding (Netherlands only) Additional information: For a search to be able to return results by SNOMED, the Epic organization must map result components to the SNOMED codes above. Use case: Search for results by procedure order type System: <Order Type OID> Category: <Order Type Abbreviation> Additional information: <Order Type OID> and <Order Type Abbreviation> will vary by Epic organization. Work with the organization to obtain these values.
code | token | Conditional | Yes | | The Observation identifier. Either category or code search parameters must be provided, but not both. The format for the code parameter is <system>|<code>. The accepted <code> values are listed below. Note: The <system> value is optional in each case, but providing it can improve performance Use case: Search for results by LOINC code System: http://loinc.org Code: <LOINC code> Use case: Search for results by other result component identifier System: <Identifier Type OID> Category: <Component ID for Identifier Type> Additional information: <Identifier Type OID> and <Component ID for Identifier Type> will vary by Epic organization. Work with the organization to obtain these values.
date | date | Optional | Yes | | Date when the specimen was obtained.
patient | reference | Conditional | Yes | | The FHIR ID for the patient associated with the Observation. Either patient or subject must be provided.

subject | reference | Conditional | Yes | | The FHIR ID for the patient associated with the Observation. Either patient or subject must be provided.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description

based-on | String | Optional | Yes | | A ServiceRequest reference for the Epic order associated with the results. Format for use in post-filtering is based-on=ServiceRequest/<ServiceRequest FHIR ID>

combo-code | token | Optional | Yes | | The code native request element is recommended over combo-code. The same results will be returned for either, but code is better for performance. This post-filter request element that has the same requirements and structure as the native request element code.

data-absent-reason | token | Optional | Yes | | The reason why normally expected content of the data element is missing. Set to "unknown" for lab result components with no values.

encounter | reference | Optional | Yes | | An Encounter reference for the encounter asscociated with the order. Format for use in post-filtering is encounter=Encounter/<Encounter FHIR ID>

has-member | reference | Optional | Yes | | An Observation reference for an isolate associated with the result. Format for use in post-filtering is has-member=Observation/<Isolate Observation FHIR ID>

issued | date | Optional | Yes | | The Observation analysis instant.

performer | reference | Optional | Yes | | A reference to the Practitioner FHIR ID for the resulting agency director. Format for use in post-filtering is performer=Practitioner/<Lab Director Practitioner FHIR ID>

specimen | String | Optional | Yes | | A Specimen reference for the specimen associated with the Observation. Format for use in post-filtering is specimen=Specimen/<Specimen FHIR ID>

status | token | Optional | Yes | | The lab status of the associated order or component. Possible values are: registered preliminary final amended corrected unknown entered-in-error cancelled pending (for organizatons in the Netherlands only) If a test resulted in an unsuccessful attempt, the status will be cancelled. For example, if a mammography is performed, but the exam images were too blurry to make clinical determination, that would be considered an unsuccessful attempt. In this example, the exam was performed, the patient was exposed to radition, but the desired clinical outcome was not achieved.

value-string | string | Optional | Yes | | The result value if the value is a string (for example: "positive", "negative", etc).

Response elements

Name | Type | Optional | Array | Cardinality | Description

Observation (R4) | Observation

Observation (R4).basedOn | Reference | | Yes | 1..* | A ServiceRequest reference for the Epic order associated with the results.

Observation (R4).category | CodeableConcept | | Yes | 1..* | The Observation type. All lab results have a category of "laboratory" System: http://terminology.hl7.org/CodeSystem/observation-category Category.code: laboratory Results for imaging orders have a category of "imaging" System: http://terminology.hl7.org/CodeSystem/observation-category Category.code: imaging All results have a category for the procedure order type System: <Order Type OID> (ends in .10.798268.30) Category.code: <Order Type Abbreviation> Additional information: <Order Type OID> and <Order Type Abbreviation> vary by Epic organization. Work with the organization to obtain these values. If doing a Read on a component Observation and the Epic organization has mapped result components to SNOMED codes, they are returned in this element as well System: http://snomed.info/sct Category.code: <SNOMED code> Additional information: For a search to be able to return SNOMED codes for result components, the Epic organization must map the result components to a SNOMED code.

Observation (R4).code | CodeableConcept | | | 1..1 | The result component identifiers. LOINC codes will have a system of http://loinc.org For other component identifiers, the system and identifiers will vary by Epic organization

Observation (R4).dataAbsentReason | CodeableConcept | | | 0..1 | The reason why content of the data element is missing. System: http://terminology.hl7.org/CodeSystem/data-absent-reason If a previous result contact on the order had result data but the most current result contact does not: code: "error" text: "Data for specified Observation no longer exists" In patient-facing contexts if the results aren't ready yet, or if the patient is missing security to view results: code: "temp-unknown" text: "Data for specified Observation is temporarily unavailabile" If the result data is missing from an Observation for any other reason: code: "unknown" text: "No value exists"

Observation (R4).derivedFrom | Reference | | Yes | 0..* | This element will not be populated in Observation.Search (Labs) responses.

Observation (R4).effectiveDateTime | dateTime | | | 1..1 | This represents the most relevant date/time for an order. This is typically the specimen collection date/time for resulted orders.

Observation (R4).encounter | Reference | | | 0..1 | The Encounter FHIR ID associated with the order.

Observation (R4).extension (CopyIndicator) (also: extension (CopyIndicator)) | Extension | | Yes | 0..1 | Indicates whether this response originates from the local system (false) or if it is from an external source (true). URL: http://nictiz.nl/fhir/StructureDefinition/ext-CopyIndicator

Observation (R4).extension (historical-fhir-id) (also: extension (historical-fhir-id)) | Extension | | Yes | 0..1 | This extension contains a historical FHIR ID in cases where the historical ID exceeded the 64-character ID length specified in the HL7 FHIR standard. This extension is only returned if the requesting client is configured to use 64-character-limited FHIR IDs as described in the FHIR ID Generation Scheme topic. This element is returned starting in the November 2025 version of Epic and in August 2025 by special update. URL: http://open.epic.com/FHIR/StructureDefinition/extension/historical-fhir-id

Observation (R4).extension (Lab-e-signature) (also: extension (Lab-e-signature)) | Extension | | Yes | 0..* | A string containing the person(s) who verified the result component and when they did so. This element will not be present in patient-facing contexts. URL: http://open.epic.com/FHIR/StructureDefinition/extension/lab-e-signature

Observation (R4).hasMember | Reference | | Yes | 0..* | Observation references for the isolates associated with microbiology results, when applicable.

Observation (R4).id | id | | | 1..1 | The Observation FHIR ID.

Observation (R4).interpretation | CodeableConcept | | Yes | 0..* | The level of abnormality for the result component. System: http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation

Observation (R4).issued | instant | | | 1..1 | The Observation analysis instant.

Observation (R4).meta | Meta | Required | | 1..1 | Metadata common to most resources.

Observation (R4).meta.lastUpdated | String | Required | | 1..1 | The instant that the result was last updated.

Observation (R4).method | CodeableConcept | | | 0..1 | This element will not be populated in Observation.Search (Labs) responses.

Observation (R4).note | Annotation | | Yes | 0..* | Comments about the result.

Observation (R4).performer | Reference | | Yes | 0..* | References to the Practitioner FHIR ID for the resulting agency director and the Organization FHIR ID for the resulting lab.

Observation (R4).referenceRange | BackboneElement | | Yes | 0..* | The reference range for the result, if applicable.

Observation (R4).referenceRange.high | Quantity | | | 0..1 | The high value of the reference range.

Observation (R4).referenceRange.low | Quantity | | | 0..1 | The low value of the reference range.
Observation (R4).referenceRange.text | String | | | 0..1 | Display version of reference range.
Observation (R4).specimen | Reference | | | 0..1 | A reference to the Specimen resource.
Observation (R4).status | code | | | 1..1 | The lab status of the associated order or component. Possible values are: registered preliminary final amended corrected unknown entered-in-error cancelled pending (for organizatons in the Netherlands only) If a test was performed but ended in an unsuccessful attempt, the status will be cancelled. For example, if a mammography is performed, and the patient was exposed to radiation, and images were captured, but the images were too blurry to make a clinical determination, that would be considered an unsuccessful attempt.
Observation (R4).subject | Reference | | | 1..1 | The Patient FHIR ID linked to the order.
Observation (R4).valueCodeableConcept | CodeableConcept | | | 0..1 | The SNOMED code for the organism associated with the result, when applicable.
Observation (R4).valueQuantity | Quantity | | | 0..1 | The result value if the result is numeric and represents a quantity.
Observation (R4).valueRange | Range | | | 0..1 | The result value if the result is structured numeric and the operator is "-".
Observation (R4).valueRatio | Ratio | | | 0..1 | The result value if the result is structured numeric and the operator is ":".
Observation (R4).valueString | string | | | 0..1 | The result value if the result is a string, or the text from the narrative or impression.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Observation?patient=eUEKdnYPKuCXlSq8WIYaPTA3&category=laboratory
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 2,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation?category=laboratory&patient=eUEKdnYPKuCXlSq8WIYaPTA3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Observation/eWilfwxg9QID48NN3HNHxVhG8H0eLgpAhiK7YzVHn.oVJvO5ZhL4
    KgFHVctbztVEorbsjpSDX-4sgAeTsTUsIjw3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/eWilfwxg9QID48NN3HNHxVhG8H0eLgpAhiK7YzVHn.oVJvO5ZhL4
    KgFHVctbztVEorbsjpSDX-4sgAeTsTUsIjw3",
      "resource": {
        "resourceType": "Observation",
        "id": "eWilfwxg9QID48NN3HNHxVhG8H0eLgpAhiK7YzVHn.oVJvO5ZhL4KgFHVctbztVEorbsjpSDX-4sgAeTsTUsIjw3",
        "meta": {
          "lastUpdated": "2022-04-29T21:54:13Z"
        },
        "basedOn": [
          {
            "reference": "ServiceRequest/eBLjU8nQwwFY4BK83wFL5scqCtdl6Qao5ETVzpO.VW2k3",
            "display": "Blood culture"
          }
        ],
        "status": "final",
        "category": [
          {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/observation-category",
                "code": "laboratory",
                "display": "Laboratory"
              }
            ],
            "text": "Laboratory"
          },
          {
            "coding": [
              {
                "system": "urn:oid:1.2.840.114350.1.13.11511.1.7.10.798268.30",
                "code": "Microbiology",
                "display": "Microbiology"
              }
            ],
            "text": "Microbiology"
          }
        ],
        "code": {
          "coding": [
            {
              "system": "urn:oid:1.2.840.114350.1.13.11511.1.7.2.768282",
              "code": "1802008",
              "display": "Blood Culture"
            }
          ],
          "text": "Blood Culture"
```

```
    },
    "subject": {
      "reference": "Patient/eUEKdnYPKuCXlSq8WIYaPTA3",
      "display": "Bugsy, Beau Dakota"
    },
    "encounter": {
      "reference": "Encounter/eKvkb8V4sgTKL9nm3MtDjXw3",
      "identifier": {
        "use": "usual",
        "system": "urn:oid:1.2.840.114350.1.13.11511.1.7.3.698084.8",
        "value": "40282"
      },
      "display": "Hospital Encounter"
    },
    "effectiveDateTime": "2022-04-29T21:53:00Z",
    "issued": "2022-04-29T21:54:13Z",
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "115329001"
        }
      ],
      "text": "Methicillin-Resistant Staphylococcus aureus"
    },
    "interpretation": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation",
            "code": "AA",
            "display": "Panic"
          }
        ],
        "text": "Panic"
      }
    ],
    "note": [
      {
        "text": "\r\n\r\nMethicillin-Resistant Staph aureus, patient may be an isolation risk."
      }
    ],
    "specimen": {
      "reference": "Specimen/eNKpY.ifDjG-raaJh60tz5w3",
      "display": "Specimen 22H-119M0006"
    },
    "hasMember": [
      {
        "reference": "Observation/eWilfwxg9QID48NN3HNHxVrWBVj0Kcd1GxT6JJOF9D8.NgYuvcuOUZTFDiDdanohh3",
        "display": "Isolate (1): STAPHYLOCOCCUS AUREUS METHICILLIN (OXACILLIN) RESISTANT"
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation/eTbKaUqpkJFl65dhjPpXaAEkUCcf-
9e1efFvc0wnFEQ68LKuB5MNeXLf2qPg3zip-4EXp9F1BhFtGy39RUCr4JQ3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/eTbKaUqpkJFl65dhjPpXaAEkUCcf-
9e1efFvc0wnFEQ68LKuB5MNeXLf2qPg3zip-4EXp9F1BhFtGy39RUCr4JQ3",
  "resource": {
    "resourceType": "Observation",
    "id": "eTbKaUqpkJFl65dhjPpXaAEkUCcf-9e1efFvc0wnFEQ68LKuB5MNeXLf2qPg3zip-4EXp9F1BhFtGy39RUCr4JQ3",
    "meta": {
      "lastUpdated": "2022-04-29T22:03:36Z"
    },
    "basedOn": [
      {
        "reference": "ServiceRequest/euV-cm3iMjdIXcE-Un9s.NIUXeGqZFzKpxCeiIcba1z03",
        "display": "SARS-CoV-2 RT PCR"
      }
    ],
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "laboratory",
            "display": "Laboratory"
          }
        ],
```

```
            "text": "Laboratory"
          },
          {
            "coding": [
              {
                "system": "urn:oid:1.2.840.114350.1.13.11511.1.7.10.798268.30",
                "code": "Lab",
                "display": "Lab"
              }
            ],
            "text": "Lab"
          }
        ],
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "94500-6",
              "display": "SARS-CoV-2 (COVID-19) RNA [Presence] in Respiratory system specimen by NAA with probe
    detection"
            },
            {
              "system": "urn:oid:1.2.840.114350.1.13.11511.1.7.2.768282",
              "code": "12300104",
              "display": "SARS COV-2 Interpretation"
            }
          ],
          "text": "SARS COV-2 Interpretation"
        },
        "subject": {
          "reference": "Patient/eUEKdnYPKuCXlSq8WIYaPTA3",
          "display": "Bugsy, Beau Dakota"
        },
        "encounter": {
          "reference": "Encounter/eKvkb8V4sgTKL9nm3MtDjXw3",
          "identifier": {
            "use": "usual",
            "system": "urn:oid:1.2.840.114350.1.13.11511.1.7.3.698084.8",
            "value": "40282"
          },
          "display": "Hospital Encounter"
        },
        "effectiveDateTime": "2022-04-29T21:53:00Z",
        "issued": "2022-04-29T22:03:36Z",
        "valueString": "Positive",
        "interpretation": [
          {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation",
                "code": "A",
                "display": "Abnormal"
              }
            ],
            "text": "Abnormal"
          }
        ],
        "specimen": {
          "reference": "Specimen/eli31DQNaAGIHyQeYD-NyHg3",
          "display": "Specimen 22H-119D0006"
        },
        "referenceRange": [
          {
            "text": "Not Detected"
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


Change log

Epic Version | Description | Type

November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change

System | URL | OID

ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4

ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90

ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94

ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2

SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

9.1.18 Observation.Search (Lines, Drains, Airways) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/970-Observation-Search-Lines-Drains-Airways-R4.html
Epic API ID | 970
FHIR version | R4
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Keywords | Flowsheets;LDA
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | GET
URL template | api/FHIR/R4/Observation?category={category}&code={code}&date={date}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=970

Description

The FHIR Observation resource defines measurements and assertions about a patient, including LDA-W (lines, drains, airways, and wound) data. There are two types of LDA-W data:

* LDA properties are documented only once. For example, a nurse might insert a line and document properties such as the insertion time, the body site, and the method of insertion. This data is searchable using the FHIR Observation resource and can be filed into Epic.

* LDA assessments are performed over time. For example, a nurse might inspect an inserted line regularly to make sure the site is clean or assess a wound to see if a new dressing should be applied. This data can be filed into Epic and, starting in the February 2022 version of Epic, is searchable using the R4 version of the FHIR Observation resource.

This resource returns only data for patients with active LDAs and wounds. When an LDA-W is completed or otherwise removed from the patient's chart, it will no longer be returned by this API.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | Either 'code' or 'category' are required. | | | In November 2021 and earlier versions, use “LDA” to search for LDA properties on a patient. Starting in February 2022, use “LDA” to search for LDA properties and LDA assessments on a patient. Use “LDA-property” to only search for LDA properties; use “LDA-assessment” to only search for LDA assessments. Either this element or code must be provided, but not both.
code | token | Either 'code' or 'category' are required. | | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. Note that the code element returns search results only for LDA properties. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization. In the App Orchard sandbox, use urn:oid:1.2.840.114350.1.13.0.1.7.2.707679.
date | date | True | | | The date range for when the LDA is active. Search parameter supported starting in Epic version February 2022.
patient | reference | Either this element or subject must be provided. If both are provided, they must match. | | | Reference to a patient resource the observation is about. Either this element or subject must be provided. If both are provided, they must match.
subject | reference | Either this element or patient must be provided. If both are provided, they must match. | | | Reference to a patient resource the observation is about. Either this element or subject must be provided. If both are provided, they must match.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
based-on | String | True | | | Unsupported
combo-code | token | True | | | LOINC code, CADSR code, or encoded flowsheet ID that is present in the code or component.code response element.
combo-data-absent-reason | token | True | | | Unsupported
combo-value-concept | token | True | | | The value of the observation if valueCodeableConcept or component.valueCodeableConcept is populated. Typically this is the case if the data type is category or custom list.
component-code | token | True | | | LOINC code, CADSR code, or encoded flowsheet ID that is present in the component.code response element.
component-data-absent-reason | token | True | | | Unsupported
component-value-concept | token | False | | | The value of the observation if component.valueCodeableConcept is populated. Typically this is the case if the data type is category or custom list.
data-absent-reason | token | True | | | Unsupported
derived-from | String | True | | | Unsupported
device | String | True | | | Unsupported
encounter | reference | True | | | The encounter associated with the target LDA observation.
focus | reference | True | | | Unsupported
has-member | reference | True | | | Unsupported
identifier | token | True | | | Unsupported
issued | date | True | | | The date and time the observation was documented.
method | token | True | | | Unsupported

part-of | String | True | | | Unsupported
performer | reference | True | | | The practitioner who filed the LDA assessment observation.
specimen | String | True | | | Unsupported
status | token | True | | | final | preliminary | amended | entered-in-error
value-concept | token | True | | | The value of the observation if valueCodeableConcept is populated. Typically this is the case if the data type is category or custom list.
value-date | date | True | | | In LDA assessments, matches to the LDA value if the data type is date.
value-string | string | True | | | Matches to valueString, if present. Refer to the valueString response element for more information.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Observation | Observation | False | | | An instance of the R4 Observation resource.
Bundle.entry.resource.Observation.category | CodeableConcept | False | Yes | | The category is always "LDA". Starting in February 2022, this element also always returns a second category which clarifies the type of LDA data returned in the response: The category “LDA-property” is used for LDA properties, which are documented only once. For example, a nurse might insert a line and document properties such as the insertion time, the body site, and the method of insertion. The category “LDA-assessment” is used for LDA assessments, which are performed over time. For example, a nurse might inspect an inserted line regularly to make sure the site is clean or assess a wound to see if a new dressing should be applied. This element can't be modified by Observation.Update.
Bundle.entry.resource.Observation.code | CodeableConcept | False | | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. This element can't be modified by Observation.Update. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.
Bundle.entry.resource.Observation.component | BackboneElement | True | Yes | | Populated only for patient-entered flowsheets.
Bundle.entry.resource.Observation.component.code | CodeableConcept | False | | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. This element can't be modified by Observation.Update. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.
Bundle.entry.resource.Observation.component.interpretation | CodeableConcept | True | Yes | | Available starting in the February 2022 version of Epic. Possible values include: “Abnormal” for observation values outside the reference range or expected norm “Significant” for observation values that a clinician has manually flagged for extra attention
Bundle.entry.resource.Observation.component.valueCodeableConcept | CodeableConcept | False | | | This element is populated if the data type is category or custom list. Only one of the value* elements is required.
Bundle.entry.resource.Observation.component.valueDateTime | String | False | | | This element is populated if the data type is category or custom list. Only one of the value* elements is required.
Bundle.entry.resource.Observation.component.valueQuantity | Quantity | False | | | This element is populated if a value has units or the data type is numeric or a percentile. Only one of the value* elements is required.
Bundle.entry.resource.Observation.component.valueString | string | False | | | This element is populated if a value doesn't have units or the data type is string. Only one of the value* elements is required.
Bundle.entry.resource.Observation.component.valueTime | String | False | | | This element is populated if the data type is time. Only one of the value* elements is required.
Bundle.entry.resource.Observation.effectiveDateTime | dateTime | True | | | Available starting in the February 2022 version of Epic for LDA assessments. This is a required element for LDA assessment observations. The date and time the observation was taken. This element can't be modified by Observation.Update.
Bundle.entry.resource.Observation.effectivePeriod | Period | True | | | Start instant and/or end instant of the LDA. This element is only returned with LDA properties.
Bundle.entry.resource.Observation.encounter | Reference | False | | | Reference to an encounter resource. This element can't be modified by Observation.Update.
Bundle.entry.resource.Observation.interpretation | CodeableConcept | True | Yes | | Available starting in the February 2022 version of Epic. Possible values include: “Abnormal” for observation values outside the reference range or expected norm “Significant” for observation values that clinician has manually flagged for extra attention
Bundle.entry.resource.Observation.issued | instant | True | | | Available starting in the February 2022 version of Epic for LDA assessments. This is a required element for LDA assessment observations. The date and time the observation was documented. This is always the current time.
Bundle.entry.resource.Observation.note | Annotation | True | Yes | | Available starting in the February 2022 version of Epic for LDA assessments. The comment associated with the LDA assessment observation value, if applicable.
Bundle.entry.resource.Observation.performer | Reference | True | Yes | | Available starting in the February 2022 version of Epic for LDA assessments. This is a required element for LDA assessment observations. The practitioner who filed the LDA assessment observation.
Bundle.entry.resource.Observation.performer.extension (also: event-performerFunction) | Extension | True | Yes | | Available starting in the February 2022 version of Epic for LDA assessments. Additional practitioners associated with the LDA assessment observation value, if applicable. Possible values include: “data entry person” for practitioners who transcribed the observation into the system “verifier” for practitioners who cosigned the observation
Bundle.entry.resource.Observation.status | code | False | | | The status of the observation. The status is always “final” for LDA properties. Starting in February 2022, possible values for LDA assessments include: “final” if the data is filed at time of retrieval “preliminary” if the data has not yet been filed at the time of retrieval “amended” if the data has been modified at time of retrieval “entered-in-error” if the data had once existed, but has since been withdrawn at time of retrieval
Bundle.entry.resource.Observation.subject | Reference | False | | | Reference to a patient resource. This element can't be modified by Observation.Update.
Bundle.entry.resource.Observation.extension (also: supportingInfo) | Extension | True | Yes | | Available starting in the February 2022 version of Epic for LDA assessments. If an LDA assessment is linked to a note written by a clinician, this is a reference to the DocumentReference resource.
Bundle.entry.resource.Observation.valueCodeableConcept | CodeableConcept | False | | | In LDA assessments, this element is populated if the data type is category or custom list. Only one of the value* elements is required.
Bundle.entry.resource.Observation.valueDateTime | dateTime | False | | | In LDA assessments, this element is populated if the data type is date. Only one of the value* elements is required.

Bundle.entry.resource.Observation.valueQuantity | Quantity | False | | | In LDA assessments, this element is populated if a value has units or the data type is numeric or a percentile. Only one of the value* elements is required.

Bundle.entry.resource.Observation.valueString | string | False | | | In LDA properties, this is a required element which returns the LDA display name. In LDA assessments, this element is populated if a value doesn't have units or if the data type is string. For image rows, a comma-delimited list of external document identifiers is returned. For assessments that include an image that does not have external document identifiers, “Images linked” is returned. Only one of the value* elements is required.

Bundle.entry.resource.Observation.valueTime | time | False | | | In LDA assessments, this element is populated if the data type is time. Only one of the value* elements is required.

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation?patient=eBFGq5CRApF1O-fv-r-9.G1kkDtJCFhZQm1YRU1ZAuuE3&category=lda
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="2"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/R4/Observation?patient=eBFGq5CRApF1O-fv-r-
    9.G1kkDtJCFhZQm1YRU1ZAuuE3&amp;category=lda"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/R4/Observation/eFPRBpmtzuDY7Twf3aBdgtg3"/>
    </link>
    <fullUrl value="https://hostname/instance/api/FHIR/R4/Observation/eFPRBpmtzuDY7Twf3aBdgtg3"/>
    <resource>
      <Observation>
        <id value="eFPRBpmtzuDY7Twf3aBdgtg3"/>
        <status value="final"/>
        <category>
          <coding>
            <system value="http://open.epic.com/FHIR/StructureDefinition/observation-category"/>
            <code value="LDA"/>
            <display value="Line, Drains, and Airways"/>
          </coding>
          <text value="Line, Drains, and Airways"/>
        </category>
        <category>
          <coding>
            <system value="http://open.epic.com/FHIR/StructureDefinition/observation-category"/>
            <code value="LDA-property"/>
            <display value="Lines, Drains, and Airways Properties"/>
          </coding>
          <text value="Lines, Drains, and Airways Properties"/>
        </category>
        <code>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679"/>
            <code value="110631"/>
            <display value="Peripheral IV"/>
          </coding>
          <coding>
            <system value="http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id"/>
            <code value="t6BvZ-T6vRrYAb8E3wEwf4Q0"/>
            <display value="Peripheral IV"/>
          </coding>
          <text value="Peripheral IV"/>
        </code>
        <subject>
          <reference value="Patient/eBFGq5CRApF1O-fv-r-9.G1kkDtJCFhZQm1YRU1ZAuuE3"/>
          <display value="Smith, John"/>
        </subject>
        <encounter>
          <reference value="Encounter/eymC2HHWDr3EXnMapj-LyHFgF.zx2pxKAengd7SoBaRE3"/>
          <display value="Office Visit"/>
        </encounter>
        <effectivePeriod>
          <start value="2022-01-17T18:30:00Z"/>
        </effectivePeriod>
        <valueString value="Peripheral IV 01/17/22 Right Hand"/>
        <component>
          <code>
            <coding>
              <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679"/>
              <code value="700"/>
              <display value="Placement Date"/>
            </coding>
            <coding>
              <system value="http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id"/>
              <code value="tfWy7Iki1auoDvR7iLy6-Hg0"/>
              <display value="Placement Date"/>
            </coding>
            <text value="Placement Date"/>
```

```
  </code>
  <valueDateTime value="2022-01-17"/>
</component>
<component>
  <code>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679"/>
      <code value="701"/>
      <display value="Placement Time"/>
    </coding>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id"/>
      <code value="tYt55O2NKZoq4zimxxUMZ6g0"/>
      <display value="Placement Time"/>
    </coding>
    <text value="Placement Time"/>
  </code>
  <valueTime value="12:30:00"/>
</component>
<component>
  <code>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679"/>
      <code value="304820"/>
      <display value="Location"/>
    </coding>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id"/>
      <code value="tdl3LyfOiLi3TKexqRYe3yA0"/>
      <display value="Location"/>
    </coding>
    <text value="Location"/>
  </code>
  <valueCodeableConcept>
    <coding>
      <code value="Hand"/>
      <display value="Hand"/>
    </coding>
    <text value="Hand"/>
  </valueCodeableConcept>
</component>
<component>
  <code>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679"/>
      <code value="304830"/>
      <display value="Orientation"/>
    </coding>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id"/>
      <code value="tjyrYFGargmdg1i-oo7qoyw0"/>
      <display value="Orientation"/>
    </coding>
    <text value="Orientation"/>
  </code>
  <valueCodeableConcept>
    <coding>
      <code value="Right"/>
      <display value="Right"/>
    </coding>
    <text value="Right"/>
  </valueCodeableConcept>
</component>
<component>
  <code>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679"/>
      <code value="1120100473"/>
      <display value="Unsuccessful Attempt Locations"/>
    </coding>
    <coding>
      <system value="http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id"/>
      <code value="tPj3wPdrR6Io3OcVoTfqmjg0"/>
      <display value="Unsuccessful Attempt Locations"/>
    </coding>
    <text value="Unsuccessful Attempt Locations"/>
  </code>
  <valueCodeableConcept>
    <coding>
      <code value="Left wrist;Right wrist"/>
      <display value="Left wrist;Right wrist"/>
    </coding>
    <text value="Left wrist;Right wrist"/>
  </valueCodeableConcept>
  <interpretation>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.7.708368.1150"/>
      <code value="SIG"/>
      <display value="Significant"/>
```

```
          </coding>
        </interpretation>
      </component>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/R4/Observation/ehkjaNxykIByHIGRfK.dE67Wy8U5t9uYfVbCGnje260i7MQONTfG
  BM9nShk5Rf0z.g.Bau7vlKK4h704ls5sNHP2HmyLqalARyCORLLS1Zsc3"/>
  </link>
  <fullUrl value="https://hostname/instance/api/FHIR/R4/Observation/ehkjaNxykIByHIGRfK.dE67Wy8U5t9uYfVbCGnje260i7MQONT
  fGBM9nShk5Rf0z.g.Bau7vlKK4h704ls5sNHP2HmyLqalARyCORLLS1Zsc3"/>
  <resource>
    <Observation>
      <id
  value="ehkjaNxykIByHIGRfK.dE67Wy8U5t9uYfVbCGnje260i7MQONTfGBM9nShk5Rf0z.g.Bau7vlKK4h704ls5sNHP2HmyLqalARyCORLLS1Zsc3
  "/>
      <extension url="http://hl7.org/fhir/StructureDefinition/observation-sequelTo">
        <valueReference>
          <reference value="Observation/eFPRBpmtzuDY7Twf3aBdgtg3"/>
          <display value="Peripheral IV 01/17/22 Right Hand"/>
        </valueReference>
      </extension>
      <status value="final"/>
      <category>
        <coding>
          <system value="http://open.epic.com/FHIR/StructureDefinition/observation-category"/>
          <code value="LDA"/>
          <display value="Line, Drains, and Airways"/>
        </coding>
        <text value="Line, Drains, and Airways"/>
      </category>
      <category>
        <coding>
          <system value="http://open.epic.com/FHIR/StructureDefinition/observation-category"/>
          <code value="LDA-assessment"/>
          <display value="Lines, Drains, and Airways Assessment"/>
        </coding>
        <text value="Lines, Drains, and Airways Assessment"/>
      </category>
      <code>
        <coding>
          <system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679"/>
          <code value="29550304840"/>
          <display value="Site Assessment"/>
        </coding>
        <coding>
          <system value="http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id"/>
          <code value="tCdGNZ80BNoU0xFUxE7pFrQ0"/>
          <display value="Site Assessment"/>
        </coding>
        <text value="Site Assessment"/>
      </code>
      <subject>
        <reference value="Patient/eBFGq5CRApF1O-fv-r-9.G1kkDtJCFhZQm1YRU1ZAuuE3"/>
        <display value="Smith, John"/>
      </subject>
      <encounter>
        <reference value="Encounter/eymC2HHWDr3EXnMapj-LyHFgF.zx2pxKAengd7SoBaRE3"/>
        <display value="Office Visit"/>
      </encounter>
      <effectiveDateTime value="2022-01-17T21:24:00Z"/>
      <issued value="2022-01-17T21:26:36Z"/>
      <performer>
        <reference value="Practitioner/eRhB5No2Z8P6jEmZfgaqHvg3"/>
        <display value="Becky Lipsitz, MD"/>
      </performer>
      <performer>
        <extension url="http://hl7.org/fhir/StructureDefinition/event-performerFunction">
          <valueCodeableConcept>
            <coding>
              <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
              <code value="VRF"/>
              <display value="verifier"/>
            </coding>
          </valueCodeableConcept>
        </extension>
        <reference value="Practitioner/e4PZNte6MtBGVltgJ30CD7A3"/>
        <display value="Mary Hale, RN"/>
      </performer>
      <performer>
        <extension url="http://hl7.org/fhir/StructureDefinition/event-performerFunction">
          <valueCodeableConcept>
            <coding>
```

```
              <system value="http://terminology.hl7.org/CodeSystem/v3-ParticipationType"/>
              <code value="ENT"/>
              <display value="data entry person"/>
            </coding>
          </valueCodeableConcept>
        </extension>
        <reference value="Practitioner/eB8EFBuK79tGFEZphFXeprg3"/>
        <display value="Henry Bratcher, MD"/>
      </performer>
      <valueCodeableConcept>
        <coding>
          <code value="Red;Tender"/>
          <display value="Red;Tender"/>
        </coding>
        <text value="Red;Tender"/>
      </valueCodeableConcept>
      <interpretation>
        <coding>
          <system value="http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation"/>
          <code value="A"/>
          <display value="Abnormal"/>
        </coding>
      </interpretation>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
  </search>
 </entry>
</Bundle>
```

Change log

Epic Version | Description | Type
August 2025 (May 2025, February 2025, November 2024 by special update) | Starting in August 2024, May 2025 with special update E11400862, February 2025 with special update E11306699, and November 2024 with special update E11211328, multi-select fields are handled separately, and if a SNOMED concept mapping exists for any individual selection, the SNOMED concept mapping is included in the FHIR response. Previously, when a user selected multiple options, the system treated the entire list as a single combined value. | Substantive Changes

9.1.19 Observation.Search (SmartData Elements) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/10121-Observation-Search-SmartData-Elements-R4.html
Epic API ID | 10121
FHIR version | R4
Categories | Clinical Diagnostics
API type | USCDI
A la carte | Industry-Standard Level 2 API
USCDI | Yes
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
HTTP method | GET
URL template | /api/FHIR/R4/Observation?category={category}&code={code}&focus={focus}&patient={patient}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10121

Description

This FHIR Observation resource returns data stored in SmartData elements with the following datatypes:

* string

* number

* date

* time

* category

* database

* element ID

* time with unit

* instant

Image and hierarchy datatypes are not returned.

SmartData elements are used to store discrete data about a patient and can be used in a wide variety of situations to store data related to the patient, a specific encounter, a note, an order, or another context. An example use case for this web service is searching through a specific patient encounter to return information related to back pain for that patient. If

information such as location of pain or type of pain is stored in SmartData elements, that information is returned.

Only SmartData elements stored in the following contexts are supported by this resource:

* Encounter
* Episode
* Note
* Order
* Organ
* Patient
* Stage
* Result
* Condition
* Problem

Request parameters

Name | Type | Optional | Array | Cardinality | Description

_count | number | Optional | Yes | | Number of results per page.

_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revInclude. All others will be ignored.

_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.

_revinclude | string | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.

category | token | Conditional | Yes | | Use smartdata to search for all SmartData Elements. Otherwise, use one of the following values to search for a specific type of SmartData Element: treatment-intervention-preference care-experience-preference Note: Category assignments are subjective and can differ across organizations. Note: Observation.Search (Care Experience Preference) (R4) uses the same searchable category of care-experience-preference.

code | token | Conditional | Yes | | Can be a SmartData element ID, a LOINC code starting in the August 2024 version of Epic, or a SNOMED code starting in the February 2025 version. The format for this search parameter is <system>|<SmartData Element ID>. The system value for LOINC: 2.16.840.1.113883.6.1 The system value for SNOMED: 2.16.840.1.113883.6.96 When searching by SmartData element ID, the system value is an object identifier (OID) that is different between Epic organizations as well as between production and non-production environments for the same organization. More information about OIDs can be found in the ID Types for APIs document. In the Vendor Services sandbox, use urn:oid:1.2.840.114350.1.13.0.1.7.2.727688. IDs should be URL encoded. For example, a SmartData Element with ID EPIC#OPH159 would encode to EPIC%23OPH159.

date | date | Optional | Yes | | Obtained date/time. If the obtained element is a period, a date that falls in the period.

focus | reference | Optional | Yes | | The FHIR ID for the context record that the SmartData element is attached to. If no focus is specified, the search will assume a patient-level search and use the value of the patient parameter.

patient | reference | Required | Yes | | The FHIR ID for the Patient the Observation is about.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description

combo-code | token | Optional | Yes | | Information about the SmartData element including the element reference and display string. Searches on either the code or component/code response elements.

combo-value-concept | token | Optional | Yes | | Searches on the component/ValueCodeableConcept response element.

component-code | token | Optional | Yes | | Information about the SmartData element including the element reference and display string. Searches on the component/code response element.

component-value-concept | token | Optional | Yes | | Searches on the component/ValueCodeableConcept response element.

has-member | reference | Optional | Yes | | References to other Observation (SmartData Element) resources.

issued | date | Optional | Yes | | The date and time the Observation was documented.

performer | reference | Optional | Yes | | Who filed the Observation.

status | token | Optional | Yes | | The value is "unknown" unless the data corresponds to a treatment intervention preference or a care experience preference, in which case the value is "final".

Response elements

Name | Type | Optional | Array | Cardinality | Description

Observation (R4) | Observation

Observation (R4).category | CodeableConcept | | Yes | 1..1 | Always includes "smartdata" coding. Additional codings of "treatment-intervention-preference" and "care-experience-preference" can be included if the element qualifies as treatment intervention preference data or care experience preference data accordingly.

Observation (R4).code | CodeableConcept | | | 1..* | The codes that represent the type of data returned. This can include a SmartData element ID or SNOMED code. Starting in the August 2024 version of Epic, LOINC codes are also supported.

Observation (R4).component | BackboneElement | | Yes | 1..* | Additional information. The number of components returned equals the number of lines in the SmartData element.

Observation (R4).component.code | CodeableConcept | | | 1..1 | Information about the SmartData element including the element reference and display string. If a SNOMED mapping for the SmartData element exists, that mapping appears here as well. The code element value varies depending on what SmartData element is passed. The system value is different between Epic organizations as well as between production and non-production environments for the same organization.

Observation (R4).component.code.text | String | | | 1..1 | Returns the line number for the component within a SmartData element.

Observation (R4).effectiveDateTime | dateTime | | | 0..1 | Clinically relevant time/time-period for observation.

Observation (R4).encounter (also: encounter) | Reference | | | | A reference to the Encounter during which this SmartData Observation was documented. Only returned for Observations recorded in an encounter context.

Observation (R4).focus | Reference | | Yes | 1..* | The FHIR ID for the context record that the SmartData element is attached to.
Observation (R4).hasMember | Reference | | Yes | 0..* | Returns nested references to other Observation (SmartData Element) resources.
Observation (R4).issued | instant | | | 1..1 | The date and time the Observation was documented.
Observation (R4).note | Annotation | | Yes | 1..* | The comments for the SmartData element. If the component and comment are saved to separate SmartData elements, the comment is returned as a separate Observation and not as a note on the component's Observation.
Observation (R4).performer | Reference | | Yes | 1..* | A reference to the Practitioner resource for the user who filed the Observation.
Observation (R4).status | code | | | 1..1 | The value is "unknown" unless the data corresponds to a treatment intervention preference or a care experience preference, in which case the value is "final".
Observation (R4).subject | Reference | | | 1..1 | The FHIR ID for the patient the Observation is about.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Observation?category=smartdata&patient=eVe.LNtp0U.2HjbCdUxvA5g3&focus=eSFXQ9FK
    vuZndGegQibaWniVhd0H4y.4AMYETyo5A6gc3
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 2,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation?category=smartdata&patient=eVe.LNtp0U.2HjbCdUxvA5g3&focu
    s=eSFXQ9FKvuZndGegQibaWniVhd0H4y.4AMYETyo5A6gc3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url":
    "https://hostname/instance/api/FHIR/R4/Observation/eBBGgszv54EfAOrdWJqYDgbqMbaIyAua1WztUUZpc720A31rLzxozrJYKb9cFnbur
    -NbpRG6U8IlGmAEUpRZi5k7UpVT9imMQsLB.NDBYxpY3"
        }
      ],
      "fullUrl":
    "https://hostname/instance/api/FHIR/R4/Observation/eBBGgszv54EfAOrdWJqYDgbqMbaIyAua1WztUUZpc720A31rLzxozrJYKb9cFnbur
    -NbpRG6U8IlGmAEUpRZi5k7UpVT9imMQsLB.NDBYxpY3",
      "resource": {
        "resourceType": "Observation",
        "id": "eBBGgszv54EfAOrdWJqYDgbqMbaIyAua1WztUUZpc720A31rLzxozrJYKb9cFnbur-
    NbpRG6U8IlGmAEUpRZi5k7UpVT9imMQsLB.NDBYxpY3",
        "status": "unknown",
        "category": [
          {
            "coding": [
              {
                "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
                "code": "smartdata",
                "display": "SmartData"
              }
            ],
            "text": "SmartData"
          }
        ],
        "code": {
          "coding": [
            {
              "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.727688",
              "code": "A32881",
              "display": "STRING"
            }
          ],
          "text": "STRING"
        },
        "subject": {
          "reference": "Patient/eVe.LNtp0U.2HjbCdUxvA5g3",
          "display": "zztest, sbz"
        },
        "focus": [
          {
            "reference": "Encounter/eSFXQ9FKvuZndGegQibaWniVhd0H4y.4AMYETyo5A6gc3",
            "display": "zztest, sbz"
          }
        ],
        "issued": "2021-04-16T20:56:23Z",
        "performer": [
          {
            "reference": "Practitioner/eIOnxJsdFcfgJ4iy716I3MA3",
            "display": "USER, EPIC"
          }
        ],
        "component": [
          {
```

```
            "code": {
              "text": "Line 1"
            },
            "valueString": "sbz"
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    },
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Observation/e-YNiuRnU8Fl-fShGQKlz4J1Dt-
    hDogVDn2VHKuKIgS5jDwf2BysMZiFe1VKz1i8j7I5qrxsEscUuYqt.HIG-8ZWIWSbA9seCPG4lnF6WPQo3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Observation/e-YNiuRnU8Fl-fShGQKlz4J1Dt-
    hDogVDn2VHKuKIgS5jDwf2BysMZiFe1VKz1i8j7I5qrxsEscUuYqt.HIG-8ZWIWSbA9seCPG4lnF6WPQo3",
      "resource": {
        "resourceType": "Observation",
        "id": "e-YNiuRnU8Fl-fShGQKlz4J1Dt-
    hDogVDn2VHKuKIgS5jDwf2BysMZiFe1VKz1i8j7I5qrxsEscUuYqt.HIG-8ZWIWSbA9seCPG4lnF6WPQo3",
        "status": "unknown",
        "category": [
          {
            "coding": [
              {
                "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
                "code": "smartdata",
                "display": "SmartData"
              }
            ],
            "text": "SmartData"
          }
        ],
        "code": {
          "coding": [
            {
              "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.727688",
              "code": "A32882",
              "display": "BOOLEAN"
            }
          ],
          "text": "BOOLEAN"
        },
        "subject": {
          "reference": "Patient/eVe.LNtp0U.2HjbCdUxvA5g3",
          "display": "zztest, sbz"
        },
        "focus": [
          {
            "reference": "Encounter/eSFXQ9FKvuZndGegQibaWniVhd0H4y.4AMYETyo5A6gc3",
            "display": "zztest, sbz"
          }
        ],
        "issued": "2021-04-16T20:56:02Z",
        "performer": [
          {
            "reference": "Practitioner/eIOnxJsdFcfgJ4iy716I3MA3",
            "display": "USER, EPIC"
          }
        ],
        "component": [
          {
            "code": {
              "text": "Line 1"
            },
            "valueBoolean": false
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


Change log

Epic Version | Description | Type
February 2025 | Observation (SmartData Elements) (R4) returns SNOMED codes without the SNOMED# prefix. | Breaking change

9.1.20 Observation.Search (Study Finding) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/10297-Observation-Search-Study-Finding-R4.html
Epic API ID | 10297
FHIR version | R4
Categories | Clinical Diagnostics
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Observation?patient={patient}&category={category}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10297

Description

This web service retrieves study result data from procedures, such as fetal and maternal measurement data from an obstetric ultrasound, myocardial finding data from a cardiology study, or mammography finding data from a radiology study.

Starting in May 2024, this web service can also retrieve study result data from audiology exams and follicle measurement data from a follicular ultrasound for fertility treatments.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | Either category or code is required. | | | Use “imaging” to search for all imaging observations. Use "exam" to search for all audiology studies.
code | token | Either category or code is required
date | date | False | | | Specify a date or date range that an Observation was resulted or recorded. Enter dates in ISO format (YYYY[-MM[-DD[THH:MM[:SS][TZ]]]]). Local server time is assumed if time zone information is not included.
patient | reference | Either patient or subject is required. | | | The reference to a patient resource.
subject | reference | Either patient or subject is required. | | | The reference to a patient resource.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
based-on | String | True | | | Order service request.
combo-code | token | True | | | Unsupported
combo-data-absent-reason | token | True | | | Unsupported
combo-value-concept | token | True | | | Unsupported
component-code | token | True | | | Currently Epic returns this element only if a patient has an obstetric ultrasound study documented that contains a gestational age based on a fetus measurement.
component-data-absent-reason | token | True | | | The reason why the expected value in the element Observation.component.value[x] is missing.
component-value-concept | token | True | | | Unsupported
data-absent-reason | token | True | | | Unsupported
derived-from | String | True | | | Unsupported
device | String | True | | | Unsupported
encounter | reference | True | | | Encounter associated with this observation value.
focus | reference | True | | | The specific fetus the observation this is about. Epic currently returns this element only if the patient has data on a fetus documented in an ultrasound study in their medical chart.
has-member | reference | True | | | SmartData element documented on the procedure. Links to its own observation that contains more specific data about the SmartData element.
identifier | token | True | | | Unsupported
issued | date | True | | | The date and time the observation was documented.
method | token | True | | | Method by which the observation was obtained. For an Estimated Fetal Weight study finding this element returns the method used to determine the Estimated Fetal Weight.
part-of | String | True | | | Unsupported
performer | reference | True | | | Unsupported
specimen | String | True | | | Unsupported
status | token | True | | | The status of the study. This should be set to final.
value-concept | token | True | | | Unsupported
value-date | date | True | | | The value of the observation, if the value is a date or period of time.
value-string | string | True | | | The value of the observation, if the value is a string, and also searches in CodeableConcept.text

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.response | Response | False | | | The FHIR response.
Bundle.entry.response.outcome | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.response.outcome.Observation | Observation | False | | | An instance of the R4 Observation resource.
Bundle.entry.response.outcome.Observation.basedOn | Reference | False | Yes | | Order service request.

Bundle.entry.response.outcome.Observation.bodySite | CodeableConcept | False | | | Contains the SNOMED code that identifies where on the body the study was performed.

Bundle.entry.response.outcome.Observation.category | CodeableConcept | False | Yes | | This is always "study-finding." Imaging observations also have the “imaging” category. Audiology studies also have the “exam” and “audiology” category values.

Bundle.entry.response.outcome.Observation.code | CodeableConcept | False | | | What type of study finding this is. If this is an Estimated Fetal Weight study finding documented on an obstetric ultrasound, then this element also includes LOINC code 11727-5.

Bundle.entry.response.outcome.Observation.component | BackboneElement | False | Yes | | Component information.

Bundle.entry.response.outcome.Observation.component.code | CodeableConcept | False | | | Currently Epic returns this element only if a patient has an obstetric ultrasound study documented that contains a gestational age based on a fetus measurement.

Bundle.entry.response.outcome.Observation.component.valueString | string | False | | | Currently Epic returns this element only if a patient has an obstetric ultrasound study documented that contains a gestational age based on a fetus measurement. This element returns the gestational age value based on the measurement.

Bundle.entry.response.outcome.Observation.effectiveDateTime | dateTime | False | | | The date and time the exam ended.

Bundle.entry.response.outcome.Observation.encounter | Reference | False | | | Encounter associated with this observation value.

Bundle.entry.response.outcome.Observation.focus | Reference | True | Yes | | The specific fetus the observation this is about. Epic currently returns this element only if the patient has data on a fetus documented in an ultrasound study in their medical chart.

Bundle.entry.response.outcome.Observation.hasMember | Reference | False | Yes | | SmartData element documented on the procedure. Links to its own observation that contains more specific data about the SmartData element.

Bundle.entry.response.outcome.Observation.issued | instant | False | | | The date and time the observation was documented.

Bundle.entry.response.outcome.Observation.method | CodeableConcept | True | | | Method by which the observation was obtained. For an Estimated Fetal Weight study finding this element returns the method used to determine the Estimated Fetal Weight.

Bundle.entry.response.outcome.Observation.referenceRange | BackboneElement | True | Yes | | Text based reference range for the observation.

Bundle.entry.response.outcome.Observation.status | code | False | | | The status of the study. This should be set to final.

Bundle.entry.response.outcome.Observation.subject | Reference | False | | | The FHIR ID for the patient the observation is about.

Bundle.entry.response.outcome.Observation.valueCodeableConcept | CodeableConcept | True | | | The coding system (SNOMED, LOINC, or Epic), specific code, and user-facing description for an observation study finding. Useful for study findings that exist in multiple coding systems.

Bundle.entry.response.outcome.Observation.valueInteger | integer | True | | | The numeric value for an observation study finding.

Bundle.entry.response.outcome.Observation.valueQuantity | Quantity | True | | | This element is populated if a value has units or the data type is numeric or a percentile.

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation?patient=epzKvfQ.SNFEmP3QlFwh6DcSbNRycEOOdMejDcAvTiFI3&category=imaging
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Observation?patient=epzKvfQ.SNFEmP3QlFwh6DcSbNRycEOOdMejDcAvTiFI3&ca
    tegory=imaging"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url":
    "https://hostname/instance/api/FHIR/R4/Observation/eJFgryT-
    06nUPbHLVFnpCw2igXjN97CZNFknSMjC5dKXF0uuTajDHLRZO.qy4c2Z83"
        }
      ],
      "fullUrl":
    "https://hostname/instance/api/FHIR/R4/Observation/eJFgryT-
    06nUPbHLVFnpCw2igXjN97CZNFknSMjC5dKXF0uuTajDHLRZO.qy4c2Z83",
      "resource": {
        "resourceType": "Observation",
        "id": "eJFgryT-06nUPbHLVFnpCw2igXjN97CZNFknSMjC5dKXF0uuTajDHLRZO.qy4c2Z83",
        "extension": [
          {
            "valueReference": {
              "reference": "Observation/enQcDqvmReQd8FpmQFhQnqqA6pUFMqIc7o4tpiuAkTlPRxfWuubz7daE48l-1DPcsHmpN-
    71wLtBoAUcVOrOfOWFRfDXjX83iOeyiuM3O7MCCxBuAT1AEI3zQh4OI9QV1I2bm1q39h1RLUjTe31jG4w3",
              "display": "FINDINGS - TESTS - OB - ULTRASOUND - MOM-SPECIFIC - GESTATIONAL SAC"
            },
            "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
          },
          {
            "valueReference": {
              "reference": "Observation/esB6B.SD-hvSTfBjzYYOTp.v4LcgfBPaYOfLLVs3L3IAKohu2YKidZl4fdONqUVXj1MjUu0DFgUnlAGg
    x7420QuPkzj1OSwIDjKbosJgEsWNPwowUoFN9WlYhWO6kIylBi41tPZqypMU-920SHuzhTA3",
              "display": "FINDINGS - TESTS - OB - ULTRASOUND - MOM-SPECIFIC - PLACENTA"
            },
            "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
          },
          {
            "valueReference": {
              "reference": "Observation/eik4yRQDgDm5wt8NsTyFB3dHj8n1qOk-Ena7e4wy-ofTSlCdl-BblYOGQCi0a3c1IgOeaUP-
```

```
ukUFtAVORfH.nsbNhIkiNevLD4ee0QdmM8E8WgKuK5pEv22TL2Pr5gFZYU1HuMR4mcVmyq6JQ2vFgwg3",
          "display": "FINDINGS - TESTS - OB - ULTRASOUND - BABY-SPECIFIC - GESTATIONAL SAC"
        },
        "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
      },
      {
        "valueReference": {
          "reference": "Observation/eu-xLmCAr2qveOdu7-1IMDN.8RjAHquZg74nl.vwjgvERBu5BCwNnJFH.Lcm3cWH6Du2KZ-
4DyTvKxxm4hX4hiAX8.-e9ufatl4LzUL5WefdkGultMFx1llj8.iVoC.C33zImchQyigOemvhC99uZ3Q3",
          "display": "FINDINGS - TESTS - OB - ULTRASOUND - BABY-SPECIFIC - YOLK SAC"
        },
        "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
      },
      {
        "valueReference": {
          "reference": "Observation/eoJiEtTMWX1L5nJLGzWvKMzU4tN99Zkpn7uzQ9yEu9p4Bj7gPRbxR685MpACniF9sHGDidYaBjOFlkoQ
Rv3b.3jGT9EVqqYpaOawqSs2dj3w6e.wSFHjdsIVxtkLhQ8dn1KCjYIOpDkZWvZv64pkDbw3",
          "display": "FINDINGS - TESTS - OB - ULTRASOUND - BABY-SPECIFIC - PRESENTATION"
        },
        "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
      },
      {
        "valueReference": {
          "reference": "Observation/ewih1pOtcFiisTOOGsvavqITupGQ2VBKJlLqjX2qtjtT65nZv9JE6fnMpm1PNeU9E1KAEy7AlLB4uMQ-
5fGudZWN80fFv0Auy7fynanPEy7En3PP2i6SrLNJ3ikxk19lLXw3SyGXgJqDHhkkoa6mgdw3",
          "display": "FINDINGS - TESTS - OB - ULTRASOUND - BABY-SPECIFIC - FETAL POLE"
        },
        "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
      },
      {
        "valueReference": {
          "reference": "Observation/e1lhXf37485X8FKVLjJ0yn1x356sxIvIQq1jjHtltCFkzggyzMcUbMdNJ-Q26HB.MRJZhdw-
CYVahAPsyR569.1T8TUUVAbAltY2K6jQZgtCboLndTN3LoYoK-20BgzyG3VOp0SZ11eitBktFJd4mIA3",
          "display": "FINDINGS - TESTS - OB - ULTRASOUND - BABY-SPECIFIC - CARDIAC ACTIVITY"
        },
        "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
      },
      {
        "valueReference": {
          "reference": "Observation/eMYLKSfTIZEpouCOo2w1VmpOZFtRwh.O5AOc8FcYtr9WWJGN3z9g2BF.hwlzj8trk5rfrViXNArWzz88
MAmy1qe67YmwKGfL8GZ08bJWHG0DcCdWrcuBEihzppYXN4imBFUspuNy4gBFLXL2J.uX0kg3",
          "display": "FINDINGS - TESTS - OB - ULTRASOUND - BABY-SPECIFIC - AMNIOTIC FLUID"
        },
        "url": "http://open.epic.com/FHIR/StructureDefinition/extension/smartdata"
      }
    ],
    "basedOn": [
      {
        "reference": "ServiceRequest/eOS.q.8Pg0GxH6e3weglmO7Wxqv.bf9-vS64qqOw08e03",
        "display": "OB US + 14 WEEKS"
      }
    ],
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
            "code": "study-finding",
            "display": "Study Finding"
          }
        ],
        "text": "Study Finding"
      }
    ],
    "code": {
      "text": "Fetus A: Fetal Anatomy"
    },
    "subject": {
      "reference": "Patient/epzKvfQ.SNFEmP3QlFwh6DcSbNRycEOOdMejDcAvTiFI3",
      "display": "Simota, Noodle"
    },
    "focus": [
      {
        "identifier": {
          "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.826983",
          "value": "1001127844"
        },
        "display": "Fetus A"
      }
    ],
    "encounter": {
      "reference": "Encounter/eLcG44-s8Jg6QtUOoGqD62BWGbgGCtXxf50ZQrK-h8.E3",
      "display": "Office Visit"
    },
    "effectiveDateTime": "2021-09-17T14:30:11Z",
    "issued": "2021-09-17T14:32:58Z"
  },
  "search": {
```

```
        "mode": "match"
      }
    }
  ]
}
```

Change log

Epic Version | Description | Type
February 2024 | SmartData Elements in the Observation (Study Finding) (R4) Resource moved from extension to hasMember | Breaking change

9.1.21 Observation.Search (Vital Signs) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/973-Observation-Search-Vital-Signs-R4.html
Epic API ID | 973
FHIR version | R4
Categories | Clinical Diagnostics
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | GET
URL template | api/FHIR/R4/Observation?patient={patient}&subject={subject}&category={category}&code={code}&date={date}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=973

Description

This web service retrieves vital sign data from a patient's chart, as well as any other non-duplicable data found in the patient's flowsheets across all encounters.

To search for Assessments, use the Observation.Search (Assessments) (R4) API instead.

Beginning in Epic version November 2026, use the Observation.Search (Flowsheets) (R4) API instead to search for other flowsheet data which are not vital signs.

The following types of vital information and corresponding LOINC codes are supported by default. How specific types of data translate from within the Epic system to FHIR might vary depending on organizational configuration:

* 2708-6 Oxygen saturation in arterial blood
* 2710-2 Oxygen saturation in capillary blood by oximetry
* 29463-7 Body weight
* 3141-9 Body weight measured
* 3150-0 Inhaled oxygen concentration
* 3151-8 Inhaled oxygen flow rate
* 59408-5 Oxygen saturation in arterial blood by pulse oximetry (SpO2)
* 8287-5 Head occipital-frontal circumference by tape measure
* 8302-2 Body height
* 8306-3 Body height --lying
* 8310-5 Body temperature
* 8462-4 Diastolic blood pressure
* 8480-6 Systolic blood pressure
* 85354-9 Blood pressure systolic and diastolic
* 8716-3 Vital signs
* 8867-4 Heart rate
* 9279-1 Respiratory rate

Remarks

To see an example of how the component.code element is structured for patient-entered flowsheet data, refer to the sample request section of the Observation.Create (Vitals) (STU3) API entry.

Request parameters

Name | Type | Optional | Array | Cardinality | Description

_count | number | Optional | Yes | | Number of results per page.

_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count, _include, and _revInclude. All others will be ignored.

_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.

_revinclude | string | Optional | Yes | | Used to include Provenance or AuditEvent resources that refer to search results, when available.

category | token | Conditional | Yes | | Use "vital-signs" to search for vitals Observations. Either category or code must be provided.

code | token | Conditional | Yes | | Formatted as <system>|<code>. If a system is not provided, the code is assumed to be a LOINC code. Valid codes and their systems include: LOINC Code System Value: http://loinc.org or no system value. Epic Encoded Flowsheet Identifier - This only works if the flowsheet is considered a vital sign. System Value: http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id Note: The encoded flowsheet identifier must be generated by IT admins at an Epic organization. Epic Internal Flowsheet Identifier - This only works if the flowsheet is considered a vital sign. System Value: Epic generated OID, varies in each Epic environment. Cancer Data Standards Registry and Repository (CADSR) Terminology System Value: http://wiki.nci.nih.gov/display/cadsr The following LOINC codes can be used for searching growth chart data: 59576-9: BMI percentile 77606-2: Weight-for-length percentile 8289-1: Head circumference percentile

date | date | Optional | Yes | | The date range for when the Observation was taken. For growth chart data, only the most recent Observation within the timeframe is returned.

patient | reference | Conditional | Yes | | Reference to a Patient resource the Observation is about. Either this parameter or subject must be provided. If both are provided, they must match.

subject | reference | Conditional | Yes | | Reference to a Patient resource the Observation is about. Either this parameter or patient must be provided. If both are provided, they must match.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description

combo-code | token | Optional | Yes | | Matches to the code or component.code element in the response.

combo-data-absent-reason | token | Optional | Yes | | The reason why the expected value in the element Observation.value* or Observation.component.value* is missing.

combo-value-concept | token | Optional | Yes | | The value or component value of the observation, if that value is a CodeableConcept. Matches to valueCodeableConcept or component.valueCodeableConcept.

component-code | token | Optional | Yes | | Matches to the component.code element in the response.

component-data-absent-reason | token | Optional | Yes | | The reason why the expected value in the element Observation.component.value* is missing.

component-value-concept | token | Optional | Yes | | The component value of the Observation, if that value is a CodeableConcept. Matches to component.valueCodeableConcept.

data-absent-reason | token | Optional | Yes | | The reason why the expected value in the element Observation.value* is missing.

encounter | reference | Optional | Yes | | Encounter associated with this Observation value, if applicable. When multiple encounters are involved, such as for growth chart percentiles, the most recent encounter must be specified.

issued | date | Optional | Yes | | The date and time the Observation was documented.

method | token | Optional | Yes | | Method by which the Observation was obtained. See the method response element for more information.

performer | reference | Optional | Yes | | Who filed the Observation. If the measurement is patient-reported, this is a reference to the Patient resource.

status | token | Optional | Yes | | The status of the Observation. Possible values include: "preliminary" - value is pended "final" - value has been filed "amended" - value has been edited "entered-in-error" - value has been deleted

value-concept | token | Optional | Yes | | The value of the Observation if it is a CodeableConcept. Matches to valueCodeableConcept.

value-date | date | Optional | Yes | | The value of the Observation, if the value is a date or period of time.

value-string | string | Optional | Yes | | The value of the Observation, if valueString is populated.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Observation (R4) | Observation

Observation (R4).bodySite | CodeableConcept | Conditional | | 0..1 | Applicable only to organizations in the Netherlands. The site on the body where the Observation was made. Used for the location where a blood pressure reading was obtained.

Observation (R4).category | CodeableConcept | | Yes | 1..1 | This element always returns "vital-signs".

Observation (R4).code | CodeableConcept | Required | | 0..* | Coded value for what is being measured. The following code systems can be returned: OID for internal Epic flowsheet ID Encoded value of the internal Epic flowsheet ID External concept mappings such as LOINC and SNOMED

Observation (R4).component | BackboneElement | | Yes | 0..* | Includes extra information about the Observation. Populated only for: Blood Pressure: Systolic (LOINC 8480-6) and diastolic (LOINC 8462-4) pressures are always returned as components if available. Can include components with a SNOMED code of 70665002 and LOINC code of 8358-4 for the cuff type or size, and a SNOMED code of 424724000 and a LOINC code of 8361-8 for cuff position. Average Blood Pressure (starting in the November 2026 version of Epic): Average systolic (LOINC 96608-5) and average diastolic (LOINC 96609-3) are always returned as components if available. Can include a component with a SNOMED code of 246432004 for number of occurrences. SpO2 can include components with LOINC code of 3151-8 for inhaled oxygen flow rate, 3150-0 for inhaled oxygen concentration, and 74206-4 for whether oxygen therapy was given. Pulse Rate can include a component with LOINC code of 44969-4 for heart rate rhythm palpitation. Weight can include a component with a code of 8352-7 for how the patient was clothed for weight reading. Patient-Entered Flowsheet Metadata Represents strict metadata if the coding sub-element is populated. The following strict metadata can be returned: 1-UUID The unique reading ID 2-Blood Glucose Meal Times 1-Before meal 2-After meal 3-Insulin Delivery Reason 1-Basal 2-Bolus 5-VO2 Max Test Type 1-Max exercise test 2-Sub-max exercise prediction 3-Non exercise prediction 6-Manually Entered - Whether or not the patient manually entered this reading into the system (as opposed to automatically entered by a device). false true Represents loose metadata if only the text sub-element is populated. UUID, Blood Glucose Meal times, Insulin Delivery Reason, and VO2 Max Test Type must have their value specified in component.valueString. Manually Entered is specified as a valueBoolean.

Observation (R4).component.code | CodeableConcept | | | 0..1 | Code for the extra information for the measurement. The following extra information can be returned for each measurement, with the associated code: Blood Pressure 8480-6: Systolic pressure 8462-4: Diastolic pressure SNOMED 70665002 / LOINC 8358-4: Cuff type or size SNOMED 424724000 / LOINC 8361-8: Cuff position Average Blood Pressure (starting in the November 2026 version of Epic) 96608-5: Systolic blood pressure mean 96609-3: Diastolic blood pressure mean SNOMED 246432004: Number of occurrences SpO2 3151-8: Inhaled oxygen flow rate 3150-0: Inhaled oxygen concentration 74206-4: Whether oxygen therapy was given Pulse Rate 44969-4: Heart rate rhythm palpitation. Weight 8352-7: Whether patient was clothed

Observation (R4).component.valueBoolean | boolean | | | 0..1 | Only used for Patient-Entered Flowsheet Metadata values. Valid values include: For Code = 6-Manually Entered false true

Observation (R4).component.valueCodeableConcept | CodeableConcept | | | 0..1 | The value as a CodeableConcept for the extra information for the measurement.

Observation (R4).component.valueQuantity | Quantity | | | 0..1 | If the extra information for the measurement is numeric, this is the numeric value. For example, it could be the flow rate (L/min) or concentration (%) for an SpO2 measurement. Starting in the November 2026 version of Epic, this is also the number of occurrences for average blood pressure.

Observation (R4).component.valueString | string | | | 0..1 | Values for the patient-entered flowsheet metadata.

Observation (R4).dataAbsentReason | CodeableConcept | | | 0..1 | Indicates that an Observation value is missing. This is always "unknown" for systolic blood pressure values in Observations for diastolic-only blood pressure flowsheets, or for diastolic blood pressure value in Observations for systolic-only blood pressure flowsheets.

Observation (R4).effectiveDateTime | dateTime | | | 0..1 | The date and time the Observation was taken. Starting in the November 2026 version of Epic, this element is omitted when effectivePeriod is used.

Observation (R4).effectivePeriod | Period | | | 0..1 | Starting in the November 2026 version of Epic, average blood pressure observations can use this element to return the begin and end instants for the average as an effective time range. If effectivePeriod is used, then effectiveDateTime is omitted.

Observation (R4).encounter | Reference | Conditional | | 1..1 | Reference to an Encounter resource that this Observation belongs to.

Observation (R4).extension (supportingInfo) (also: extension (supportingInfo)) | Extension | Conditional | Yes | 0..* | If the measurement is linked to a note written by a clinician, this returns a reference to the DocumentReference resource. URL: http://hl7.org/fhir/StructureDefinition/workflow-supportingInfo

Observation (R4).extension (temperature-in-fahrenheit) (also: extension (temperature-in-fahrenheit)) | Extension | | Yes | 0..1 | If the measurement is a temperature value, this element contains the temperature value quantity in Fahrenheit. This element is excluded for other data types. URL: http://open.epic.com/FHIR/STU3/StructureDefinition/temperature-in-fahrenheit

Observation (R4).extension (temperature-in-fahrenheit).valueQuantity | Quantity | | | 1..1 | The temperature value in degrees Fahrenheit [degF]. This represents the same temperature measurement as Observation.valueQuantity, which can use different units.

Observation (R4).extension (template-id) (also: extension (template-id)) | Extension | | Yes | 0..1 | Starting in the May 2024 version of Epic, or February 2024 with a special update, this is the specific flowsheet template (a collection of flowsheet rows and groups) to which data was filed. URL: http://open.epic.com/FHIR/StructureDefinition/extension/template-id

Observation (R4).extension (template-id).valueIdentifier | Identifier | | | 1..1 | The system and value for the flowsheet template.

Observation (R4).extension (template-id).valueIdentifier.system | String | | | 1..1 | The OID for the Epic internal flowsheet template identifier. This varies based on the specific Epic environment.

Observation (R4).extension (template-id).valueIdentifier.value | String | | | 1..1 | The Epic internal identifier of the flowsheet template.

Observation (R4).interpretation | CodeableConcept | | Yes | 0..* | Returns a system of http://terminology.hl7.org/CodeSystem/v3-ObservationInterpretation and value of "A" for vitals with an abnormal status. Returns an environment-specific OID system with a value of "SIG" for vitals with a significant status. In Netherlands locales starting in the November 2022 release of Epic, these interpretation values can also be returned, alongside a system value of http://snomed.info/sct: For heart rate profile Observations, possible values include: SNOMED code 271636001 (Heart regular) for normal heart rate readings SNOMED code 248650006 (Heart irregular) for abnormal heart rate readings For blood glucose profile Observations, possible values include: SNOMED code 281302008 (Above reference range) for blood glucose readings higher than the reference range SNOMED code 11896004 (Intermediate) for blood glucose readings within the reference range (inclusive of the reference range values) SNOMED code 281300000 (Below reference range) for blood glucose readings below the reference range

Observation (R4).issued | instant | | | 1..1 | The date and time the Observation was documented.

Observation (R4).method | CodeableConcept | | | 0..1 | For patient-entered flowsheets, this element is returned with a text value of "Patient-entered". In Netherlands locales with blood pressure vitals Observations, SNOMED (system http://snomed.info/sct) codes can be returned: Blood pressure taken with an invasive method: 10179008 Blood pressure taken with a non-invasive method: 22762002

Observation (R4).note | Annotation | | Yes | 0..1 | The comment associated with the value.

Observation (R4).performer | Reference | | Yes | 1..* | Reference to a FHIR resource for who filed the Observation. If the vital was filed by a patient, this references a Patient resource. Otherwise, it references a Practitioner resource.

Observation (R4).performer.extension (performerFunction) (also: extension (performerFunction)) | Extension | | Yes | 0..1 | Additional Practitioners associated with the Observation value, if applicable. Possible values include: "data entry person" for Practitioners who transcribed the Observation into the system "verifier" for Practitioners who cosigned the Observation URL: http://hl7.org/fhir/StructureDefinition/event-performerFunction

Observation (R4).referenceRange | BackboneElement | Conditional | Yes | 0..1 | Applicable to organizations in the Netherlands starting in the November 2022 version of Epic. This is populated only for blood glucose readings. This element indicates the abnormal minimum and maximum values defined for the blood glucose reading.

Observation (R4).referenceRange.high | Quantity | | | 0..1 | Applicable to organizations in the Netherlands starting in the November 2022 version of Epic. The higher reference limit for blood glucose readings.

Observation (R4).referenceRange.low | Quantity | | | 0..1 | Applicable to organizations in the Netherlands starting in the November 2022 version of Epic. The lower reference limit for blood glucose readings.

Observation (R4).status | code | | | 1..1 | This element has the following possible values: "preliminary" - value is pended "final" - value has been filed "amended" - value has been edited "entered-in-error" - value has been deleted

Observation (R4).subject | Reference | Required | | 1..1 | The FHIR ID for the patient the Observation is about.

Observation (R4).valueCodeableConcept | CodeableConcept | | | 0..1 | This element is populated if the data type is a category or custom list. Only one of the value* elements is returned.

Observation (R4).valueDateTime | dateTime | | | 0..1 | This element is populated if data type is date. Only one of the value* elements is returned.

Observation (R4).valueQuantity | Quantity | | | 0..1 | This element is populated if a value has units or the data type is numeric. Only one of the value* elements is returned.

Observation (R4).valueString | string | | | 0..1 | This element is populated if a value doesn't have units or the data type is string. Only one of the value* elements is returned.

Observation (R4).valueTime | time | | | 0..1 | This element is populated if data type is time. Only one of the value* elements is returned.

Sample request

```
GET https://hostname/instance/api/FHIR/R4/Observation?category=vital-signs&patient=e63wRTbPfr1p8UW81d8Seiw3
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 2,
```

```
"link": [
  {
    "relation": "self",
    "url": "https://hostname/instance/api/FHIR/R4/Observation?category=vital-signs&patient=e63wRTbPfr1p8UW81d8Seiw3"
  }
],
"entry": [
  {
    "link": [
      {
        "relation": "self",
        "url":
  "https://hostname/instance/api/FHIR/R4/Observation/envjcVAhuFtXhXNFIg1Dr-2-
  8diVcq3BOMcZpbjYOC7JAJ1pPzK0v1075T4XMHL.83"
      }
    ],
    "fullUrl":
  "https://hostname/instance/api/FHIR/R4/Observation/envjcVAhuFtXhXNFIg1Dr-2-
  8diVcq3BOMcZpbjYOC7JAJ1pPzK0v1075T4XMHL.83",
    "resource": {
      "resourceType": "Observation",
      "id": "envjcVAhuFtXhXNFIg1Dr-2-8diVcq3BOMcZpbjYOC7JAJ1pPzK0v1075T4XMHL.83",
      "status": "final",
      "category": [
        {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/observation-category",
              "code": "vital-signs",
              "display": "Vital Signs"
            }
          ],
          "text": "Vital Signs"
        }
      ],
      "code": {
        "coding": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.0.1.7.2.707679",
            "code": "5",
            "display": "BP"
          },
          {
            "system": "http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id",
            "code": "tBdNYepLeojPG60x7nUx9kQ0",
            "display": "BP"
          },
          {
            "system": "urn:oid:1.2.246.537.6.96",
            "code": "8462-4"
          },
          {
            "system": "urn:oid:1.2.246.537.6.96",
            "code": "8480-6"
          },
          {
            "system": "http://loinc.org",
            "code": "55284-4",
            "display": "Blood pressure systolic and diastolic"
          },
          {
            "system": "http://loinc.org",
            "code": "85354-9",
            "display": "Blood pressure panel with all children optional"
          },
          {
            "system": "http://loinc.org",
            "code": "8716-3",
            "display": "Vital signs"
          },
          {
            "system": "http://snomed.info/sct",
            "code": "1081000119105",
            "display": "Opioid dependence, on agonist therapy (disorder)"
          },
          {
            "system": "http://www.ama-assn.org/go/cpt",
            "code": "27130",
            "display": "Arthroplasty, acetabular and proximal femoral prosthetic replacement (total hip arthroplasty),
   with or without autograft or allograft"
          }
        ],
        "text": "BP"
      },
      "subject": {
        "reference": "Patient/e63wRTbPfr1p8UW81d8Seiw3",
        "display": "Mychart, Teddy"
      },
      "encounter": {
```

```
      "reference": "Encounter/e8hBVXYLK9-otx9sWEP7xJQ3",
      "identifier": {
        "use": "usual",
        "system": "urn:oid:1.2.840.114350.1.13.0.1.7.3.698084.8",
        "value": "1875"
      },
      "display": "Office Visit"
    },
    "effectiveDateTime": "2010-06-12T17:20:00Z",
    "issued": "2010-06-12T17:20:27Z",
    "performer": [
      {
        "reference": "Practitioner/eU2bhJHhLPs8.VF8LzIGbIw3",
        "display": "Nurse Family Medicine, RN"
      }
    ],
    "component": [
      {
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "8480-6",
              "display": "Systolic blood pressure"
            }
          ],
          "text": "Systolic blood pressure"
        },
        "valueQuantity": {
          "value": 135,
          "unit": "mm[Hg]",
          "system": "http://unitsofmeasure.org",
          "code": "mm[Hg]"
        }
      },
      {
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "8462-4",
              "display": "Diastolic blood pressure"
            }
          ],
          "text": "Diastolic blood pressure"
        },
        "valueQuantity": {
          "value": 85,
          "unit": "mm[Hg]",
          "system": "http://unitsofmeasure.org",
          "code": "mm[Hg]"
        }
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url":
"https://hostname/instance/api/FHIR/R4/Observation/e1hj2YagJLq39h3qfCS3B9i-
EZIOMO.aHMy2kKGxyGuJhQTs0rkYDCOmRCi8IX.P13"
    }
  ],
  "fullUrl":
"https://hostname/instance/api/FHIR/R4/Observation/e1hj2YagJLq39h3qfCS3B9i-
EZIOMO.aHMy2kKGxyGuJhQTs0rkYDCOmRCi8IX.P13",
  "resource": {
    "resourceType": "Observation",
    "id": "e1hj2YagJLq39h3qfCS3B9i-EZIOMO.aHMy2kKGxyGuJhQTs0rkYDCOmRCi8IX.P13",
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "vital-signs",
            "display": "Vital Signs"
          }
        ],
        "text": "Vital Signs"
      }
    ],
    "code": {
      "coding": [
```

```
        {
          "system": "urn:oid:1.2.840.114350.1.13.0.1.7.2.707679",
          "code": "5",
          "display": "BP"
        },
        {
          "system": "http://open.epic.com/FHIR/StructureDefinition/observation-flowsheet-id",
          "code": "tBdNYepLeojPG60x7nUx9kQ0",
          "display": "BP"
        },
        {
          "system": "urn:oid:1.2.246.537.6.96",
          "code": "8462-4"
        },
        {
          "system": "urn:oid:1.2.246.537.6.96",
          "code": "8480-6"
        },
        {
          "system": "http://loinc.org",
          "code": "55284-4",
          "display": "Blood pressure systolic and diastolic"
        },
        {
          "system": "http://loinc.org",
          "code": "85354-9",
          "display": "Blood pressure panel with all children optional"
        },
        {
          "system": "http://loinc.org",
          "code": "8716-3",
          "display": "Vital signs"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "1081000119105",
          "display": "Opioid dependence, on agonist therapy (disorder)"
        },
        {
          "system": "http://www.ama-assn.org/go/cpt",
          "code": "27130",
          "display": "Arthroplasty, acetabular and proximal femoral prosthetic replacement (total hip arthroplasty),
with or without autograft or allograft"
        }
      ],
      "text": "BP"
    },
    "subject": {
      "reference": "Patient/e63wRTbPfr1p8UW81d8Seiw3",
      "display": "Mychart, Teddy"
    },
    "encounter": {
      "reference": "Encounter/ebxvM3lGd503DIAQma30uzg3",
      "identifier": {
        "use": "usual",
        "system": "urn:oid:1.2.840.114350.1.13.0.1.7.3.698084.8",
        "value": "1882"
      },
      "display": "Office Visit"
    },
    "effectiveDateTime": "2010-04-21T16:40:00Z",
    "issued": "2010-04-21T16:40:26Z",
    "performer": [
      {
        "reference": "Practitioner/eU2bhJHhLPs8.VF8LzIGbIw3",
        "display": "Nurse Family Medicine, RN"
      }
    ],
    "component": [
      {
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "8480-6",
              "display": "Systolic blood pressure"
            }
          ],
          "text": "Systolic blood pressure"
        },
        "valueQuantity": {
          "value": 138,
          "unit": "mm[Hg]",
          "system": "http://unitsofmeasure.org",
          "code": "mm[Hg]"
        }
      },
      {
        "code": {
```

```
            "coding": [
              {
                "system": "http://loinc.org",
                "code": "8462-4",
                "display": "Diastolic blood pressure"
              }
            ],
            "text": "Diastolic blood pressure"
          },
          "valueQuantity": {
            "value": 88,
            "unit": "mm[Hg]",
            "system": "http://unitsofmeasure.org",
            "code": "mm[Hg]"
          }
        }
      ]
    },
    "search": {
      "mode": "match"
    }
  }
 ]
}
```

Change log

Epic Version | Description | Type
May 2024 (February 2024 by special update) | A new “template-id” extension has been added. If vitals data was filed using the Observation.Create API, this extension is returned and includes the ID of the flowsheet template that vitals data was filed to. | Substantive Changes
August 2024 | Observation (Vitals) (R4) no longer returns Functional Status, Disability Status, Mental/Cognitive Status, or Social Determinants of Health (SDOH) assessments. | Breaking Change
August 2024 | Observation Sub-Resource Security Overrides Encoded Flowsheet ID | Breaking change
August 2025 (May 2025, February 2025, November 2024 by special update) | Starting in August 2024, May 2025 with special update E11400862, February 2025 with special update E11306699, and November 2024 with special update E11211328, multi-select fields are handled separately, and if a SNOMED concept mapping exists for any individual selection, the SNOMED concept mapping is included in the FHIR response. Previously, when a user selected multiple options, the system treated the entire list as a single combined value. | Substantive Changes

9.1.22 Observation.Update (Lines, Drains, Airways) (R4)

File: site/apis/09-Clinical-Diagnostics/1-R4/974-Observation-Update-Lines-Drains-Airways-R4.html
Epic API ID | 974
FHIR version | R4
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Keywords | Flowsheets;LDA
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2019
HTTP method | PUT
URL template | api/FHIR/R4/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=974

Description

The FHIR Observation resource defines measurements and assertions about a patient, including LDA-W (lines, drains, airways, and wound) data. Observation.Update allows the client to modify existing LDA-W data in the database, including LDA properties and LDA assessments.

This resource uses an E-Tag and an If-Match HTTP header.

If the update is successful, HTTP status code 200 will be returned. If the update is not successful, an operational outcome will be returned.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | An instance of the R4 Observation resource.
Observation.category | CodeableConcept | False | Yes | | The category is always "LDA". This element can't be modified by Observation.Update.
Observation.code | CodeableConcept | False | | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. This element can't be modified by Observation.Update. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.
Observation.component | BackboneElement | False | Yes | | Populated only for patient-entered flowsheets.

Observation.component.code | CodeableConcept | False | | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.

Observation.component.extension (also: LDA-Property-Comment) | Extension | False | Yes | | Comment for the property row.

Observation.component.valueQuantity | Quantity | False | | | Value of the property row.

Observation.effectiveDateTime | dateTime | False | | | The prioritized instant. This element can't be modified by Observation.Update.

Observation.effectivePeriod | Period | False | | | Start instant and/or end instant of the LDA.

Observation.encounter | Reference | False | | | Reference to an encounter resource. This element can't be modified by Observation.Update.

Observation.note | Annotation | False | Yes | | The comment. Limited to 254 characters in the August 2025 version of Epic and later. Limited to 60 characters in earlier versions of Epic.

Observation.extension (also: sequelTo) | Extension | False | Yes | | sequelTo extension that specifies the parent LDA FHIR ID.

Observation.status | code | True | | | The status of the observation. Only a value of "final" is supported. We do not support filing data that isn't finalized.

Observation.subject | Reference | False | | | Reference to a patient resource. This element can't be modified by Observation.Update.

Observation.valueQuantity | Quantity | False | | | Value of the assessment. For image rows, a comma-delimited list of external DCS IDs is required.

Observation.valueString | string | False | | | LDA display name.

Sample request

```
https://hostname/instance/api/FHIR/R4/Observation/ePgszz-JpoXhhBHDW-tMoBw3
<Observation>
<id value="ePgszz-JpoXhhBHDW-tMoBw3" />
<category>
<coding>
<system value="http://open.epic.com/FHIR/StructureDefinition/observation-category" />
<code value="LDA" />
<display value="Line, Drains, and Airways" />
</coding>
<text value="Line, Drains, and Airways" />
</category>
<code>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679" />
<code value="4578" />
<display value="FLOQA Line LDA" />
</coding>
<text value="FLOQA Line LDA" />
</code>
<subject>
<reference value="Patient/eZO66HeGXB6VCJemL-ciSKA3" />
<display value="Meiko Lufhir" />
</subject>
<encounter>
<reference value="Encounter/e.R4dWbUDqF01.Z6BEIfEkZecZklxAhL82tS5wzzlD0o3" />
</encounter>
<effectivePeriod>
<start value="2019-09-04T21:42:00Z" />
</effectivePeriod>
<component>
<code>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679" />
<code value="700" />
<display value="Placement Date" />
</coding>
<text value="Placement Date" />
</code>
<valueDateTime value="2019-09-04" />
</component>
<component>
<code>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679" />
<code value="701" />
<display value="Placement Time" />
</coding>
<text value="Placement Time" />
</code>
<valueTime value="16:42:00" />
</component>
<component>
<code>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679" />
<code value="4570" />
<display value="FLOQA Region Property Row" />
</coding>
<text value="FLOQA Region Property Row" />
</code>
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.696784.98010" />
<code value="Arm" />
<display value="Arm" />
```

```
</coding>
<text value="Arm" />
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679" />
<code value="4571" />
<display value="FLOQA Anterior/Posterior Property Row" />
</coding>
<text value="FLOQA Anterior/Posterior Property Row" />
</code>
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.696784.98013" />
<code value="Ant" />
<display value="Anterior" />
</coding>
<text value="Anterior" />
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679" />
<code value="4572" />
<display value="FLOQA Left/Right Property Row" />
</coding>
<text value="FLOQA Left/Right Property Row" />
</code>
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.5325.1.7.4.696784.98011" />
<code value="R" />
<display value="Right" />
</coding>
<text value="Right" />
</valueCodeableConcept>
</component>
</Observation>
```

Sample response

```
HTTP/1.1 200 OK
Cache-Control: no-cache,no-store
Pragma: no-cache
Content-Type: application/xml; charset=utf-8
Expires: -1
Server: Microsoft-IIS/8.5
ServerMetrics: {"BlockReads":0,"BlockWrites":21,"BlocksAllocated":3,"DBTime":160,"ECFNetworkTime":1,"ECFRequestCount":0,
    "ECFRequestTime":161,"GREF":14106,"JournalEntries":111,"LocksFailed":0,"LocksGranted":35,"MCommands":198854,"MemoryD
    ifference":0,"WorkflowEventBlockReads":0,"WorkflowEventDBTime":160,"WorkflowEventECFNetworkTime":0,"WorkflowEventECF
    RequestCount":0,"WorkflowEventGREF":14106}
X-AspNet-Version: 4.0.30319
X-Powered-By: ASP.NET
Date: Thu, 05 Sep 2019 21:57:41 GMT
Content-Length: 0
```

Change log

Epic Version | Description | Type

November 2022 | Epic has updated the behavior of certain modifier elements in FHIR write APIs to ensure the data in the incoming request body matches the data filed to the database and to comply with the HL7 FHIR specification. Previously, the modifier element was ignored in certain situations, and an undocumented default value was saved in its place. Now, if the modifier element is included and is set to an unaccepted value, a fatal error logs, preventing the request from filing to the database. The FHIR request returns an OperationOutcome with the invalid element and what value is accepted for that element. Element Accepted Incoming Value Default Value status final final | Substantive change

Element | Accepted Incoming Value | Default Value

status | final | final

9.2 Clinical Diagnostics: STU3

File: site/apis/09-Clinical-Diagnostics/2-STU3/index.html

* DiagnosticReport.Read (Results) (STU3)
* DiagnosticReport.Search (Results) (STU3)
* Observation.$lastn (Activities of Daily Living) (STU3)
* Observation.$lastn (Labs) (STU3)
* Observation.$lastn (Vitals) (STU3)
* Observation.Create (LDA-W) (STU3)
* Observation.Create (Vitals) (STU3)
* Observation.Read (Activities of Daily Living) (STU3)

* Observation.Read (Core Characteristics) (STU3)
* Observation.Read (Labs) (STU3)
* Observation.Read (LDA-W) (STU3)
* Observation.Read (Vitals) (STU3)
* Observation.Search (Activities of Daily Living) (STU3)
* Observation.Search (Core Characteristics) (STU3)
* Observation.Search (Labs) (STU3)
* Observation.Search (LDA-W) (STU3)
* Observation.Search (Vitals) (STU3)
* Observation.Update (LDA-W) (STU3)

9.2.1 DiagnosticReport.Read (Results) (STU3)

File: site/apis/09-Clinical-Diagnostics/2-STU3/842-DiagnosticReport-Read-Results-STU3.html
Epic API ID | 842
FHIR version | STU3
Categories | Clinical Diagnostics
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Laboratory;Tests;Values;Results
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2018
HTTP method | GET
URL template | api/FHIR/STU3/DiagnosticReport/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=842

Description

The DiagnosticReport resource represents an Epic resulted, patient-related, non-medication order and data associated with the order and the patient.

The read interaction enables the look up of a DiagnosticReport resource by a constant server ID. The read interaction allows clients to store only the server ID, and with a single request, retrieve the most up-to-date result data on a patient. Read interactions typically begin with a client having previously established a relationship, often through querying for DiagnosticReports through the search interaction.

Starting in February 2024, DiagnosticReport can return EKG data using the Midmark third-party integration. If the EKG order collected heart rate and blood pressure data as well, this data is returned as Observations contained within the report.

Remarks

Heart rate and blood pressure data associated with a Midmark result are represented as full Vital Signs Observations within the DiagnosticReport. Unlike other Observations associated with the order, these are contained resources, and are only accessible embedded within the DiagnosticReport. These resources are associated with the following LOINC codes:

* Heart rate: 8867-4
* Diastolic blood pressure: 8462-4
* Systolic blood pressure: 8480-6

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The DiagnosticReport FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
DiagnosticReport | DiagnosticReport | False | | | An instance of the STU3 DiagnosticReport resource.
DiagnosticReport.category | CodeableConcept | False | | | The service category.
DiagnosticReport.code | CodeableConcept | False | | | The Name/Code for this diagnostic report.
DiagnosticReport.context | Reference | False | | | Encounter when the test was ordered. This will be either a FHIR Encounter reference in the encounter.reference element, or an encounter CSN identifier in the encounter.identifier element.
DiagnosticReport.effectiveDateTime | String | False | | | Clinically relevant time/time period for report.
DiagnosticReport.identifier | Identifier | False | Yes | | The business identifier for the report.
DiagnosticReport.issued | String | False | | | DateTime this version was released.

DiagnosticReport.performer | Performer | False | Yes | | Participants in producing the report. This is always a reference to the Practitioner resource.
DiagnosticReport.result | Reference | False | Yes | | Observations, simple or complex nested groups.
DiagnosticReport.status | String | False | | | The status of the report. You can find the list of values here: http://hl7.org/fhir/diagnostic-report-status.
DiagnosticReport.subject | Reference | False | | | The subject of the report.

Sample request

```
https://hostname/instance/api/FHIR/STU3/DiagnosticReport/e5E.109CBAdWihXIpCUsydSyV8eEIWsMB0mCIYZtfmdY3
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="2"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/STU3/DiagnosticReport?patient=efvHwbc1k1CQ9XjM1zvvefQ3&amp;date=2017&
    amp;code=57021-8"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
      <url
    value="https://hostname/instance/api/FHIR/STU3/DiagnosticReport/e5E.109CBAdWihXIpCUsydSyV8eEIWsMB0mCIYZtfmdY3"/>
    </link>
    <fullUrl
    value="https://hostname/instance/api/FHIR/STU3/DiagnosticReport/e5E.109CBAdWihXIpCUsydSyV8eEIWsMB0mCIYZtfmdY3"/>
    <resource>
      <DiagnosticReport>
        <id value="e5E.109CBAdWihXIpCUsydSyV8eEIWsMB0mCIYZtfmdY3"/>
        <identifier>
          <use value="official"/>
          <type>
            <coding>
              <system value="http://hl7.org/fhir/identifier-type"/>
              <code value="PLAC"/>
              <display value="Placer Identifier"/>
            </coding>
            <text value="Placer Identifier"/>
          </type>
          <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.798268"/>
          <value value="4916783"/>
        </identifier>
        <status value="final"/>
        <category>
          <coding>
            <system value="1.2.840.114350.1.13.861.1.7.10.798268.30"/>
            <code value="Lab"/>
          </coding>
          <text value="Lab"/>
        </category>
        <code>
          <coding>
            <system value="http://loinc.org"/>
            <code value="57021-8"/>
            <display value="CBC W Auto Differential panel - Blood"/>
          </coding>
          <text value="CBC"/>
        </code>
        <subject>
          <reference value="https://hostname/instance/api/FHIR/STU3/Patient/efvHwbc1k1CQ9XjM1zvvefQ3"/>
          <display value="Fhir D Cds Jr., Esq."/>
        </subject>
        <effectiveDateTime value="2017-03-15T22:29:00Z"/>
        <issued value="2017-03-15T22:29:00Z"/>
        <performer>
          <actor>
            <reference value="https://hostname/instance/api/FHIR/STU3/Practitioner/egCj4Z7EG1mplI4xYrONFRQ3"/>
            <display value="Starter Provider"/>
          </actor>
        </performer>
        <result>
          <reference
    value="https://hostname/instance/api/FHIR/STU3/Observation/etuoDL5b9cxK7rouHziDHo3bjQmpwoJSahIBuVQ5gga03"/>
          <display value="Narrative"/>
        </result>
      </DiagnosticReport>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
  <entry>
    <link>
      <relation value="self"/>
      <url
    value="https://hostname/instance/api/FHIR/STU3/DiagnosticReport/eUA5TYCR5UjuJ6-vwQ0bvhbv9WZEMJQyI9LfiBa0H74M3"/>
    </link>
```

```
    <fullUrl
    value="https://hostname/instance/api/FHIR/STU3/DiagnosticReport/eUA5TYCR5UjuJ6-vwQ0bvhbv9WZEMJQyI9LfiBa0H74M3"/>
    <resource>
      <DiagnosticReport>
        <id value="eUA5TYCR5UjuJ6-vwQ0bvhbv9WZEMJQyI9LfiBa0H74M3"/>
        <identifier>
          <use value="official"/>
          <type>
            <coding>
              <system value="http://hl7.org/fhir/identifier-type"/>
              <code value="PLAC"/>
              <display value="Placer Identifier"/>
            </coding>
            <text value="Placer Identifier"/>
          </type>
          <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.798268"/>
          <value value="4916781"/>
        </identifier>
        <identifier>
          <use value="official"/>
          <type>
            <coding>
              <system value="http://hl7.org/fhir/identifier-type"/>
              <code value="FILL"/>
              <display value="Filler Identifier"/>
            </coding>
            <text value="Filler Identifier"/>
          </type>
          <system value="urn:oid:1.2.840.114350.1.13.861.1.7.3.798268.320"/>
          <value value="CDS201703151729"/>
        </identifier>
        <status value="final"/>
        <category>
          <coding>
            <system value="1.2.840.114350.1.13.861.1.7.10.798268.30"/>
            <code value="Lab"/>
          </coding>
          <text value="Lab"/>
        </category>
        <code>
          <coding>
            <system value="http://loinc.org"/>
            <code value="57021-8"/>
            <display value="CBC W Auto Differential panel - Blood"/>
          </coding>
          <text value="CBC"/>
        </code>
        <subject>
          <reference value="https://hostname/instance/api/FHIR/STU3/Patient/efvHwbc1k1CQ9XjM1zvvefQ3"/>
          <display value="Fhir D Cds Jr., Esq."/>
        </subject>
        <effectiveDateTime value="2017-03-15T22:29:00Z"/>
        <issued value="2017-03-15T22:29:00Z"/>
        <performer>
          <actor>
            <reference value="https://hostname/instance/api/FHIR/STU3/Practitioner/egCj4Z7EG1mplI4xYrONFRQ3"/>
            <display value="Starter Provider"/>
          </actor>
        </performer>
        <result>
          <reference
    value="https://hostname/instance/api/FHIR/STU3/Observation/emGEihY4EcbjdZ51YjFEFNM.iIW6ZLw1U7F50ay3zGxo3"/>
          <display value="Narrative"/>
        </result>
      </DiagnosticReport>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
</Bundle>
```

Change log

Epic Version | Description | Type
November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4
ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90
ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94
ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

9.2.2 DiagnosticReport.Search (Results) (STU3)

File: site/apis/09-Clinical-Diagnostics/2-STU3/843-DiagnosticReport-Search-Results-STU3.html
Epic API ID | 843
FHIR version | STU3
Categories | Clinical Diagnostics
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Laboratory;Tests;Values;Results
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2018
HTTP method | GET
URL template | api/FHIR/STU3/DiagnosticReport?patient={patient}&code={code}&date={date}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=843

Description

The DiagnosticReport FHIR resource represents resulted diagnostic orders and relevant data associated with the order and the patient. Supported diagnostic services include laboratory, radiology, and cardiology. Within DiagnosticReport, the Observation FHIR resource returns individual components from laboratory tests and text-based content, such as narratives or impressions, from radiology and cardiology results. This resource returns textual results and discrete result components. It can return lab, imaging, and cardiology results. It can also return micro and EKG and pathology results that are not a scan. Scans filed into Epic are not returned by this resource.

The DiagnosticReport resource corresponds to the following Common Clinical Data Set (CCDS) Elements: Laboratory Tests.

The Search interaction enables the client to query for all diagnostic reports that have been filed for a given patient. The client, having established the patient in question, now wishes to retrieve all documented test level result data for the patient, and can further refine the search to specify a date or a LOINC code for the DiagnosticReport resources in question.

Starting in February 2024, DiagnosticReport can return EKG data using the Midmark third-party integration. If the EKG order collected heart rate and blood pressure data as well, this data is returned as Observations contained within the report.

Some healthcare activities could be classified as either a diagnostic service or a procedure (e.g. endoscopic procedures, biopsies, etc.). Individual health systems can define which activities they classify for the Procedure vs. DiagnosticReport resource. As a result, you may see some procedures returned as a DiagnosticReport resource at some health systems, but as a Procedure resource at others.

Note:

Remarks

Text based result content, such as narratives, impressions, or addendums, will be represented as related Observations rather than in the text element inherited by DomainResource. If the results from a Search return medication orders, an error will be returned for those orders.

Heart rate and blood pressure data associated with a Midmark result are represented as full Vital Signs Observations within the DiagnosticReport. Unlike other Observations associated with the order, these are contained resources, and are only accessible embedded within the DiagnosticReport. These resources are associated with the following LOINC codes:

* Heart rate: 8867-4

* Diastolic blood pressure: 8462-4

* Systolic blood pressure: 8480-6

Request parameters

Name | Type | Optional | Array | Cardinality | Description
code | String | True | | | The code for the report as a whole, as opposed to codes for the atomic results, which are the names on the observation resource referred to from the result. Starting in the November 2021 version of Epic, or the August 2021 version with special updates, this can be any code type associated with the procedures, including LOINC, CPT, SNOMED, or other code types mapped to a procedure in Epic. In earlier versions, this can only be a LOINC code. This parameter is formatted with or without a system specified (code=[code] or code=[system]|[code]). If specifying a system, the corresponding code must be provided.
date | String | True | | | The date or date range the order was resulted.
patient | String | Either patient or subject must be provided. If both are provided, they must reference the same patient. | | | The subject of the report if it is a patient.
subject | String | Either patient or subject must be provided. If both are provided, they must reference the same patient. | | | The subject of the report if it is a patient.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single DiagnosticReport returned from the search.
Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.DiagnosticReport | DiagnosticReport | False | | | An instance of the STU3 DiagnosticReport resource.
Bundle.entry.resource.DiagnosticReport.category | CodeableConcept | False | | | The service category.
Bundle.entry.resource.DiagnosticReport.code | CodeableConcept | False | | | The Name/Code for this diagnostic report.
Bundle.entry.resource.DiagnosticReport.context | Reference | False | | | Encounter when the test was ordered. This will be either a FHIR Encounter reference in the encounter.reference element, or an encounter CSN identifier in the encounter.identifier element.
Bundle.entry.resource.DiagnosticReport.effectiveDateTime | String | False | | | Clinically relevant time/time period for report.
Bundle.entry.resource.DiagnosticReport.identifier | Identifier | False | Yes | | Additional identifiers for the DiagnosticReport.
Bundle.entry.resource.DiagnosticReport.identifier.system | String | False | | | The namespace for the identifier value.
Bundle.entry.resource.DiagnosticReport.identifier.type | CodeableConcept | False | | | Description of the identifier.
Bundle.entry.resource.DiagnosticReport.identifier.use | String | False | | | How the identifier is used.
Bundle.entry.resource.DiagnosticReport.identifier.value | String | False | | | The identifier value.
Bundle.entry.resource.DiagnosticReport.issued | String | False | | | DateTime this version was released.
Bundle.entry.resource.DiagnosticReport.performer | Performer | False | Yes | | Participants in producing the report. This is always a reference to the Practitioner resource.
Bundle.entry.resource.DiagnosticReport.result | Reference | False | Yes | | Observations, simple or complex nested groups.
Bundle.entry.resource.DiagnosticReport.status | String | False | | | The status of the report. You can find the list of values here: http://hl7.org/fhir/diagnostic-report-status.
Bundle.entry.resource.DiagnosticReport.subject | Reference | False | | | The subject of the report

Sample request

```
https://hostname/instance/api/FHIR/STU3/DIAGNOSTICREPORT?patient=efvHwbc1k1CQ9XjM1zvvefQ3&date=2017&code=57021-8
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
<type value="searchset"/>
<total value="2"/>
<link>
<relation value="self"/>
<url value="https://hostname/instance/api/FHIR/STU3/DiagnosticReport?patient=efvHwbc1k1CQ9XjM1zvvefQ3&amp;date=2017&amp;
    code=57021-8"
/>
</link>
<entry>
<link>
<relation value="self"/>
<url value="https://hostname/instance/api/FHIR/STU3/DiagnosticReport/e5E.109CBAdWihXIpCUsydSyV8eEIWsMB0mCIYZtfmdY3"
/>
</link>
<fullUrl
    value="https://hostname/instance/api/FHIR/STU3/DiagnosticReport/e5E.109CBAdWihXIpCUsydSyV8eEIWsMB0mCIYZtfmdY3"/>
<resource>
<DiagnosticReport>
<id value="e5E.109CBAdWihXIpCUsydSyV8eEIWsMB0mCIYZtfmdY3"/>
<identifier>
<use value="official"/>
<type>
<coding>
<system value="http://hl7.org/fhir/identifier-type"/>
<code value="PLAC"/>
<display value="Placer Identifier"/>
</coding>
<text value="Placer Identifier"/>
</type>
<system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.798268"/>
<value value="4916783"/>
</identifier>
<status value="final"/>
<category>
<coding>
<system value="1.2.840.114350.1.13.861.1.7.10.798268.30"/>
<code value="Lab"/>
</coding>
<text value="Lab"/>
</category>
<code>
<coding>
<system value="http://loinc.org"/>
<code value="57021-8"/>
<display value="CBC W Auto Differential panel - Blood"/>
</coding>
<text value="CBC"/>
</code>
<subject>
<reference value="https://hostname/instance/api/FHIR/STU3/Patient/efvHwbc1k1CQ9XjM1zvvefQ3"/>
<display value="Fhir D Cds Jr., Esq."/>
</subject>
<effectiveDateTime value="2017-03-15T22:29:00Z"/>
<issued value="2017-03-15T22:29:00Z"/>
<performer>
```

```
<actor>
<reference value="https://hostname/instance/api/FHIR/STU3/Practitioner/egCj4Z7EG1mplI4xYrONFRQ3"/>
<display value="Starter Provider"/>
</actor>
</performer>
<result>
<reference value="https://hostname/instance/api/FHIR/STU3/Observation/etuoDL5b9cxK7rouHziDHo3bjQmpwoJSahIBuVQ5gga03"/>
<display value="Narrative"/>
</result>
</DiagnosticReport>
</resource>
<search>
<mode value="match"/>
</search>
</entry>
<entry>
<link>
<relation value="self"/>
<url value="https://hostname/instance/api/FHIR/STU3/DiagnosticReport/eUA5TYCR5UjuJ6-vwQ0bvhbv9WZEMJQyI9LfiBa0H74M3"
/>
</link>
<fullUrl
    value="https://hostname/instance/api/FHIR/STU3/DiagnosticReport/eUA5TYCR5UjuJ6-vwQ0bvhbv9WZEMJQyI9LfiBa0H74M3"/>
<resource>
<DiagnosticReport>
<id value="eUA5TYCR5UjuJ6-vwQ0bvhbv9WZEMJQyI9LfiBa0H74M3"/>
<identifier>
<use value="official"/>
<type>
<coding>
<system value="http://hl7.org/fhir/identifier-type"/>
<code value="PLAC"/>
<display value="Placer Identifier"/>
</coding>
<text value="Placer Identifier"/>
</type>
<system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.798268"/>
<value value="4916781"/>
</identifier>
<identifier>
<use value="official"/>
<type>
<coding>
<system value="http://hl7.org/fhir/identifier-type"/>
<code value="FILL"/>
<display value="Filler Identifier"/>
</coding>
<text value="Filler Identifier"/>
</type>
<system value=""https://open.epic.com/FHIR/861/order-accession-number/ALB%20LAB",/>
<value value="CDS201703151729"/>
</identifier>
<status value="final"/>
<category>
<coding>
<system value="1.2.840.114350.1.13.861.1.7.10.798268.30"/>
<code value="Lab"/>
</coding>
<text value="Lab"/>
</category>
<code>
<coding>
<system value="http://loinc.org"/>
<code value="57021-8"/>
<display value="CBC W Auto Differential panel - Blood"/>
</coding>
<text value="CBC"/>
</code>
<subject>
<reference value="https://hostname/instance/api/FHIR/STU3/Patient/efvHwbc1k1CQ9XjM1zvvefQ3"/>
<display value="Fhir D Cds Jr., Esq."/>
</subject>
<effectiveDateTime value="2017-03-15T22:29:00Z"/>
<issued value="2017-03-15T22:29:00Z"/>
<performer>
<actor>
<reference value="https://hostname/instance/api/FHIR/STU3/Practitioner/egCj4Z7EG1mplI4xYrONFRQ3"/>
<display value="Starter Provider"/>
</actor>
</performer>
<result>
<reference value="https://hostname/instance/api/FHIR/STU3/Observation/emGEihY4EcbjdZ51YjFEFNM.iIW6ZLw1U7F50ay3zGxo3"/>
<display value="Narrative"/>
</result>
</DiagnosticReport>
</resource>
<search>
<mode value="match"/>
</search>
```

```
</entry>
</Bundle>
```

Change log

Epic Version | Description | Type
November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4
ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90
ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94
ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

9.2.3 Observation.$lastn (Activities of Daily Living) (STU3)

File: site/apis/09-Clinical-Diagnostics/2-STU3/10282-Observation-lastn-Activities-of-Daily-Living-STU3.html
Epic API ID | 10282
FHIR version | STU3
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/STU3/Observation/$lastn?patient={patient}&category={category}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10282

Description

The Observation (Activities of Daily Living) resource returns information about a patient's last known physical and mental capabilities such as walking, bathing, and eating. Data represents the last known functional and mental statuses documented in flowsheets or inpatient goals per encounter. Information returned is further restricted to records specified by the Epic organization in profile variables.

The $lastn interaction shares much with the standard Search interaction, but serves a more specific function by returning the most recent results of a particular category for a patient. If multiple instances of the same assessment or goal have been recorded for the patient, only the most recent is returned by $lastn. The following SNOMED categories are supported; both codes are currently always returned for activies of daily living observations:

* 118228005-Functional finding

* 384821006-Mental state, behavior and/or psychosocial function finding

For example, if the category is 384821006, the $lastn operation will return each of the most recent mental state, behavior and/or psychosocial functional findings documented in flowsheets or inpatient goals for the patient, such as a neurological assessment or a goal to alleviate depressive symptoms.

If multiple assessments or goals have the same effective datetime, and are also the most recent, $lastn returns all instances that have that datetime.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | String | False | | | One of the following SNOMED codes: 118228005-Functional finding 384821006-Mental state, behavior and/or psychosocial function finding
patient | String | False
subject | String | Either patient or subject must be provided. If both are provided they must reference the same patient. | | | The reference to a patient resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Observation | Observation | False | | | A single Observation FHIR resource.
Bundle.entry.resource.Observation.category | CodeableConcept | False | Yes | | Type of observation. Always includes SNOMED codes "118228005" and "384821006", and custom code "functional-mental status."
Bundle.entry.resource.Observation.code | CodeableConcept | False | | | The type of observation. Values mapped. If no mapping exists, only the text value is returned.

Bundle.entry.resource.Observation.comment | string | True | | | Comments about the observation. A comment if flowsheet data or the associated note if inpatient goals. If using cascading flowsheets, the note element contains information about the additional flowsheet, if applicable, instead of the note. For example, a nurse documents flowsheet “daily activities” as abnormal, which opens flowsheet “mobility.” The nurse documents the value “highly mobile” in the flowsheet. The note element for “daily activities” is returned as “daily activities;highly mobile” if flowsheet “mobility” isn’t included in the associated profile variable.

Bundle.entry.resource.Observation.effectivePeriod | Period | True | | | Clinically relevant time period for the observation

Bundle.entry.resource.Observation.meta | Meta | True | | | The document metadata.

Bundle.entry.resource.Observation.meta.profile | String | True | | | Available in Netherlands environments only. This element shows that this API conforms to the profile connected to one of the following canonical urls: * http://nictiz.nl/fhir/StructureDefinition/zib-LivingSituation * http://nictiz.nl/fhir/StructureDefinition/zib-DrugUse * http://nictiz.nl/fhir/StructureDefinition/zib-AlcoholUse * http://nictiz.nl/fhir/StructureDefinition/zib-TobaccoUse * http://nictiz.nl/fhir/StructureDefinition/zib-FunctionalOrMentalStatus

Bundle.entry.resource.Observation.status | code | False | | | Always "final"

Bundle.entry.resource.Observation.subject | Reference | False | | | Who the observation is about. Reference to the patient resource.

Bundle.entry.resource.Observation.valueCodeableConcept | CodeableConcept | True | | | Result of the observation. Values mapped. If no mapping exists, only the text value is returned.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Observation/$lastn?patient=em2zwhHegmZEu39N4dUEIYA3&category=384821006
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 3,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/STU3/Observation/$lastn?category=http://snomed.info/sct%7C384821006&pat
    ient=eeiBNKuRR84uD0nIFn73BGQ3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/STU3/Observation/euvO8iq4aYWriTr-
    MA7S-IiEXfytgR18RyIFf.KiIXD8KUnmIzSncsOvCLXboRq4H3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/STU3/Observation/euvO8iq4aYWriTr-
    MA7S-IiEXfytgR18RyIFf.KiIXD8KUnmIzSncsOvCLXboRq4H3",
      "resource": {
        "resourceType": "Observation",
        "id": "euvO8iq4aYWriTr-MA7S-IiEXfytgR18RyIFf.KiIXD8KUnmIzSncsOvCLXboRq4H3",
        "status": "final",
        "category": [
          {
            "coding": [
              {
                "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
                "code": "functional-mental-status",
                "display": "Functional and Mental Status"
              }
            ]
          },
          {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "118228005",
                "display": "Functional finding"
              }
            ]
          },
          {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "384821006",
                "display": "Mental state, behavior and/or psychosocial function finding"
              }
            ]
          }
        ],
        "code": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "118228005",
              "display": "Functional finding (finding)"
            },
            {
              "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.737879",
```

```
          "code": "1585",
          "display": "BH LTG ALLEVIATE DEPRESSIVE SYMPTOMS"
        }
      ],
      "text": "LTG: Alleviate depressive symptoms and return to previous level of effective functioning"
    },
    "subject": {
      "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eeiBNKuRR84uD0nIFn73BGQ3",
      "display": "Bobby, Barbara"
    },
    "effectivePeriod": {
      "start": "2021-11-01",
      "end": "2021-11-01"
    },
    "valueCodeableConcept": {
      "text": "Adequate for Discharge"
    }
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/STU3/Observation/euvO8iq4aYWriTr-
MA7S-IiEXfytgR18RyIFf.KiIXD8KG14zOC9nOVXkhcDRj7PE3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/STU3/Observation/euvO8iq4aYWriTr-
MA7S-IiEXfytgR18RyIFf.KiIXD8KG14zOC9nOVXkhcDRj7PE3",
  "resource": {
    "resourceType": "Observation",
    "id": "euvO8iq4aYWriTr-MA7S-IiEXfytgR18RyIFf.KiIXD8KG14zOC9nOVXkhcDRj7PE3",
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
            "code": "functional-mental-status",
            "display": "Functional and Mental Status"
          }
        ]
      },
      {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "118228005",
            "display": "Functional finding"
          }
        ]
      },
      {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "384821006",
            "display": "Mental state, behavior and/or psychosocial function finding"
          }
        ]
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "118228005",
          "display": "Functional finding (finding)"
        },
        {
          "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.737879",
          "code": "1585",
          "display": "BH LTG ALLEVIATE DEPRESSIVE SYMPTOMS"
        }
      ],
      "text": "LTG: Alleviate depressive symptoms and return to previous level of effective functioning"
    },
    "subject": {
      "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eeiBNKuRR84uD0nIFn73BGQ3",
      "display": "Bobby, Barbara"
    },
    "effectivePeriod": {
      "start": "2021-11-01",
      "end": "2021-11-01"
    },
    "valueCodeableConcept": {
```

```
          "text": "Met"
        }
      },
      "search": {
        "mode": "match"
      }
    },
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/STU3/Observation/euvO8iq4aYWriTr-
    MA7S-Il5ox3kYrwWIIhmETp0-epdIyFbxCFQ867ahG4ZXLBDH3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/STU3/Observation/euvO8iq4aYWriTr-
    MA7S-Il5ox3kYrwWIIhmETp0-epdIyFbxCFQ867ahG4ZXLBDH3",
      "resource": {
        "resourceType": "Observation",
        "id": "euvO8iq4aYWriTr-MA7S-Il5ox3kYrwWIIhmETp0-epdIyFbxCFQ867ahG4ZXLBDH3",
        "status": "final",
        "category": [
          {
            "coding": [
              {
                "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
                "code": "functional-mental-status",
                "display": "Functional and Mental Status"
              }
            ]
          },
          {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "118228005",
                "display": "Functional finding"
              }
            ]
          },
          {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "384821006",
                "display": "Mental state, behavior and/or psychosocial function finding"
              }
            ]
          }
        ],
        "code": {
          "coding": [
            {
              "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679",
              "code": "305390",
              "display": "R NEURO (WDL) -SIMPLE"
            }
          ],
          "text": "Neuro (WDL)"
        },
        "subject": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eeiBNKuRR84uD0nIFn73BGQ3",
          "display": "Bobby, Barbara"
        },
        "effectivePeriod": {
          "start": "2021-10-27T15:47:00Z"
        },
        "valueCodeableConcept": {
          "text": "Exceptions to WDL"
        },
        "comment": "Level of Consciousness: New agitation; Orientation Level: Unable to assess; Cognition: Appropriate
    judgement; Speech: Appropriate for developmental age; Facial Symmetry: Right facial drooping; Swallow: Able to
    swallow solids and liquids without difficulty; Pupil Assessment : Yes; Hand Grasp/Motor Function/Sensation
    Assessment: Grasp; Neuro Symptoms: Anxiety; Agitation; Neuro Additional Assessments: Babinski reflexes; Orthotic
    Devices Applied: Back brace"
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


Change log

Change Log

Epic Version | Description | Type
February 2026 | Observation.$lastn Sub-Resourcing Scope Enforced | Breaking change

9.2.4 Observation.$lastn (Labs) (STU3)

File: site/apis/09-Clinical-Diagnostics/2-STU3/10188-Observation-lastn-Labs-STU3.html
Epic API ID | 10188
FHIR version | STU3
Categories | Clinical Diagnostics, Labs
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/STU3/Observation/$lastn?patient={patient}&category={category}&max={max}&subject={subject}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10188

Description

The Observation (Labs) resource represents component level data for lab results. Observation is limited to reading microbiology sensitivities (method level, isolate level, and sensitivity level), lab component results, and textual (narrative and impression) Observations. To identify all lab procedures performed for a patient, a client might consider using the DiagnosticReport resource instead, which will contain pointers to one or more Observation resources that contain the component results. If you are familiar with HL7v2, DiagnosticReport is analogous to an OBR segment, where Observation is like an OBX segment.

For example, consider a CBC (Complete Blood Count) test. The overall CBC test would be represented as a DiagnosticReport resource. The component results (white blood cell count, red blood cell count, hemoglobin, platelet count, etc.) would each be returned in independent Observation resources.

The $lastn interaction shares much with the standard Search interaction, but serves a more specific function by returning the most recent lab results of a particular category for a patient. For example, if the category is hematology, the $lastn operation will return each of the most recent lab results for hematology, such as the most recent white blood cell count, the most recent hemoglobin count, the most recent platelet count, and so on. This is distinct from using the Search interaction in that $lastn will return the most recent results from each subtype of a category, whereas a Search interaction returns all of the hematology results ever recorded for a patient that meet the specified search parameters, including historical results that might be superseded by more recent results.

$lastn supports searching only by certain SNOMED code categories. Result components in Epic must be mapped to one of the supported SNOMED codes in order to be returned as part of a $lastn response. If a client wants to search on a specific result component ID or LOINC code, the client should instead use the Search interaction, which supports code as a search parameter.

The category element is populated with SNOMED codes. These codes are based on a mapping an internal ID to external SNOMED codes. The following SNOMED codes are supported, and must be mapped in the External Mapping Utility:

* 252275004-Hematology (Hematologie)
* 275711006-Chemistry (Klinische chemie)
* 68793005-Serology (Serologie/ immunologie)
* 395124008-Virology (Virologie)
* 314076009-Toxicology (Toxicologie)
* 19851009-Microbiology (Microbiologie)
* 405825005-Molecular genetics (Moleculaire genetica)

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | String | False | | | One of the following SNOMED codes that are mapped to result components (with Dutch translation): 252275004-Hematology (Hematologie) 275711006-Chemistry (Klinische chemie) 68793005-Serology (Serologie/ immunologie) 395124008-Virology (Virologie) 314076009-Toxicology (Toxicologie) 19851009-Microbiology (Microbiologie) 405825005-Molecular genetics (Moleculaire genetica)
max | String | False | | | The maximum amount of responses for each subtype of the category. If not set, the default is 1, which means one response for each subtype. Note that your response might return more responses per subtype than configured if multiple components have the same effective datetime.
patient | String | Either patient or subject must be provided. If both are provided, they must reference the same patient. | | | The reference to a patient resource.
subject | String | Either patient or subject must be provided. If both are provided, they must reference the same patient. | | | The reference to a patient resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.

Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Observation | Observation | False | | | A single Observation FHIR resource.
Bundle.entry.resource.Observation.basedOn | Reference | True | Yes | | The procedure or service this observation is associated with.
Bundle.entry.resource.Observation.category | CodeableConcept | False | Yes | | The Observation type. Always "laboratory". This also includes a code entry for the result type, using the ResultTypeCodelist and Order Type values. Starting in February 2022 and in November 2021 with a special update, Netherlands-specific SNOMED code 49581000146104-Laboratory Test Finding (Finding) is also returned.
Bundle.entry.resource.Observation.code | CodeableConcept | False | | | Code of the Observation. Observation Grouper Code Description Component The component name Method The procedure code Isolate The isolate LOINC code Narrative or Impression Set to "Narrative" or "Impression" Addendum Set to "Addendum" or "Unsigned Addendum" Sensitivity The antibiotic code
Observation Grouper | Code Description
Component | The component name
Method | The procedure code
Isolate | The isolate LOINC code
Narrative or Impression | Set to "Narrative" or "Impression"
Addendum | Set to "Addendum" or "Unsigned Addendum"
Sensitivity | The antibiotic code
Bundle.entry.resource.Observation.comment | string | True | | | Comments about the result.
Bundle.entry.resource.Observation.context | Reference | True | | | The Encounter FHIR ID linked to the order.
Bundle.entry.resource.Observation.effectiveDateTime | dateTime | False | | | The prioritized instant. This is the instant the results were last updated.
Bundle.entry.resource.Observation.id | id | False | | | The Observation FHIR ID.
Bundle.entry.resource.Observation.interpretation | CodeableConcept | True | | | The level of abnormality. If the value isn't mapped we will just send the text value in the Codeable concept. Relevant for sensitivities.
Bundle.entry.resource.Observation.issued | instant | True | | | The observation instant.
Bundle.entry.resource.Observation.method | CodeableConcept | True | | | SNOMED code mapped to procedure.
Bundle.entry.resource.Observation.performer | Reference | True | Yes | | A reference to the resulting agency's director's Practitioner resource.
Bundle.entry.resource.Observation.referenceRange | BackboneElement | True | Yes | | The reference range for the result, if applicable.
Bundle.entry.resource.Observation.referenceRange.high | Quantity | True | | | The high value of the reference range.
Bundle.entry.resource.Observation.referenceRange.low | Quantity | True | | | The low value of the reference range.
Bundle.entry.resource.Observation.referenceRange.text | string | True | | | Display version of reference range.
Bundle.entry.resource.Observation.related | BackboneElement | True | Yes | | Resource related to this observation.
Bundle.entry.resource.Observation.status | code | False | | | The status of the lab order. For organizations in the Netherlands, this element can also include a TestResultStatus extension that indicates the test result status.
Bundle.entry.resource.Observation.subject | Reference | False | | | The Patient FHIR ID linked to the order.
Bundle.entry.resource.Observation.valueCodeableConcept | CodeableConcept | True | | | This element is populated only if a SNOMED code is associated with the component.
Bundle.entry.resource.Observation.valueQuantity | Quantity | True | | | The result value if the result is numeric and represents a quantity.
Bundle.entry.resource.Observation.valueRange | Range | True | | | This element is populated only if the result is structured numeric and the operator is "-".
Bundle.entry.resource.Observation.valueRatio | Ratio | True | | | This element is populated only if the result is structured numeric and the operator is ":".
Bundle.entry.resource.Observation.valueString | string | True | | | The result value if the result is a string.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Observation/$lastn?patient=em2zwhHegmZEu39N4dUEIYA3&category=252275004&max=1
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 3,
  "link": [
    {
      "relation": "self",
      "url":
    "https://hostname/instance/api/FHIR/STU3/Observation/$lastn?patient=em2zwhHegmZEu39N4dUEIYA3&category=252275004&max=
    1"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url":
    "https://hostname/instance/api/FHIR/STU3/Observation/eVNtTeUk7rGw8ANthUDzdf.IrotX5l7bNlJwQLYSK81YjjyRGrTdmi2y5J4gLwM
    vA3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/STU3/Observation/eVNtTeUk7rGw8ANthUDzdf.IrotX5l7bNlJwQLYSK81YjjyRGr
    Tdmi2y5J4gLwMvA3",
      "resource": {
        "resourceType": "Observation",
        "id": "eVNtTeUk7rGw8ANthUDzdf.IrotX5l7bNlJwQLYSK81YjjyRGrTdmi2y5J4gLwMvA3",
        "basedOn": [
          {
            "reference":
    "https://hostname/instance/api/FHIR/STU3/ProcedureRequest/enAUwSSpKZSC0.xEsuSdgiyfoQ5WW55FjygL7YREC4QU3"
```

```
      }
    ],
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "252275004",
            "display": "Hematology test (procedure)"
          }
        ],
        "text": "Hematology test (procedure)"
      },
      {
        "coding": [
          {
            "system": "http://hl7.org/fhir/observation-category",
            "code": "laboratory",
            "display": "Laboratory"
          }
        ],
        "text": "Laboratory"
      },
      {
        "coding": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.10.798268.30",
            "code": "Lab Only",
            "display": "Lab Only"
          }
        ],
        "text": "Lab Only"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "51880-3",
          "display": "Packed erythrocytes units given [#]"
        },
        {
          "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.768282",
          "code": "4088",
          "display": "DKF RBC"
        }
      ],
      "text": "DKF RBC"
    },
    "subject": {
      "reference": "https://hostname/instance/api/FHIR/STU3/Patient/em2zwhHegmZEu39N4dUEIYA3",
      "display": "Friend, Caly F Sr."
    },
    "context": {
      "reference": "https://hostname/instance/api/FHIR/STU3/Encounter/eFtZP-IbR5djPIoNQQ3OdRprHr0M24BDnnqEFHojvXxM3"
    },
    "effectiveDateTime": "2021-08-19T05:00:00Z",
    "issued": "2021-08-20T00:49:40Z",
    "valueQuantity": {
      "value": 55
    },
    "specimen": {
      "reference": "https://hostname/instance/api/FHIR/STU3/Specimen/eNb1mwVj6uN31C-NyO3MEVQ8ZzPMmt2TLw1YoBpuZzb83"
    }
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url":
"https://hostname/instance/api/FHIR/STU3/Observation/evCqoUPrtZkCH6qUADRgQN5HW6VoqP5lR9xiEvQrqnI4UOB5UOMPkeSbw3SgEng
8e3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/STU3/Observation/evCqoUPrtZkCH6qUADRgQN5HW6VoqP5lR9xiEvQrqnI4UOB5UO
MPkeSbw3SgEng8e3",
  "resource": {
    "resourceType": "Observation",
    "id": "evCqoUPrtZkCH6qUADRgQN5HW6VoqP5lR9xiEvQrqnI4UOB5UOMPkeSbw3SgEng8e3",
    "basedOn": [
      {
        "reference": "https://hostname/instance/api/FHIR/STU3/ProcedureRequest/edR4M6xA4LHj0f4AbJih049zj5EXI2bQz-
qjtESdTXB43"
      }
```

```
    ],
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "252275004",
            "display": "Hematology test (procedure)"
          }
        ],
        "text": "Hematology test (procedure)"
      },
      {
        "coding": [
          {
            "system": "http://hl7.org/fhir/observation-category",
            "code": "laboratory",
            "display": "Laboratory"
          }
        ],
        "text": "Laboratory"
      },
      {
        "coding": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.10.798268.30",
            "code": "Lab Only",
            "display": "Lab Only"
          }
        ],
        "text": "Lab Only"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "51880-3",
          "display": "Packed erythrocytes units given [#]"
        },
        {
          "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.768282",
          "code": "4088",
          "display": "DKF RBC"
        }
      ],
      "text": "DKF RBC"
    },
    "subject": {
      "reference": "https://hostname/instance/api/FHIR/STU3/Patient/em2zwhHegmZEu39N4dUEIYA3",
      "display": "Friend, Caly F Sr."
    },
    "context": {
      "reference": "https://hostname/instance/api/FHIR/STU3/Encounter/eFtZP-IbR5djPIoNQQ3OdRprHr0M24BDnnqEFHojvXxM3"
    },
    "effectiveDateTime": "2021-08-19T05:00:00Z",
    "issued": "2021-08-20T13:57:31Z",
    "valueQuantity": {
      "value": 65
    },
    "specimen": {
      "reference": "https://hostname/instance/api/FHIR/STU3/Specimen/eOuz3qX98XXdgPq733Lzq9Y-b-Imb3vb9ykpOnfbmL3Y3"
    }
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/STU3/Observation/eq8Gn54WaD--x1mNu9Wk3Lxyk-
YeLpaWeSnuswQGoNsZlaS90E8aZ7pSy9jeMG2pz3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/STU3/Observation/eq8Gn54WaD--x1mNu9Wk3Lxyk-
YeLpaWeSnuswQGoNsZlaS90E8aZ7pSy9jeMG2pz3",
  "resource": {
    "resourceType": "Observation",
    "id": "eq8Gn54WaD--x1mNu9Wk3Lxyk-YeLpaWeSnuswQGoNsZlaS90E8aZ7pSy9jeMG2pz3",
    "basedOn": [
      {
        "reference":
"https://hostname/instance/api/FHIR/STU3/ProcedureRequest/eGan955RZsh6ekCes.1d63b7Fl6G5EFN3XYuo2mufg603"
      }
    ],
    "status": "final",
```

```
        "category": [
          {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "252275004",
                "display": "Hematology test (procedure)"
              }
            ],
            "text": "Hematology test (procedure)"
          },
          {
            "coding": [
              {
                "system": "http://hl7.org/fhir/observation-category",
                "code": "laboratory",
                "display": "Laboratory"
              }
            ],
            "text": "Laboratory"
          },
          {
            "coding": [
              {
                "system": "urn:oid:1.2.840.114350.1.13.861.1.7.10.798268.30",
                "code": "Lab Only",
                "display": "Lab Only"
              }
            ],
            "text": "Lab Only"
          }
        ],
        "code": {
          "coding": [
            {
              "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.768282",
              "code": "4089",
              "display": "DKF WBC"
            }
          ],
          "text": "DKF WBC"
        },
        "subject": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Patient/em2zwhHegmZEu39N4dUEIYA3",
          "display": "Friend, Caly F Sr."
        },
        "context": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Encounter/eFtZP-IbR5djPIoNQQ3OdRprHr0M24BDnnqEFHojvXxM3"
        },
        "effectiveDateTime": "2021-08-20T05:00:00Z",
        "issued": "2021-08-20T13:58:25Z",
        "valueQuantity": {
          "value": 47
        },
        "specimen": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Specimen/eZz7xyJBknac39vAN-anFt6jgCNJwE0K-aWXFeqRLeUg3"
        }
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


Change log

Change Log

Epic Version | Description | Type
February 2026 | Observation.$lastn Sub-Resourcing Scope Enforced | Breaking change

9.2.5 Observation.$lastn (Vitals) (STU3)

File: site/apis/09-Clinical-Diagnostics/2-STU3/10484-Observation-lastn-Vitals-STU3.html
Epic API ID | 10484
FHIR version | STU3
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/STU3/Observation/$lastn?category={category}&code={code}&max={max}&patient={patient}&subject={subject}
Supports OAuth 2.0 | True

Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10484

Description

The Observation (Vitals) resource represents component-level data for vital signs.

The $lastn operation is much like the search interaction, but serves a more specific function by returning the most recent vitals readings of each type of reading for a patient. For example, if the query is for the last 3 readings (via max=3 parameter) of height, weight, and blood pressure (via the code parameter),

This is different from other FHIR requests (like using the _count parameter or searching on category) because this operation is guaranteed to return results from each code. While simply getting the 5 most recent vitals readings might return only temperature readings, for example, use of the $lastn operation ensures that the most recent vitals readings for each code are returned (height, weight, blood pressure).

This is useful to obtain summary data pertaining to a category. While a normal Observation search is useful for getting all readings, the $lastn operation is ideal for getting an up-to-date and comprehensive snapshot of a patient that covers all relevant readings.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | String | True | | | Set to ‘vital-signs’ to search for vitals observations. If not set, but a code in the supported valueset is supplied, ‘vital-signs’ will be assumed.
code | String | False | | | Code of observation. * LOINC 85354-9 Blood pressure systolic and diastolic * LOINC 29463-7 Body weight * LOINC 8302-2 Body height * LOINC 8308-9 Body height standing * LOINC 8306-3 Body height laying * LOINC 8867-4 Heart rate * LOINC 8310-5 Body temperature * LOINC 59408-5 Oxygen saturation * LOINC 8893-0 Pulse rate * LOINC 14770-2 Fasting glucose * LOINC 14743-9 General glucose * LOINC 14760-3 Post-meal glucose
max | String | True | | | The maximum number of responses for each code. If not set, the default is 1, which means one response for each subtype. The number of response entries returned for each code can exceed the max in the case of identical time stamps and FLO to code mapping. They can also be less than max if enough matching results don’t exist.
patient | String | One of patient or subject is required. If both are provided, they must match. | | | The reference to a patient resource.
subject | String | One of patient or subject is required. If both are provided, they must match. | | | The reference to a patient resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Observation | Observation | False | | | A single Observation FHIR resource.
Bundle.entry.resource.Observation.bodySite | CodeableConcept | False | | | Applicable only to organizations in the Netherlands starting in Epic version August 2021. The site on the body where the observation was made. Used for the location where a blood pressure reading was obtained.
Bundle.entry.resource.Observation.category | CodeableConcept | False | Yes | | This element always returns "vital-signs". Starting in November 2022 in Netherlands locales, this element also returns SNOMED code 49581000146104 (Laboratory test finding (finding) when returning observations linked to the blood glucose profile.
Bundle.entry.resource.Observation.code | CodeableConcept | False | | | A LOINC code. Either this element or category must be provided, but not both.
Bundle.entry.resource.Observation.comment | string | True | | | Comments about the observation value.
Bundle.entry.resource.Observation.component | BackboneElement | True | Yes
Bundle.entry.resource.Observation.component.code | CodeableConcept | True | | | This element is used for observations filed by patients to patient-entered flowsheets. Metadata related to the observation. Represents strict metadata if the coding sub-element is populated. The following strict metadata are supported: 1-UUID The unique reading ID. 2-Blood Glucose Meal Times 1-Before meal 2-After meal 3-Insulin Delivery Reason 1-Basal 2-Bolus Represents loose metadata if only the text sub-element is populated. Applicable to organizations in the Netherlands starting in November 2022: LOINC code 74206-4 (Oxygen therapy was given [NTDS]) LOINC code 44969-4 (Heart rate rhythm palpation)
Bundle.entry.resource.Observation.component.valueCodeableConcept | CodeableConcept | True | | | Applicable to organizations in the Netherlands starting in November 2022: Codes from CodeSystem “v2 Yes/no indicator” indicate whether extra oxygen was given at the time of an oxygen saturation reading SNOMED codes from PolsRegelmatigheidCodelijst indicate the regularity of the pulse rate at the time of a pulse reading
Bundle.entry.resource.Observation.dataAbsentReason | CodeableConcept | True | | | Available starting in the August 2021 version of Epic. Indicates that an observation value is missing. Will always be “unknown” for systolic blood pressure values in observations for diastolic-only blood pressure flowsheets, or for diastolic blood pressure value in observations for systolic-only blood pressure flowsheets.
Bundle.entry.resource.Observation.effectiveDateTime | dateTime | False | | | The date and time the observation was taken.
Bundle.entry.resource.Observation.issued | instant | False | | | The date and time the observation was documented. This is always the current time.
Bundle.entry.resource.Observation.meta | Meta | True | | | The document metadata.
Bundle.entry.resource.Observation.meta.profile | String | True | | | Available in Netherlands environments only. This element shows that this API conforms to the profile connected to one of the following canonical urls: http://nictiz.nl/fhir/StructureDefinition/zib-BloodPressure http://nictiz.nl/fhir/StructureDefinition/zib-BodyWeight http://nictiz.nl/fhir/StructureDefinition/zib-BodyHeight http://nictiz.nl/fhir/StructureDefinition/zib-BodyTemperature http://nictiz.nl/fhir/StructureDefinition/zib-OxygenSaturation http://nictiz.nl/fhir/StructureDefinition/zib-HeartRate http://nictiz.nl/fhir/StructureDefinition/zib-PulseRate http://nictiz.nl/fhir/StructureDefinition/vitalsign-bloodglucose
Bundle.entry.resource.Observation.method | CodeableConcept | True | | | Method by which the observation was obtained. Possible values include: "Patient-entered" for patient-entered flowsheet vital signs Applicable only to organizations in the Netherlands starting in August 2021: SNOMED code 10179008 (invasive) for blood pressure readings using an invasive method Applicable only to organizations in the Netherlands starting in August 2021: SNOMED code 22762002 (non-invasive) for blood pressure readings not using an invasive method Starting in November 2022 in Netherlands locales: SNOMED codes from the TemperatuurTypeCodelijst for the specific anatomical location used for body temperature readings Starting in

November 2022 in Netherlands locales: SNOMED codes from the HartslagMeetMethodeCodelijst for the specific method used for heart rate readings Starting in November 2022 in Netherlands locales: SNOMED codes from the TestmethodeCodelijst for the specific method used for blood glucose readings

Bundle.entry.resource.Observation.performer | Reference | False | Yes | | Who filed the observation.

Bundle.entry.resource.Observation.referenceRange | BackboneElement | True | Yes

Bundle.entry.resource.Observation.referenceRange.low | Quantity | True | | | Applicable to organizations in the Netherlands starting in November 2022. The lower reference limit for blood glucose readings.

Bundle.entry.resource.Observation.status | code | False | | | The status of the observation. This is set to amended or final depending on whether the field has been corrected.

Bundle.entry.resource.Observation.subject | Reference | False | | | The FHIR ID for the patient the observation is about.

Bundle.entry.resource.Observation.valueCodeableConcept | CodeableConcept | False | | | This element is populated if the data type is category or custom list. Only one of the value* elements is required.

Bundle.entry.resource.Observation.valueDateTime | String | False | | | This element is populated if the data type is date. Only one of the value* elements is required.

Bundle.entry.resource.Observation.valueQuantity | Quantity | False | | | This element is populated if a value has units or the data type is numeric. Only one of the value* elements is required.

Bundle.entry.resource.Observation.valueString | string | False | | | This element is populated if a value doesn't have units or the data type is string. Only one of the value[x] elements is required.

Bundle.entry.resource.Observation.valueTime | String | False | | | This element is populated if the data type is time. Only one of the value* elements is required.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Observation/$lastn?patient=eivR2DL-WAG8Nxx3xBYUu7A3&code=85354-9&max=3
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://vs-icx18-25/Interconnect-FHIR-Current-Stage1-Primary-
    Unsecured/api/FHIR/STU3/Observation/$lastn?patient=eivR2DL-WAG8Nxx3xBYUu7A3&code=85354-9&max=3"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://vs-icx18-25/Interconnect-FHIR-Current-Stage1-Primary-
    Unsecured/api/FHIR/STU3/Observation/fEEClzpado6uQbQeeQXu58V.tYaw9h.u-0ug6QUx30.E4"
        }
      ],
      "fullUrl": "https://vs-icx18-25/Interconnect-FHIR-Current-Stage1-Primary-
    Unsecured/api/FHIR/STU3/Observation/fEEClzpado6uQbQeeQXu58V.tYaw9h.u-0ug6QUx30.E4",
      "resource": {
        "resourceType": "Observation",
        "id": "fEEClzpado6uQbQeeQXu58V.tYaw9h.u-0ug6QUx30.E4",
        "status": "final",
        "category": [
          {
            "coding": [
              {
                "system": "http://hl7.org/fhir/observation-category",
                "code": "vital-signs",
                "display": "Vital Signs"
              }
            ],
            "text": "Vital Signs"
          }
        ],
        "code": {
          "coding": [
            {
              "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.707679",
              "code": "5",
              "display": "BP"
            },
            {
              "system": "http://open.epic.com/FHIR/STU3/StructureDefinition/observation-flowsheet-id",
              "code": "tK.3GdpCnBjtUSc8i9O-JVA0",
              "display": "BP"
            },
            {
              "system": "urn:oid:1.2.246.537.6.96",
              "code": "8462-4"
            },
            {
              "system": "urn:oid:1.2.246.537.6.96",
              "code": "8480-6"
            },
```

```
        {
          "system": "http://loinc.org",
          "code": "55284-4",
          "display": "Blood pressure systolic and diastolic"
        },
        {
          "system": "http://loinc.org",
          "code": "85354-9",
          "display": "Blood pressure panel with all children optional"
        },
        {
          "system": "http://loinc.org",
          "code": "8716-3",
          "display": "Vital signs"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "168406009",
          "display": "Severe dyskaryosis on cervical smear cannot exclude invasive carcinoma (finding)"
        },
        {
          "system": "http://snomed.info/sct",
          "code": "275919002",
          "display": "Weight loss advised (situation)"
        }
      ],
      "text": "BP"
    },
    "subject": {
      "reference": "https://vs-icx18-25/Interconnect-FHIR-Current-Stage1-Primary-
Unsecured/api/FHIR/STU3/Patient/eivR2DL-WAG8Nxx3xBYUu7A3",
      "display": "Patient, Sample Test"
    },
    "context": {
      "reference": "https://vs-icx18-25/Interconnect-FHIR-Current-Stage1-Primary-
Unsecured/api/FHIR/STU3/Encounter/evlDwBsdbEdI1ujNYz3-H3B9PsDnv9.UzSAWV37F5e8A3",
      "identifier": {
        "use": "usual",
        "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.3.698084.8",
        "value": "10000407505"
      },
      "display": "Office Visit"
    },
    "effectiveDateTime": "2022-05-09T12:00:00Z",
    "issued": "2022-05-09T14:59:38Z",
    "performer": [
      {
        "reference": "https://vs-icx18-25/Interconnect-FHIR-Current-Stage1-Primary-
Unsecured/api/FHIR/STU3/Practitioner/el47x-56LP8CI1u4MDhQJaA3",
        "display": "Doctor Sample, MD"
      }
    ],
    "component": [
      {
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "8480-6",
              "display": "Systolic blood pressure"
            }
          ],
          "text": "Systolic blood pressure"
        },
        "valueQuantity": {
          "value": 120,
          "unit": "mm[Hg]",
          "system": "http://unitsofmeasure.org",
          "code": "mm[Hg]"
        }
      },
      {
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "8462-4",
              "display": "Diastolic blood pressure"
            }
          ],
          "text": "Diastolic blood pressure"
        },
        "valueQuantity": {
          "value": 80,
          "unit": "mm[Hg]",
          "system": "http://unitsofmeasure.org",
          "code": "mm[Hg]"
        }
      }
```

```
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```

Change log

Change Log

Epic Version | Description | Type
February 2026 | Observation.$lastn Sub-Resourcing Scope Enforced | Breaking change

9.2.6 Observation.Create (LDA-W) (STU3)

File: site/apis/09-Clinical-Diagnostics/2-STU3/897-Observation-Create-LDA-W-STU3.html
Epic API ID | 897
FHIR version | STU3
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Keywords | Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2018
HTTP method | POST
URL template | api/FHIR/STU3/Observation
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=897

Description

The FHIR Observation resource defines measurements and assertions about a patient, including LDA-W (lines, drains, airways, and wound) data.

This web service cannot write to closed encounters. Ensure that your app doesn't need to write to closed encounters before you begin development.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | A single Observation FHIR resource.
Observation.category | CodeableConcept | False | Yes | | "LDA" to create LDA or LDA assessment.
Observation.code | CodeableConcept | False | | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. This element can't be modified by Observation.Update. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.
Observation.comment | string | False | | | Comment for the assessement. Limited to 254 characters in the August 2025 version of Epic and later. Limited to 60 characters in earlier versions of Epic.
Observation.component | BackboneElement | True | Yes
Observation.component.code | CodeableConcept | False | | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. This element can't be modified by Observation.Update. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.
Observation.component.extension (also: LDA-Property-Comment) | Extension | True | Yes | | Comment for the property row.
Observation.component.valueQuantity | Quantity | False | | | Value of the property row.
Observation.context | Reference | False | | | Reference to an Encounter resource.
Observation.effectivePeriod | Period | True | | | Start instant and/or end instant of the LDA.
Observation.related | BackboneElement | False | Yes
Observation.related.target | Reference | False | | | The parent LDA observation FHIR ID. This element can't be modified by Observation.Update.
Observation.related.type | code | True
Observation.status | code | True | | | The status of the observation. Only a value of "final" is supported. We do not support filing data that isn't finalized.
Observation.subject | Reference | False | | | Reference to a patient resource.
Observation.valueQuantity | Quantity | False | | | Value of the assessment. For image rows, a comma-delimited list of external document IDs is required. valueString, valueCodeableConcept, valueTime, and valueDateTime are also accepted.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | A single Observation FHIR resource.

Observation.id | id | False | | | The FHIR ID of the newly created observation resource. Optionally, the client can set the return preference header to "representation", and the server will return the entire resource in a format similar to Observation.Read.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Observation/
<Observation>
<category>
<coding>
<system value="http://open.epic.com/FHIR/STU3/StructureDefinition/observation-category" />
<code value="LDA" />
<display value="Line, Drains, and Airways" />
</coding>
<text value="Line, Drains, and Airways" />
</category>
<code>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679" />
<code value="42674" />
<display value="HZL Wound LDA" />
</coding>
<text value="HZL Wound LDA" />
</code>
<subject>
<reference value="https://hostname/instance/api/FHIR/STU3/Patient/ew9Okvb9gsZXjzlK7YzqP1w3" />
<display value="Haruka Lufhir" />
</subject>
<context>
<reference value="https://hostname/instance/api/FHIR/STU3/Encounter/e5wRUdV7oHoL8VOeT.CdLnVfN7YsByGgW1GzM51mMDEw3" />
</context>
<effectivePeriod>
<start value="2018-11-01T18:10:00Z" />
</effectivePeriod>
<component>
<code>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679" />
<code value="4572" />
<display value="FLOQA Left/Right Property Row" />
</coding>
<text value="FLOQA Left/Right Property Row" />
</code>
<valueCodeableConcept>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.861.1.7.4.696784.98011" />
<code value="R" />
<display value="Right" />
</coding>
<text value="Right" />
</valueCodeableConcept>
</component>
<component>
<code>
<coding>
<system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679" />
<code value="4577" />
<display value="FLOQA String Property Row" />
</coding>
<text value="FLOQA String Property Row" />
</code>
<valueString value="Haruka's wound" />
</component>
</Observation>
```

Sample response

```
HTTP/1.1 201 Created
```

Change log

Epic Version | Description | Type

November 2022 | Epic has updated the behavior of certain modifier elements in FHIR write APIs to ensure the data in the incoming request body matches the data filed to the database and to comply with the HL7 FHIR specification. Previously, the modifier element was ignored in certain situations, and an undocumented default value was saved in its place. Now, if the modifier element is included and is set to an unaccepted value, a fatal error logs, preventing the request from filing to the database. The FHIR request returns an OperationOutcome with the invalid element and what value is accepted for that element. Element Accepted Incoming Value Default Value status final final | Substantive change

Element | Accepted Incoming Value | Default Value

status | final | final

9.2.7 Observation.Create (Vitals) (STU3)

File: site/apis/09-Clinical-Diagnostics/2-STU3/496-Observation-Create-Vitals-STU3.html

Epic API ID | 496

FHIR version | STU3

Categories | Clinical Diagnostics

API type | Industry-Standard
USCDI | No
Keywords | Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2018
HTTP method | POST
URL template | api/FHIR/STU3/Observation
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=496

Description

The FHIR Observation.Create (Vitals) resource can file to all non-duplicable flowsheet rows, including vital signs. This resource can file vital signs for all flowsheets.

When a patient-facing app uses this API, it writes to patient-entered flowsheets. Before that can be done, a user in Epic must place an order to create the episode and patient entered flowsheets that will be filed to. Clinician or backend apps using this API will file to the same flowsheets that clinicians can document in.

An error is returned if a reading already exists. If an ID is included in the request, an error is returned. The server assigns an ID to the newly created resource.

Alternatives: For patient-entered information, there is an equivalent Incoming HL7v2 Incoming Clinical Documentation Flowsheet Data - Patient Entered Interface. For non-patient-entered information, there is an equivalent Incoming HL7v2 Incoming Clinical Documentation Flowsheet Data Interface.

Encounter Considerations: For non-Patient contexts, there are several limitations with this web service compared to the HL7v2 Interface alternative.

* Cannot write to closed encounters.

* Cannot write to future appointments.

* Cannot create an encounter on the fly.

* Cannot match to an encounter without an Encounter FHIR ID (i.e. fuzzy match).

Patient-context Considerations: This service can only file one reading at a time. For patient-entered information, this can trigger multiple notifications to the provider. If this is a concern, consider using the HL7v2 Interface alternative, which can file multiple readings at a time.

Filing Readings to Same Time-stamp This service can only file a reading if one doesn't already exist at the same timestamp. You may run into this issue when using the Try It case if you do not change the pre-populated timestamp. The HL7v2 Interface alternative is able to overwrite and update the existing value at the same timestamp.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | A single Observation FHIR resource.
Observation.category | CodeableConcept | True | Yes | | This element supports only "vital-signs". If multiple category values are provided, they are filed if vital-signs is presented. The default is vital-signs.
Observation.code | CodeableConcept | False | | | The flowsheet (FLO) ID, encoded flowsheet ID, or LOINC codes to flowsheet mapping. What is being measured. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization. If a customer has provided you with an encoded flowsheet ID, use http://open.epic.com/FHIR/STU3/StructureDefinition/observation-flowsheet-id
Observation.comment | string | True | | | The comment. Applies only to the provider-facing context. Limited to 254 characters in the August 2025 version of Epic and later. Limited to 60 characters in earlier versions of Epic.
Observation.component | BackboneElement | True | Yes | | Populated only for patient-entered flowsheets.
Observation.component.code | CodeableConcept | True | | | This element is used for observations filed by patients to patient-entered flowsheets. Metadata related to the observation. Represents strict metadata if the coding sub-element is populated. The following strict metadata are supported: 1-UUID The unique reading ID. 2-Blood Glucose Meal Times 1-Before meal 2-After meal 3-Insulin Delivery Reason 1-Basal 2-Bolus 6-Manually Entered 0-False 1-True Represents loose metadata if only the text sub-element is populated. UUID, Blood Glucose Meal times, Insulin Delivery Reason, and Manually Entered must have their value specified in component.valueString.
Observation.component.code.coding | Coding | False | Yes | | Coding value
Observation.component.code.coding.code | String | True | | | For patient-entered blood pressure: 8480-6 for Systolic and 8462-4 for Diastolic For strict metadata: The following strict metadata items are supported. Customers will need to set up specific compatible flowsheet rows that can ingest this metadata. Use the Integer for the code value. 1: UUID 2: Blood Glucose Meal Times 5: VO2 Max Test Type 6: Manually Entered UUID, Blood Glucose Meal times, and V02 Max Test Type must have their value specified in component.valueString. Manually Entered must have its value specified in component.valueBoolean
Observation.component.code.coding.display | String | True | | | Display of concept
Observation.component.code.coding.system | String | Required if including component.code.coding.code | | | For patient-entered blood pressure: http://loinc.org. For strict metadata: In the sandbox, use "urn:oid:1.2.840.114350.1.13.0.1.7.4.708368.1310". See ID Types for APIs for more info.
Observation.component.valueQuantity | Quantity | Required for splitting systolic and diastolic readings. | | | Systolic or diastolic value for patient-entered blood pressure.
Observation.component.valueString | String | Required for certain metadata items. | | | Used for the strict metadata items UUID, Blood Glucose Meal Times, VO2 Max Test Type, and Manually Entered Metadata code valueString 1: UUID A UUID representing the unique reading ID 2: Blood Glucose

Meal Times 1: Before meal 2: After meal (Use the integers as the string) 5: VO2 Max Test Type 1: Max Exercise Test 2: Sub-max exercise prediction 3: Non exercise prediction (Use the integers as the string) 6: Manually Entered 0: false 1: true Either the integers "0" or "1" or the strings "false"/"true" are acceptable

Metadata code | valueString

1: UUID | A UUID representing the unique reading ID

2: Blood Glucose Meal Times | 1: Before meal 2: After meal (Use the integers as the string)

5: VO2 Max Test Type | 1: Max Exercise Test 2: Sub-max exercise prediction 3: Non exercise prediction (Use the integers as the string)

6: Manually Entered | 0: false 1: true Either the integers "0" or "1" or the strings "false"/"true" are acceptable

Observation.context | Reference | False | | | The encounter that this observation belongs to. Applies only to the provider-facing context.

Observation.performer | Reference | False | Yes | | Who filed the observation. For the provider-facing context, the default is the signed-in user.

Observation.status | code | True | | | The status of the observation. Only a value of "final" is supported. We do not support filing data that isn't finalized.

Observation.subject | Reference | True | | | The patient FHIR ID for whom the observation is about. For the patient-facing context, the default is the user signed in.

Observation.valueCodeableConcept | CodeableConcept | False | | | This element is populated if the data type is a custom list or category type.

Observation.valueQuantity | Quantity | False | | | This element is populated if a value has units or the data type is numeric. Only one of value[x] is required.

Observation.valueString | string | False | | | This element is populated if a value doesn't have units or the data type is a string. Only one of value[x] is required.

Observation.valueTime | String | False | | | Use this element if data type is time. Only one of value[x] is required.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Observation | Observation | False | | | A single Observation FHIR resource.

Observation.category | CodeableConcept | False | Yes | | This element always returns "vital-signs".

Observation.code | CodeableConcept | False | | | The flowsheet (FLO) ID, encoded flowsheet ID, or LOINC codes to flowsheet mapping. What is being measured. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.

Observation.comment | string | True | | | Comments about the observation value.

Observation.component | BackboneElement | True | Yes | | Populated only for blood pressure observation.

Observation.component.code | CodeableConcept | False | | | Codeable concept.

Observation.component.code.coding | Coding | False | Yes | | Coding value

Observation.component.code.coding.code | String | False | | | For patient-entered blood pressure: 8480-6 for Systolic and 8462-4 for Diastolic For strict metadata: The following strict metadata items are supported. Customers will need to set up specific compatible flowsheet rows that can ingest this metadata. Use the Integer for the code value. 1: UUID 2: Blood Glucose Meal Times 5: VO2 Max Test Type 6: Manually Entered Each metadata item must have their value specified in component.valueString.

Observation.component.code.coding.display | String | False | | | Display of concept

Observation.component.code.coding.system | String | False | | | For patient-entered blood pressure: http://loinc.org. For strict metadata: In the sandbox, use "urn:oid:1.2.840.114350.1.13.0.1.7.4.708368.1310". See ID Types for APIs for more info.

Observation.component.valueQuantity | Quantity | False | | | Systolic or diastolic value for patient-entered blood pressure.

Observation.context | Reference | False | | | The encounter this observation belongs to. Applies only to the provider-facing context.

Observation.effectiveDateTime | dateTime | False | | | The date and time the observation was taken.

Observation.issued | instant | False | | | The date and time the observation was documented. This is always the current time.

Observation.method | CodeableConcept | True | | | This is set to "Patient-entered" for PEF vital signs.

Observation.performer | Reference | False | Yes | | Who filed the observation.

Observation.status | code | False | | | This element is always set to Final.

Observation.subject | Reference | False | | | The FHIR ID for the patient the observation is about.

Observation.valueCodeableConcept | CodeableConcept | False | | | This element is populated if the data type is category or custom list. Only one of the value* elements is required.

Observation.valueDateTime | String | False | | | This element is populated if data type date. Only one of value[x] is required.

Observation.valueQuantity | Quantity | False | | | This element is populated if a value has units or the data type is numeric. Only one of the value* elements is required.

Observation.valueString | string | False | | | This element is populated if a value doesn't have units or the data type is string. Only one of the value[x] elements is required.

Observation.valueTime | String | False | | | This element is populated if data type time. Only one of value[x] is required.

Sample request

```
-----------------------------------------------
Strict Metadata in Component Element:
-----------------------------------------------
<Observation>
<category><coding><system value="http://hl7.org/fhir/observation-category" /><code value="vital-signs" /><display
    value="Vital Signs" /></coding><text value="Vital Signs" /></category>
<code><coding><system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679" /><code value="2162" /><display value="After
    Lunch Glucose" /></coding><text value="After Lunch Glucose" /></code>
<subject><reference value="https://hostname/instance/api/FHIR/STU3/Patient/eMMKGfU7AJS5yjSRNjZdLxA3" /><display
    value="Meiko Lufhir Jr." /></subject>
<effectiveDateTime value="2017-08-04T17:42:00Z" />
<performer><reference value="https://hostname/instance/api/FHIR/STU3/Patient/eMMKGfU7AJS5yjSRNjZdLxA3" /><display
    value="Meiko Lufhir Jr." /></performer>
<valueQuantity><value value="6.3" /><unit value="mmol/L" /></valueQuantity>
<component>
<code>
<coding><system value="system=urn:oid:1.2.840.114350.1.13.861.1.7.2.708368"/><code value="2"/><display value="Blood
```

```
    Glucose Meal Time"/></coding>
</code>
<valueString value="2"/>
</component>
</Observation>
-----------------------------------------------
Multiple Strict Metadata in Component Element:
-----------------------------------------------
<Observation>
<category><coding><system value="http://hl7.org/fhir/observation-category" /><code value="vital-signs" /><display
    value="Vital Signs" /></coding><text value="Vital Signs" /></category>
<code><coding><system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679" /><code value="2162" /><display value="After
    Lunch Glucose" /></coding><text value="After Lunch Glucose" /></code>
<subject><reference value="https://hostname/instance/api/FHIR/STU3/Patient/eMMKGfU7AJS5yjSRNjZdLxA3" /><display
    value="Meiko Lufhir Jr." /></subject>
<effectiveDateTime value="2017-08-04T17:42:00Z" />
<performer><reference value="https://hostname/instance/api/FHIR/STU3/Patient/eMMKGfU7AJS5yjSRNjZdLxA3" /><display
    value="Meiko Lufhir Jr." /></performer>
<valueQuantity><value value="6.3" /><unit value="mmol/L" /></valueQuantity>
<component>
<code>
<coding><system value="system=urn:oid:1.2.840.114350.1.13.861.1.7.2.708368"/><code value="2"/><display value="Blood
    Glucose Meal Time"/></coding>
</code>
<valueString value="2"/>
</component>
<component>
<code>
<coding><system value="system=urn:oid:1.2.840.114350.1.13.861.1.7.2.708368"/><code value="1"/><display
    value="UUID"/></coding>
</code>
<valueString value="APLID09234920324"/>
</component>
</Observation>
-----------------------------------------------
Loose Metadata in Component Element:
-----------------------------------------------
<Observation>
<category><coding><system value="http://hl7.org/fhir/observation-category" /><code value="vital-signs" /><display
    value="Vital Signs" /></coding><text value="Vital Signs" /></category>
<code><coding><system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679" /><code value="2162" /><display value="After
    Lunch Glucose" /></coding><text value="After Lunch Glucose" /></code>
<subject><reference value="https://hostname/instance/api/FHIR/STU3/Patient/eMMKGfU7AJS5yjSRNjZdLxA3" /><display
    value="Meiko Lufhir Jr." /></subject>
<effectiveDateTime value="2017-08-04T17:42:00Z" />
<performer><reference value="https://hostname/instance/api/FHIR/STU3/Patient/eMMKGfU7AJS5yjSRNjZdLxA3" /><display
    value="Meiko Lufhir Jr." /></performer>
<valueQuantity><value value="6.3" /><unit value="mmol/L" /></valueQuantity>
<component>
<code>
<text>"Long Metadata Key for Glucose Meal Time that was not defined beforehand"</text>
</code>
<valueString value="After meal"/>
</component>
</Observation>
```

Sample response

```
HTTP/1.1 201 Created
Cache-Control: no-cache
Pragma: no-cache
Content-Type: application/xml; charset=utf-8
Expires: -1
Location: https://hostname/instance/api/FHIR/STU3/Observation/ejeZM7uM4zxl8gj7D9AqwEvM4HOV.6mxzXCMDzJynl8hNLOMksvJObZBaS
    y4R8WA1SGfaNubTGwUJVmepQDQ3pw3
Server: Microsoft-IIS/8.5
ServerMetrics: [{"create^FHIRXOBSSTU3CRTE":{"BlockReads":83,"BlockWrites":51,"BlocksAllocated":2,"DBTime":145,"ECFNetwor
    kTime":2,"ECFRequestCount":0,"ECFRequestTime":147,"GREF":7154,"JournalEntries":168,"LocksFailed":0,"LocksGranted":42
    ,"MCommands":85498,"MemoryDifference":471040,"WorkflowEventBlockReads":83,"WorkflowEventDBTime":145,"WorkflowEventEC
    FNetworkTime":0,"WorkflowEventECFRequestCount":0,"WorkflowEventGREF":7154}}]
X-AspNet-Version: 4.0.30319
X-Powered-By: ASP.NET
Date: Mon, 22 Jan 2018 19:39:23 GMT
Content-Length: 0
```

Change log

Epic Version | Description | Type

November 2022 | Epic has updated the behavior of certain modifier elements in FHIR write APIs to ensure the data in the incoming request body matches the data filed to the database and to comply with the HL7 FHIR specification. Previously, the modifier element was ignored in certain situations, and an undocumented default value was saved in its place. Now, if the modifier element is included and is set to an unaccepted value, a fatal error logs, preventing the request from filing to the database. The FHIR request returns an OperationOutcome with the invalid element and what value is accepted for that element. Element Accepted Incoming Value Default Value status final final | Substantive change

Element | Accepted Incoming Value | Default Value

status | final | final

9.2.8 Observation.Read (Activities of Daily Living) (STU3)

File: site/apis/09-Clinical-Diagnostics/2-STU3/10124-Observation-Read-Activities-of-Daily-Living-STU3.html
Epic API ID | 10124
FHIR version | STU3
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2021
HTTP method | GET
URL template | /api/FHIR/STU3/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10124

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | A single Observation FHIR resource.
Observation.category | CodeableConcept | False | Yes | | Type of observation. Always includes SNOMED codes “118228005” and “384821006”, and custom code “functional-mental status.”
Observation.code | CodeableConcept | False | | | The type of observation. Values mapped. If no mapping exists, only the text value is returned.
Observation.comment | string | True | | | Comments about the observation. A comment if flowsheet data or the associated note if inpatient goals. If using cascading flowsheets, the note element contains information about the additional flowsheet, if applicable, instead of the note. For example, a nurse documents flowsheet “daily activities” as abnormal, which opens flowsheet “mobility.” The nurse documents the value “highly mobile” in the flowsheet. The note element for “daily activities” is returned as “daily activities;highly mobile” if flowsheet “mobility” isn’t included in the Epic organization's configuration.
Observation.effectivePeriod | Period | True | | | Clinically relevant time-period for the observation
Observation.meta | Meta | True | | | The document metadata.
Observation.meta.profile | String | True | | | Available in Netherlands environments only. This element shows that this API conforms to the profile connected to one of the following canonical urls: * http://nictiz.nl/fhir/StructureDefinition/zib-LivingSituation * http://nictiz.nl/fhir/StructureDefinition/zib-DrugUse * http://nictiz.nl/fhir/StructureDefinition/zib-AlcoholUse * http://nictiz.nl/fhir/StructureDefinition/zib-TobaccoUse * http://nictiz.nl/fhir/StructureDefinition/zib-FunctionalOrMentalStatus
Observation.status | code | False | | | Always "final"
Observation.subject | Reference | False | | | Who the observation is about. Reference to the patient resource.
Observation.valueCodeableConcept | CodeableConcept | True | | | Result of the observation. Values mapped. If not mapping exists, only the text value is returned.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Observation/eK.YYu4CGX.bgDi.i48Ry0rwjGC3r1K2R6gMLL-cvPyTD.BkmsblugeJXH0AR7lq43
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "eK.YYu4CGX.bgDi.i48Ry0rwjGC3r1K2R6gMLL-cvPyTD.BkmsblugeJXH0AR7lq43",
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
          "code": "functional-mental-status",
          "display": "Functional and Mental Status"
        }
      ]
    },
    {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "118228005",
          "display": "Functional finding"
        }
      ]
    },
    {
      "coding": [
```

```
        {
          "system": "http://snomed.info/sct",
          "code": "384821006",
          "display": "Mental state, behavior and/or psychosocial function finding"
        }
      ]
    }
  ],
  "code": {
    "text": "Body Movement in Desired Range"
  },
  "subject": {
    "reference": "https://hostname/instance/api/FHIR/STU3/Patient/enoCp373txQSghUte78RDIQ3",
    "display": "FHIR, Automation"
  },
  "effectivePeriod": {
    "start": "2021-05-04",
    "end": "2021-05-07"
  },
  "valueCodeableConcept": {
    "text": "Progressing"
  }
}
```

9.2.9 Observation.Read (Core Characteristics) (STU3)

File: site/apis/09-Clinical-Diagnostics/2-STU3/853-Observation-Read-Core-Characteristics-STU3.html
Epic API ID | 853
FHIR version | STU3
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Keywords | Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2018
HTTP method | GET
URL template | /api/FHIR/STU3/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=853

Description

This FHIR Observation resource returns a core characteristic such as gestational age or birth weight given the associated FHIR ID.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | The read interaction uses the given Observation FHIR ID to retrieve the most up-to-date information about core characteristics.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | A single Observation FHIR resource.
Observation.category | CodeableConcept | False | Yes | | Always "core-characteristics".
Observation.issued | instant | False | | | The instant when this observation was last edited.
Observation.performer | Reference | False | Yes | | Who documented the observation.
Observation.status | code | False | | | Always set to "final".
Observation.subject | Reference | False | | | The patient.
Observation.valueQuantity | Quantity | False | | | The actual value of the observation.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Observation/ejeZM7uM4zxl8gj7D9AqwEtd5Ne3k4Sa0OLeJlrU4aP03
```

Sample response

```
<Observation xmlns="http://hl7.org/fhir">
  <id value="ejeZM7uM4zxl8gj7D9AqwEtd5Ne3k4Sa0OLeJlrU4aP03"/>
  <status value="final"/>
  <category>
    <coding>
      <system value="http://open.epic.com/FHIR/STU3/StructureDefinition/observation-category"/>
      <code value="core-characteristics"/>
      <display value="Core Characteristics"/>
    </coding>
    <text value="Core Characteristics"/>
```

```
  </category>
  <code>
    <coding>
      <system value="http://loinc.org"/>
      <code value="76516-4"/>
      <display value="Gestational age at birth"/>
    </coding>
    <text value="Gestational age at birth"/>
  </code>
  <subject>
    <reference value="https://hostname/instance/api/FHIR/STU3/Patient/eMMKGfU7AJS5yjSRNjZdLxA3"/>
    <display value="Meiko Lufhir Jr., PhD"/>
  </subject>
  <issued value="2018-02-15T22:33:12Z"/>
  <performer>
    <reference value="https://hostname/instance/api/FHIR/STU3/Practitioner/efv90i4pFs9fOm7QQhHOUHw3"/>
    <display value="Amber Hxl, MD"/>
  </performer>
  <valueQuantity>
    <value value="214"/>
    <unit value="d"/>
    <system value="http://unitsofmeasure.org"/>
    <code value="d"/>
  </valueQuantity>
</Observation>
```

9.2.10 Observation.Read (Labs) (STU3)

File: site/apis/09-Clinical-Diagnostics/2-STU3/494-Observation-Read-Labs-STU3.html
Epic API ID | 494
FHIR version | STU3
Categories | Clinical Diagnostics, Labs
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | BMI percentile per age and sex for youth 2-20;Weights for age per length and sex;Occipital-frontal circumference for children < 3 years old;Vital Signs;Diastolic blood pressure;Systolic blood pressure;Body height;Body weight;Heart rate;Respiratory rate;Body temperature;Pulse oximetry;Inhaled oxygen concentration;Diagnostic Reports;Imaging;Laboratory;Pathology;Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2018
HTTP method | GET
URL template | api/FHIR/STU3/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=494

Description

The Observation (Labs) resource returns component level data for lab results. Observation is not normally used on its own to access lab results, because it is limited to reading microbiology sensitivities (method level, isolate level, and sensitivity level), textual (narrative and impression) Observations, and certain imaging Observations (such as Midmark ECG results). Instead, a system queries the DiagnosticReport resource to identify all lab procedures performed for a patient. The DiagnosticReport results contain pointers to one or more Observation resources that contain the component results. If you are familiar with HL7v2, DiagnosticReport is analogous to an OBR segment, where Observation is like an OBX segment.

For example, consider a CBC (Complete Blood Count) test. The overall CBC test would be represented as a DiagnosticReport resource. The component results (white blood cell count, red blood cell count, hemoglobin, platelet count, and so on) would each be returned in independent Observation resources.

Most systems expect to receive a LOINC code identifying each component. An Observation resource will be returned for a component result even if a LOINC code is not found, however the Observation.code property will be empty.

Some labs, such as pathology or cytology, often file a PDF, or a link to a PDF or other document as the result. We do not currently support returning PDF content or other reference pointer content via FHIR, however the test will be discoverable via a DiagnosticReport search.

For microbiology labs, the DiagnosticReport resource represents the culture. It includes references to Observation for each isolate. The isolates then either reference an Observation for each method, or, if there is no method provided by the lab system, then the isolate will reference Observations for each sensitivity result. If methods are provided by the lab, then the method Observations will reference the sensitivity results.

The two potential structures are outlined below.

Microbiology results without methods:

Culture (DiagnosticReport)

Isolate A (Observation)

* Sensitivity results for antibiotic W (Observation)

* Sensitivity results for antibiotic X (Observation)

Isolate B (Observation)

* Sensitivity results for antibiotic W (Observation)

* Sensitivity results for antibiotic X (Observation)

Microbiology results with methods:

Culture (DiagnosticReport)

Isolate A (Observation)

* Method 1 (Observation) Sensitivity results for antibiotic W (Observation) b. Sensitivity results for antibiotic X (Observation)

* Method 2 (Observation) Sensitivity results for antibiotic W (Observation) b. Sensitivity results for antibiotic X (Observation)

Isolate B (Observation)

* Method 1 (Observation) Sensitivity results for antibiotic W (Observation) b. Sensitivity results for antibiotic X (Observation)

* Method 2 (Observation) Sensitivity results for antibiotic W (Observation) b. Sensitivity results for antibiotic X (Observation)

The category element is populated with SNOMED codes. In November 2021 and later, these codes are based on a mapping from component records to external SNOMED codes. From August 2021 to May 2021, this applies only to Netherlands environments and these codes are based on the value in the Order Type item in Epic. The following SNOMED codes are supported, and must be mapped in the External Mapping Utility:

* 252275004-Hematology (Hematologie)

* 275711006-Chemistry (Klinische chemie)

* 68793005-Serology (Serologie/ immunologie)

* 395124008-Virology (Virologie)

* 314076009-Toxicology (Toxicologie)

* 19851009-Microbiology (Microbiologie)

* 405825005-Molecular genetics (Moleculaire genetica)

Starting in February 2024, this sub-resource is also able to draw from data components linked to Midmark ECG orders. The code in these orders is linked to discrete items on the order (or other associated) records, corresponding to concepts with LOINC codes. In these instances, the category of the Observation is associated with the “Imaging” category as well as the “Laboratory” category. The discrete elements are tied to the following LOINC codes:

* PR interval- 8625-6

* P Wave axis- 8626-4

* QRS duration- 8633-0

* QRS wave axis- 8632-2

* QT interval- 8634-8

* QTC interval- 8636-3

* T wave axis- 8638-9

* Diagnostic statements- 8601-7

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | Required | | | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation (STU3) | Observation | | | | A single Observation FHIR resource.
Observation (STU3).category | CodeableConcept | | Yes | 1..* | The Observation type. Always "laboratory". For organizations in the Netherlands only, includes a category entry for the result type, using the ResultTypeCodelist.
Observation (STU3).code | CodeableConcept | | | 1..1 | Code of the Observation. Observation Grouper Code Description Component The component name Method The procedure code Isolate The isolate LOINC code Narrative or Impression Set to "Narrative" or "Impression" Addendum Set to "Addendum" or "Unsigned Addendum" Sensitivity The antibiotic code
Observation Grouper | Code Description
Component | The component name
Method | The procedure code
Isolate | The isolate LOINC code
Narrative or Impression | Set to "Narrative" or "Impression"
Addendum | Set to "Addendum" or "Unsigned Addendum"
Sensitivity | The antibiotic code
Observation (STU3).comment | string | | | 0..1 | Comments about the result.
Observation (STU3).context | Reference | | | 0..1 | The Encounter FHIR ID linked to the order.
Observation (STU3).dataAbsentReason | CodeableConcept | | | 0..1 | The reason why normally expected content of the data element is missing. Set to "unknown" for lab result components with no values.

Observation (STU3).effectiveDateTime | dateTime | | | 1..1 | The prioritized instant. This is the instant the results were last updated.

Observation (STU3).extension (historical-fhir-id) (also: extension (historical-fhir-id)) | Extension | | Yes | 0..1 | This extension contains a historical FHIR ID in cases where the historical ID exceeded the 64-character ID length specified in the HL7 FHIR standard. This extension is only returned if the requesting client is configured to use 64-character-limited FHIR IDs as described in the FHIR ID Generation Scheme topic. This element is returned starting in the November 2025 version of Epic and in August 2025 by special update. URL: http://open.epic.com/FHIR/StructureDefinition/extension/historical-fhir-id

Observation (STU3).extension (Lab-e-signature) (also: extension (Lab-e-signature)) | Extension | | Yes | 1..* | The signatures on a component. This is returned only for components with signatures as a string. URL: http://open.epic.com/FHIR/StructureDefinition/extension/lab-e-signature

Observation (STU3).id | id | | | 1..1 | The Observation FHIR ID.

Observation (STU3).interpretation | CodeableConcept | | | 0..1 | The level of abnormality. If the value isn't mapped we will just send the text value in the Codeable concept. Relevant for sensitivities.

Observation (STU3).issued | instant | | | 0..1 | The observation instant.

Observation (STU3).meta | Meta | | | 0..1 | The document metadata.

Observation (STU3).meta.profile | String | | Yes | 0..1 | Available in Netherlands environments only. This element shows that this API conforms to the profile connected to the following canonical url: http://nictiz.nl/fhir/StructureDefinition/zib-LaboratoryTestResult-Observation

Observation (STU3).method | CodeableConcept | | | 0..1 | SNOMED code mapped to procedure in Epic.

Observation (STU3).performer | Reference | | Yes | 0..* | A reference to the resulting agency's director's Practitioner resource.

Observation (STU3).referenceRange | BackboneElement | | Yes | 0..* | The reference range for the result, if applicable.

Observation (STU3).referenceRange.high | Quantity | | | 0..1 | The high value of the reference range.

Observation (STU3).referenceRange.low | Quantity | | | 0..1 | The low value of the reference range.

Observation (STU3).referenceRange.text | string | | | 0..1 | Display version of reference range.

Observation (STU3).related | BackboneElement | | Yes | 0..* | Resource related to this observation.

Observation (STU3).related.target | Reference | | | 0..1 | Isolate level Observation FHIR ID.

Observation (STU3).related.type | code | | | 0..1 | This element is always set to "has-member".

Observation (STU3).specimen | Reference | | | 0..1 | A reference to the Specimen resource.

Observation (STU3).status | code | | | 1..1 | The status of the lab order. For organizations in the Netherlands, this element can also include a TestResultStatus extension that indicates the test result status.

Observation (STU3).subject | Reference | | | 1..1 | The Patient FHIR ID linked to the order.

Observation (STU3).valueCodeableConcept | CodeableConcept | | | 0..1 | This element is populated only if a SNOMED code is associated with the component.

Observation (STU3).valueQuantity | Quantity | | | 0..1 | The result value if the result is numeric and represents a quantity.

Observation (STU3).valueRange | Range | | | 0..1 | The result value if the result is structured numeric and the operator is "-".

Observation (STU3).valueRatio | Ratio | | | 0..1 | The result value if the result is structured numeric and the operator is ":".

Observation (STU3).valueString | string | | | 0..1 | The result value if the result is a string.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Observation/eAMXBdyxr2HSrK7BSIJoPna14R45Ho9Cwgyy5qfxFTqJraA5OWLQFNaXFUhKX.yTF3
```

Sample response


```
{
  "resourceType": "Observation",
  "id": "eAMXBdyxr2HSrK7BSIJoPna14R45Ho9Cwgyy5qfxFTqJraA5OWLQFNaXFUhKX.yTF3",
  "basedOn": [
    {
      "reference": "https://hostname/instance/api/FHIR/STU3/ProcedureRequest/e88uk8v7SSbyNXOiSR9-moxFQZin7nVZSFmyIN-
    HJDho3"
    }
  ],
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "252275004",
          "display": "Hematology test (procedure)"
        }
      ],
      "text": "Hematology test (procedure)"
    },
    {
      "coding": [
        {
          "system": "http://hl7.org/fhir/observation-category",
          "code": "laboratory",
          "display": "Laboratory"
        }
      ],
      "text": "Laboratory"
    },
    {
      "coding": [
        {
          "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.10.798268.30",
          "code": "Lab",
          "display": "Lab"
        }
      ],
      "text": "Lab"
    }
```

```
  ],
  "code": {
    "coding": [
      {
        "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.5.737384.188",
        "code": "68"
      },
      {
        "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.5.737384.431",
        "code": "AF-GLUKOOSI"
      },
      {
        "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.5.737384.430",
        "code": "1459"
      },
      {
        "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.768282",
        "code": "68",
        "display": "GLUCOSE"
      }
    ],
    "text": "GLUCOSE"
  },
  "subject": {
    "reference": "https://hostname/instance/api/FHIR/STU3/Patient/e3E2Rkizy.WmzItzk1PbZAQ3",
    "display": "Marshall, Fhir One"
  },
  "context": {
    "reference": "https://hostname/instance/api/FHIR/STU3/Encounter/eorXVhQrtx.V6AYpjcoaYBMEpqd5Y8s7Eq8Mr4Ltn.2U3"
  },
  "effectiveDateTime": "2021-09-02T18:41:00Z",
  "issued": "2021-09-02T19:10:25Z",
  "performer": [
    {
      "reference": "https://hostname/instance/api/FHIR/STU3/Practitioner/ecw01Sml5VFYLaE2Mp3-hgQ3",
      "display": "Lab Admin Tpr"
    }
  ],
  "valueQuantity": {
    "value": 95,
    "unit": "mmol/L",
    "system": "http://unitsofmeasure.org",
    "code": "mmol/L"
  },
  "specimen": {
    "reference": "https://hostname/instance/api/FHIR/STU3/Specimen/eiWBs9u0Cb1fNJ5IgL3cxqgWFghRXa9VQggFtlJ3i2303"
  },
  "referenceRange": [
    {
      "low": {
        "value": 80,
        "unit": "mmol/L",
        "system": "http://unitsofmeasure.org",
        "code": "mmol/L"
      },
      "high": {
        "value": 110,
        "unit": "mmol/L",
        "system": "http://unitsofmeasure.org",
        "code": "mmol/L"
      },
      "text": "80 - 110 mmol/L"
    }
  ]
}
```


Change log

Epic Version | Description | Type

November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change

System | URL | OID

ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4

ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90

ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94

ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2

SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

9.2.11 Observation.Read (LDA-W) (STU3)

File: site/apis/09-Clinical-Diagnostics/2-STU3/898-Observation-Read-LDA-W-STU3.html

Epic API ID | 898

FHIR version | STU3
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Keywords | Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2018
HTTP method | GET
URL template | api/FHIR/STU3/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=898

Description

The FHIR Observation resource defines measurements and assertions about a patient, including LDA-W (lines, drains, airways, and wound) data.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | The read interaction uses the given Observation FHIR ID to retrieve the most up-to-date information about a LDA (details and properties) or a LDA assessment.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | A single Observation FHIR resource.
Observation.category | CodeableConcept | False | Yes | | The category is always "LDA". This element can't be modified by Observation.Update.
Observation.code | CodeableConcept | False | | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. This element can't be modified by Observation.Update. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.
Observation.comment | string | True | | | Comment for the assessment.
Observation.component | BackboneElement | False | Yes
Observation.component.code | CodeableConcept | False | | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.
Observation.component.extension (also: LDA-Property-Comment) | Extension | True | Yes | | Comment for the property row.
Observation.component.valueQuantity | Quantity | False | | | Value of the property row.
Observation.component.valueString | String | False | | | Value of the property row.
Observation.context | Reference | False | | | Reference to an encounter resource. This element can't be modified by Observation.Update.
Observation.effectiveDateTime | dateTime | False | | | The prioritized instant. This element can't be modified by Observation.Update.
Observation.effectivePeriod | Period | False | | | Start instant and/or end instant of the LDA.
Observation.issued | instant | False | | | Instant the assessment was documented.
Observation.performer | Reference | False | Yes | | Reference to a practitioner resource.
Observation.related | BackboneElement | False | Yes | | This element can't be modified by the Observation.Update.
Observation.related.target | Reference | False | | | The parent LDA observation FHIR ID. This element can't be modified by Observation.Update.
Observation.related.type | code | True | | | Always "sequel-to". This element can't be modified by Observation.Update.
Observation.status | code | False | | | The status is always "final".
Observation.subject | Reference | False | | | Reference to a patient resource. This element can't be modified by Observation.Update.
Observation.valueQuantity | Quantity | False | | | Value of the assessment. For image rows, a comma-delimited list of external DCS IDs is required.
Observation.valueString | string | False | | | LDA display name.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Observation/e1CQQb8SWGwdSTExJM-VUIp6Inqjs4Q.8fKCtINxMo1tf2qhgHdLqmpqoMmEJU58r3
```

Sample response

```
<Observation xmlns="http://hl7.org/fhir">
  <id value="e1CQQb8SWGwdSTExJM-VUIp6Inqjs4Q.8fKCtINxMo1tf2qhgHdLqmpqoMmEJU58r3"/>
  <status value="final"/>
  <category>
    <coding>
      <system value="http://open.epic.com/FHIR/STU3/StructureDefinition/observation-category"/>
      <code value="LDA"/>
      <display value="Line, Drains, and Airways"/>
    </coding>
    <text value="Line, Drains, and Airways"/>
  </category>
  <code>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
```

```
      <code value="42675"/>
      <display value="Wound Length (cm)"/>
    </coding>
    <coding>
      <system value="http://snomed.info/sct"/>
      <code value="401238003"/>
      <display value="Length of wound (observable entity)"/>
    </coding>
    <text value="Wound Length (cm)"/>
  </code>
  <subject>
    <reference value="https://hostname/instance/api/FHIR/STU3/Patient/ej4nFxzySJAysjMuv4rTBNA3"/>
    <display value="Meiko Lufhir"/>
  </subject>
  <context>
    <reference value="https://hostname/instance/api/FHIR/STU3/Encounter/eKzsYGMB9rxAHlrtpROR1iBa2CWjmdcgpdgv8cDE4ZCQ3"/>
  </context>
  <effectiveDateTime value="2018-10-18T16:00:00Z"/>
  <issued value="2018-10-18T20:49:05Z"/>
  <performer>
    <reference value="https://hostname/instance/api/FHIR/STU3/Practitioner/e89p8GCwg7wvke-vxglQ0WA3"/>
    <display value="Amber Hxl, MD"/>
  </performer>
  <valueQuantity>
    <value value="2"/>
    <unit value="cm"/>
  </valueQuantity>
  <comment value="Not severe"/>
  <related>
    <type value="sequel-to"/>
    <target>
      <reference value="https://hostname/instance/api/FHIR/STU3/Observation/ebhUU2Y8Wnz8PEsBIRUDfBw3"/>
      <display value="HZL Wound LDA Meiko's wound Left Ankle"/>
    </target>
  </related>
</Observation>
```

Change log

Epic Version | Description | Type

August 2025 (May 2025, February 2025, November 2024 by special update) | Starting in August 2024, May 2025 with special update E11400862, February 2025 with special update E11306699, and November 2024 with special update E11211328, multi-select fields are handled separately, and if a SNOMED concept mapping exists for any individual selection, the SNOMED concept mapping is included in the FHIR response. Previously, when a user selected multiple options, the system treated the entire list as a single combined value. | Substantive Changes

9.2.12 Observation.Read (Vitals) (STU3)

File: site/apis/09-Clinical-Diagnostics/2-STU3/497-Observation-Read-Vitals-STU3.html

Epic API ID | 497

FHIR version | STU3

Categories | Clinical Diagnostics

API type | USCDI

A la carte | Industry-Standard Level 1 API

USCDI | Yes

Keywords | BMI percentile per age and sex for youth 2-20;Weights for age per length and sex;Occipital-frontal circumference for children < 3 years old;Vital Signs;Diastolic blood pressure;Systolic blood pressure;Body height;Body weight;Heart rate;Respiratory rate;Body temperature;Pulse oximetry;Inhaled oxygen concentration;Diagnostic Reports;Imaging;Laboratory;Pathology;Flowsheets

Available in inpatient and other contexts | Yes

ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base

ONC certification versions | Epic 2018

HTTP method | GET

URL template | api/FHIR/STU3/Observation/{ID}

Supports OAuth 2.0 | True

Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients

Performs scope validation | False

Source | https://fhir.epic.com/Specifications?api=497

Description

The FHIR Observation resource defines measurements and assertions about a patient, including vital signs, laboratory data, imaging results, devices measurements, clinical assessment tools, personal characteristics, social history, and core characteristics.

Remarks

To see an example of how the component.code element is structured for patient-entered flowsheet data, refer to the sample request section of the Observation.Create (Vitals) (STU3) API entry.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | token | False | | | The observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | A single Observation FHIR resource.
Observation.bodySite | CodeableConcept | False | | | Applicable only to organizations in the Netherlands starting in Epic version August 2021. The site on the body where the observation was made. Used for the location where a blood pressure reading was obtained.
Observation.category | CodeableConcept | False | Yes | | This element always returns "vital-signs". Starting in November 2022 in Netherlands locales, this element also returns SNOMED code 49581000146104 (Laboratory test finding (finding) when returning observations linked to the blood glucose profile.
Observation.code | CodeableConcept | False | | | The flowsheet (FLO) ID, encoded flowsheet ID, or LOINC codes to flowsheet mapping. What is being measured. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.
Observation.comment | string | True | | | Comments about the observation value.
Observation.component | BackboneElement | False | Yes
Observation.component.code | CodeableConcept | True | | | This element is used for observations filed by patients to patient-entered flowsheets. Metadata related to the observation. Represents strict metadata if the coding sub-element is populated. The following strict metadata are supported: 1-UUID The unique reading ID. 2-Blood Glucose Meal Times 1-Before meal 2-After meal 3-Insulin Delivery Reason 1-Basal 2-Bolus 6-Manually Entered 0-False 1-True Represents loose metadata if only the text sub-element is populated. UUID, Blood Glucose Meal times, Insulin Delivery Reason, and Manually Entered must have their value specified in component.valueString. Applicable to organizations in the Netherlands starting in November 2022: LOINC code 74206-4 (Oxygen therapy was given [NTDS]) LOINC code 44969-4 (Heart rate rhythm palpation)
Observation.component.valueCodeableConcept | CodeableConcept | True | | | Applicable to organizations in the Netherlands starting in November 2022: Codes from CodeSystem “v2 Yes/no indicator” indicate whether extra oxygen was given at the time of an oxygen saturation reading SNOMED codes from PolsRegelmatigheidCodelijst indicate the regularity of the pulse rate at the time of a pulse reading
Observation.dataAbsentReason | CodeableConcept | True | | | Available starting in the August 2021 version of Epic. Indicates that an observation value is missing. Will always be “unknown” for systolic blood pressure values in observations for diastolic-only blood pressure flowsheets, or for diastolic blood pressure value in observations for systolic-only blood pressure flowsheets.
Observation.effectiveDateTime | dateTime | False | | | The date and time the observation was taken.
Observation.interpretation | CodeableConcept | False | | | Available starting in November 2022 in Netherlands locales. For heart rate profile observations, possible values include: SNOMED code 271636001 (Heart regular) for normal heart rate readings SNOMED code 248650006 (Heart irregular) for abnormal heart rate readings For blood glucose profile observations, possible values include: SNOMED code 281302008 (Above reference range) for blood glucose readings higher than the reference range SNOMED code 11896004 (Intermediate) for blood glucose readings within the reference range (inclusive of the reference range values) SNOMED code 281300000 (Below reference range) for blood glucose readings below the reference range
Observation.issued | instant | False | | | The date and time the observation was documented. This is always the current time.
Observation.meta | Meta | True | | | The document metadata.
Observation.meta.profile | String | True | | | Available in Netherlands environments only. This element shows that this API conforms to the profile connected to one of the following canonical urls: http://nictiz.nl/fhir/StructureDefinition/zib-BloodPressure http://nictiz.nl/fhir/StructureDefinition/zib-BodyWeight http://nictiz.nl/fhir/StructureDefinition/zib-BodyHeight http://nictiz.nl/fhir/StructureDefinition/zib-BodyTemperature http://nictiz.nl/fhir/StructureDefinition/zib-OxygenSaturation http://nictiz.nl/fhir/StructureDefinition/zib-HeartRate http://nictiz.nl/fhir/StructureDefinition/zib-PulseRate http://nictiz.nl/fhir/StructureDefinition/vitalsign-bloodglucose
Observation.method | CodeableConcept | True | | | Method by which the observation was obtained. Possible values include: "Patient-entered" for patient-entered flowsheet vital signs Applicable only to organizations in the Netherlands starting in August 2021: SNOMED code 10179008 (invasive) for blood pressure readings using an invasive method Applicable only to organizations in the Netherlands starting in August 2021: SNOMED code 22762002 (non-invasive) for blood pressure readings not using an invasive method Starting in November 2022 in Netherlands locales: SNOMED codes from the TemperatuurTypeCodelijst for the specific anatomical location used for body temperature readings Starting in November 2022 in Netherlands locales: SNOMED codes from the HartslagMeetMethodeCodelijst for the specific method used for heart rate readings Starting in November 2022 in Netherlands locales: SNOMED codes from the TestmethodeCodelijst for the specific method used for blood glucose readings
Observation.performer | Reference | False | Yes | | Who filed the observation.
Observation.referenceRange | BackboneElement | False | Yes
Observation.referenceRange.high | Quantity | True | | | Applicable to organizations in the Netherlands starting in November 2022. The higher reference limit for blood glucose readings.
Observation.referenceRange.low | Quantity | True | | | Applicable to organizations in the Netherlands starting in November 2022. The lower reference limit for blood glucose readings.
Observation.status | code | False | | | The status of the observation. This is set to amended or final depending on whether the field has been corrected.
Observation.subject | Reference | False | | | The FHIR ID for the patient the observation is about.
Observation.valueCodeableConcept | CodeableConcept | False | | | This element is populated if the data type is category or custom list. Only one of the value* elements is required.
Observation.valueDateTime | String | False | | | This element is populated if the data type is date. Only one of the value* elements is required.
Observation.valueQuantity | Quantity | False | | | This element is populated if a value has units or the data type is numeric. Only one of the value* elements is required.
Observation.valueString | string | False | | | This element is populated if a value doesn't have units or the data type is string. Only one of the value[x] elements is required.
Observation.valueTime | String | False | | | This element is populated if the data type is time. Only one of the value* elements is required.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Observation/ejeZM7uM4zxl8gj7D9AqwEvM4HOV.6mxzXCMDzJynl8jllTO7qcPDcDtpwpDoGvQJ2dv
    .Ddi1YFzyN2wDgbESRw3
```

Sample response

```
<Observation xmlns="http://hl7.org/fhir">
  <category>
    <coding>
      <system value="http://hl7.org/fhir/observation-category"/>
      <code value="vital-signs"/>
      <display value="Vital Signs"/>
    </coding>
    <text value="Vital Signs"/>
  </category>
  <code>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
      <code value="557395"/>
      <display value="HXL Weight"/>
    </coding>
    <coding>
      <system value="http://loinc.org"/>
      <code value="3141-9"/>
      <display value="Body weight Measured"/>
    </coding>
    <coding>
      <system value="http://loinc.org"/>
      <code value="8716-3"/>
      <display value="Vital signs"/>
    </coding>
    <text value="HXL Weight"/>
  </code>
  <subject>
    <reference value="https://hostname/instance/api/FHIR/STU3/Patient/eMMKGfU7AJS5yjSRNjZdLxA3"/>
    <display value="Meiko Lufhir Jr., PhD"/>
  </subject>
  <effectiveDateTime value="2017-07-26T17:42:00Z"/>
  <performer>
    <reference value="https://hostname/instance/api/FHIR/STU3/Patient/eMMKGfU7AJS5yjSRNjZdLxA3"/>
    <display value="Meiko Lufhir Jr., PhD"/>
  </performer>
  <valueQuantity>
    <value value="150"/>
    <unit value="lb"/>
  </valueQuantity>
  <id value="ejeZM7uM4zxl8gj7D9AqwEvM4HOV.6mxzXCMDzJynl8jllTO7qcPDcDtpwpDoGvQJ2dv.Ddi1YFzyN2wDgbESRw3"/>
  <status value="final"/>
  <issued value="2017-07-26T17:42:38Z"/>
  <method>
    <text value="Patient-entered"/>
  </method>
</Observation>
```

Change log

Epic Version | Description | Type

August 2025 (May 2025, February 2025, November 2024 by special update) | Starting in August 2024, May 2025 with special update E11400862, February 2025 with special update E11306699, and November 2024 with special update E11211328, multi-select fields are handled separately, and if a SNOMED concept mapping exists for any individual selection, the SNOMED concept mapping is included in the FHIR response. Previously, when a user selected multiple options, the system treated the entire list as a single combined value. | Substantive Changes

9.2.13 Observation.Search (Activities of Daily Living) (STU3)

File: site/apis/09-Clinical-Diagnostics/2-STU3/10125-Observation-Search-Activities-of-Daily-Living-STU3.html

Epic API ID | 10125

FHIR version | STU3

Categories | Clinical Diagnostics

API type | Industry-Standard

USCDI | No

Available in inpatient and other contexts | Yes

ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base

ONC certification versions | August 2021

HTTP method | GET

URL template | /api/FHIR/STU3/Observation?category={category}&patient={patient}&subject={subject}

Supports OAuth 2.0 | True

Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients

Performs scope validation | False

Source | https://fhir.epic.com/Specifications?api=10125

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description

category | token | False | | | Use "functional-mental-status", http://snomed.info/sct|384821006, and/or http://snomed.info/sct|118228005 to search for activities of daily living.

patient | reference | Either patient or subject must be provided. If both are provided, they must reference the same patient. | | | The reference to a patient resource.
subject | reference | Either patient or subject must be provided. If both are provided, they must reference the same patient. | | | The reference to a patient resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Observation | Observation | False | | | A single Observation FHIR resource.
Bundle.entry.resource.Observation.category | CodeableConcept | False | Yes | | Type of observation. Always includes SNOMED codes “118228005” and “384821006”, and custom code “functional-mental status.”
Bundle.entry.resource.Observation.code | CodeableConcept | False | | | The type of observation. Values mapped. If no mapping exists, only the text value is returned.
Bundle.entry.resource.Observation.comment | string | True | | | Comments about the observation . A comment if flowsheet data or the associated note if inpatient goals. If using cascading flowsheets, the note element contains information about the additional flowsheet, if applicable, instead of the note. For example, a nurse documents flowsheet “daily activities” as abnormal, which opens flowsheet “mobility.” The nurse documents the value “highly mobile” in the flowsheet. The note element for “daily activities” is returned as “daily activities;highly mobile” if flowsheet “mobility” isn’t included in the Epic organization's configuration.
Bundle.entry.resource.Observation.effectivePeriod | Period | True | | | Clinically relevant time period for the observation
Bundle.entry.resource.Observation.status | code | False | | | Always "final"
Bundle.entry.resource.Observation.subject | Reference | False | | | Who the observation is about. Reference to the patient resource.
Bundle.entry.resource.Observation.valueCodeableConcept | CodeableConcept | True | | | Result of the observation. Values mapped. If no mapping exists, only the text value is returned.

Sample request

```
https://hostname/instance/api/FHIR/stu3/Observation?patient=enoCp373txQSghUte78RDIQ3&category=http://snomed.info/sct|384
    821006
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 2,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/STU3/Observation?patient=enoCp373txQSghUte78RDIQ3&category=http://snome
    d.info/sct%7C384821006,http://snomed.info/sct%7C118228005"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/STU3/Observation/eK.YYu4CGX.bgDi.i48Ry0qqp31Vci5cpqDiXY17qo-
    lcGEO.J6Z8OJbctD0XBFWB3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/STU3/Observation/eK.YYu4CGX.bgDi.i48Ry0qqp31Vci5cpqDiXY17qo-
    lcGEO.J6Z8OJbctD0XBFWB3",
      "resource": {
        "resourceType": "Observation",
        "id": "eK.YYu4CGX.bgDi.i48Ry0qqp31Vci5cpqDiXY17qo-lcGEO.J6Z8OJbctD0XBFWB3",
        "status": "final",
        "category": [
          {
            "coding": [
              {
                "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
                "code": "functional-mental-status",
                "display": "Functional and Mental Status"
              }
            ]
          },
          {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "118228005",
                "display": "Functional finding"
              }
            ]
          },
          {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "384821006",
```

```
              "display": "Mental state, behavior and/or psychosocial function finding"
            }
          ]
        }
      ],
      "code": {
        "text": "Neuro (WDL)"
      },
      "subject": {
        "reference": "https://hostname/instance/api/FHIR/STU3/Patient/enoCp373txQSghUte78RDIQ3",
        "display": "FHIR, Automation"
      },
      "effectivePeriod": {
        "start": "2021-05-04T20:00:00Z"
      },
      "valueCodeableConcept": {
        "text": "Exceptions to WDL"
      },
      "comment": "Level of Consciousness: New agitation; Orientation Level: Disoriented to place; Cognition:
  Impulsive"
    },
    "search": {
      "mode": "match"
    }
  },
  {
    "link": [
      {
        "relation": "self",
        "url": "https://hostname/instance/api/FHIR/STU3/Observation/eK.YYu4CGX.bgDi.i48Ry0rwjGC3r1K2R6gMLL-
  cvPyTD.BkmsblugeJXH0AR7lq43"
      }
    ],
    "fullUrl": "https://hostname/instance/api/FHIR/STU3/Observation/eK.YYu4CGX.bgDi.i48Ry0rwjGC3r1K2R6gMLL-
  cvPyTD.BkmsblugeJXH0AR7lq43",
    "resource": {
      "resourceType": "Observation",
      "id": "eK.YYu4CGX.bgDi.i48Ry0rwjGC3r1K2R6gMLL-cvPyTD.BkmsblugeJXH0AR7lq43",
      "status": "final",
      "category": [
        {
          "coding": [
            {
              "system": "http://open.epic.com/FHIR/StructureDefinition/observation-category",
              "code": "functional-mental-status",
              "display": "Functional and Mental Status"
            }
          ]
        },
        {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "118228005",
              "display": "Functional finding"
            }
          ]
        },
        {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "384821006",
              "display": "Mental state, behavior and/or psychosocial function finding"
            }
          ]
        }
      ],
      "code": {
        "text": "Body Movement in Desired Range"
      },
      "subject": {
        "reference": "https://hostname/instance/api/FHIR/STU3/Patient/enoCp373txQSghUte78RDIQ3",
        "display": "FHIR, Automation"
      },
      "effectivePeriod": {
        "start": "2021-05-04",
        "end": "2021-05-07"
      },
      "valueCodeableConcept": {
        "text": "Progressing"
      }
    },
    "search": {
      "mode": "match"
    }
  }
]
}
```

9.2.14 Observation.Search (Core Characteristics) (STU3)

File: site/apis/09-Clinical-Diagnostics/2-STU3/854-Observation-Search-Core-Characteristics-STU3.html
Epic API ID | 854
FHIR version | STU3
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Keywords | Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | August 2018
HTTP method | GET
URL template | /api/FHIR/STU3/Observation?patient={patient}&category={category}&code={code}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=854

Description

This FHIR Observation resource returns core characteristics for a patient including gestational age and birth weight.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | Either this or code must be provided, not both. | | | Use "core-characteristics" to search for characteristics observations.
code | token | Either this or category must be provided, not both. | | | LOINC code of observation. Can be 76516-4 for gestational age or 8339-4 for birth weight.
patient | reference | False | | | Reference to a patient resource. If both patient and subject are provided, they must match.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.response | Response | False
Bundle.entry.response.outcome | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.response.outcome.Observation | Observation | False | | | A single Observation FHIR resource.
Bundle.entry.response.outcome.Observation.category | CodeableConcept | False | Yes | | Always "core-characteristics".
Bundle.entry.response.outcome.Observation.issued | instant | False | | | The instant when this observation was last edited.
Bundle.entry.response.outcome.Observation.performer | Reference | False | Yes | | Who documented the observation.
Bundle.entry.response.outcome.Observation.status | code | False | | | Always set to "final".
Bundle.entry.response.outcome.Observation.subject | Reference | False | | | The patient.
Bundle.entry.response.outcome.Observation.valueQuantity | Quantity | False | | | The actual value of the observation.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Observation?patient=eMMKGfU7AJS5yjSRNjZdLxA3&category=core-characteristics
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="2"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/STU3/Observation?patient=eMMKGfU7AJS5yjSRNjZdLxA3&amp;category=core-
    characteristics"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/STU3/Observation/ejeZM7uM4zxl8gj7D9AqwEtd5Ne3k4Sa0OLeJlrU4aP03"/>
    </link>
    <fullUrl value="https://hostname/instance/api/FHIR/STU3/Observation/ejeZM7uM4zxl8gj7D9AqwEtd5Ne3k4Sa0OLeJlrU4aP03"/>
    <resource>
      <Observation>
        <id value="ejeZM7uM4zxl8gj7D9AqwEtd5Ne3k4Sa0OLeJlrU4aP03"/>
        <status value="final"/>
        <category>
          <coding>
            <system value="http://open.epic.com/FHIR/STU3/StructureDefinition/observation-category"/>
            <code value="core-characteristics"/>
            <display value="Core Characteristics"/>
          </coding>
          <text value="Core Characteristics"/>
        </category>
        <code>
```

```
        <coding>
          <system value="http://loinc.org"/>
          <code value="76516-4"/>
          <display value="Gestational age at birth"/>
        </coding>
        <text value="Gestational age at birth"/>
      </code>
      <subject>
        <reference value="https://hostname/instance/api/FHIR/STU3/Patient/eMMKGfU7AJS5yjSRNjZdLxA3"/>
        <display value="Meiko Lufhir Jr., PhD"/>
      </subject>
      <issued value="2018-02-15T22:33:12Z"/>
      <performer>
        <reference value="https://hostname/instance/api/FHIR/STU3/Practitioner/efv90i4pFs9fOm7QQhHOUHw3"/>
        <display value="Amber Hxl, MD"/>
      </performer>
      <valueQuantity>
        <value value="214"/>
        <unit value="d"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="d"/>
      </valueQuantity>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
  </search>
 </entry>
 <entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/STU3/Observation/ejeZM7uM4zxl8gj7D9AqwEjHpi80iv1UoezjBFqs.VoI3"/>
  </link>
  <fullUrl value="https://hostname/instance/api/FHIR/STU3/Observation/ejeZM7uM4zxl8gj7D9AqwEjHpi80iv1UoezjBFqs.VoI3"/>
  <resource>
    <Observation>
      <id value="ejeZM7uM4zxl8gj7D9AqwEjHpi80iv1UoezjBFqs.VoI3"/>
      <status value="final"/>
      <category>
        <coding>
          <system value="http://open.epic.com/FHIR/STU3/StructureDefinition/observation-category"/>
          <code value="core-characteristics"/>
          <display value="Core Characteristics"/>
        </coding>
        <text value="Core Characteristics"/>
      </category>
      <code>
        <coding>
          <system value="http://loinc.org"/>
          <code value="8339-4"/>
          <display value="Birth weight measured"/>
        </coding>
        <text value="Birth weight measured"/>
      </code>
      <subject>
        <reference value="https://hostname/instance/api/FHIR/STU3/Patient/eMMKGfU7AJS5yjSRNjZdLxA3"/>
        <display value="Meiko Lufhir Jr., PhD"/>
      </subject>
      <issued value="2018-02-07T20:42:54Z"/>
      <performer>
        <reference value="https://hostname/instance/api/FHIR/STU3/Practitioner/egCj4Z7EG1mplI4xYrONFRQ3"/>
        <display value="Starter Provider"/>
      </performer>
      <valueQuantity>
        <value value="5"/>
        <unit value="kg"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="kg"/>
      </valueQuantity>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
  </search>
 </entry>
</Bundle>
```


9.2.15 Observation.Search (Labs) (STU3)

File: site/apis/09-Clinical-Diagnostics/2-STU3/495-Observation-Search-Labs-STU3.html
Epic API ID | 495
FHIR version | STU3
Categories | Clinical Diagnostics, Labs
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes

Keywords | BMI percentile per age and sex for youth 2-20;Weights for age per length and sex;Occipital-frontal circumference for children < 3 years old;Vital Signs;Diastolic blood pressure;Systolic blood pressure;Body height;Body weight;Heart rate;Respiratory rate;Body temperature;Pulse oximetry;Inhaled oxygen concentration;Diagnostic Reports;Imaging;Laboratory;Pathology;Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2018
HTTP method | GET
URL template | api/FHIR/STU3/Observation?patient={patient}&subject={subject}&category={category}&code={code}&date={date}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=495

Description

The Observation (Labs) resource returns component level data for lab results. Observation is not normally used on its own to access lab results, because it is limited to reading microbiology sensitivities (method level, isolate level, and sensitivity level) and textual (narrative and impression) Observations. Instead, a system queries the DiagnosticReport resource to identify all lab procedures performed for a patient. The DiagnosticReport results will contain pointers to one or more Observation resources that contain the component results. If you are familiar with HL7v2, DiagnosticReport is analogous to an OBR segment, where Observation is like an OBX segment.

For example, consider a CBC (Complete Blood Count) test. The overall CBC test would be represented as a DiagnosticReport resource. The component results (white blood cell count, red blood cell count, hemoglobin, platelet count, and so on) would each be returned in independent Observation resources.

Most systems expect to receive a LOINC code identifying each component. An Observation resource will be returned for a component result even if a LOINC code is not found, however the Observation.code property will be empty.

Some labs, such as pathology or cytology, often file a PDF, or a link to a PDF or other document as the result. We do not currently support returning PDF content or other reference pointer content via FHIR, however the test will be discoverable via a DiagnosticReport search.

When processing values, be sure to consider all elements. Classification of value is subject to change from system to system and version to version.

For microbiology labs, the DiagnosticReport resource represents the culture. It includes references to Observation for each isolate. The isolates may then either reference an Observation for each method, or, if there is no method provided by the lab system, then the isolate will reference Observations for each sensitivity result. If methods are provided by the lab, then the method Observations will reference the sensitivity results.

The two potential structures are outlined below.

Microbiology results without methods:

Culture (DiagnosticReport)

Isolate A (Observation)

* Sensitivity results for antibiotic W (Observation)

* Sensitivity results for antibiotic X (Observation)

Isolate B (Observation)

* Sensitivity results for antibiotic W (Observation)

* Sensitivity results for antibiotic X (Observation)

Microbiology results with methods:

Culture (DiagnosticReport)

Isolate A (Observation)

* Method 1 (Observation) Sensitivity results for antibiotic W (Observation) b. Sensitivity results for antibiotic X (Observation)

* Method 2 (Observation) Sensitivity results for antibiotic W (Observation) b. Sensitivity results for antibiotic X (Observation)

Isolate B (Observation)

* Method 1 (Observation) Sensitivity results for antibiotic W (Observation) b. Sensitivity results for antibiotic X (Observation)

* Method 2 (Observation) Sensitivity results for antibiotic W (Observation) b. Sensitivity results for antibiotic X (Observation)

Request parameters

Name | Type | Optional | Array | Cardinality | Description
_count | number | Optional | Yes | | Number of results per page.
_id | token | Optional | Yes | | FHIR resource IDs for the desired resources. Can be used in combination with _count and _include. All others will be ignored.
_include | string | Optional | Yes | | Used to indicate which referenced resources to include in the search bundle.
category | token | Conditional | Yes | | Use "laboratory" to search for lab observations.
code | token | Conditional | Yes | | The LOINC code of the Observation.
date | date | Optional | Yes | | Date when the specimen was obtained.

patient | reference | Conditional | Yes | | The reference to a patient resource.
subject | reference | Conditional | Yes | | The reference to a patient resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation (STU3) | Observation | | | | A single Observation FHIR resource.
Observation (STU3).basedOn | Reference | | Yes | 0..* | The procedure or service this observation is associated with.
Observation (STU3).category | CodeableConcept | | Yes | 1..* | The Observation type. Always "laboratory". For organizations in the Netherlands only, includes a category entry for the result type, using the ResultTypeCodelist.
Observation (STU3).code | CodeableConcept | | | 1..1 | Code of the Observation. Observation Grouper Code Description Component The component name Method The procedure code Isolate The isolate LOINC code Narrative or Impression Set to "Narrative" or "Impression" Addendum Set to "Addendum" or "Unsigned Addendum" Sensitivity The antibiotic code
Observation Grouper | Code Description
Component | The component name
Method | The procedure code
Isolate | The isolate LOINC code
Narrative or Impression | Set to "Narrative" or "Impression"
Addendum | Set to "Addendum" or "Unsigned Addendum"
Sensitivity | The antibiotic code
Observation (STU3).comment | string | | | 0..1 | Comments about the result.
Observation (STU3).context | Reference | | | 0..1 | The Encounter FHIR ID linked to the order.
Observation (STU3).dataAbsentReason | CodeableConcept | | | 0..1 | The reason why normally expected content of the data element is missing. Set to "unknown" for lab result components with no values.
Observation (STU3).effectiveDateTime | dateTime | | | 1..1 | The prioritized instant. This is the instant the results were last updated.
Observation (STU3).extension (CopyIndicator) (also: extension (CopyIndicator)) | Extension | | Yes | 0..1 | Indicates whether this response originates from the local system (false) or if it is from an external source (true). URL: http://nictiz.nl/fhir/StructureDefinition/ext-CopyIndicator
Observation (STU3).extension (historical-fhir-id) (also: extension (historical-fhir-id)) | Extension | | Yes | 0..1 | This extension contains a historical FHIR ID in cases where the historical ID exceeded the 64-character ID length specified in the HL7 FHIR standard. This extension is only returned if the requesting client is configured to use 64-character-limited FHIR IDs as described in the FHIR ID Generation Scheme topic. This element is returned starting in the November 2025 version of Epic and in August 2025 by special update. URL: http://open.epic.com/FHIR/StructureDefinition/extension/historical-fhir-id
Observation (STU3).extension (Lab-e-signature) (also: extension (Lab-e-signature)) | Extension | | Yes | 1..* | The signatures on a component. This is returned only for components with signatures as a string. URL: http://open.epic.com/FHIR/StructureDefinition/extension/lab-e-signature
Observation (STU3).id | id | | | 1..1 | The Observation FHIR ID.
Observation (STU3).interpretation | CodeableConcept | | | 0..1 | The level of abnormality. If the value isn't mapped we will just send the text value in the Codeable concept. Relevant for sensitivities.
Observation (STU3).issued | instant | | | 0..1 | The observation instant.
Observation (STU3).method | CodeableConcept | | | 0..1 | SNOMED code mapped to procedure in Epic.
Observation (STU3).performer | Reference | | Yes | 0..* | References to the Practitioner FHIR ID for the resulting agency director and the Organization FHIR ID for the resulting lab.
Observation (STU3).referenceRange | BackboneElement | | Yes | 0..* | The reference range for the result, if applicable.
Observation (STU3).referenceRange.high | Quantity | | | 0..1 | The high value of the reference range.
Observation (STU3).referenceRange.low | Quantity | | | 0..1 | The low value of the reference range.
Observation (STU3).referenceRange.text | string | | | 0..1 | Display version of reference range.
Observation (STU3).related | BackboneElement | | Yes | 0..* | Resource related to this observation.
Observation (STU3).specimen | Reference | | | 0..1 | A reference to the Specimen resource.
Observation (STU3).status | code | | | 1..1 | The status of the lab order. For organizations in the Netherlands, this element can also include a TestResultStatus extension that indicates the test result status.
Observation (STU3).subject | Reference | | | 1..1 | The Patient FHIR ID linked to the order.
Observation (STU3).valueCodeableConcept | CodeableConcept | | | 0..1 | This element is populated only if a SNOMED code is associated with the component.
Observation (STU3).valueQuantity | Quantity | | | 0..1 | The result value if the result is numeric and represents a quantity.
Observation (STU3).valueRange | Range | | | 0..1 | The result value if the result is structured numeric and the operator is "-".
Observation (STU3).valueRatio | Ratio | | | 0..1 | The result value if the result is structured numeric and the operator is ":".
Observation (STU3).valueString | string | | | 0..1 | The result value if the result is a string.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Observation?patient=e3E2Rkizy.WmzItzk1PbZAQ3&category=laboratory&_count=1
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "next",
      "url": "https://hostname/instance/api/FHIR/STU3/Observation?patient=e3E2Rkizy.WmzItzk1PbZAQ3&category=laboratory&_
    count=1&sessionID=08-57AED5E4149711ECBEA4D2068206269D"
    }
  ],
  "entry": [
    {
      "link": [
```

```
    {
      "relation": "self",
      "url":
"https://hostname/instance/api/FHIR/STU3/Observation/eAMXBdyxr2HSrK7BSIJoPna14R45Ho9Cwgyy5qfxFTqJraA5OWLQFNaXFUhKX.y
TF3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/STU3/Observation/eAMXBdyxr2HSrK7BSIJoPna14R45Ho9Cwgyy5qfxFTqJraA5OW
LQFNaXFUhKX.yTF3",
  "resource": {
    "resourceType": "Observation",
    "id": "eAMXBdyxr2HSrK7BSIJoPna14R45Ho9Cwgyy5qfxFTqJraA5OWLQFNaXFUhKX.yTF3",
    "basedOn": [
      {
        "reference":
"https://hostname/instance/api/FHIR/STU3/ProcedureRequest/e88uk8v7SSbyNXOiSR9-moxFQZin7nVZSFmyIN-HJDho3"
      }
    ],
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "252275004",
            "display": "Hematology test (procedure)"
          }
        ],
        "text": "Hematology test (procedure)"
      },
      {
        "coding": [
          {
            "system": "http://hl7.org/fhir/observation-category",
            "code": "laboratory",
            "display": "Laboratory"
          }
        ],
        "text": "Laboratory"
      },
      {
        "coding": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.10.798268.30",
            "code": "Lab",
            "display": "Lab"
          }
        ],
        "text": "Lab"
      }
    ],
    "code": {
      "coding": [
        {
          "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.5.737384.188",
          "code": "68"
        },
        {
          "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.5.737384.431",
          "code": "AF-GLUKOOSI"
        },
        {
          "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.5.737384.430",
          "code": "1459"
        },
        {
          "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.768282",
          "code": "68",
          "display": "GLUCOSE"
        }
      ],
      "text": "GLUCOSE"
    },
    "subject": {
      "reference": "https://hostname/instance/api/FHIR/STU3/Patient/e3E2Rkizy.WmzItzk1PbZAQ3",
      "display": "Marshall, Fhir One"
    },
    "context": {
      "reference": "https://hostname/instance/api/FHIR/STU3/Encounter/eorXVhQrtx.V6AYpjcoaYBMEpqd5Y8s7Eq8Mr4Ltn.2U3"
    },
    "effectiveDateTime": "2021-09-02T18:41:00Z",
    "issued": "2021-09-02T19:10:25Z",
    "performer": [
      {
        "reference": "https://hostname/instance/api/FHIR/STU3/Practitioner/ecw01Sml5VFYLaE2Mp3-hgQ3",
        "display": "Lab Admin Tpr"
      }
    ],
    "valueQuantity": {
```

```
          "value": 95,
          "unit": "mmol/L",
          "system": "http://unitsofmeasure.org",
          "code": "mmol/L"
        },
        "specimen": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Specimen/eiWBs9u0Cb1fNJ5IgL3cxqgWFghRXa9VQggFtlJ3i2303"
        },
        "referenceRange": [
          {
            "low": {
              "value": 80,
              "unit": "mmol/L",
              "system": "http://unitsofmeasure.org",
              "code": "mmol/L"
            },
            "high": {
              "value": 110,
              "unit": "mmol/L",
              "system": "http://unitsofmeasure.org",
              "code": "mmol/L"
            },
            "text": "80 - 110 mmol/L"
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


Change log

Epic Version | Description | Type
November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4
ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90
ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94
ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

9.2.16 Observation.Search (LDA-W) (STU3)

File: site/apis/09-Clinical-Diagnostics/2-STU3/899-Observation-Search-LDA-W-STU3.html
Epic API ID | 899
FHIR version | STU3
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Keywords | Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2018
HTTP method | GET
URL template | api/FHIR/STU3/Observation?patient={patient}&subject={subject}&code={code}&category={category}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=899

Description

The FHIR Observation resource defines measurements and assertions about a patient, including LDA-W (lines, drains, airways, and wound) data.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | Either this or code must be provided. | | | If provided, always set to "LDA".
code | token | Either this element or category must be provided, but not both. | | | LOINC code, CADSR code, or encoded flowsheet ID. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.

patient | reference | Either patient or subject must be provided. If both are provided, they must reference the same patient. | | | The reference to a patient resource.
subject | reference | Either patient or subject must be provided. If both are provided, they must reference the same patient. | | | The reference to a patient resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Observation | Observation | False | | | A single Observation FHIR resource.
Bundle.entry.resource.Observation.category | CodeableConcept | False | Yes | | The category is always "LDA". This element can't be modified by Observation.Update.
Bundle.entry.resource.Observation.code | CodeableConcept | False | | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. This element can't be modified by Observation.Update. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.
Bundle.entry.resource.Observation.component | BackboneElement | True | Yes
Bundle.entry.resource.Observation.component.code | CodeableConcept | False | | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. This element can't be modified by Observation.Update. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.
Bundle.entry.resource.Observation.component.valueQuantity | Quantity | False | | | Value of the property row.
Bundle.entry.resource.Observation.context | Reference | False | | | Reference to a patient resource. This element can't be modified by Observation.Update.
Bundle.entry.resource.Observation.effectivePeriod | Period | True | | | Start instant and/or end instant of the LDA.
Bundle.entry.resource.Observation.status | code | False | | | The status is always "final".
Bundle.entry.resource.Observation.subject | Reference | False | | | Reference to a patient resource. This element can't be modified by Observation.Update.
Bundle.entry.resource.Observation.valueString | string | False | | | LDA display name.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Observation?patient=ej4nFxzySJAysjMuv4rTBNA3&category=lda
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="2"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/STU3/Observation?patient=ej4nFxzySJAysjMuv4rTBNA3&amp;category=lda"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/STU3/Observation/ebhUU2Y8Wnz8PEsBIRUDfBw3"/>
    </link>
    <fullUrl value="https://hostname/instance/api/FHIR/STU3/Observation/ebhUU2Y8Wnz8PEsBIRUDfBw3"/>
    <resource>
      <Observation>
        <id value="ebhUU2Y8Wnz8PEsBIRUDfBw3"/>
        <status value="final"/>
        <category>
          <coding>
            <system value="http://open.epic.com/FHIR/STU3/StructureDefinition/observation-category"/>
            <code value="LDA"/>
            <display value="Line, Drains, and Airways"/>
          </coding>
          <text value="Line, Drains, and Airways"/>
        </category>
        <code>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
            <code value="42674"/>
            <display value="HZL Wound LDA"/>
          </coding>
          <text value="HZL Wound LDA"/>
        </code>
        <subject>
          <reference value="https://hostname/instance/api/FHIR/STU3/Patient/ej4nFxzySJAysjMuv4rTBNA3"/>
          <display value="Meiko Lufhir"/>
        </subject>
        <context>
          <reference
    value="https://hostname/instance/api/FHIR/STU3/Encounter/eKzsYGMB9rxAHlrtpROR1iBa2CWjmdcgpdgv8cDE4ZCQ3"/>
        </context>
        <effectivePeriod>
          <start value="2018-10-18T18:12:00Z"/>
        </effectivePeriod>
        <valueString value="HZL Wound LDA Meiko's wound Left Ankle"/>
```

```
      <component>
        <code>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
            <code value="700"/>
            <display value="Placement Date"/>
          </coding>
          <text value="Placement Date"/>
        </code>
        <valueDateTime value="2018-10-18"/>
      </component>
      <component>
        <code>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
            <code value="701"/>
            <display value="Placement Time"/>
          </coding>
          <text value="Placement Time"/>
        </code>
        <valueTime value="13:12:00"/>
      </component>
      <component>
        <code>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
            <code value="4570"/>
            <display value="FLOQA Region Property Row"/>
          </coding>
          <text value="FLOQA Region Property Row"/>
        </code>
        <valueCodeableConcept>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.861.1.7.4.696784.98010"/>
            <code value="Ankle"/>
            <display value="Ankle"/>
          </coding>
          <text value="Ankle"/>
        </valueCodeableConcept>
      </component>
      <component>
        <code>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
            <code value="4572"/>
            <display value="FLOQA Left/Right Property Row"/>
          </coding>
          <text value="FLOQA Left/Right Property Row"/>
        </code>
        <valueCodeableConcept>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.861.1.7.4.696784.98011"/>
            <code value="L"/>
            <display value="Left"/>
          </coding>
          <text value="Left"/>
        </valueCodeableConcept>
      </component>
      <component>
        <code>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
            <code value="4577"/>
            <display value="FLOQA String Property Row"/>
          </coding>
          <text value="FLOQA String Property Row"/>
        </code>
        <valueString value="Meiko's wound"/>
      </component>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/STU3/Observation/eI62g4JblrChJAFb3Op3OrQ3"/>
  </link>
  <fullUrl value="https://hostname/instance/api/FHIR/STU3/Observation/eI62g4JblrChJAFb3Op3OrQ3"/>
  <resource>
    <Observation>
      <id value="eI62g4JblrChJAFb3Op3OrQ3"/>
      <status value="final"/>
      <category>
        <coding>
          <system value="http://open.epic.com/FHIR/STU3/StructureDefinition/observation-category"/>
          <code value="LDA"/>
```

```
        <display value="Line, Drains, and Airways"/>
      </coding>
      <text value="Line, Drains, and Airways"/>
    </category>
    <code>
      <coding>
        <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
        <code value="911396000"/>
        <display value="Wound"/>
      </coding>
      <text value="Wound"/>
    </code>
    <subject>
      <reference value="https://hostname/instance/api/FHIR/STU3/Patient/ej4nFxzySJAysjMuv4rTBNA3"/>
      <display value="Meiko Lufhir"/>
    </subject>
    <context>
      <reference
value="https://hostname/instance/api/FHIR/STU3/Encounter/eKzsYGMB9rxAHlrtpROR1iBa2CWjmdcgpdgv8cDE4ZCQ3"/>
    </context>
    <effectivePeriod>
      <start value="2018-10-29T17:59:00Z"/>
    </effectivePeriod>
    <valueString value="Wound 10/29/18 Abrasion Foot Left;Posterior"/>
    <component>
      <code>
        <coding>
          <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
          <code value="700"/>
          <display value="Placement Date"/>
        </coding>
        <text value="Placement Date"/>
      </code>
      <valueDateTime value="2018-10-29"/>
    </component>
    <component>
      <code>
        <coding>
          <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
          <code value="701"/>
          <display value="Placement Time"/>
        </coding>
        <text value="Placement Time"/>
      </code>
      <valueTime value="12:59:00"/>
    </component>
    <component>
      <code>
        <coding>
          <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
          <code value="911303740"/>
          <display value="Wound Location Orientation"/>
        </coding>
        <text value="Wound Location Orientation"/>
      </code>
      <valueCodeableConcept>
        <coding>
          <code value="Left;Posterior"/>
          <display value="Left;Posterior"/>
        </coding>
        <text value="Left;Posterior"/>
      </valueCodeableConcept>
    </component>
    <component>
      <code>
        <coding>
          <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
          <code value="911396060"/>
          <display value="Location"/>
        </coding>
        <text value="Location"/>
      </code>
      <valueCodeableConcept>
        <coding>
          <code value="Foot"/>
          <display value="Foot"/>
        </coding>
        <text value="Foot"/>
      </valueCodeableConcept>
    </component>
    <component>
      <code>
        <coding>
          <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
          <code value="911400634"/>
          <display value="Wound Type"/>
        </coding>
        <text value="Wound Type"/>
      </code>
```

```
          <valueCodeableConcept>
            <coding>
              <code value="Abrasion"/>
              <display value="Abrasion"/>
            </coding>
            <text value="Abrasion"/>
          </valueCodeableConcept>
        </component>
      </Observation>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
</Bundle>
```

Change log

Epic Version | Description | Type
August 2025 (May 2025, February 2025, November 2024 by special update) | Starting in August 2024, May 2025 with special update E11400862, February 2025 with special update E11306699, and November 2024 with special update E11211328, multi-select fields are handled separately, and if a SNOMED concept mapping exists for any individual selection, the SNOMED concept mapping is included in the FHIR response. Previously, when a user selected multiple options, the system treated the entire list as a single combined value. | Substantive Changes

9.2.17 Observation.Search (Vitals) (STU3)

File: site/apis/09-Clinical-Diagnostics/2-STU3/498-Observation-Search-Vitals-STU3.html
Epic API ID | 498
FHIR version | STU3
Categories | Clinical Diagnostics
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | BMI percentile per age and sex for youth 2-20;Weights for age per length and sex;Occipital-frontal circumference for children < 3 years old;Vital Signs;Diastolic blood pressure;Systolic blood pressure;Body height;Body weight;Heart rate;Respiratory rate;Body temperature;Pulse oximetry;Inhaled oxygen concentration;Diagnostic Reports;Imaging;Laboratory;Pathology;Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2018
HTTP method | GET
URL template | api/FHIR/STU3/Observation?patient={patient}&subject={subject}&code={code}&date={date}&category={category}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=498

Description

This web service will retrieve vital sign data from a patient's chart, as well as any other non-duplicable data found in the patient's flowsheets across all encounters.

Remarks

To see an example of how the component.code element is structured for patient-entered flowsheet data, refer to the sample request section of the Observation.Create (Vitals) (STU3) API entry.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | token | False | | | Use "vital-signs" to search for vitals observations. Either this element or code must be provided, but not both.
code | token | False | | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.
date | date | True | | | The date range for when the observation was taken.
patient | reference | False | | | Reference to a patient resource the observation is about. Either this element or subject must be provided. If both are provided, they must match.
subject | reference | False | | | Reference to a patient resource the observation is about. Either this element or patient must be provided. If both are provided, they must match.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Observation | Observation | False | | | A single Observation FHIR resource.

Bundle.entry.resource.Observation.bodySite | CodeableConcept | False | | | Applicable only to organizations in the Netherlands starting in Epic version August 2021. The site on the body where the observation was made. Used for the location where a blood pressure reading was obtained.

Bundle.entry.resource.Observation.category | CodeableConcept | False | Yes | | This element always returns "vital-signs". Starting in November 2022 in Netherlands locales, this element also returns SNOMED code 49581000146104 (Laboratory test finding (finding) when returning observations linked to the blood glucose profile.

Bundle.entry.resource.Observation.code | CodeableConcept | False | | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.

Bundle.entry.resource.Observation.comment | string | True | | | Comments about the observation value.

Bundle.entry.resource.Observation.component | BackboneElement | True | Yes

Bundle.entry.resource.Observation.component.code | CodeableConcept | True | | | This element is used for observations filed by patients to patient-entered flowsheets. Metadata related to the observation. Represents strict metadata if the coding sub-element is populated. The following strict metadata are supported: 1-UUID The unique reading ID. 2-Blood Glucose Meal Times 1-Before meal 2-After meal 3-Insulin Delivery Reason 1-Basal 2-Bolus 6-Manually Entered 0-False 1-True Represents loose metadata if only the text sub-element is populated. UUID, Blood Glucose Meal times, Insulin Delivery Reason, and Manually Entered must have their value specified in component.valueString. Applicable to organizations in the Netherlands starting in November 2022: LOINC code 74206-4 (Oxygen therapy was given [NTDS]) LOINC code 44969-4 (Heart rate rhythm palpation)

Bundle.entry.resource.Observation.component.valueCodeableConcept | CodeableConcept | True | | | Applicable to organizations in the Netherlands starting in November 2022: Codes from CodeSystem “v2 Yes/no indicator” indicate whether extra oxygen was given at the time of an oxygen saturation reading SNOMED codes from PolsRegelmatigheidCodelijst indicate the regularity of the pulse rate at the time of a pulse reading

Bundle.entry.resource.Observation.dataAbsentReason | CodeableConcept | True | | | Available starting in the August 2021 version of Epic. Indicates that an observation value is missing. Will always be “unknown” for systolic blood pressure values in observations for diastolic-only blood pressure flowsheets, or for diastolic blood pressure value in observations for systolic-only blood pressure flowsheets.

Bundle.entry.resource.Observation.effectiveDateTime | dateTime | False | | | The date and time the observation was taken.

Bundle.entry.resource.Observation.issued | instant | False | | | The date and time the observation was documented. This is always the current time.

Bundle.entry.resource.Observation.meta | Meta | True | | | The document metadata.

Bundle.entry.resource.Observation.meta.profile | String | True | | | Available in Netherlands environments only. This element shows that this API conforms to the profile connected to one of the following canonical urls: http://nictiz.nl/fhir/StructureDefinition/zib-BloodPressure http://nictiz.nl/fhir/StructureDefinition/zib-BodyWeight http://nictiz.nl/fhir/StructureDefinition/zib-BodyHeight http://nictiz.nl/fhir/StructureDefinition/zib-BodyTemperature http://nictiz.nl/fhir/StructureDefinition/zib-OxygenSaturation http://nictiz.nl/fhir/StructureDefinition/zib-HeartRate http://nictiz.nl/fhir/StructureDefinition/zib-PulseRate http://nictiz.nl/fhir/StructureDefinition/vitalsign-bloodglucose

Bundle.entry.resource.Observation.method | CodeableConcept | True | | | Method by which the observation was obtained. Possible values include: "Patient-entered" for patient-entered flowsheet vital signs Applicable only to organizations in the Netherlands starting in August 2021: SNOMED code 10179008 (invasive) for blood pressure readings using an invasive method Applicable only to organizations in the Netherlands starting in August 2021: SNOMED code 22762002 (non-invasive) for blood pressure readings not using an invasive method Starting in November 2022 in Netherlands locales: SNOMED codes from the TemperatuurTypeCodelijst for the specific anatomical location used for body temperature readings Starting in November 2022 in Netherlands locales: SNOMED codes from the HartslagMeetMethodeCodelijst for the specific method used for heart rate readings Starting in November 2022 in Netherlands locales: SNOMED codes from the TestmethodeCodelijst for the specific method used for blood glucose readings

Bundle.entry.resource.Observation.performer | Reference | False | Yes | | Who filed the observation.

Bundle.entry.resource.Observation.referenceRange | BackboneElement | True | Yes

Bundle.entry.resource.Observation.referenceRange.low | Quantity | True | | | Applicable to organizations in the Netherlands starting in November 2022. The lower reference limit for blood glucose readings.

Bundle.entry.resource.Observation.status | code | False | | | The status of the observation. This is set to amended or final depending on whether the field has been corrected.

Bundle.entry.resource.Observation.subject | Reference | False | | | The FHIR ID for the patient the observation is about.

Bundle.entry.resource.Observation.valueCodeableConcept | CodeableConcept | False | | | This element is populated if the data type is category or custom list. Only one of the value* elements is required.

Bundle.entry.resource.Observation.valueDateTime | String | False | | | This element is populated if the data type is date. Only one of the value* elements is required.

Bundle.entry.resource.Observation.valueQuantity | Quantity | False | | | This element is populated if a value has units or the data type is numeric. Only one of the value* elements is required.

Bundle.entry.resource.Observation.valueString | string | False | | | This element is populated if a value doesn't have units or the data type is string. Only one of the value[x] elements is required.

Bundle.entry.resource.Observation.valueTime | String | False | | | This element is populated if the data type is time. Only one of the value* elements is required.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Observation?patient=eMMKGfU7AJS5yjSRNjZdLxA3&category=vital-signs
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="2"/>
  <link>
    <relation value="next"/>
    <url value="https://hostname/instance/api/FHIR/STU3/Observation?patient=eMMKGfU7AJS5yjSRNjZdLxA3&amp;category=vital-
    signs&amp;_count=2&amp;sessionID=13-9E9D08B2FFAC11E78C52460231621F93"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/STU3/Observation/eXtL1gYZT1-
    YF8BWqSpPaWluHLQXjBRTCHvqgL14AuVK5zH7kVB43ACpHtlJSy7siSMMsQ8rLIWHMT940WUO.-w3"/>
    </link>
```

```
  <fullUrl value="https://hostname/instance/api/FHIR/STU3/Observation/eXtL1gYZT1-
  YF8BWqSpPaWluHLQXjBRTCHvqgL14AuVK5zH7kVB43ACpHtlJSy7siSMMsQ8rLIWHMT940WUO.-w3"/>
  <resource>
    <Observation>
      <category>
        <coding>
          <system value="http://hl7.org/fhir/observation-category"/>
          <code value="vital-signs"/>
          <display value="Vital Signs"/>
        </coding>
        <text value="Vital Signs"/>
      </category>
      <code>
        <coding>
          <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
          <code value="5"/>
          <display value="BP"/>
        </coding>
        <coding>
          <system value="http://loinc.org"/>
          <code value="55284-4"/>
          <display value="Blood pressure systolic and diastolic"/>
        </coding>
        <coding>
          <system value="http://loinc.org"/>
          <code value="8716-3"/>
          <display value="Vital signs"/>
        </coding>
        <text value="BP"/>
      </code>
      <subject>
        <reference value="https://hostname/instance/api/FHIR/STU3/Patient/eMMKGfU7AJS5yjSRNjZdLxA3"/>
        <display value="Meiko Lufhir Jr., PhD"/>
      </subject>
      <effectiveDateTime value="2017-11-08T17:00:00Z"/>
      <performer>
        <reference value="https://hostname/instance/api/FHIR/STU3/Practitioner/efv90i4pFs9fOm7QQhHOUHw3"/>
        <display value="Amber Hxl, MD"/>
      </performer>
      <id value="eXtL1gYZT1-YF8BWqSpPaWluHLQXjBRTCHvqgL14AuVK5zH7kVB43ACpHtlJSy7siSMMsQ8rLIWHMT940WUO.-w3"/>
      <status value="final"/>
      <issued value="2017-11-08T20:03:36Z"/>
      <component>
        <code>
          <coding>
            <system value="http://loinc.org"/>
            <code value="8480-6"/>
            <display value="Systolic blood pressure"/>
          </coding>
          <text value="Systolic blood pressure"/>
        </code>
        <valueQuantity>
          <value value="120"/>
          <unit value="mm[Hg]"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="mm[Hg]"/>
        </valueQuantity>
      </component>
      <component>
        <code>
          <coding>
            <system value="http://loinc.org"/>
            <code value="8462-4"/>
            <display value="Systolic blood pressure"/>
          </coding>
          <text value="Diastolic blood pressure"/>
        </code>
        <valueQuantity>
          <value value="90"/>
          <unit value="mm[Hg]"/>
          <system value="http://unitsofmeasure.org"/>
          <code value="mm[Hg]"/>
        </valueQuantity>
      </component>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/STU3/Observation/ejeZM7uM4zxl8gj7D9AqwEvM4HOV.6mxzXCMDzJynl8hNLOMks
  vJObZBaSy4R8WA1SGfaNubTGwUJVmepQDQ3pw3"/>
  </link>
  <fullUrl value="https://hostname/instance/api/FHIR/STU3/Observation/ejeZM7uM4zxl8gj7D9AqwEvM4HOV.6mxzXCMDzJynl8hNLOM
  ksvJObZBaSy4R8WA1SGfaNubTGwUJVmepQDQ3pw3"/>
  <resource>
```

```
      <Observation>
        <category>
          <coding>
            <system value="http://hl7.org/fhir/observation-category"/>
            <code value="vital-signs"/>
            <display value="Vital Signs"/>
          </coding>
          <text value="Vital Signs"/>
        </category>
        <code>
          <coding>
            <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
            <code value="557395"/>
            <display value="HXL Weight"/>
          </coding>
          <coding>
            <system value="http://loinc.org"/>
            <code value="3141-9"/>
            <display value="Body weight Measured"/>
          </coding>
          <coding>
            <system value="http://loinc.org"/>
            <code value="8716-3"/>
            <display value="Vital signs"/>
          </coding>
          <text value="HXL Weight"/>
        </code>
        <subject>
          <reference value="https://hostname/instance/api/FHIR/STU3/Patient/eMMKGfU7AJS5yjSRNjZdLxA3"/>
          <display value="Meiko Lufhir Jr., PhD"/>
        </subject>
        <effectiveDateTime value="2018-01-22T12:30:00Z"/>
        <performer>
          <reference value="https://hostname/instance/api/FHIR/STU3/Patient/eMMKGfU7AJS5yjSRNjZdLxA3"/>
          <display value="Meiko Lufhir Jr., PhD"/>
        </performer>
        <valueQuantity>
          <value value="128"/>
          <unit value="lb"/>
        </valueQuantity>
        <id value="ejeZM7uM4zxl8gj7D9AqwEvM4HOV.6mxzXCMDzJynl8hNLOMksvJObZBaSy4R8WA1SGfaNubTGwUJVmepQDQ3pw3"/>
        <status value="final"/>
        <issued value="2018-01-22T19:39:23Z"/>
        <method>
          <text value="Patient-entered"/>
        </method>
      </Observation>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
</Bundle>
```

Change log

Epic Version | Description | Type
August 2025 (May 2025, February 2025, November 2024 by special update) | Starting in August 2024, May 2025 with special update E11400862, February 2025 with special update E11306699, and November 2024 with special update E11211328, multi-select fields are handled separately, and if a SNOMED concept mapping exists for any individual selection, the SNOMED concept mapping is included in the FHIR response. Previously, when a user selected multiple options, the system treated the entire list as a single combined value. | Substantive Changes

9.2.18 Observation.Update (LDA-W) (STU3)

File: site/apis/09-Clinical-Diagnostics/2-STU3/900-Observation-Update-LDA-W-STU3.html
Epic API ID | 900
FHIR version | STU3
Categories | Clinical Diagnostics
API type | Industry-Standard
USCDI | No
Keywords | Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | May 2019
HTTP method | PUT
URL template | api/FHIR/STU3/Observation
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=900

Description

The FHIR Observation resource defines measurements and assertions about a patient, including LDA-W (lines, drains, airways, and wound) data. Observation.Update allows the client to modify existing LDA-W data in the database, including LDA properties and LDA assessments.

This resource uses an E-Tag and an If-Match HTTP header.

If the update is successful, HTTP status code 200 will be returned. If the update is not successful, an operational outcome will be returned.

Request parameters

Name | Type | Optional | Array | Cardinality | Description

Observation | Observation | False | | | A single Observation FHIR resource.

Observation.category | CodeableConcept | False | Yes | | The category is always "LDA". This element can't be modified by Observation.Update.

Observation.code | CodeableConcept | False | | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. This element can't be modified by Observation.Update. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.

Observation.comment | string | True | | | Comment for the assessement. Limited to 254 characters in the August 2025 version of Epic and later. Limited to 60 characters in earlier versions of Epic.

Observation.component | BackboneElement | True | Yes | | LOINC code, CADSR code, or encoded flowsheet ID. Either this element or category must be provided, but not both. The code element value varies depending upon what is passed (for example, a flowsheet ID vs. a LOINC code). If using a flowsheet ID, the system value is different between Epic organizations, and it is also different between production and non-production environments for the same organization.

Observation.component.extension (also: LDA-Property-Comment) | Extension | False | Yes | | Comment for the property row.

Observation.component.valueQuantity | Quantity | False | | | Value of the property row.

Observation.context | Reference | False | | | Reference to an encounter resource. This element can't be modified by Observation.Update.

Observation.effectiveDateTime | dateTime | False | | | The prioritized instant. This element can't be modified by Observation.Update.

Observation.effectivePeriod | Period | True | | | Start instant and/or end instant of the LDA.

Observation.related | BackboneElement | False | Yes | | This element can't be modified by the Observation.Update.

Observation.related.target | Reference | False | | | The parent LDA observation FHIR ID. This element can't be modified by Observation.Update.

Observation.related.type | code | True

Observation.status | code | True | | | The status of the observation. Only a value of "final" is supported. We do not support filing data that isn't finalized.

Observation.subject | Reference | False | | | Reference to a patient resource. This element can't be modified by Observation.Update.

Observation.valueQuantity | Quantity | False | | | Value of the assessment. For image rows, a comma-delimited list of external DCS IDs is required.

Observation.valueString | string | False | | | LDA display name.

Sample request

```
<Observation>
  <category>
    <coding>
      <system value="http://open.epic.com/FHIR/STU3/StructureDefinition/observation-category"/>
      <code value="LDA"/>
      <display value="Line, Drains, and Airways"/>
    </coding>
    <text value="Line, Drains, and Airways"/>
  </category>
  <code>
    <coding>
      <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
      <code value="42674"/>
      <display value="HZL Wound LDA"/>
    </coding>
    <text value="HZL Wound LDA"/>
  </code>
  <subject>
    <reference value="https://hostname/instance/api/FHIR/STU3/Patient/ew9Okvb9gsZXjzlK7YzqP1w3"/>
    <display value="Haruka Lufhir"/>
  </subject>
  <context>
    <reference value="https://hostname/instance/api/FHIR/STU3/Encounter/e5wRUdV7oHoL8VOeT.CdLnVfN7YsByGgW1GzM51mMDEw3"/>
  </context>
  <component>
    <code>
      <coding>
        <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
        <code value="700"/>
        <display value="Placement Date"/>
      </coding>
      <text value="Placement Date"/>
    </code>
    <valueDateTime value="2018-10-29"/>
  </component>
  <component>
    <code>
      <coding>
        <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
        <code value="701"/>
        <display value="Placement Time"/>
      </coding>
      <text value="Placement Time"/>
```

```
    </code>
    <valueTime value="08:00:00"/>
  </component>
  <component>
    <code>
      <coding>
        <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
        <code value="4572"/>
        <display value="FLOQA Left/Right Property Row"/>
      </coding>
      <text value="FLOQA Left/Right Property Row"/>
    </code>
    <valueCodeableConcept>
      <coding>
        <system value="urn:oid:1.2.840.114350.1.13.861.1.7.4.696784.98011"/>
        <code value="L"/>
        <display value="Left"/>
      </coding>
      <text value="Left"/>
    </valueCodeableConcept>
  </component>
  <component>
    <code>
      <coding>
        <system value="urn:oid:1.2.840.114350.1.13.861.1.7.2.707679"/>
        <code value="4577"/>
        <display value="FLOQA String Property Row"/>
      </coding>
      <text value="FLOQA String Property Row"/>
    </code>
    <valueString value="Haruka's painful wound"/>
  </component>
</Observation>
```

Sample response

```
HTTP/1.1 200 OK
```

Change log

Epic Version | Description | Type

November 2022 | Epic has updated the behavior of certain modifier elements in FHIR write APIs to ensure the data in the incoming request body matches the data filed to the database and to comply with the HL7 FHIR specification. Previously, the modifier element was ignored in certain situations, and an undocumented default value was saved in its place. Now, if the modifier element is included and is set to an unaccepted value, a fatal error logs, preventing the request from filing to the database. The FHIR request returns an OperationOutcome with the invalid element and what value is accepted for that element. Element Accepted Incoming Value Default Value status final final | Substantive change

Element | Accepted Incoming Value | Default Value

status | final | final

9.3 Clinical Diagnostics: DSTU2

File: site/apis/09-Clinical-Diagnostics/3-DSTU2/index.html

* DiagnosticReport.Read (Results) (DSTU2)
* DiagnosticReport.Search (Results) (DSTU2)
* Observation.Read (Labs) (DSTU2)
* Observation.Read (Vitals) (DSTU2)
* Observation.Search (Labs) (DSTU2)
* Observation.Search (Vitals) (DSTU2)

9.3.1 DiagnosticReport.Read (Results) (DSTU2)

File: site/apis/09-Clinical-Diagnostics/3-DSTU2/10-DiagnosticReport-Read-Results-DSTU2.html

Epic API ID | 10

FHIR version | DSTU2

Categories | Clinical Diagnostics

API type | USCDI

A la carte | Industry-Standard Level 1 API

USCDI | Yes

Keywords | Laboratory;Tests;Values;Results

Available in inpatient and other contexts | Yes

ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base

ONC certification versions | Epic 2017

HTTP method | GET

URL template | api/FHIR/DSTU2/DiagnosticReport/{ID}

Supports OAuth 2.0 | True

Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients

Performs scope validation | False

Source | https://fhir.epic.com/Specifications?api=10

Description

The DiagnosticReport FHIR resource represents resulted diagnostic orders and relevant data associated with the order and the patient. Supported diagnostic services include laboratory, radiology, and cardiology. If you are interested in querying for individual components from laboratory tests and text-based content, such as narratives or impressions, from radiology and cardiology results, these are returned in DiagnosticReport STU3 and R4 rather than DiagnosticReport DSTU2.

The DiagnosticReport resource corresponds to the following Common Clinical Data Set (CCDS) Elements: Laboratory Tests.

Note:

Remarks

Text based result content, such as narrative or impressions, will be represented as related Observations rather than in the text element inherited by DomainResource. If the results from a Search return medication orders, an error will be returned for those orders.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The DiagnosticReport FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
DiagnosticReport | DiagnosticReport | True | | | An instance of the DSTU2 DiagnosticReport resource.
DiagnosticReport.category | CodeableConcept | False | | | The service category.
DiagnosticReport.code | CodeableConcept | False | | | The Name/Code for this diagnostic report.
DiagnosticReport.effectiveDateTime | DateTime | True | | | Clinically relevant time/time period for report.
DiagnosticReport.identifier | Identifier | False | Yes | | The business identifier for the report.
DiagnosticReport.issued | DateTime | True | | | DateTime this version was released.
DiagnosticReport.performer | ReferencedResource | False | | | Participants in producing the report. This is always a reference to the Practitioner resource.
DiagnosticReport.result | ReferencedResource | False | Yes | | Observations, simple or complex nested groups.
DiagnosticReport.status | String | False | | | The status of the report. You can find the list of values here: http://hl7.org/fhir/diagnostic-report-status.
DiagnosticReport.subject | ReferencedResource | False | | | The subject of the report.

Sample request

```
https://hostname/instance/api/FHIR/DSTU2/DiagnosticReport/ToZrehgqN3LTiFeKDB7yFJgB
```

Sample response

```
<DiagnosticReport xmlns="http://hl7.org/fhir">
  <status value="Final result"/>
  <effectiveDateTime value="2016-02-16T17:30:00"/>
  <issued value="2016-02-16T19:35:00"/>
  <id value="ToZrehgqN3LTiFeKDB7yFJgB"/>
  <identifier>
    <use value="official"/>
    <system value="Epic"/>
    <value value="991774"/>
    <type>
      <text value="Placer Identifier"/>
      <coding>
        <system value="http://hl7.org/fhir/ValueSet/identifier-type"/>
        <code value="PLAC"/>
        <display value="Placer Identifier"/>
      </coding>
    </type>
  </identifier>
  <code>
    <text value="AMYLASE - Final result"/>
    <coding>
      <system value="http://loinc.org"/>
      <code value="1798-8"/>
      <display value="Amylase [Enzymatic activity/volume] in Serum or Plasma"/>
    </coding>
  </code>
  <subject>
    <display value="Fiona Hailey Ingrid Robertson"/>
    <reference value="https://hostname/instance/api/FHIR/DSTU2/Patient/T9tA6WZi2ctFj0t10bq679Z84UsQZI6rpZh1RjWOsuAYB"/>
  </subject>
  <performer>
    <display value="Helen Claire Wilson, MD"/>
    <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/TgEfSWOatltrvRWYMDUWo8AB"/>
  </performer>
  <result>
    <display value="Component: AMYLASE [U/L] IN SER/PLAS"/>
    <reference value="https://hostname/instance/api/FHIR/DSTU2/Observation/TkDOY1zCTUF-
```

```
    fwD9Nek3Vw92qgkvQaqp.wwvK842G7jIB"/>
  </result>
</DiagnosticReport>
```

9.3.2 DiagnosticReport.Search (Results) (DSTU2)

File: site/apis/09-Clinical-Diagnostics/3-DSTU2/11-DiagnosticReport-Search-Results-DSTU2.html
Epic API ID | 11
FHIR version | DSTU2
Categories | Clinical Diagnostics
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | Laboratory;Tests;Values;Results
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2017
HTTP method | GET
URL template | api/FHIR/DSTU2/DiagnosticReport?patient={patient}&subject={subject}&code={code}&date={date}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=11

Description

The DiagnosticReport FHIR resource represents resulted diagnostic orders and relevant data associated with the order and the patient. Supported diagnostic services include laboratory, radiology, and cardiology. If you are interested in querying for individual components from laboratory tests and text-based content, such as narratives or impressions, from radiology and cardiology results, these are returned in DiagnosticReport STU3 and R4 rather than DiagnosticReport DSTU2.

The DiagnosticReport resource corresponds to the following Common Clinical Data Set (CCDS) Elements: Laboratory Tests.

Note:

Remarks

Text based result content, such as narratives, impressions, or addendums, will be represented as related Observations rather than in the text element inherited by DomainResource. If the results from a Search return medication orders, an error will be returned for those orders.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
code | String | True | | | LOINC code of procedure(s).
date | String | True | | | Date when obtained.
patient | String | False | | | The patient FHIR ID.
subject | String | False | | | The patient FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | True | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single DiagnosticReport returned from the search.
Bundle.entry.resource | Resource | True | | | A single resource returned as part of this bundle.
Bundle.entry.resource.DiagnosticReport | DiagnosticReport | True | | | An instance of the DSTU2 DiagnosticReport resource.
Bundle.entry.resource.DiagnosticReport.code | CodeableConcept | True | | | LOINC code for this report.
Bundle.entry.resource.DiagnosticReport.effectiveDateTime | DateTime | False | | | User prioritized instant.
Bundle.entry.resource.DiagnosticReport.identifier | Identifier | True | Yes | | Identifiers used to refer to this report.
Bundle.entry.resource.DiagnosticReport.identifier.type | CodeableConcept | True | | | Will be set to "PLAC" for placer identifier and "FILL" for filler identifier.
Bundle.entry.resource.DiagnosticReport.identifier.use | String | False | | | Signifies the use of the report. Will be set to "official" or "usual.
Bundle.entry.resource.DiagnosticReport.identifier.value | String | True | | | Value of the identifier.
Bundle.entry.resource.DiagnosticReport.issued | DateTime | False | | | Date when diagnostic report was released (result contact date/time).
Bundle.entry.resource.DiagnosticReport.performer | ReferencedResource | False | | | The diagnostic service that is responsible for issuing the report.
Bundle.entry.resource.DiagnosticReport.result | ReferencedResource | False | Yes | | All observations related to this diagnostic report.
Bundle.entry.resource.DiagnosticReport.status | String | False | | | The status of the report. You can find the list of values here: http://hl7.org/fhir/diagnostic-report-status.
Bundle.entry.resource.DiagnosticReport.subject | ReferencedResource | False | | | The patient.

Sample request

```
https://hostname/instance/api/FHIR/DSTU2/DiagnosticReport?patient=T9tA6WZi2ctFj0t10bq679Z84UsQZI6rpZh1RjWOsuAYB
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="5"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/DiagnosticReport?patient=T9tA6WZi2ctFj0t10bq679Z84UsQZI6rpZh1Rj
    WOsuAYB"/>
  </link>
  <entry>
    <fullUrl value="https://hostname/instance/api/FHIR/DSTU2/DiagnosticReport/T8O0O6RlSva82VyURBzePmgB"/>
    <link>
      <relation value="self"/>
      <url value="https://hostname/instance/api/FHIR/DSTU2/DiagnosticReport/T8O0O6RlSva82VyURBzePmgB"/>
    </link>
    <resource>
      <DiagnosticReport>
        <status value="Final result"/>
        <effectiveDateTime value="2016-01-06T15:15:00"/>
        <issued value="2016-01-06T21:45:00"/>
        <id value="T8O0O6RlSva82VyURBzePmgB"/>
        <identifier>
          <use value="official"/>
          <system value="Epic"/>
          <value value="991775"/>
          <type>
            <text value="Placer Identifier"/>
            <coding>
              <system value="http://hl7.org/fhir/ValueSet/identifier-type"/>
              <code value="PLAC"/>
              <display value="Placer Identifier"/>
            </coding>
          </type>
        </identifier>
        <code>
          <text value="BASIC METABOLIC PANEL - Final result"/>
          <coding>
            <system value="http://loinc.org"/>
            <code value="70219-1"/>
            <display value="Basic metabolic 2008 panel with ionized calcium - Serum or Plasma"/>
          </coding>
        </code>
        <subject>
          <display value="Fiona Hailey Ingrid Robertson"/>
          <reference
    value="https://hostname/instance/api/FHIR/DSTU2/Patient/T9tA6WZi2ctFj0t10bq679Z84UsQZI6rpZh1RjWOsuAYB"/>
        </subject>
        <performer>
          <display value="Helen Claire Wilson, MD"/>
          <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/TgEfSWOatltrvRWYMDUWo8AB"/>
        </performer>
        <result>
          <display value="Component: SODIUM [MMOL/L] IN SER/PLAS"/>
          <reference
    value="https://hostname/instance/api/FHIR/DSTU2/Observation/TmE5Cs5-FJWIWP8XzMKHDeAdh1BGisFUTqpElz3vP3BUB"/>
        </result>
        <result>
          <display value="Component: POTASSIUM [MMOL/L] IN SER/PLAS"/>
          <reference
    value="https://hostname/instance/api/FHIR/DSTU2/Observation/TmE5Cs5-FJWIWP8XzMKHDeB3.nSiF.WaGBlZGM3yV-48B"/>
        </result>
        <result>
          <display value="Component: CHLORIDE [MMOL/L] IN SER/PLAS"/>
          <reference
    value="https://hostname/instance/api/FHIR/DSTU2/Observation/TmE5Cs5-FJWIWP8XzMKHDeEhtqKJAq8ZjfKGSdBLPF9wB"/>
        </result>
        <result>
          <display value="Component: CARBON DIOXIDE, TOTAL [MMOL/L] IN SER/PLAS"/>
          <reference
    value="https://hostname/instance/api/FHIR/DSTU2/Observation/TmE5Cs5-FJWIWP8XzMKHDeN0UfkD3sIwHhZEj7-qoDa8B"/>
        </result>
        <result>
          <display value="Component: UREA NITROGEN [MG/DL] IN SER/PLAS"/>
          <reference
    value="https://hostname/instance/api/FHIR/DSTU2/Observation/TmE5Cs5-FJWIWP8XzMKHDeHFZZvgWViT1SfaRpQXQyLwB"/>
        </result>
        <result>
          <display value="Component: CREATININE [MG/DL] IN SER/PLAS"/>
          <reference
    value="https://hostname/instance/api/FHIR/DSTU2/Observation/TmE5Cs5-FJWIWP8XzMKHDeNzlcwoPfIGPJvq6ZvJOKHIB"/>
        </result>
        <result>
          <display value="Component: GLUCOSE [MG/DL] IN SER/PLAS"/>
          <reference
    value="https://hostname/instance/api/FHIR/DSTU2/Observation/TmE5Cs5-FJWIWP8XzMKHDeHut1WCfqMBNdm0i210kBTsB"/>
        </result>
        <result>
          <display value="Component: CALCIUM [MG/DL] IN SER/PLAS"/>
          <reference
    value="https://hostname/instance/api/FHIR/DSTU2/Observation/TmE5Cs5-FJWIWP8XzMKHDeNh40NgXPwrq5rJO-E8OG34B"/>
```

```
      </result>
      <result>
        <display value="Component: ANION GAP 3 IN SER/PLAS"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Observation/TmE5Cs5-FJWIWP8XzMKHDeE5k.sQJ0CKr9XCNMH6Y94cB"/>
      </result>
      <result>
        <display value="Component: GLOMERULAR FILTRATION RATE ML/MIN/1.73 SQ M.PREDICTED"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Observation/TmE5Cs5-FJWIWP8XzMKHDeJMyrKFbaxtDmiG8N4cFM7UB"/>
      </result>
    </DiagnosticReport>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <fullUrl value="https://hostname/instance/api/FHIR/DSTU2/DiagnosticReport/T.WjBOaU4G3BcwyZSxPkG4gB"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/DiagnosticReport/T.WjBOaU4G3BcwyZSxPkG4gB"/>
  </link>
  <resource>
    <DiagnosticReport>
      <status value="Final result"/>
      <effectiveDateTime value="2016-02-16T14:15:00"/>
      <issued value="2016-02-16T21:56:00"/>
      <id value="T.WjBOaU4G3BcwyZSxPkG4gB"/>
      <identifier>
        <use value="official"/>
        <system value="Epic"/>
        <value value="991776"/>
        <type>
          <text value="Placer Identifier"/>
          <coding>
            <system value="http://hl7.org/fhir/ValueSet/identifier-type"/>
            <code value="PLAC"/>
            <display value="Placer Identifier"/>
          </coding>
        </type>
      </identifier>
      <code>
        <text value="BASIC METABOLIC PANEL - Final result"/>
        <coding>
          <system value="http://loinc.org"/>
          <code value="70219-1"/>
          <display value="Basic metabolic 2008 panel with ionized calcium - Serum or Plasma"/>
        </coding>
      </code>
      <subject>
        <display value="Fiona Hailey Ingrid Robertson"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/T9tA6WZi2ctFj0t10bq679Z84UsQZI6rpZh1RjWOsuAYB"/>
      </subject>
      <performer>
        <display value="Helen Claire Wilson, MD"/>
        <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/TgEfSWOatltrvRWYMDUWo8AB"/>
      </performer>
      <result>
        <display value="Component: SODIUM [MMOL/L] IN SER/PLAS"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Observation/Txo93A2q4nARSXpYmJke011td32xEmOKxeq6sD6at6CcB"/>
      </result>
      <result>
        <display value="Component: POTASSIUM [MMOL/L] IN SER/PLAS"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Observation/Txo93A2q4nARSXpYmJke01-juD3gQbzjul5Slbd7pBEEB"/>
      </result>
      <result>
        <display value="Component: CHLORIDE [MMOL/L] IN SER/PLAS"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Observation/Txo93A2q4nARSXpYmJke015hKxt9KLeA5KPIUzdoK2mgB"/>
      </result>
      <result>
        <display value="Component: CARBON DIOXIDE, TOTAL [MMOL/L] IN SER/PLAS"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Observation/Txo93A2q4nARSXpYmJke01.cIB8y7Cfh22P5Fn4ut4VgB"/>
      </result>
      <result>
        <display value="Component: UREA NITROGEN [MG/DL] IN SER/PLAS"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Observation/Txo93A2q4nARSXpYmJke0144C0qUy4dXdZAtdsZh4ryEB"/>
      </result>
      <result>
        <display value="Component: CREATININE [MG/DL] IN SER/PLAS"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Observation/Txo93A2q4nARSXpYmJke012VJbxU9K6MErslhHc2YAjQB"/>
      </result>
```

```
      <result>
        <display value="Component: GLUCOSE [MG/DL] IN SER/PLAS"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Observation/Txo93A2q4nARSXpYmJke01wmK3bmv.CBxIs6wmhGDTt0B"/>
      </result>
      <result>
        <display value="Component: CALCIUM [MG/DL] IN SER/PLAS"/>
        <reference value="https://hostname/instance/api/FHIR/DSTU2/Observation/Txo93A2q4nARSXpYmJke012hRX6s.2iGfqcU-
  bCjQ29wB"/>
      </result>
      <result>
        <display value="Component: ANION GAP 3 IN SER/PLAS"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Observation/Txo93A2q4nARSXpYmJke019aQG9b0cR1ArTVmSmu6wt8B"/>
      </result>
      <result>
        <display value="Component: GLOMERULAR FILTRATION RATE ML/MIN/1.73 SQ M.PREDICTED"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Observation/Txo93A2q4nARSXpYmJke017ejwBgJzhk9e9eDBht-gDgB"/>
      </result>
    </DiagnosticReport>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <fullUrl value="https://hostname/instance/api/FHIR/DSTU2/DiagnosticReport/Twv82wknQ25J-YYKLIXMSxAB"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/DiagnosticReport/Twv82wknQ25J-YYKLIXMSxAB"/>
  </link>
  <resource>
    <DiagnosticReport>
      <status value="Final result"/>
      <effectiveDateTime value="2016-02-03T14:15:00"/>
      <issued value="2016-02-03T17:20:00"/>
      <id value="Twv82wknQ25J-YYKLIXMSxAB"/>
      <identifier>
        <use value="official"/>
        <system value="Epic"/>
        <value value="991772"/>
        <type>
          <text value="Placer Identifier"/>
          <coding>
            <system value="http://hl7.org/fhir/ValueSet/identifier-type"/>
            <code value="PLAC"/>
            <display value="Placer Identifier"/>
          </coding>
        </type>
      </identifier>
      <code>
        <text value="AMYLASE - Final result"/>
        <coding>
          <system value="http://loinc.org"/>
          <code value="1798-8"/>
          <display value="Amylase [Enzymatic activity/volume] in Serum or Plasma"/>
        </coding>
      </code>
      <subject>
        <display value="Fiona Hailey Ingrid Robertson"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/T9tA6WZi2ctFj0t10bq679Z84UsQZI6rpZh1RjWOsuAYB"/>
      </subject>
      <performer>
        <display value="Helen Claire Wilson, MD"/>
        <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/TgEfSWOatltrvRWYMDUWo8AB"/>
      </performer>
      <result>
        <display value="Component: AMYLASE [U/L] IN SER/PLAS"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Observation/T9Anxpw7ouctvl65w0z7uLykGta0R.IbFLtsUurHajo0B"/>
      </result>
    </DiagnosticReport>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <fullUrl value="https://hostname/instance/api/FHIR/DSTU2/DiagnosticReport/TV3h8wDr6Mnnd89wDSQiX1AB"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/DiagnosticReport/TV3h8wDr6Mnnd89wDSQiX1AB"/>
  </link>
  <resource>
    <DiagnosticReport>
      <status value="Final result"/>
      <effectiveDateTime value="2016-02-05T17:45:00"/>
```

```
      <issued value="2016-02-05T21:45:00"/>
      <id value="TV3h8wDr6Mnnd89wDSQiX1AB"/>
      <identifier>
        <use value="official"/>
        <system value="Epic"/>
        <value value="991773"/>
        <type>
          <text value="Placer Identifier"/>
          <coding>
            <system value="http://hl7.org/fhir/ValueSet/identifier-type"/>
            <code value="PLAC"/>
            <display value="Placer Identifier"/>
          </coding>
        </type>
      </identifier>
      <code>
        <text value="AMYLASE - Final result"/>
        <coding>
          <system value="http://loinc.org"/>
          <code value="1798-8"/>
          <display value="Amylase [Enzymatic activity/volume] in Serum or Plasma"/>
        </coding>
      </code>
      <subject>
        <display value="Fiona Hailey Ingrid Robertson"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/T9tA6WZi2ctFj0t10bq679Z84UsQZI6rpZh1RjWOsuAYB"/>
      </subject>
      <performer>
        <display value="Helen Claire Wilson, MD"/>
        <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/TgEfSWOatltrvRWYMDUWo8AB"/>
      </performer>
      <result>
        <display value="Component: AMYLASE [U/L] IN SER/PLAS"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Observation/TAoipQXgi26J.P7sog-3jmKJK0P2TpeVkwkAFcJhI5.8B"/>
      </result>
    </DiagnosticReport>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <fullUrl value="https://hostname/instance/api/FHIR/DSTU2/DiagnosticReport/ToZrehgqN3LTiFeKDB7yFJgB"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/DiagnosticReport/ToZrehgqN3LTiFeKDB7yFJgB"/>
  </link>
  <resource>
    <DiagnosticReport>
      <status value="Final result"/>
      <effectiveDateTime value="2016-02-16T17:30:00"/>
      <issued value="2016-02-16T19:35:00"/>
      <id value="ToZrehgqN3LTiFeKDB7yFJgB"/>
      <identifier>
        <use value="official"/>
        <system value="Epic"/>
        <value value="991774"/>
        <type>
          <text value="Placer Identifier"/>
          <coding>
            <system value="http://hl7.org/fhir/ValueSet/identifier-type"/>
            <code value="PLAC"/>
            <display value="Placer Identifier"/>
          </coding>
        </type>
      </identifier>
      <code>
        <text value="AMYLASE - Final result"/>
        <coding>
          <system value="http://loinc.org"/>
          <code value="1798-8"/>
          <display value="Amylase [Enzymatic activity/volume] in Serum or Plasma"/>
        </coding>
      </code>
      <subject>
        <display value="Fiona Hailey Ingrid Robertson"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/T9tA6WZi2ctFj0t10bq679Z84UsQZI6rpZh1RjWOsuAYB"/>
      </subject>
      <performer>
        <display value="Helen Claire Wilson, MD"/>
        <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/TgEfSWOatltrvRWYMDUWo8AB"/>
      </performer>
      <result>
        <display value="Component: AMYLASE [U/L] IN SER/PLAS"/>
        <reference value="https://hostname/instance/api/FHIR/DSTU2/Observation/TkDOY1zCTUF-
  fwD9Nek3Vw92qgkvQaqp.wwvK842G7jIB"/>
```

```
        </result>
      </DiagnosticReport>
    </resource>
    <search>
      <mode value="match"/>
    </search>
  </entry>
  <entry>
    <resource>
      <OperationOutcome>
        <issue>
          <severity value="warning"/>
          <details value="This response includes information available to the authorized user at the time of the
    request. It may not contain the entire record available in the system."/>
          <code>
            <text value="informational"/>
            <coding>
              <system value="http://hl7.org/fhir/issue-type"/>
              <code value="informational"/>
              <display value="informational"/>
            </coding>
          </code>
        </issue>
      </OperationOutcome>
    </resource>
    <search>
      <mode value="outcome"/>
    </search>
  </entry>
</Bundle>
```

9.3.3 Observation.Read (Labs) (DSTU2)

File: site/apis/09-Clinical-Diagnostics/3-DSTU2/27-Observation-Read-Labs-DSTU2.html
Epic API ID | 27
FHIR version | DSTU2
Categories | Clinical Diagnostics, Labs
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | BMI percentile per age and sex for youth 2-20;Weights for age per length and sex;Occipital-frontal circumference for children < 3 years old;Vital Signs;Diastolic blood pressure;Systolic blood pressure;Body height;Body weight;Heart rate;Respiratory rate;Body temperature;Pulse oximetry;Inhaled oxygen concentration;Diagnostic Reports;Imaging;Laboratory;Pathology;Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2015
HTTP method | GET
URL template | api/FHIR/DSTU2/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=27

Description

The FHIR Observation resource defines information about a measurement or simple assertion made about a patient. The below documentation describes how the Observation resource is used to describe analyte level result information or sensitivities and isolates from a microbiology analysis.

The Observation resource corresponds to the following Common Clinical Data Set (CCDS) Elements: Laboratory Values/Results and Laboratory Tests.

The Read interaction enables the look up of an Observation resource by a constant server ID. Clients can store only the server ID, and with a single request, retrieve the most up-to-date information about the observation. Read interactions typically begin with a client having previously established a relationship, often through querying for observations through the search interaction.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | The Observation FHIR ID.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | A single Observation FHIR resource.
Observation.category | CodeableConcept | False | | | The Observation type. Always "laboratory".
Observation.code | CodeableConcept | False | | | LOINC code for Observation. If no LOINC code is mapped, only "text" will be returned.
Observation.comments | String | True | | | Comments about the result.

Observation.effectiveDateTime | String | False | | | The prioritized instant. This is the instant the results were last updated. This element can't be modified by Observation.Update.
Observation.id | String | False | | | The Observation FHIR ID.
Observation.interpretation | CodeableConcept | True | | | The level of abnormality. If the value isn't mapped we will just send the text value in the Codeable concept.
Observation.performer | ReferencedResource | True | Yes | | A reference to the Practitioner resource for the director of the associated resulting agency.
Observation.referenceRange | ReferenceRange | True | Yes | | Reference range (if available).
Observation.referenceRange.high | Quantity | True | | | High range, if relevant.
Observation.referenceRange.low | Quantity | True | | | Low range, if relevant.
Observation.referenceRange.text | String | True | | | Textual reference range.
Observation.related | Related | True | Yes | | Related isolate level Observation.
Observation.related.target | ReferencedResource | False | | | Observation FHIR ID for the isolate.
Observation.related.type | String | True | | | Always "has-member".
Observation.status | String | False | | | The status of the lab order.
Observation.subject | ReferencedResource | False | | | The patient FHIR ID associated with the order.
Observation.valueCodeableConcept | CodeableConcept | True | | | This element is populated only if a SNOMED code is associated with the component.
Observation.valueQuantity | Quantity | True | | | A result value. This element is populated only if the result is structured numeric and represents a quantity.
Observation.valueQuantity.code | String | True | | | Unit's coded value.
Observation.valueQuantity.comparator | String | True | | | How to understand the value. Find a full list of potential values here.
Observation.valueQuantity.system | String | True | | | Applies to quantity unit's code.
Observation.valueQuantity.unit | String | True | | | Quantity's unit.
Observation.valueQuantity.value | Decimal | True | | | Numeric value of the result.
Observation.valueString | String | True | | | A result value. This element is populated only if the result is a string.

Sample request

```
https://hostname/instance/api/FHIR/DSTU2/Observation/T7W3P4gfS128KkAvYVaU1em6mVb6XttPcLHC4B8CxU7QB
```

Sample response

```
<Observation xmlns="http://hl7.org/fhir">
  <effectiveDateTime value="2016-02-16T21:56:00Z"/>
  <status value="final"/>
  <id value="T7W3P4gfS128KkAvYVaU1em6mVb6XttPcLHC4B8CxU7QB"/>
  <code>
    <text value="ANION GAP 3 IN SER/PLAS"/>
    <coding>
      <system value="http://loinc.org"/>
      <code value="1863-0"/>
      <display value="Anion gap 4 in Serum or Plasma"/>
    </coding>
  </code>
  <valueQuantity>
    <value value="21"/>
    <unit value="mmol/L"/>
    <code value="mmol/L"/>
    <system value="http://unitsofmeasure.org"/>
  </valueQuantity>
  <subject>
    <display value="Fiona Hailey Ingrid Robertson"/>
    <reference value="https://hostname/instance/api/FHIR/DSTU2/Patient/Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB"/>
  </subject>
  <referenceRange>
    <text value="30 mmol/L"/>
    <high>
      <value value="30"/>
      <unit value="mmol/L"/>
      <code value="mmol/L"/>
      <system value="http://unitsofmeasure.org"/>
    </high>
  </referenceRange>
</Observation>
```

9.3.4 Observation.Read (Vitals) (DSTU2)

File: site/apis/09-Clinical-Diagnostics/3-DSTU2/449-Observation-Read-Vitals-DSTU2.html
Epic API ID | 449
FHIR version | DSTU2
Categories | Clinical Diagnostics
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | BMI percentile per age and sex for youth 2-20;Weights for age per length and sex;Occipital-frontal circumference for children < 3 years old;Vital Signs;Diastolic blood pressure;Systolic blood pressure;Body height;Body weight;Heart rate;Respiratory rate;Body temperature;Pulse oximetry;Inhaled oxygen concentration;Diagnostic Reports;Imaging;Laboratory;Pathology;flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base

ONC certification versions | Epic 2015
HTTP method | GET
URL template | api/FHIR/DSTU2/Observation/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=449

Description

The FHIR Observation resource defines information about a measurement or simple assertion made about a patient. This web service will retrieve vital sign data from a patient's chart. This includes any data found in the patient's flowsheets across all encounters. This resource can be queried by a combination of patient ID and vital sign LOINC code(s). The below documentation describes how the Observation resource is used to provide measurements gathered as a vital sign or flowsheet data point.

The Observation resource corresponds to the following Common Clinical Data Set (CCDS) Elements: Vital Signs.

The Read interaction enables the look up of an Observation resource by a constant server ID. Clients can store only the server ID, and with a single request, retrieve the most up-to-date information about the observation. Read interactions typically begin with a client having previously established a relationship, often through querying for observations through the search interaction.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Observation | Observation | False | | | A single Observation FHIR resource.
Observation.bodySite | CodeableConcept | False | | | Body site the observation relates to.
Observation.code | CodeableConcept | False | | | A codeable concept.
Observation.comments | String | False | | | Comments about the observation value.
Observation.dataAbsentReason | CodeableConcept | False | | | Why is the result missing - only present when no value is returned.
Observation.effectiveDateTime | String | False | | | Date the observation applies to.
Observation.encounter | ReferencedResource | False | | | Encounter this observation was made during.
Observation.identifier | Identifier | True | Yes | | Unique Id for this observation.
Observation.interpretation | CodeableConcept | False | | | High, low, normal.
Observation.issued | DateTime | True | | | Date/Time this result was documented or made available.
Observation.performer | ReferencedResource | False | Yes | | Who did the observation.
Observation.referenceRange | ReferenceRange | True | Yes
Observation.referenceRange.age | Range | False | | | Age range.
Observation.referenceRange.high | Quantity | False | | | High range.
Observation.referenceRange.low | Quantity | False | | | Low range.
Observation.referenceRange.meaning | CodeableConcept | False | | | Meaning of the reference range.
Observation.referenceRange.text | String | False | | | Text based reference range.
Observation.related | Related | True | Yes
Observation.related.target | ReferencedResource | False | | | Observation that is related to this one.
Observation.related.type | String | False | | | How is this related to the observation.
Observation.status | String | False | | | Status of the observation - hard coded to either Amended or Final depending on if the field has been corrected or not. Deleted observations will not be returned.
Observation.subject | ReferencedResource | False | | | Who this was about.
Observation.valueQuantity | Quantity | False | | | Only one value[x] element will be returned - when a value has units we will use valueQuantity.
Observation.valueString | String | False | | | Only one value[x] element will be returned - when we do not have a unit we will use valueString.

Sample request

```
http://hostname/instance/api/FHIR/DSTU2/Observation/TzAezIZwcmkd582mzwBjyZuxdcyvHO42ACKhsr-Y7ACIn4US0-AyeZdwm3eQonaC1B
```

Sample response

```
<Observation xmlns="http://hl7.org/fhir">
  <effectiveDateTime value="2016-01-13T21:29:00Z"/>
  <status value="final"/>
  <id value="TzAezIZwcmkd582mzwBjyZuxdcyvHO42ACKhsr-Y7ACIn4US0-AyeZdwm3eQonaC1B"/>
  <code>
    <text value="Body temperature"/>
    <coding>
      <system value="http://loinc.org"/>
      <code value="8310-5"/>
      <display value="Body temperature"/>
    </coding>
  </code>
  <valueQuantity>
    <value value="37.3"/>
    <unit value="Cel"/>
    <code value="Cel"/>
    <system value="http://unitsofmeasure.org"/>
  </valueQuantity>
  <subject>
    <display value="John Observation Doe"/>
```

```
    <reference value="https://hostname/instance/api/FHIR/DSTU2/Patient/ToueZw1.F0ZmssWl-YvEOkzXHFyPqpsv.7zUXEqQjQ2QB"/>
  </subject>
  <performer>
    <display value="John Doe, MD"/>
    <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/TiOmkdUgh7pJl0-mpSeeFtQB"/>
  </performer>
</Observation>
```

9.3.5 Observation.Search (Labs) (DSTU2)

File: site/apis/09-Clinical-Diagnostics/3-DSTU2/28-Observation-Search-Labs-DSTU2.html
Epic API ID | 28
FHIR version | DSTU2
Categories | Clinical Diagnostics, Labs
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | BMI percentile per age and sex for youth 2-20;Weights for age per length and sex;Occipital-frontal circumference for children < 3 years old;Vital Signs;Diastolic blood pressure;Systolic blood pressure;Body height;Body weight;Heart rate;Respiratory rate;Body temperature;Pulse oximetry;Inhaled oxygen concentration;Diagnostic Reports;Imaging;Laboratory;Pathology;Flowsheets
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2015
HTTP method | GET
URL template | api/FHIR/DSTU2/Observation?code={code}&date={date}&patient={patient}&subject={subject}&category={category}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=28

Description

The Observation (Labs) resource returns component level data for lab results. Observation is not normally used on its own to access lab results, because it is limited to reading microbiology sensitivities (method level, isolate level, and sensitivity level) and textual (narrative and impression) Observations. Instead, a system queries the DiagnosticReport resource to identify all lab procedures performed for a patient. The DiagnosticReport results will contain pointers to one or more Observation resources that contain the component results. If you are familiar with HL7v2, DiagnosticReport is analogous to an OBR segment, where Observation is like an OBX segment.

For example, consider a CBC (Complete Blood Count) test. The overall CBC test would be represented as a DiagnosticReport resource. The component results (white blood cell count, red blood cell count, hemoglobin, platelet count, etc.) would each be returned in independent Observation resources.

Most systems expect to receive a LOINC code identifying each component. An Observation resource will be returned for a component result even if a LOINC code is not found, however the Observation.code property will be empty.

Some labs, such as pathology or cytology, often file a PDF, or a link to a PDF or other document as the result. We do not currently support returning PDF content or other reference pointer content via FHIR, however the test will be discoverable via a DiagnosticReport search.

For microbiology labs, the DiagnosticReport resource represents the culture. It includes references to Observation for each isolate. The isolates may then either reference an Observation for each method, or, if there is no method provided by the lab system, then the isolate will reference Observations for each sensitivity result. If methods are provided by the lab, then the method Observations will reference the sensitivity results.

The two potential structures are outlined below.

Microbiology results without methods:

* Culture (DiagnosticReport)

* Isolate A (Observation)

* Sensitivity results for antibiotic W (Observation)

* Sensitivity results for antibiotic X (Observation)

* Isolate B (Observation)

* Sensitivity results for antibiotic W (Observation)

* Sensitivity results for antibiotic X (Observation)

Microbiology results with methods:

* Culture (DiagnosticReport)

* Isolate A (Observation)

* Method 1 (Observation) a. Sensitivity results for antibiotic W (Observation) b. Sensitivity results for antibiotic X (Observation)

* Method 2 (Observation) a. Sensitivity results for antibiotic W (Observation) b. Sensitivity results for antibiotic X (Observation)

* Isolate B (Observation)

* Method 1 (Observation) a. Sensitivity results for antibiotic W (Observation) b. Sensitivity results for antibiotic X (Observation)
* Method 2 (Observation) a. Sensitivity results for antibiotic W (Observation) b. Sensitivity results for antibiotic X (Observation)

Remarks

When searching, a patient resource ID and LOINC code must be supplied.

Additionally, individual microbiology sensitivities can only be viewed when referenced directly and cannot be returned on their own with an Observation search. Instead, consider using the DiagnosticReport resource for this purpose, which will reference one or more observation resources containing the microbiology details . The Observation resource will still return microbiology-related concepts when executing a Read with the appropriate ID, for instance, the ones returned by the DiagnosticReport.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
category | String | Either this or code must be provided, but not both. | | | Use "laboratory" to search for lab observations.
code | String | Either this or category must be provided, but not both. | | | The LOINC code of the Observation.
date | String | True | | | Date when the specimen was obtained.
patient | String | Either patient or subject must be provided. If both are provided, they must reference the same patient. | | | The reference to a patient resource.
subject | String | Either patient or subject must be provided. If both are provided, they must reference the same patient. | | | The reference to a patient resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | Resource | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Observation | Observation | False | | | A single Observation FHIR resource.
Bundle.entry.resource.Observation.code | CodeableConcept | False | | | LOINC code for Observation. If no LOINC code is mapped, only "text" will be returned.
Bundle.entry.resource.Observation.comments | String | True | | | Comments about the result.
Bundle.entry.resource.Observation.effectiveDateTime | String | False | | | The prioritized instant. This element can't be modified by Observation.Update.
Bundle.entry.resource.Observation.id | String | False | | | The Observation FHIR ID.
Bundle.entry.resource.Observation.interpretation | CodeableConcept | True | | | The level of abnormality. If the value isn't mapped we will just send the text value in the Codeable concept.
Bundle.entry.resource.Observation.performer | ReferencedResource | True | Yes | | A reference to the resulting agency's director's Practitioner resource.
Bundle.entry.resource.Observation.referenceRange | ReferenceRange | True | Yes | | Reference range (if available).
Bundle.entry.resource.Observation.referenceRange.high | Quantity | True | | | High range, if relevant.
Bundle.entry.resource.Observation.referenceRange.low | Quantity | True | | | Low range, if relevant.
Bundle.entry.resource.Observation.referenceRange.text | String | True | | | Textual reference range.
Bundle.entry.resource.Observation.related | Related | True | Yes | | Related isolate level Observation.
Bundle.entry.resource.Observation.status | String | False | | | The status of the lab order.
Bundle.entry.resource.Observation.subject | ReferencedResource | False | | | The Patient FHIR ID associated with the order.
Bundle.entry.resource.Observation.valueCodeableConcept | CodeableConcept | True | | | This element is populated only if a SNOMED code is associated with the component.
Bundle.entry.resource.Observation.valueQuantity | Quantity | True | | | A result value. This element is populated only if the result is structured numeric and represents a quantity.
Bundle.entry.resource.Observation.valueQuantity.code | String | True | | | Unit's coded value.
Bundle.entry.resource.Observation.valueQuantity.comparator | String | True | | | How to understand the value. Find a full list of potential values here.
Bundle.entry.resource.Observation.valueQuantity.system | String | True | | | Applies to quantity unit's code.
Bundle.entry.resource.Observation.valueQuantity.unit | String | True | | | Quantity's unit.
Bundle.entry.resource.Observation.valueQuantity.value | Decimal | True | | | Numeric value of the result.
Bundle.entry.resource.Observation.valueString | String | True | | | A result value. This element is populated only if the result is a string.
Bundle.total | Int32 | True | | | Total number of resources that matched the search criteria.

Sample request

```
https://hostname/instance/api/FHIR/DSTU2/Observation?patient=eTtVpZg2-.J.yFIP1qJbK4A3&category=laboratory
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
  <type value="searchset"/>
  <total value="23"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Observation?patient=Tb19ikuzxOh6YglUN28cuQ-
    d.3AjOduTYFVxlRM82bqEB&amp;code=1798-8,2951-2,2823-3,2075-0,2028-9,3094-0,2160-0,2345-7,17861-6,1863-0,48642-3"/>
  </link>
  <entry>
    <link>
      <relation value="self"/>
```

```
    <url value="https://hostname/instance/api/FHIR/DSTU2/Observation/T7W3P4gfS128KkAvYVaU1em6mVb6XttPcLHC4B8CxU7QB"/>
  </link>
  <resource>
    <Observation>
      <effectiveDateTime value="2016-02-16T14:15:00Z"/>
      <status value="final"/>
      <id value="T7W3P4gfS128KkAvYVaU1em6mVb6XttPcLHC4B8CxU7QB"/>
      <code>
        <text value="ANION GAP 3 IN SER/PLAS"/>
        <coding>
          <system value="http://loinc.org"/>
          <code value="1863-0"/>
          <display value="Anion gap 4 in Serum or Plasma"/>
        </coding>
      </code>
      <valueQuantity>
        <value value="21"/>
        <unit value="mmol/L"/>
        <code value="mmol/L"/>
        <system value="http://unitsofmeasure.org"/>
      </valueQuantity>
      <subject>
        <display value="Fiona Hailey Ingrid Robertson"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB"/>
      </subject>
      <referenceRange>
        <text value="30 mmol/L"/>
        <high>
          <value value="30"/>
          <unit value="mmol/L"/>
          <code value="mmol/L"/>
          <system value="http://unitsofmeasure.org"/>
        </high>
      </referenceRange>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Observation/TgnZLCpvPpFSE4m47wp.eeV81sXgfu.Epcg4eYtTK90YB"/>
  </link>
  <resource>
    <Observation>
      <effectiveDateTime value="2016-01-06T15:15:00Z"/>
      <status value="final"/>
      <id value="TgnZLCpvPpFSE4m47wp.eeV81sXgfu.Epcg4eYtTK90YB"/>
      <code>
        <text value="ANION GAP 3 IN SER/PLAS"/>
        <coding>
          <system value="http://loinc.org"/>
          <code value="1863-0"/>
          <display value="Anion gap 4 in Serum or Plasma"/>
        </coding>
      </code>
      <valueQuantity>
        <value value="20"/>
        <unit value="mmol/L"/>
        <code value="mmol/L"/>
        <system value="http://unitsofmeasure.org"/>
      </valueQuantity>
      <subject>
        <display value="Fiona Hailey Ingrid Robertson"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB"/>
      </subject>
      <referenceRange>
        <text value="30 mmol/L"/>
        <high>
          <value value="30"/>
          <unit value="mmol/L"/>
          <code value="mmol/L"/>
          <system value="http://unitsofmeasure.org"/>
        </high>
      </referenceRange>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Observation/TuHolDRplT8vByfjh7ltjfeQcN06N50yvxDd1ijKuQPYB"/>
```

```
  </link>
  <resource>
    <Observation>
      <effectiveDateTime value="2016-02-16T17:30:00Z"/>
      <status value="final"/>
      <id value="TuHolDRplT8vByfjh7ltjfeQcN06N50yvxDd1ijKuQPYB"/>
      <code>
        <text value="AMYLASE [U/L] IN SER/PLAS"/>
        <coding>
          <system value="http://loinc.org"/>
          <code value="1798-8"/>
          <display value="Amylase [Enzymatic activity/volume] in Serum or Plasma"/>
        </coding>
      </code>
      <valueQuantity>
        <value value="101"/>
        <unit value="U/L"/>
        <code value="U/L"/>
        <system value="http://unitsofmeasure.org"/>
      </valueQuantity>
      <subject>
        <display value="Fiona Hailey Ingrid Robertson"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB"/>
      </subject>
      <referenceRange>
        <text value="25 - 110 U/L"/>
        <low>
          <value value="25"/>
          <unit value="U/L"/>
          <code value="U/L"/>
          <system value="http://unitsofmeasure.org"/>
        </low>
        <high>
          <value value="110"/>
          <unit value="U/L"/>
          <code value="U/L"/>
          <system value="http://unitsofmeasure.org"/>
        </high>
      </referenceRange>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Observation/Tr4HWBRC5emIecTysb-3jzIa2yXlYTaDqTJMSOHBoFNAB"/>
  </link>
  <resource>
    <Observation>
      <effectiveDateTime value="2016-02-05T17:45:00Z"/>
      <status value="final"/>
      <id value="Tr4HWBRC5emIecTysb-3jzIa2yXlYTaDqTJMSOHBoFNAB"/>
      <code>
        <text value="AMYLASE [U/L] IN SER/PLAS"/>
        <coding>
          <system value="http://loinc.org"/>
          <code value="1798-8"/>
          <display value="Amylase [Enzymatic activity/volume] in Serum or Plasma"/>
        </coding>
      </code>
      <valueQuantity>
        <value value="104"/>
        <unit value="U/L"/>
        <code value="U/L"/>
        <system value="http://unitsofmeasure.org"/>
      </valueQuantity>
      <subject>
        <display value="Fiona Hailey Ingrid Robertson"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB"/>
      </subject>
      <referenceRange>
        <text value="25 - 110 U/L"/>
        <low>
          <value value="25"/>
          <unit value="U/L"/>
          <code value="U/L"/>
          <system value="http://unitsofmeasure.org"/>
        </low>
        <high>
          <value value="110"/>
          <unit value="U/L"/>
          <code value="U/L"/>
          <system value="http://unitsofmeasure.org"/>
        </high>
```

```
      </referenceRange>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Observation/TCaYEKn6H1cEzh1Ac28C4eZzk0jiRkW4rCpI-XuKJptgB"/>
  </link>
  <resource>
    <Observation>
      <effectiveDateTime value="2016-02-03T14:15:00Z"/>
      <status value="final"/>
      <id value="TCaYEKn6H1cEzh1Ac28C4eZzk0jiRkW4rCpI-XuKJptgB"/>
      <code>
        <text value="AMYLASE [U/L] IN SER/PLAS"/>
        <coding>
          <system value="http://loinc.org"/>
          <code value="1798-8"/>
          <display value="Amylase [Enzymatic activity/volume] in Serum or Plasma"/>
        </coding>
      </code>
      <valueQuantity>
        <value value="115"/>
        <unit value="U/L"/>
        <code value="U/L"/>
        <system value="http://unitsofmeasure.org"/>
      </valueQuantity>
      <subject>
        <display value="Fiona Hailey Ingrid Robertson"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB"/>
      </subject>
      <referenceRange>
        <text value="25 - 110 U/L"/>
        <low>
          <value value="25"/>
          <unit value="U/L"/>
          <code value="U/L"/>
          <system value="http://unitsofmeasure.org"/>
        </low>
        <high>
          <value value="110"/>
          <unit value="U/L"/>
          <code value="U/L"/>
          <system value="http://unitsofmeasure.org"/>
        </high>
      </referenceRange>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Observation/T7W3P4gfS128KkAvYVaU1enPvJEfnD82OKMVVu8wDfpIB"/>
  </link>
  <resource>
    <Observation>
      <effectiveDateTime value="2016-02-16T14:15:00Z"/>
      <status value="final"/>
      <id value="T7W3P4gfS128KkAvYVaU1enPvJEfnD82OKMVVu8wDfpIB"/>
      <code>
        <text value="UREA NITROGEN [MG/DL] IN SER/PLAS"/>
        <coding>
          <system value="http://loinc.org"/>
          <code value="3094-0"/>
          <display value="Urea nitrogen [Mass/volume] in Serum or Plasma"/>
        </coding>
      </code>
      <valueQuantity>
        <value value="16"/>
        <unit value="mg/dL"/>
        <code value="mg/dL"/>
        <system value="http://unitsofmeasure.org"/>
      </valueQuantity>
      <subject>
        <display value="Fiona Hailey Ingrid Robertson"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB"/>
      </subject>
      <referenceRange>
        <text value="4 - 21 mg/dL"/>
        <low>
```

```
          <value value="4"/>
          <unit value="mg/dL"/>
          <code value="mg/dL"/>
          <system value="http://unitsofmeasure.org"/>
        </low>
        <high>
          <value value="21"/>
          <unit value="mg/dL"/>
          <code value="mg/dL"/>
          <system value="http://unitsofmeasure.org"/>
        </high>
      </referenceRange>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Observation/TgnZLCpvPpFSE4m47wp.eebE9gzTqnTNHc2QZ0GsU9tYB"/>
  </link>
  <resource>
    <Observation>
      <effectiveDateTime value="2016-01-06T15:15:00Z"/>
      <status value="final"/>
      <id value="TgnZLCpvPpFSE4m47wp.eebE9gzTqnTNHc2QZ0GsU9tYB"/>
      <code>
        <text value="UREA NITROGEN [MG/DL] IN SER/PLAS"/>
        <coding>
          <system value="http://loinc.org"/>
          <code value="3094-0"/>
          <display value="Urea nitrogen [Mass/volume] in Serum or Plasma"/>
        </coding>
      </code>
      <valueQuantity>
        <value value="14"/>
        <unit value="mg/dL"/>
        <code value="mg/dL"/>
        <system value="http://unitsofmeasure.org"/>
      </valueQuantity>
      <subject>
        <display value="Fiona Hailey Ingrid Robertson"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB"/>
      </subject>
      <referenceRange>
        <text value="4 - 21 mg/dL"/>
        <low>
          <value value="4"/>
          <unit value="mg/dL"/>
          <code value="mg/dL"/>
          <system value="http://unitsofmeasure.org"/>
        </low>
        <high>
          <value value="21"/>
          <unit value="mg/dL"/>
          <code value="mg/dL"/>
          <system value="http://unitsofmeasure.org"/>
        </high>
      </referenceRange>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Observation/T7W3P4gfS128KkAvYVaU1epauxknhuTe22vDovA1Cv8oB"/>
  </link>
  <resource>
    <Observation>
      <effectiveDateTime value="2016-02-16T14:15:00Z"/>
      <status value="final"/>
      <id value="T7W3P4gfS128KkAvYVaU1epauxknhuTe22vDovA1Cv8oB"/>
      <code>
        <text value="CARBON DIOXIDE, TOTAL [MMOL/L] IN SER/PLAS"/>
        <coding>
          <system value="http://loinc.org"/>
          <code value="2028-9"/>
          <display value="Carbon dioxide, total [Moles/volume] in Serum or Plasma"/>
        </coding>
      </code>
      <valueQuantity>
        <value value="19"/>
        <unit value="mmol/L"/>
```

```
        <code value="mmol/L"/>
        <system value="http://unitsofmeasure.org"/>
      </valueQuantity>
      <subject>
        <display value="Fiona Hailey Ingrid Robertson"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB"/>
      </subject>
      <referenceRange>
        <text value="13 - 22 mmol/L"/>
        <low>
          <value value="13"/>
          <unit value="mmol/L"/>
          <code value="mmol/L"/>
          <system value="http://unitsofmeasure.org"/>
        </low>
        <high>
          <value value="22"/>
          <unit value="mmol/L"/>
          <code value="mmol/L"/>
          <system value="http://unitsofmeasure.org"/>
        </high>
      </referenceRange>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Observation/TgnZLCpvPpFSE4m47wp.eeWh6ClhEomytvJfaw.MVmyEB"/>
  </link>
  <resource>
    <Observation>
      <effectiveDateTime value="2016-01-06T15:15:00Z"/>
      <status value="final"/>
      <id value="TgnZLCpvPpFSE4m47wp.eeWh6ClhEomytvJfaw.MVmyEB"/>
      <code>
        <text value="CARBON DIOXIDE, TOTAL [MMOL/L] IN SER/PLAS"/>
        <coding>
          <system value="http://loinc.org"/>
          <code value="2028-9"/>
          <display value="Carbon dioxide, total [Moles/volume] in Serum or Plasma"/>
        </coding>
      </code>
      <valueQuantity>
        <value value="17"/>
        <unit value="mmol/L"/>
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      </subject>
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        <high>
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          <code value="2075-0"/>
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        <high>
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          <code value="17861-6"/>
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          <code value="48642-3"/>
          <display value="Glomerular filtration rate/1.73 sq M predicted among non-blacks by Creatinine-based formula
  (MDRD)"/>
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  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Observation/TgnZLCpvPpFSE4m47wp.eeQ4PaLw9Y2k4HGsIFph4UXoB"/>
  </link>
  <resource>
    <Observation>
      <effectiveDateTime value="2016-01-06T15:15:00Z"/>
      <status value="final"/>
      <id value="TgnZLCpvPpFSE4m47wp.eeQ4PaLw9Y2k4HGsIFph4UXoB"/>
      <code>
        <text value="GLOMERULAR FILTRATION RATE ML/MIN/1.73 SQ M.PREDICTED"/>
        <coding>
          <system value="http://loinc.org"/>
          <code value="48642-3"/>
          <display value="Glomerular filtration rate/1.73 sq M predicted among non-blacks by Creatinine-based formula
  (MDRD)"/>
        </coding>
      </code>
      <valueQuantity>
        <value value="99.0"/>
        <unit value="mL/min/1.73m*2"/>
      </valueQuantity>
      <subject>
        <display value="Fiona Hailey Ingrid Robertson"/>
        <reference
  value="https://hostname/instance/api/FHIR/DSTU2/Patient/Tb19ikuzxOh6YglUN28cuQ-d.3AjOduTYFVxlRM82bqEB"/>
      </subject>
      <referenceRange>
        <text value="60.0 mL/min/1.73m*2"/>
        <low>
          <value value="60"/>
          <unit value="mL/min/1.73m*2"/>
        </low>
      </referenceRange>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
  </search>
 </entry>
 <entry>
  <resource>
    <OperationOutcome>
      <issue>
        <severity value="warning"/>
        <details value="This response includes information available to the authorized user at the time of the
  request. It may not contain the entire record available in the system."/>
        <code>
          <text value="informational"/>
          <coding>
            <system value="http://hl7.org/fhir/issue-type"/>
            <code value="informational"/>
            <display value="informational"/>
          </coding>
        </code>
      </issue>
    </OperationOutcome>
  </resource>
  <search>
    <mode value="outcome"/>
  </search>
 </entry>
</Bundle>
```


9.3.6 Observation.Search (Vitals) (DSTU2)

File: site/apis/09-Clinical-Diagnostics/3-DSTU2/450-Observation-Search-Vitals-DSTU2.html
Epic API ID | 450
FHIR version | DSTU2
Categories | Clinical Diagnostics
API type | USCDI
A la carte | Industry-Standard Level 1 API
USCDI | Yes
Keywords | BMI percentile per age and sex for youth 2-20;Weights for age per length and sex;Occipital-frontal circumference for children < 3 years old;Vital Signs;Diastolic blood pressure;Systolic blood pressure;Body height;Body weight;Heart rate;Respiratory rate;Body temperature;Pulse oximetry;Inhaled oxygen concentration;Diagnostic Reports;Imaging;Laboratory;Pathology;flowsheets
Available in inpatient and other contexts | Yes

ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2015
HTTP method | GET
URL template | /api/FHIR/DSTU2/Observation?code={code}&patient={patient}&date={date}&subject={subject}&category={category}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=450

Description

The FHIR Observation resource defines information about a measurement or simple assertion made about a patient. This web service will retrieve vital sign data from a patient's chart. This includes any data found in the patient's flowsheets across all encounters. This resource can be queried by a combination of patient ID and vital sign LOINC code(s). The below documentation describes how the Observation resource is used to provide measurements gathered as a vital sign or flowsheet data point.

The Observation resource corresponds to the following Common Clinical Data Set (CCDS) Elements: Vital Signs.

The Search interaction enables the client to query for all observations of a specific name or list of names that have been filed for a given patient. The client, having established the patient in question, now wishes to retrieve observations describing documented flowsheet data for a given type of vital sign for the patient. Clients can further refine the search to specify a date for the Observation resources in question.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
_id | String | True | | | List of Observation Read Ids.
category | String | False | | | Search for Observation resources for a valid category (vital-signs, social-history, laboratory). Both patient and category (or code) are required for a valid patient-based search.
code | String | True | | | LOINC code for the type of observation - translated to a flowsheet id by EXM mapping.
date | String | True | | | Date or date range for which the observation applies.
subject | String | True | | | FHIR Patient ID

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | Resource | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Observation | Observation | False | | | A single Observation FHIR resource.
Bundle.entry.resource.Observation.bodySite | CodeableConcept | False | | | Body site the observation relates to.
Bundle.entry.resource.Observation.comments | String | False | | | Comments about the observation value.
Bundle.entry.resource.Observation.dataAbsentReason | CodeableConcept | False | | | Why is the result missing - only present when no value is returned.
Bundle.entry.resource.Observation.effectiveDateTime | String | False | | | Date the observation applies to.
Bundle.entry.resource.Observation.encounter | ReferencedResource | False | | | Encounter this observation was made during.
Bundle.entry.resource.Observation.identifier | Identifier | True | Yes | | Unique Id for this observation.
Bundle.entry.resource.Observation.interpretation | CodeableConcept | False | | | High, low, normal.
Bundle.entry.resource.Observation.issued | DateTime | True | | | Date/Time this result was documented or made available.
Bundle.entry.resource.Observation.performer | ReferencedResource | False | Yes | | Who did the observation.
Bundle.entry.resource.Observation.referenceRange | ReferenceRange | True | Yes
Bundle.entry.resource.Observation.referenceRange.age | Range | False | | | Age Range
Bundle.entry.resource.Observation.referenceRange.high | Quantity | False | | | High range.
Bundle.entry.resource.Observation.referenceRange.low | Quantity | False | | | Low range.
Bundle.entry.resource.Observation.referenceRange.meaning | CodeableConcept | False | | | Meaning of the reference range.
Bundle.entry.resource.Observation.referenceRange.text | String | False | | | Text based reference range
Bundle.entry.resource.Observation.related | Related | True | Yes
Bundle.entry.resource.Observation.related.target | ReferencedResource | False | | | Observation that is related to this one.
Bundle.entry.resource.Observation.related.type | String | False | | | How is this related to the observation.
Bundle.entry.resource.Observation.status | String | False | | | Status of the observation - hard coded to either Amended or Final depending on if the field has been corrected or not. Deleted observations will not be returned.
Bundle.entry.resource.Observation.subject | ReferencedResource | False | | | Who this was about.
Bundle.entry.resource.Observation.valueQuantity | Quantity | False | | | Only one value[x] element will be returned - when a value has units we will use valueQuantity.
Bundle.entry.resource.Observation.valueString | String | False | | | Only one value[x] element will be returned - when we do not have a unit we will use valueString.

Sample request

```
http://hostname/instance/api/FHIR/DSTU2/Observation?patient=ToueZw1.F0ZmssWl-YvEOkzXHFyPqpsv.7zUXEqQjQ2QB&code=8310-5
```

Sample response

```
<Bundle xmlns="http://hl7.org/fhir">
```

```
<type value="searchset"/>
<total value="2"/>
<link>
  <relation value="self"/>
  <url value="https://hostname/instance/api/FHIR/DSTU2/Observation?patient=ToueZw1.F0ZmssWl-
  YvEOkzXHFyPqpsv.7zUXEqQjQ2QB&amp;code=8310-5"/>
</link>
<entry>
  <fullUrl value="https://hostname/instance/api/FHIR/DSTU2/Observation/TzAezIZwcmkd582mzwBjyZuxdcyvHO42ACKhsr-
  Y7ACL4V3jV4HAu2DIXALoDIpXVB"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Observation/TzAezIZwcmkd582mzwBjyZuxdcyvHO42ACKhsr-
  Y7ACL4V3jV4HAu2DIXALoDIpXVB"/>
  </link>
  <resource>
    <Observation>
      <effectiveDateTime value="2016-01-13T21:34:00Z"/>
      <status value="final"/>
      <id value="TzAezIZwcmkd582mzwBjyZuxdcyvHO42ACKhsr-Y7ACL4V3jV4HAu2DIXALoDIpXVB"/>
      <code>
        <text value="Body temperature"/>
        <coding>
          <system value="http://loinc.org"/>
          <code value="8310-5"/>
          <display value="Body temperature"/>
        </coding>
      </code>
      <valueQuantity>
        <value value="37.4"/>
        <unit value="Cel"/>
        <system value="http://unitsofmeasure.org"/>
        <code value="Cel"/>
      </valueQuantity>
      <subject>
        <display value="John Observation Doe"/>
        <reference value="https://hostname/instance/api/FHIR/DSTU2/Patient/ToueZw1.F0ZmssWl-
  YvEOkzXHFyPqpsv.7zUXEqQjQ2QB"/>
      </subject>
      <performer>
        <display value="John Doe, MD"/>
        <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/TiOmkdUgh7pJl0-mpSeeFtQB"/>
      </performer>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
  </search>
</entry>
<entry>
  <fullUrl value="https://hostname/instance/api/FHIR/DSTU2/Observation/TzAezIZwcmkd582mzwBjyZuxdcyvHO42ACKhsr-
  Y7ACIn4US0-AyeZdwm3eQonaC1B"/>
  <link>
    <relation value="self"/>
    <url value="https://hostname/instance/api/FHIR/DSTU2/Observation/TzAezIZwcmkd582mzwBjyZuxdcyvHO42ACKhsr-
  Y7ACIn4US0-AyeZdwm3eQonaC1B"/>
  </link>
  <resource>
    <Observation>
      <effectiveDateTime value="2016-01-13T21:29:00Z"/>
      <status value="final"/>
      <id value="TzAezIZwcmkd582mzwBjyZuxdcyvHO42ACKhsr-Y7ACIn4US0-AyeZdwm3eQonaC1B"/>
      <code>
        <text value="Body temperature"/>
        <coding>
          <system value="http://loinc.org"/>
          <code value="8310-5"/>
          <display value="Body temperature"/>
        </coding>
      </code>
      <valueQuantity>
        <value value="37.3"/>
        <unit value="Cel"/>
        <code value="Cel"/>
        <system value="http://unitsofmeasure.org"/>
      </valueQuantity>
      <subject>
        <display value="John Observation Doe"/>
        <reference value="https://hostname/instance/api/FHIR/DSTU2/Patient/ToueZw1.F0ZmssWl-
  YvEOkzXHFyPqpsv.7zUXEqQjQ2QB"/>
      </subject>
      <performer>
        <display value="John Doe, MD"/>
        <reference value="https://hostname/instance/api/FHIR/DSTU2/Practitioner/TiOmkdUgh7pJl0-mpSeeFtQB"/>
      </performer>
    </Observation>
  </resource>
  <search>
    <mode value="match"/>
```

```
    </search>
  </entry>
</Bundle>
```